Complex Analytic and Differential Geometry

Jean-Pierre Demailly

Universite de Grenoble I
Institut Fourier, UMR 5582 du CNRS
38402 Saint-Martin d'Heres, France

Version of Thursday June 21, 2012

Contents

I Complex Differential Calculus and Pseudoconvexity **4**
§ 1 Differential Calculus on Manifolds . . . 4
§ 2 Currents on Differentiable Manifolds . . . 10
§ 3 Holomorphic Functions and Complex Manifolds . . . 17
§ 4 Subharmonic Functions . . . 27
§ 5 Plurisubharmonic Functions . . . 36
§ 6 Domains of Holomorphy and Stein Manifolds . . . 42
§ 7 Pseudoconvex Open Sets in $\mathbb{C}^n$ . . . 50
§ 8 Exercises . . . 57

II Coherent Sheaves and Analytic Spaces **62**
§ 1 Presheaves and Sheaves . . . 62
§ 2 The Local Ring of Germs of Analytic Functions . . . 76
§ 3 Coherent Sheaves . . . 80
§ 4 Complex Analytic Sets. Local Properties . . . 87
§ 5 Complex Spaces . . . 98
§ 6 Analytic Cycles and Meromorphic Functions . . . 101
§ 7 Normal Spaces and Normalization . . . 106
§ 8 Holomorphic Mappings and Extension Theorems . . . 112
§ 9 Complex Analytic Schemes . . . 117
§ 10 Bimeromorphic maps, Modifications and Blow-ups . . . 121
§ 11 Exercises . . . 122

III Positive Currents and Lelong Numbers **125**

IV Sheaf Cohomology and Spectral Sequences **126**

V Vector Bundles **127**

VI Hodge Theory **128**
§ 1 Differential Operators on Vector Bundles . . . 128
§ 2 Formalism of PseudoDifferential Operators . . . 130

VII Positive Vector Bundles and Vanishing Theorems **132**

VIII L^2 Estimates on Pseudoconvex Manifolds **133**

IX Finiteness Theorems for q-Convex Spaces and Stein Spaces **134**

References **135**

Chapter I

Complex Differential Calculus and Pseudoconvexity

This introductive chapter is mainly a review of the basic tools and concepts which will be employed in the rest of the book: differential forms, currents, holomorphic and plurisubharmonic functions, holomorphic convexity and pseudoconvexity. Our study of holomorphic convexity is principally concentrated here on the case of domains in $\mathbb{C}^n$. The more powerful machinery needed for the study of general complex varieties (sheaves, positive currents, hermitian differential geometry) will be introduced in Chapters II to V. Although our exposition pretends to be almost self-contained, the reader is assumed to have at least a vague familiarity with a few basic topics, such as differential calculus, measure theory and distributions, holomorphic functions of one complex variable, ... Most of the necessary background can be found in the books of [Rudin 1966] and [Warner 1971]; the basics of distribution theory can be found in Chapter I of [Hörmander 1963]. On the other hand, the reader who has already some knowledge of complex analysis in several variables should probably bypass this chapter.

§ 1. Differential Calculus on Manifolds

§ 1.A. Differentiable Manifolds

The notion of manifold is a natural extension of the notion of submanifold defined by a set of equations in $\mathbb{R}^n$. However, as already observed by Riemann during the 19th century, it is important to define the notion of a manifold in a flexible way, without necessarily requiring that the underlying topological space is embedded in an affine space. The precise formal definition was first introduced by H. Weyl in [Weyl 1913].

Let $m \in \mathbb{N}$ and $k \in \mathbb{N} \cup \{\infty, \omega\}$. We denote by $\mathscr{C}^k$ the class of functions which are k-times differentiable with continuous derivatives if $k \neq \omega$, and by $\mathscr{C}^\omega$ the class of real analytic functions. A *differentiable manifold* M of real dimension m and of class $\mathscr{C}^k$ is a topological space (which we shall always assume Hausdorff and separable, i.e. possessing a countable basis of the topology), equipped with an atlas of class $\mathscr{C}^k$ with values in $\mathbb{R}^m$. An *atlas* of class $\mathscr{C}^k$ is a collection of homeomorphisms $\tau_\alpha : U_\alpha \longrightarrow V_\alpha$, $\alpha \in I$, called *differentiable charts*, such that $(U_\alpha)_{\alpha \in I}$ is an open covering of M and V_α an open subset of $\mathbb{R}^m$, and such that for all $\alpha, \beta \in I$ the *transition map*

$$\tau_{\alpha\beta} = \tau_\alpha \circ \tau_\beta^{-1} : \tau_\beta(U_\alpha \cap U_\beta) \longrightarrow \tau_\alpha(U_\alpha \cap U_\beta) \tag{1.1}$$

is a $\mathscr{C}^k$ diffeomorphism from an open subset of V_β onto an open subset of V_α (see Fig. 1). Then the components $\tau_\alpha(x) = (x_1^\alpha, \ldots, x_m^\alpha)$ are called the *local coordinates* on U_α defined by the chart τ_α ; they are related by the transition relation $x^\alpha = \tau_{\alpha\beta}(x^\beta)$.

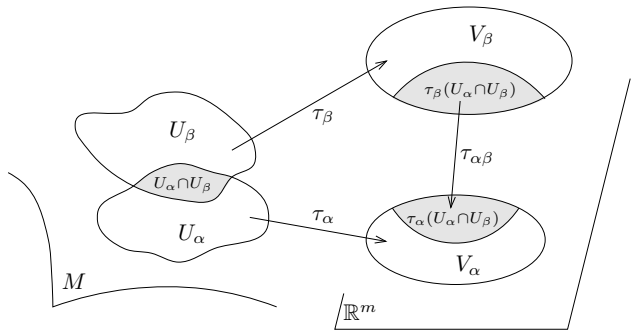

Fig. I-1 Charts and transition maps

If $\Omega \subset M$ is open and $s \in \mathbb{N} \cup \{\infty, \omega\}$, $0 \leqslant s \leqslant k$, we denote by $C^s(\Omega, \mathbb{R})$ the set of functions f of class C^s on Ω, i.e. such that $f \circ \tau_\alpha^{-1}$ is of class C^s on $\tau_\alpha(U_\alpha \cap \Omega)$ for each α ; if Ω is not open, $C^s(\Omega, \mathbb{R})$ is the set of functions which have a C^s extension to some neighborhood of Ω.

A *tangent vector* ξ at a point $a \in M$ is by definition a differential operator acting on functions, of the type

$$C^1(\Omega, \mathbb{R}) \ni f \longmapsto \xi \cdot f = \sum_{1 \leqslant j \leqslant m} \xi_j \, \frac{\partial f}{\partial x_j}(a)$$

in any local coordinate system $(x_1, \ldots, x_m)$ on an open set $\Omega \ni a$. We then simply write $\xi = \sum \xi_j \, \partial/\partial x_j$. For every $a \in \Omega$, the n-tuple $(\partial/\partial x_j)_{1 \leqslant j \leqslant m}$ is therefore a basis of the *tangent space* to M at a, which we denote by $T_{M,a}$. The *differential* of a function f at a is the linear form on $T_{M,a}$ defined by

$$df_a(\xi) = \xi \cdot f = \sum \xi_j \, \partial f/\partial x_j(a), \qquad \forall \xi \in T_{M,a}.$$

In particular $dx_j(\xi) = \xi_j$ and we may consequently write $df = \sum (\partial f/\partial x_j) dx_j$. From this, we see that $(dx_1, \ldots, dx_m)$ is the dual basis of $(\partial/\partial x_1, \ldots, \partial/\partial x_m)$ in the cotangent space $T^\star_{M,a}$. The disjoint unions $T_M = \bigcup_{x \in M} T_{M,x}$ and $T^\star_M = \bigcup_{x \in M} T^\star_{M,x}$ are called the *tangent* and *cotangent bundles* of M.

If ξ is a vector field of class C^s over Ω, that is, a map $x \mapsto \xi(x) \in T_{M,x}$ such that $\xi(x) = \sum \xi_j(x) \, \partial/\partial x_j$ has C^s coefficients, and if η is another vector field of class C^s with $s \geqslant 1$, the *Lie bracket* $[\xi, \eta]$ is the vector field such that

$$[\xi, \eta] \cdot f = \xi \cdot (\eta \cdot f) - \eta \cdot (\xi \cdot f). \tag{1.2}$$

In coordinates, it is easy to check that

$$[\xi, \eta] = \sum_{1 \leqslant j,k \leqslant m} \Big(\xi_j \frac{\partial \eta_k}{\partial x_j} - \eta_j \frac{\partial \xi_k}{\partial x_j} \Big) \frac{\partial}{\partial x_k}. \tag{1.3}$$

§ 1.B. Differential Forms

A differential form u of degree p, or briefly a p-form over M, is a map u on M with values $u(x) \in \Lambda^p T^\star_{M,x}$. In a coordinate open set $\Omega \subset M$, a differential p-form can be written

$$u(x) = \sum_{|I|=p} u_I(x)\, dx_I,$$

where $I = (i_1, \ldots, i_p)$ is a multi-index with integer components, $i_1 < \ldots < i_p$ and $dx_I := dx_{i_1} \wedge \ldots \wedge dx_{i_p}$. The notation $|I|$ stands for the number of components of I, and is read *length* of I. For all integers $p = 0, 1, \ldots, m$ and $s \in \mathbb{N} \cup \{\infty\}$, $s \leqslant k$, we denote by $C^s(M, \Lambda^p T^\star_M)$ the space of differential p-forms of class C^s, i.e. with C^s coefficients u_I. Several natural operations on differential forms can be defined.

§ 1.B.1. *Wedge Product.* If $v(x) = \sum v_J(x)\, dx_J$ is a q-form, the *wedge product* of u and v is the form of degree $(p+q)$ defined by

$$u \wedge v(x) = \sum_{|I|=p, |J|=q} u_I(x) v_J(x)\, dx_I \wedge dx_J. \tag{1.4}$$

§ 1.B.2. *Contraction by a tangent vector.* A p-form u can be viewed as an antisymmetric p-linear form on T_M. If $\xi = \sum \xi_j\, \partial/\partial x_j$ is a tangent vector, we define the *contraction* $\xi \lrcorner\, u$ to be the differential form of degree $p-1$ such that

$$(\xi \lrcorner\, u)(\eta_1, \ldots, \eta_{p-1}) = u(\xi, \eta_1, \ldots, \eta_{p-1}) \tag{1.5}$$

for all tangent vectors η_j. Then $(\xi, u) \longmapsto \xi \lrcorner\, u$ is bilinear and we find easily

$$\frac{\partial}{\partial x_j} \lrcorner\, dx_I = \begin{cases} 0 & \text{if } \ j \notin I, \\ (-1)^{l-1} dx_{I \smallsetminus \{j\}} & \text{if } \ j = i_l \in I. \end{cases}$$

A simple computation based on the above formula shows that contraction by a tangent vector is a *derivation*, i.e.

$$\xi \lrcorner\, (u \wedge v) = (\xi \lrcorner\, u) \wedge v + (-1)^{\deg u} u \wedge (\xi \lrcorner\, v). \tag{1.6}$$

§ 1.B.3. *Exterior derivative.* This is the differential operator

$$d : \mathscr{C}^s(M, \Lambda^p T^\star_M) \longrightarrow \mathscr{C}^{s-1}(M, \Lambda^{p+1} T^\star_M)$$

defined in local coordinates by the formula

$$du = \sum_{|I|=p,\, 1 \leqslant k \leqslant m} \frac{\partial u_I}{\partial x_k}\, dx_k \wedge dx_I. \tag{1.7}$$

Alternatively, one can define du by its action on arbitrary vector fields $\xi_0, \ldots, \xi_p$ on M. The formula is as follows

$$\begin{aligned} du(\xi_0, \ldots, \xi_p) = &\sum_{0 \leqslant j \leqslant p} (-1)^j \xi_j \cdot u(\xi_0, \ldots, \widehat{\xi_j}, \ldots, \xi_p) \\ &+ \sum_{0 \leqslant j < k \leqslant p} (-1)^{j+k} u([\xi_j, \xi_k], \xi_0, \ldots, \widehat{\xi_j}, \ldots, \widehat{\xi_k}, \ldots, \xi_p). \end{aligned} \tag{1.7$'$}$$

The reader will easily check that (1.7) actually implies (1.7′). The advantage of (1.7′) is that it does not depend on the choice of coordinates, thus du is intrinsically defined. The two basic properties of the exterior derivative (again left to the reader) are:

$$d(u \wedge v) = du \wedge v + (-1)^{\deg u} u \wedge dv, \qquad (\textit{Leibnitz' rule}) \tag{1.8}$$

$$d^2 = 0. \tag{1.9}$$

A form u is said to be *closed* if $du = 0$ and *exact* if u can be written $u = dv$ for some form v.

§ 1.B.4. *De Rham Cohomology Groups.* Recall that a cohomological complex $K^\bullet = \bigoplus_{p\in\mathbb{Z}}$ is a collection of modules K^p over some ring, equipped with differentials, i.e., linear maps $d^p : K^p \to K^{p+1}$ such that $d^{p+1} \circ d^p = 0$. The *cocycle, coboundary* and *cohomology modules* $Z^p(K^\bullet)$, $B^p(K^\bullet)$ and $H^p(K^\bullet)$ are defined respectively by

$$\begin{cases} Z^p(K^\bullet) = \operatorname{Ker} d^p : K^p \to K^{p+1}, & Z^p(K^\bullet) \subset K^p, \\ B^p(K^\bullet) = \operatorname{Im} d^{p-1} : K^{p-1} \to K^p, & B^p(K^\bullet) \subset Z^p(K^\bullet) \subset K^p, \\ H^p(K^\bullet) = Z^p(K^\bullet)/B^p(K^\bullet). \end{cases} \tag{1.10}$$

Now, let M be a differentiable manifold, say of class $\mathscr{C}^\infty$ for simplicity. The *De Rham complex* of M is defined to be the complex $K^p = \mathscr{C}^\infty(M, \Lambda^p T^\star_M)$ of smooth differential forms, together with the exterior derivative $d^p = d$ as differential, and $K^p = \{0\}$, $d^p = 0$ for $p < 0$. We denote by $Z^p(M, \mathbb{R})$ the cocycles (closed p-forms) and by $B^p(M, \mathbb{R})$ the coboundaries (exact p-forms). By convention $B^0(M, \mathbb{R}) = \{0\}$. The *De Rham cohomology group* of M in degree p is

$$H^p_{\mathrm{DR}}(M, \mathbb{R}) = Z^p(M, \mathbb{R})/B^p(M, \mathbb{R}). \tag{1.11}$$

When no confusion with other types of cohomology groups may occur, we sometimes denote these groups simply by $H^p(M, \mathbb{R})$. The symbol $\mathbb{R}$ is used here to stress that we are considering real valued p-forms; of course one can introduce a similar group $H^p_{\mathrm{DR}}(M, \mathbb{C})$ for complex valued forms, i.e. forms with values in $\mathbb{C} \otimes \Lambda^p T^\star_M$. Then $H^p_{\mathrm{DR}}(M, \mathbb{C}) = \mathbb{C} \otimes H^p_{\mathrm{DR}}(M, \mathbb{R})$ is the complexification of the real De Rham cohomology group. It is clear that $H^0_{\mathrm{DR}}(M, \mathbb{R})$ can be identified with the space of locally constant functions on M, thus

$$H^0_{\mathrm{DR}}(M, \mathbb{R}) = \mathbb{R}^{\pi_0(X)},$$

where $\pi_0(X)$ denotes the set of connected components of M.

Similarly, we introduce the De Rham cohomology groups with compact support

$$H^p_{\mathrm{DR},c}(M, \mathbb{R}) = Z^p_c(M, \mathbb{R})/B^p_c(M, \mathbb{R}), \tag{1.12}$$

associated with the De Rham complex $K^p = \mathscr{C}^\infty_c(M, \Lambda^p T^\star_M)$ of smooth differential forms with compact support.

§ 1.B.5. *Pull-Back.* If $F : M \longrightarrow M'$ is a differentiable map to another manifold M', $\dim_{\mathbb{R}} M' = m'$, and if $v(y) = \sum v_J(y)\, dy_J$ is a differential p-form on M', the pull-back $F^\star v$ is the differential p-form on M obtained after making the substitution $y = F(x)$ in v, i.e.

$$F^\star v(x) = \sum v_I\big(F(x)\big)\, dF_{i_1} \wedge \ldots \wedge dF_{i_p}. \tag{1.13}$$

If we have a second map $G : M' \longrightarrow M''$ and if w is a differential form on M'', then $F^\star(G^\star w)$ is obtained by means of the substitutions $z = G(y)$, $y = F(x)$, thus

$$F^\star(G^\star w) = (G \circ F)^\star w. \tag{1.14}$$

Moreover, we always have $d(F^\star v) = F^\star(dv)$. It follows that the pull-back $F^\star$ is closed if v is closed and exact if v is exact. Therefore $F^\star$ induces a morphism on the quotient spaces

$$F^\star : H^p_{\mathrm{DR}}(M', \mathbb{R}) \longrightarrow H^p_{\mathrm{DR}}(M, \mathbb{R}). \tag{1.15}$$

§ 1.C. Integration of Differential Forms

A manifold M is *orientable* if and only if there exists an atlas (τ_α) such that all transition maps $\tau_{\alpha\beta}$ preserve the orientation, i.e. have positive jacobian determinants. Suppose that M is oriented, that is, equipped with such an atlas. If $u(x) = f(x_1, \ldots, x_m)\, dx_1 \wedge \ldots \wedge dx_m$ is a continuous form of maximum degree $m = \dim_{\mathbb{R}} M$, with compact support in a coordinate open set Ω, we set

$$\int_M u = \int_{\mathbb{R}^m} f(x_1, \ldots, x_m)\, dx_1 \ldots dx_m. \tag{1.16}$$

By the change of variable formula, the result is independent of the choice of coordinates, provided we consider only coordinates corresponding to the given orientation. When u is an arbitrary form with compact support, the definition of $\int_M u$ is easily extended by means of a partition of unity with respect to coordinate open sets covering $\operatorname{Supp} u$. Let $F : M \longrightarrow M'$ be a diffeomorphism between oriented manifolds and v a volume form on M'. The change of variable formula yields

$$\int_M F^\star v = \pm \int_{M'} v \tag{1.17}$$

according whether F preserves orientation or not.

We now state Stokes' formula, which is basic in many contexts. Let K be a compact subset of M with piecewise C^1 boundary. By this, we mean that for each point $a \in \partial K$ there are coordinates $(x_1, \ldots, x_m)$ on a neighborhood V of a, centered at a, such that

$$K \cap V = \big\{x \in V\,;\, x_1 \leqslant 0, \ldots, x_l \leqslant 0\big\}$$

for some index $l \geqslant 1$. Then $\partial K \cap V$ is a union of smooth hypersurfaces with piecewise C^1 boundaries:

$$\partial K \cap V = \bigcup_{1 \leqslant j \leqslant l} \big\{x \in V\,;\, x_1 \leqslant 0, \ldots, x_j = 0, \ldots, x_l \leqslant 0\big\}.$$

At points of ∂K where $x_j = 0$, then $(x_1, \ldots, \widehat{x_j}, , \ldots, x_m)$ define coordinates on ∂K. We take the orientation of ∂K given by these coordinates or the opposite one, according to the sign $(-1)^{j-1}$. For any differential form u of class C^1 and degree $m-1$ on M, we then have

(1.18) Stokes' formula. $$\int_{\partial K} u = \int_K du.$$

The formula is easily checked by an explicit computation when u has compact support in V: indeed if $u = \sum_{1\leqslant j\leqslant n} u_j\, dx_1 \wedge \ldots \wedge \widehat{dx_j} \wedge \ldots \wedge dx_m$ and $\partial_j K \cap V$ is the part of $\partial K \cap V$ where $x_j = 0$, a partial integration with respect to x_j yields

$$\int_{\partial_j K\cap V} u_j\, dx_1 \wedge \ldots \widehat{dx_j} \ldots dx_m = \int_V \frac{\partial u_j}{\partial x_j}\, dx_1 \wedge \ldots dx_m,$$
$$\int_{\partial K\cap V} u = \sum_{1\leqslant j\leqslant m} (-1)^{j-1} \int_{\partial_j K\cap V} u_j\, dx_1 \wedge \ldots \widehat{dx_j} \ldots \wedge dx_m = \int_V du.$$

The general case follows by a partition of unity. In particular, if u has compact support in M, we find $\int_M du = 0$ by choosing $K \supset \operatorname{Supp} u$.

§ 1.D. Homotopy Formula and Poincaré Lemma

Let u be a differential form on $[0,1]\times M$. For $(t,x)\in[0,1]\times M$, we write

$$u(t,x) = \sum_{|I|=p} u_I(t,x)\, dx_I + \sum_{|J|=p-1} \widetilde{u}_J(t,x)\, dt \wedge dx_J.$$

We define an operator

$$K : C^s([0,1]\times M, \Lambda^p T^\star_{[0,1]\times M}) \longrightarrow C^s(M, \Lambda^{p-1} T^\star_M)$$
$$Ku(x) = \sum_{|J|=p-1} \Big(\int_0^1 \widetilde{u}_J(t,x)\, dt\Big) dx_J \tag{1.19}$$

and say that Ku is the form obtained by integrating u along $[0,1]$. A computation of the operator $dK + Kd$ shows that all terms involving partial derivatives $\partial\widetilde{u}_J/\partial x_k$ cancel, hence

$$Kdu + dKu = \sum_{|I|=p} \Big(\int_0^1 \frac{\partial u_I}{\partial t}(t,x)\, dt\Big) dx_I = \sum_{|I|=p} \big(u_I(1,x) - u_I(0,x)\big) dx_I,$$
$$Kdu + dKu = i_1^\star u - i_0^\star u, \tag{1.20}$$

where $i_t : M \to [0,1]\times M$ is the injection $x \mapsto (t,x)$.

(1.21) Corollary. *Let $F, G : M \longrightarrow M'$ be $\mathscr{C}^\infty$ maps. Suppose that F, G are smoothly homotopic, i.e. that there exists a $\mathscr{C}^\infty$ map $H : [0,1]\times M \longrightarrow M'$ such that $H(0,x) = F(x)$ and $H(1,x) = G(x)$. Then*

$$F^\star = G^\star : H^p_{\mathrm{DR}}(M', \mathbb{R}) \longrightarrow H^p_{\mathrm{DR}}(M, \mathbb{R}).$$

Proof. If v is a p-form on M', then

$$\begin{aligned} G^\star v - F^\star v &= (H\circ i_1)^\star v - (H\circ i_0)^\star v = i_1^\star(H^\star v) - i_0^\star(H^\star v) \\ &= d(KH^\star v) + KH^\star(dv) \end{aligned}$$

by (1.20) applied to $u = H^\star v$. If v is closed, then $F^\star v$ and $G^\star v$ differ by an exact form, so they define the same class in $H^p_{\mathrm{DR}}(M, \mathbb{R})$. $\square$

(1.22) Corollary. *If the manifold M is contractible, i.e. if there is a smooth homotopy $H : [0,1] \times M \to M$ from a constant map $F : M \to \{x_0\}$ to $G = \mathrm{Id}_X$, then $H^0_{\mathrm{DR}}(M, \mathbb{R}) = \mathbb{R}$ and $H^p_{\mathrm{DR}}(M, \mathbb{R}) = 0$ for $p \geqslant 1$.*

Proof. $F^\star$ is clearly zero in degree $p \geqslant 1$, while $F^\star : H^0_{\mathrm{DR}}(M, \mathbb{R}) \xrightarrow{\simeq} \mathbb{R}$ is induced by the evaluation map $u \mapsto u(x_0)$. The conclusion then follows from the equality $F^\star = G^\star = \mathrm{Id}$ on cohomology groups. $\square$

(1.23) Poincaré lemma. *Let $\Omega \subset \mathbb{R}^m$ be a starshaped open set. If a form $v = \sum v_I dx_I \in C^s(\Omega, \Lambda^p T^\star_\Omega)$, $p \geqslant 1$, satisfies $dv = 0$, there exists a form $u \in C^s(\Omega, \Lambda^{p-1} T^\star_\Omega)$ such that $du = v$.*

Proof. Let $H(t, x) = tx$ be the homotopy between the identity map $\Omega \to \Omega$ and the constant map $\Omega \to \{0\}$. By the above formula

$$d(KH^\star v) = G^\star v - F^\star v = \begin{cases} v - v(0) & \text{if } \ p = 0, \\ v & \text{if } \ p \geqslant 1. \end{cases}$$

Hence $u = KH^\star v$ is the $(p-1)$-form we are looking for. An explicit computation based on (1.19) easily gives

$$u(x) = \sum_{\substack{|I|=p \\ 1 \leqslant k \leqslant p}} \Big(\int_0^1 t^{p-1} v_I(tx)\, dt \Big) (-1)^{k-1} x_{i_k} dx_{i_1} \wedge \ldots \widehat{dx_{i_k}} \ldots \wedge dx_{i_p}. \tag{1.23}$$

§ 2. Currents on Differentiable Manifolds

§ 2.A. Definition and Examples

Let M be a $\mathscr{C}^\infty$ differentiable manifold, $m = \dim_{\mathbb{R}} M$. All the manifolds considered in § 2 will be assumed to be oriented. We first introduce a topology on the space of differential forms $\mathscr{C}^s(M, \Lambda^p T^\star_M)$. Let $\Omega \subset M$ be a coordinate open set and u a p-form on M, written $u(x) = \sum u_I(x)\, dx_I$ on Ω. To every compact subset $L \subset \Omega$ and every integer $s \in \mathbb{N}$, we associate a seminorm

$$p^s_L(u) = \sup_{x \in L} \max_{|I|=p, |\alpha| \leqslant s} |D^\alpha u_I(x)|, \tag{2.1}$$

where $\alpha = (\alpha_1, \ldots, \alpha_m)$ runs over $\mathbb{N}^m$ and $D^\alpha = \partial^{|\alpha|} / \partial x_1^{\alpha_1} \ldots \partial x_m^{\alpha_m}$ is a derivation of order $|\alpha| = \alpha_1 + \cdots + \alpha_m$. This type of multi-index, which will always be denoted by Greek letters, should not be confused with multi-indices of the type $I = (i_1, \ldots, i_p)$ introduced in § 1.

(2.2) Definition. *We introduce as follows spaces of p-forms on manifolds.*

a) *We denote by* $\mathscr{E}^p(M)$ $\big($*resp.* ${}^s\mathscr{E}^p(M)\big)$ *the space* $\mathscr{C}^\infty(M, \Lambda^p T^\star_M)$ $\big($*resp. the space* $\mathscr{C}^s(M, \Lambda^p T^\star_M)\big)$*, equipped with the topology defined by all seminorms* p^s_L *when* s*,* L*,* Ω *vary* (*resp. when* L*,* Ω *vary*).

b) *If* $K \subset M$ *is a compact subset,* $\mathscr{D}^p(K)$ *will denote the subspace of elements* $u \in \mathscr{E}^p(M)$ *with support contained in* K*, together with the induced topology;* $\mathscr{D}^p(M)$ *will stand for the set of all elements with compact support, i.e.* $\mathscr{D}^p(M) := \bigcup_K \mathscr{D}^p(K)$.

c) *The spaces of* C^s*-forms* ${}^s\mathscr{D}^p(K)$ *and* ${}^s\mathscr{D}^p(M)$ *are defined similarly.*

Since our manifolds are assumed to be separable, the topology of $\mathscr{E}^p(M)$ can be defined by means of a countable set of seminorms p^s_L, hence $\mathscr{E}^p(M)$ (and likewise ${}^s\mathscr{E}^p(M)$) is a Fréchet space. The topology of ${}^s\mathscr{D}^p(K)$ is induced by any finite set of seminorms $p^s_{K_j}$ such that the compact sets K_j cover K ; hence ${}^s\mathscr{D}^p(K)$ is a Banach space. It should be observed however that $\mathscr{D}^p(M)$ is not a Fréchet space; in fact $\mathscr{D}^p(M)$ is dense in $\mathscr{E}^p(M)$ and thus non complete for the induced topology. According to [De Rham 1955] spaces of *currents* are defined as the topological duals of the above spaces, in analogy with the usual definition of distributions.

(2.3) Definition. *The space of currents of dimension* p (*or degree* $m-p$) *on* M *is the space* $\mathscr{D}'_p(M)$ *of linear forms* T *on* $\mathscr{D}^p(M)$ *such that the restriction of* T *to all subspaces* $\mathscr{D}^p(K)$*,* $K \subset\subset M$*, is continuous. The degree is indicated by raising the index, hence we set*

$$\mathscr{D}'^{m-p}(M) = \mathscr{D}'_p(M) := \textit{topological dual } \big(\mathscr{D}^p(M)\big)'.$$

The space ${}^s\mathscr{D}'_p(M) = {}^s\mathscr{D}'^{m-p}(M) := \big({}^s\mathscr{D}^p(M)\big)'$ *is defined similarly and is called the space of currents of order* s *on* M.

In the sequel, we let $\langle T, u\rangle$ be the pairing between a current T and a *test form* $u \in \mathscr{D}^p(M)$. It is clear that ${}^s\mathscr{D}'_p(M)$ can be identified with the subspace of currents $T \in \mathscr{D}'_p(M)$ which are continuous for the seminorm p^s_K on $\mathscr{D}^p(K)$ for every compact set K contained in a coordinate patch Ω. The *support* of T, denoted $\operatorname{Supp} T$, is the smallest closed subset $A \subset M$ such that the restriction of T to $\mathscr{D}^p(M \smallsetminus A)$ is zero. The topological dual $\mathscr{E}'_p(M)$ can be identified with the set of currents of $\mathscr{D}'_p(M)$ with compact support: indeed, let T be a linear form on $\mathscr{E}^p(M)$ such that

$$|\langle T, u\rangle| \leqslant C \max\{p^s_{K_j}(u)\}$$

for some $s \in \mathbb{N}$, $C \geqslant 0$ and a finite number of compact sets K_j ; it follows that $\operatorname{Supp} T \subset \bigcup K_j$. Conversely let $T \in \mathscr{D}'_p(M)$ with support in a compact set K. Let K_j be compact patches such that K is contained in the interior of $\bigcup K_j$ and $\psi \in \mathscr{D}(M)$ equal to 1 on K with $\operatorname{Supp}\psi \subset \bigcup K_j$. For $u \in \mathscr{E}^p(M)$, we define $\langle T, u\rangle = \langle T, \psi u\rangle$; this is independent of ψ and the resulting T is clearly continuous on $\mathscr{E}^p(M)$. The terminology used for the dimension and degree of a current is justified by the following two examples.

(2.4) Example. Let $Z \subset M$ be a closed oriented submanifold of M of dimension p and class C^1 ; Z may have a boundary ∂Z. The *current of integration* over Z, denoted $[Z]$, is

defined by

$$\langle [Z], u\rangle = \int_Z u, \qquad u \in {}^0\mathscr{D}^p(M).$$

It is clear that $[Z]$ is a current of order 0 on M and that $\operatorname{Supp}[Z] = Z$. Its dimension is $p = \dim Z$.

(2.5) Example. If f is a differential form of degree q on M with L^1_{loc} coefficients, we can associate to f the current of dimension $m-q$:

$$\langle T_f, u\rangle = \int_M f \wedge u, \qquad u \in {}^0\mathscr{D}^{m-q}(M).$$

T_f is of degree q and of order 0. The correspondence $f \longmapsto T_f$ is injective. In the same way L^1_{loc} functions on $\mathbb{R}^m$ are identified to distributions, we will identify f with its image $T_f \in {}^0\mathscr{D}'^q(M) = {}^0\mathscr{D}'_{m-q}(M)$.

§ 2.B. Exterior Derivative and Wedge Product

§ 2.B.1. *Exterior Derivative.* Many of the operations available for differential forms can be extended to currents by simple duality arguments. Let $T \in {}^s\mathscr{D}'^q(M) = {}^s\mathscr{D}'_{m-p}(M)$. The *exterior derivative*

$$dT \in {}^{s+1}\mathscr{D}'^{q+1}(M) = {}^{s+1}\mathscr{D}'_{m-q-1}$$

is defined by

$$\langle dT, u\rangle = (-1)^{q+1}\,\langle T, du\rangle, \qquad u \in {}^{s+1}\mathscr{D}^{m-q-1}(M). \tag{2.6}$$

The continuity of the linear form dT on ${}^{s+1}\mathscr{D}^{m-q-1}(M)$ follows from the continuity of the map $d : {}^{s+1}\mathscr{D}^{m-q-1}(K) \longrightarrow {}^s\mathscr{D}^{m-q}(K)$. For all forms $f \in {}^1\mathscr{E}^q(M)$ and $u \in \mathscr{D}^{m-q-1}(M)$, Stokes' formula implies

$$0 = \int_M d(f \wedge u) = \int_M df \wedge u + (-1)^q\, f \wedge du,$$

thus in example (2.5) one actually has $dT_f = T_{df}$ as it should be. In example (2.4), another application of Stokes' formula yields $\int_Z du = \int_{\partial Z} u$, therefore $\langle [Z], du\rangle = \langle [\partial Z], u\rangle$ and

$$d[Z] = (-1)^{m-p+1}[\partial Z]. \tag{2.7}$$

§ 2.B.2. *Wedge Product.* For $T \in {}^s\mathscr{D}'^q(M)$ and $g \in {}^s\mathscr{E}^r(M)$, the wedge product $T \wedge g \in {}^s\mathscr{D}'^{q+r}(M)$ is defined by

$$\langle T \wedge g, u\rangle = \langle T, g \wedge u\rangle, \qquad u \in {}^s\mathscr{D}^{m-q-r}(M). \tag{2.8}$$

This definition is licit because $u \mapsto g \wedge u$ is continuous in the C^s-topology. The relation

$$d(T \wedge g) = dT \wedge g + (-1)^{\deg T} T \wedge dg$$

is easily verified from the definitions.

(2.9) Proposition. *Let $(x_1, \ldots, x_m)$ be a coordinate system on an open subset $\Omega \subset M$. Every current $T \in {}^s\mathscr{D}'^q(M)$ of degree q can be written in a unique way*

$$T = \sum_{|I|=q} T_I \, dx_I \quad \text{on } \Omega,$$

where T_I are distributions of order s on Ω, considered as currents of degree 0.

Proof. If the result is true, for all $f \in {}^s\mathscr{D}^0(\Omega)$ we must have

$$\langle T, f\, dx_{\complement I}\rangle = \langle T_I, dx_I \wedge f\, dx_{\complement I}\rangle = \varepsilon(I, \complement I)\, \langle T_I, f\, dx_1 \wedge \ldots \wedge dx_m\rangle,$$

where $\varepsilon(I, \complement I)$ is the signature of the permutation $(1, \ldots, m) \longmapsto (I, \complement I)$. Conversely, this can be taken as a definition of the coefficient T_I :

$$T_I(f) = \langle T_I, f\, dx_1 \wedge \ldots \wedge dx_m\rangle := \varepsilon(I, \complement I)\, \langle T, f\, dx_{\complement I}\rangle, \quad f \in {}^s\mathscr{D}^0(\Omega). \tag{2.10}$$

Then T_I is a distribution of order s and it is easy to check that $T = \sum T_I\, dx_I$. □

In particular, currents of order 0 on M can be considered as differential forms with measure coefficients. In order to unify the notations concerning forms and currents, we set

$$\langle T, u\rangle = \int_M T \wedge u$$

whenever $T \in {}^s\mathscr{D}'_p(M) = {}^s\mathscr{D}'^{m-p}(M)$ and $u \in {}^s\mathscr{E}^p(M)$ are such that $\operatorname{Supp} T \cap \operatorname{Supp} u$ is compact. This convention is made so that the notation becomes compatible with the identification of a form f to the current T_f.

§ 2.C. Direct and Inverse Images

§ 2.C.1. *Direct Images.* Assume now that M_1, M_2 are oriented differentiable manifolds of respective dimensions m_1, m_2, and that

$$F : M_1 \longrightarrow M_2 \tag{2.11}$$

is a $\mathscr{C}^\infty$ map. The pull-back morphism

$${}^s\mathscr{D}^p(M_2) \longrightarrow {}^s\mathscr{E}^p(M_1), \qquad u \longmapsto F^\star u \tag{2.12}$$

is continuous in the C^s topology and we have $\operatorname{Supp} F^\star u \subset F^{-1}(\operatorname{Supp} u)$, but in general $\operatorname{Supp} F^\star u$ is not compact. If $T \in {}^s\mathscr{D}'_p(M_1)$ is such that the restriction of F to $\operatorname{Supp} T$ is *proper*, i.e. if $\operatorname{Supp} T \cap F^{-1}(K)$ is compact for every compact subset $K \subset M_2$, then the linear form $u \longmapsto \langle T, F^\star u\rangle$ is well defined and continuous on ${}^s\mathscr{D}^p(M_2)$. There exists therefore a unique current denoted $F_\star T \in {}^s\mathscr{D}'_p(M_2)$, called *the direct image* of T by F, such that

$$\langle F_\star T, u\rangle = \langle T, F^\star u\rangle, \quad \forall u \in {}^s\mathscr{D}^p(M_2). \tag{2.13}$$

We leave the straightforward proof of the following properties to the reader.

(2.14) Theorem. *For every $T \in {}^s\mathscr{D}'_p(M_1)$ such that $F_{\restriction \operatorname{Supp} T}$ is proper, the direct image $F_\star T \in {}^s\mathscr{D}'_p(M_2)$ is such that*

a) $\operatorname{Supp} F_\star T \subset F(\operatorname{Supp} T)$;

b) $d(F_\star T) = F_\star(dT)$;

c) $F_\star(T \wedge F^\star g) = (F_\star T) \wedge g, \quad \forall g \in {}^s\mathscr{C}^q(M_2, \mathbb{R})$;

d) *If* $G : M_2 \longrightarrow M_3$ *is a* $\mathscr{C}^\infty$ *map such that* $(G \circ F)_{\restriction \operatorname{Supp} T}$ *is proper, then*

$$G_\star(F_\star T) = (G \circ F)_\star T.$$

(2.15) Special case. Assume that F is a submersion, i.e. that F is surjective and that for every $x \in M_1$ the differential map $d_x F : T_{M_1,x} \longrightarrow T_{M_2,F(x)}$ is surjective. Let g be a differential form of degree q on M_1, with L^1_{loc} coefficients, such that $F_{\restriction \operatorname{Supp} g}$ is proper. We claim that $F_\star g \in {}^0\mathscr{D}'_{m_1-q}(M_2)$ is the form of degree $q - (m_1 - m_2)$ obtained from g by integration along the fibers of F, also denoted

$$F_\star g(y) = \int_{z \in F^{-1}(y)} g(z).$$

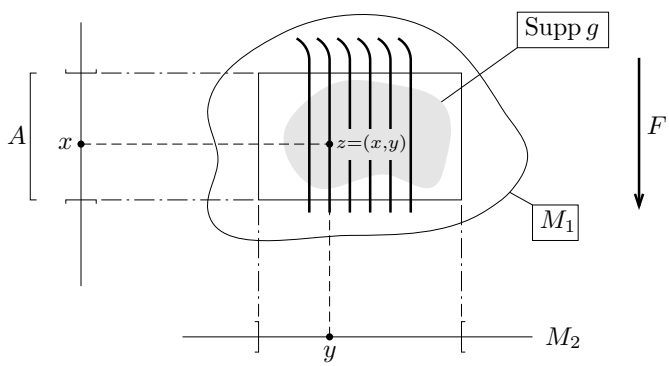

Fig. I.2 Local description of a submersion as a projection.

In fact, this assertion is equivalent to the following generalized form of Fubini's theorem:

$$\int_{M_1} g \wedge F^\star u = \int_{y \in M_2} \Big(\int_{z \in F^{-1}(y)} g(z) \Big) \wedge u(y), \qquad \forall u \in {}^0\mathscr{D}^{m_1-q}(M_2).$$

By using a partition of unity on M_1 and the constant rank theorem, the verification of this formula is easily reduced to the case where $M_1 = A \times M_2$ and $F = \mathrm{pr}_2$, cf. Fig. 2. The fibers $F^{-1}(y) \simeq A$ have to be oriented in such a way that the orientation of M_1 is the product of the orientation of A and M_2. Let us write $r = \dim A = m_1 - m_2$ and let $z = (x, y) \in A \times M_2$ be any point of M_1. The above formula becomes

$$\int_{A \times M_2} g(x, y) \wedge u(y) = \int_{y \in M_2} \Big(\int_{x \in A} g(x, y) \Big) \wedge u(y),$$

where the direct image of g is computed from $g = \sum g_{I,J}(x,y)\, dx_I \wedge dy_J$, $|I| + |J| = q$, by the formula

$$F_\star g(y) = \int_{x\in A} g(x,y) = \sum_{|J|=q-r} \Big(\int_{x\in A} g_{(1,\ldots,r),J}(x,y)\, dx_1 \wedge \ldots \wedge dx_r \Big) dy_J. \tag{2.16}$$

In this situation, we see that $F_\star g$ has L^1_{loc} coefficients on M_2 if g is L^1_{loc} on M_1, and that the map $g \longmapsto F_\star g$ is continuous in the C^s topology.

(2.17) Remark. If $F : M_1 \longrightarrow M_2$ is a diffeomorphism, then we have $F_\star g = \pm (F^{-1})^\star g$ according whether F preserves the orientation or not. In fact formula (1.17) gives

$$\langle F_\star g, u\rangle = \int_{M_1} g \wedge F^\star u = \pm \int_{M_2} (F^{-1})^\star (g \wedge F^\star u) = \pm \int_{M_2} (F^{-1})^\star g \wedge u.$$

§ 2.C.2. *Inverse Images.* Assume that $F : M_1 \longrightarrow M_2$ is a submersion. As a consequence of the continuity statement after (2.16), one can always define the inverse image $F^\star T \in {}^s\mathscr{D}'^q(M_1)$ of a current $T \in {}^s\mathscr{D}'^q(M_2)$ by

$$\langle F^\star T, u\rangle = \langle T, F_\star u\rangle, \qquad u \in {}^s\mathscr{D}^{q+m_1-m_2}(M_1).$$

Then $\dim F^\star T = \dim T + m_1 - m_2$ and Th. 2.14 yields the formulas:

$$d(F^\star T) = F^\star(dT), \qquad F^\star(T \wedge g) = F^\star T \wedge F^\star g, \qquad \forall g \in {}^s\mathscr{D}^\bullet(M_2). \tag{2.18}$$

Take in particular $T = [Z]$, where Z is an oriented C^1-submanifold of M_2. Then $F^{-1}(Z)$ is a submanifold of M_1 and has a natural orientation given by the isomorphism

$$T_{M_1,x}/T_{F^{-1}(Z),x} \longrightarrow T_{M_2,F(x)}/T_{Z,F(x)},$$

induced by $d_x F$ at every point $x \in Z$. We claim that

$$F^\star[Z] = [F^{-1}(Z)]. \tag{2.19}$$

Indeed, we have to check that $\int_Z F_\star u = \int_{F^{-1}(Z)} u$ for every $u \in {}^s\mathscr{D}^\bullet(M_1)$. By using a partition of unity on M_1, we may again assume $M_1 = A \times M_2$ and $F = \mathrm{pr}_2$. The above equality can be written

$$\int_{y\in Z} F_\star u(y) = \int_{(x,y)\in A\times Z} u(x,y).$$

This follows precisely from (2.16) and Fubini's theorem.

§ 2.C.3. *Weak Topology.* The weak topology on $\mathscr{D}'_p(M)$ is the topology defined by the collection of seminorms $T \longmapsto |\langle T, f\rangle|$ for all $f \in \mathscr{D}^p(M)$. With respect to the weak topology, all the operations

$$T \longmapsto dT, \quad T \longmapsto T \wedge g, \quad T \longmapsto F_\star T, \quad T \longmapsto F^\star T \tag{2.20}$$

defined above are continuous. A set $B \subset \mathscr{D}'_p(M)$ is bounded for the weak topology (weakly bounded for short) if and only if $\langle T, f\rangle$ is bounded when T runs over B, for every fixed $f \in \mathscr{D}^p(M)$. The standard Banach-Alaoglu theorem implies that every weakly bounded closed subset $B \subset \mathscr{D}'_p(M)$ is weakly compact.

§ 2.D. Tensor Products, Homotopies and Poincaré Lemma

§ 2.D.1. *Tensor Products.* If S, T are currents on manifolds M, M' there exists a unique current on $M \times M'$, denoted $S \otimes T$ and defined in a way analogous to the tensor product of distributions, such that for all $u \in \mathscr{D}^\bullet(M)$ and $v \in \mathscr{D}^\bullet(M')$

$$\langle S \otimes T, \mathrm{pr}_1^\star u \wedge \mathrm{pr}_2^\star v\rangle = (-1)^{\deg T \deg u} \langle S, u\rangle \, \langle T, v\rangle. \tag{2.21}$$

One verifies easily that $d(S \otimes T) = dS \otimes T + (-1)^{\deg S} S \otimes dT$.

§ 2.D.2. *Homotopy Formula.* Assume that $H : [0,1] \times M_1 \longrightarrow M_2$ is a $\mathscr{C}^\infty$ homotopy from $F(x) = H(0,x)$ to $G(x) = H(1,x)$ and that $T \in \mathscr{D}'_\bullet(M_1)$ is a current such that $H_{\restriction [0,1]\times \mathrm{Supp}\, T}$ is proper. If $[0,1]$ is considered as the current of degree 0 on $\mathbb{R}$ associated to its characteristic function, we find $d[0,1] = \delta_0 - \delta_1$, thus

$$\begin{aligned} d\big(H_\star([0,1] \otimes T)\big) &= H_\star(\delta_0 \otimes T - \delta_1 \otimes T + [0,1] \otimes dT) \\ &= F_\star T - G_\star T + H_\star([0,1] \otimes dT). \end{aligned}$$

Therefore we obtain the *homotopy formula*

$$F_\star T - G_\star T = d\big(H_\star([0,1] \otimes T)\big) - H_\star([0,1] \otimes dT). \tag{2.22}$$

When T is closed, i.e. $dT = 0$, we see that $F_\star T$ and $G_\star T$ are cohomologous on M_2, i.e. they differ by an exact current dS.

§ 2.D.3. *Regularization of Currents.* Let $\rho \in \mathscr{C}^\infty(\mathbb{R}^m)$ be a function with support in $B(0,1)$, such that $\rho(x)$ depends only on $|x| = (\sum |x_i|^2)^{1/2}$, $\rho \geqslant 0$ and $\int_{\mathbb{R}^m} \rho(x)\, dx = 1$. We associate to ρ the family of functions (ρ_ε) such that

$$\rho_\varepsilon(x) = \frac{1}{\varepsilon^m}\, \rho\Big(\frac{x}{\varepsilon}\Big), \quad \mathrm{Supp}\, \rho_\varepsilon \subset B(0,\varepsilon), \quad \int_{\mathbb{R}^m} \rho_\varepsilon(x)\, dx = 1. \tag{2.23}$$

We shall refer to this construction by saying that (ρ_ε) is a *family of smoothing kernels.* For every current $T = \sum T_I\, dx_I$ on an open subset $\Omega \subset \mathbb{R}^m$, the family of smooth forms

$$T \star \rho_\varepsilon = \sum_I (T_I \star \rho_\varepsilon)\, dx_I,$$

defined on $\Omega_\varepsilon = \{x \in \mathbb{R}^m \ ; \ d(x, \complement\Omega) > \varepsilon\}$, converges weakly to T as ε tends to 0. Indeed, $\langle T \star \rho_\varepsilon, f\rangle = \langle T, \rho_\varepsilon \star f\rangle$ and $\rho_\varepsilon \star f$ converges to f in $\mathscr{D}^p(\Omega)$ with respect to all seminorms p_K^s.

§ 2.D.4. *Poincaré Lemma for Currents.* Let $T \in {}^s\mathscr{D}'^q(\Omega)$ be a closed current on an open set $\Omega \subset \mathbb{R}^m$. We first show that T is cohomologous to a smooth form. In fact, let $\psi \in \mathscr{C}^\infty(\mathbb{R}^m)$ be a cut-off function such that $\operatorname{Supp}\psi \subset \overline{\Omega}$, $0 < \psi \leqslant 1$ and $|d\psi| \leqslant 1$ on Ω. For any vector $v \in B(0,1)$ we set

$$F_v(x) = x + \psi(x)v.$$

Since $x \mapsto \psi(x)v$ is a contraction, F_v is a diffeomorphism of $\mathbb{R}^m$ which leaves $\complement\Omega$ invariant pointwise, so $F_v(\Omega) = \Omega$. This diffeomorphism is homotopic to the identity through the homotopy $H_v(t,x) = F_{tv}(x) : [0,1] \times \Omega \longrightarrow \Omega$ which is proper for every v. Formula (2.22) implies

$$(F_v)_\star T - T = d\big((H_v)_\star([0,1] \otimes T)\big).$$

After averaging with a smoothing kernel $\rho_\varepsilon(v)$ we get $\Theta - T = dS$ where

$$\Theta = \int_{B(0,\varepsilon)} (F_v)_\star T\, \rho_\varepsilon(v)\, dv, \qquad S = \int_{B(0,\varepsilon)} (H_v)_\star([0,1] \otimes T)\, \rho_\varepsilon(v)\, dv.$$

Then S is a current of the same order s as T and Θ is smooth. Indeed, for $u \in \mathscr{D}^p(\Omega)$ we have

$$\langle \Theta, u\rangle = \langle T, u_\varepsilon\rangle \quad \text{ where } \; u_\varepsilon(x) = \int_{B(0,\varepsilon)} F_v^\star u(x)\, \rho_\varepsilon(v)\, dv \;;$$

we can make a change of variable $z = F_v(x) \Leftrightarrow v = \psi(x)^{-1}(z - x)$ in the last integral and perform derivatives on ρ_ε to see that each seminorm $p_K^t(u_\varepsilon)$ is controlled by the sup norm of u. Thus Θ and all its derivatives are currents of order 0, so Θ is smooth. Now we have $d\Theta = 0$ and by the usual Poincaré lemma (1.23) applied to Θ we obtain

(2.24) Theorem. *Let $\Omega \subset \mathbb{R}^m$ be a starshaped open subset and $T \in {}^s\mathscr{D}'^q(\Omega)$ a current of degree $q \geqslant 1$ and order s such that $dT = 0$. There exists a current $S \in {}^s\mathscr{D}'^{q-1}(\Omega)$ of degree $q-1$ and order $\leqslant s$ such that $dS = T$ on Ω.* □

§ 3. Holomorphic Functions and Complex Manifolds

§ 3.A. Cauchy Formula in One Variable

We start by recalling a few elementary facts in one complex variable theory. Let $\Omega \subset \mathbb{C}$ be an open set and let $z = x + \mathrm{i}y$ be the complex variable, where $x, y \in \mathbb{R}$. If f is a function of class C^1 on Ω, we have

$$df = \frac{\partial f}{\partial x}\, dx + \frac{\partial f}{\partial y}\, dy = \frac{\partial f}{\partial z}\, dz + \frac{\partial f}{\partial \overline{z}}\, d\overline{z}$$

with the usual notations

$$\frac{\partial}{\partial z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} - \mathrm{i}\frac{\partial}{\partial y}\Big), \qquad \frac{\partial}{\partial \overline{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + \mathrm{i}\frac{\partial}{\partial y}\Big). \tag{3.1}$$

The function f is holomorphic on Ω if df is $\mathbb{C}$-linear, that is, $\partial f/\partial \overline{z} = 0$.

(3.2) Cauchy formula. *Let $K \subset \mathbb{C}$ be a compact set with piecewise C^1 boundary ∂K. Then for every $f \in C^1(K, \mathbb{C})$*

$$f(w) = \frac{1}{2\pi \mathrm{i}} \int_{\partial K} \frac{f(z)}{z-w}\, dz - \int_K \frac{1}{\pi(z-w)} \frac{\partial f}{\partial \overline{z}}\, d\lambda(z), \qquad w \in K^\circ$$

where $d\lambda(z) = \frac{\mathrm{i}}{2} dz \wedge d\overline{z} = dx \wedge dy$ is the Lebesgue measure on $\mathbb{C}$.

Proof. Assume for simplicity $w = 0$. As the function $z \mapsto 1/z$ is locally integrable at $z = 0$, we get

$$\begin{aligned}
\int_K \frac{1}{\pi z} \frac{\partial f}{\partial \overline{z}}\, d\lambda(z) &= \lim_{\varepsilon \to 0} \int_{K \smallsetminus D(0,\varepsilon)} \frac{1}{\pi z} \frac{\partial f}{\partial \overline{z}} \frac{\mathrm{i}}{2} dz \wedge d\overline{z} \\
&= \lim_{\varepsilon \to 0} \int_{K \smallsetminus D(0,\varepsilon)} d\Big[\frac{1}{2\pi \mathrm{i}} f(z) \frac{dz}{z}\Big] \\
&= \frac{1}{2\pi \mathrm{i}} \int_{\partial K} f(z) \frac{dz}{z} - \lim_{\varepsilon \to 0} \frac{1}{2\pi \mathrm{i}} \int_{\partial D(0,\varepsilon)} f(z) \frac{dz}{z}
\end{aligned}$$

by Stokes' formula. The last integral is equal to $\frac{1}{2\pi} \int_0^{2\pi} f(\varepsilon e^{\mathrm{i}\theta})\, d\theta$ and converges to $f(0)$ as ε tends to 0. □

When f is holomorphic on Ω, we get the usual Cauchy formula

$$f(w) = \frac{1}{2\pi \mathrm{i}} \int_{\partial K} \frac{f(z)}{z-w}\, dz, \qquad w \in K^\circ, \tag{3.3}$$

from which many basic properties of holomorphic functions can be derived: power and Laurent series expansions, Cauchy residue formula, ... Another interesting consequence is:

(3.4) Corollary. *The L^1_{loc} function $E(z) = 1/\pi z$ is a fundamental solution of the operator $\partial/\partial\overline{z}$ on $\mathbb{C}$, i.e. $\partial E/\partial\overline{z} = \delta_0$ (Dirac measure at 0). As a consequence, if v is a distribution with compact support in $\mathbb{C}$, then the convolution $u = (1/\pi z) \star v$ is a solution of the equation $\partial u/\partial\overline{z} = v$.*

Proof. Apply (3.2) with $w = 0$, $f \in \mathscr{D}(\mathbb{C})$ and $K \supset \operatorname{Supp} f$, so that $f = 0$ on the boundary ∂K and $f(0) = \langle 1/\pi z, -\partial f/\partial\overline{z}\rangle$. □

(3.5) Remark. It should be observed that this formula cannot be used to solve the equation $\partial u/\partial\overline{z} = v$ when $\operatorname{Supp} v$ is not compact; moreover, if $\operatorname{Supp} v$ is compact, a solution u with compact support need not always exist. Indeed, we have a necessary condition

$$\langle v, z^n\rangle = -\langle u, \partial z^n/\partial\overline{z}\rangle = 0$$

for all integers $n \geqslant 0$. Conversely, when the necessary condition $\langle v, z^n\rangle = 0$ is satisfied, the canonical solution $u = (1/\pi z) \star v$ has compact support: this is easily seen by means of the power series expansion $(w-z)^{-1} = \sum z^n w^{-n-1}$, if we suppose that $\operatorname{Supp} v$ is contained in the disk $|z| < R$ and that $|w| > R$.

§ 3.B. Holomorphic Functions of Several Variables

Let $\Omega \subset \mathbb{C}^n$ be an open set. A function $f : \Omega \to \mathbb{C}$ is said to be holomorphic if f is continuous and separately holomorphic with respect to each variable, i.e. $z_j \mapsto f(\ldots, z_j, \ldots)$ is holomorphic when $z_1, \ldots, z_{j-1}, z_{j+1}, \ldots, z_n$ are fixed. The set of holomorphic functions on Ω is a ring and will be denoted $\mathscr{O}(\Omega)$. We first extend the Cauchy formula to the case of polydisks. The open polydisk $D(z_0, R)$ of center $(z_{0,1}, \ldots, z_{0,n})$ and (multi)radius $R = (R_1, , \ldots, R_n)$ is defined as the product of the disks of center $z_{0,j}$ and radius $R_j > 0$ in each factor $\mathbb{C}$:

$$D(z_0, R) = D(z_{0,1}, R_1) \times \ldots \times D(z_{0,n}, R_n) \subset \mathbb{C}^n. \tag{3.6}$$

The *distinguished boundary* of $D(z_0, R)$ is by definition the product of the boundary circles

$$\Gamma(z_0, R) = \Gamma(z_{0,1}, R_1) \times \ldots \times \Gamma(z_{0,n}, R_n). \tag{3.7}$$

It is important to observe that the distinguished boundary is smaller than the topological boundary $\partial D(z_0, R) = \bigcup_j \{z \in \overline{D}(z_0, R)\,;\, |z_j - z_{0,j}| = R_j\}$ when $n \geqslant 2$. By induction on n, we easily get the

(3.8) Cauchy formula on polydisks. *If $\overline{D}(z_0, R)$ is a closed polydisk contained in Ω and $f \in \mathscr{O}(\Omega)$, then for all $w \in D(z_0, R)$ we have*

$$f(w) = \frac{1}{(2\pi \mathrm{i})^n} \int_{\Gamma(z_0,R)} \frac{f(z_1, \ldots, z_n)}{(z_1 - w_1) \ldots (z_n - w_n)}\, dz_1 \ldots dz_n. \qquad \square$$

The expansion $(z_j - w_j)^{-1} = \sum (w_j - z_{0,j})^{\alpha_j} (z_j - z_{0,j})^{-\alpha_j - 1}$, $\alpha_j \in \mathbb{N}$, $1 \leqslant j \leqslant n$, shows that f can be expanded as a convergent power series $f(w) = \sum_{\alpha \in \mathbb{N}^n} a_\alpha (w - z_0)^\alpha$ over the polydisk $D(z_0, R)$, with the standard notations $z^\alpha = z_1^{\alpha_1} \ldots z_n^{\alpha_n}$, $\alpha! = \alpha_1! \ldots \alpha_n!$ and with

$$a_\alpha = \frac{1}{(2\pi \mathrm{i})^n} \int_{\Gamma(z_0,R)} \frac{f(z_1, \ldots, z_n)\, dz_1 \ldots dz_n}{(z_1 - z_{0,1})^{\alpha_1 + 1} \ldots (z_n - z_{0,n})^{\alpha_n + 1}} = \frac{f^{(\alpha)}(z_0)}{\alpha!}. \tag{3.9}$$

As a consequence, f is holomorphic over Ω if and only if f is $\mathbb{C}$-analytic. Arguments similar to the one variable case easily yield the

(3.10) Analytic continuation theorem. *If Ω is connected and if there exists a point $z_0 \in \Omega$ such that $f^{(\alpha)}(z_0) = 0$ for all $\alpha \in \mathbb{N}^n$, then $f = 0$ on Ω.* $\square$

Another consequence of (3.9) is the *Cauchy inequality*

$$|f^{(\alpha)}(z_0)| \leqslant \frac{\alpha!}{R^\alpha} \sup_{\Gamma(z_0,R)} |f|, \qquad \overline{D}(z_0, R) \subset \Omega, \tag{3.11}$$

From this, it follows that every bounded holomorphic function on $\mathbb{C}^n$ is constant (Liouville's theorem), and more generally, every holomorphic function F on $\mathbb{C}^n$ such that

$|F(z)| \leqslant A(1+|z|)^B$ with suitable constants $A, B \geqslant 0$ is in fact a polynomial of total degree $\leqslant B$.

We endow $\mathscr{O}(\Omega)$ with the topology of uniform convergence on compact sets $K \subset\subset \Omega$, that is, the topology induced by $C^0(\Omega, \mathbb{C})$. Then $\mathscr{O}(\Omega)$ is closed in $C^0(\Omega, \mathbb{C})$. The Cauchy inequalities (3.11) show that all derivations D^α are continuous operators on $\mathscr{O}(\Omega)$ and that any sequence $f_j \in \mathscr{O}(\Omega)$ that is uniformly bounded on all compact sets $K \subset\subset \Omega$ is locally equicontinuous. By Ascoli's theorem, we obtain

(3.12) Montel's theorem. *Every locally uniformly bounded sequence (f_j) in $\mathscr{O}(\Omega)$ has a convergent subsequence $(f_{j(\nu)})$.*

In other words, bounded subsets of the Fréchet space $\mathscr{O}(\Omega)$ are relatively compact (a Fréchet space possessing this property is called a Montel space).

§ 3.C. Differential Calculus on Complex Analytic Manifolds

A *complex analytic manifold* X of dimension $\dim_{\mathbb{C}} X = n$ is a differentiable manifold equipped with a holomorphic atlas (τ_α) with values in $\mathbb{C}^n$; this means by definition that the transition maps $\tau_{\alpha\beta}$ are holomorphic. The tangent spaces $T_{X,x}$ then have a natural complex vector space structure, given by the coordinate isomorphisms

$$d\tau_\alpha(x) : T_{X,x} \longrightarrow \mathbb{C}^n, \qquad U_\alpha \ni x\,;$$

the induced complex structure on $T_{X,x}$ is indeed independent of α since the differentials $d\tau_{\alpha\beta}$ are $\mathbb{C}$-linear isomorphisms. We denote by $T_X^{\mathbb{R}}$ the underlying real tangent space and by $J \in \mathrm{End}(T_X^{\mathbb{R}})$ the *almost complex structure*, i.e. the operator of multiplication by $\mathrm{i} = \sqrt{-1}$. If $(z_1, \ldots, z_n)$ are complex analytic coordinates on an open subset $\Omega \subset X$ and $z_k = x_k + \mathrm{i}y_k$, then $(x_1, y_1, \ldots, x_n, y_n)$ define real coordinates on Ω, and $T^{\mathbb{R}}_{X\restriction\Omega}$ admits $(\partial/\partial x_1,\ \partial/\partial y_1,\ \ldots,\ \partial/\partial x_n,\ \partial/\partial y_n)$ as a basis; the almost complex structure is given by $J(\partial/\partial x_k) = \partial/\partial y_k$, $J(\partial/\partial y_k) = -\partial/\partial x_k$. The complexified tangent space $\mathbb{C} \otimes T_X = \mathbb{C} \otimes_{\mathbb{R}} T_X^{\mathbb{R}} = T_X^{\mathbb{R}} \oplus \mathrm{i}T_X^{\mathbb{R}}$ splits into conjugate complex subspaces which are the eigenspaces of the complexified endomorphism $\mathrm{Id} \otimes J$ associated to the eigenvalues i and $-\mathrm{i}$. These subspaces have respective bases

$$\frac{\partial}{\partial z_k} = \frac{1}{2}\Big(\frac{\partial}{\partial x_k} - \mathrm{i}\frac{\partial}{\partial y_k}\Big), \qquad \frac{\partial}{\partial \overline{z}_k} = \frac{1}{2}\Big(\frac{\partial}{\partial x_k} + \mathrm{i}\frac{\partial}{\partial y_k}\Big), \qquad 1 \leqslant k \leqslant n \tag{3.13}$$

and are denoted $T^{1,0}X$ (*holomorphic vectors* or *vectors of type* $(1,0)$) and $T^{0,1}X$ (*antiholomorphic vectors* or *vectors of type* $(0,1)$). The subspaces $T^{1,0}X$ and $T^{0,1}X$ are canonically isomorphic to the complex tangent space T_X (with complex structure J) and its conjugate $\overline{T_X}$ (with conjugate complex structure $-J$), via the $\mathbb{C}$-linear embeddings

$$\begin{aligned} T_X &\longrightarrow T_X^{1,0} \subset \mathbb{C} \otimes T_X, & \overline{T_X} &\longrightarrow T_X^{0,1} \subset \mathbb{C} \otimes T_X \\ \xi &\longmapsto \frac{1}{2}(\xi - \mathrm{i}J\xi), & \xi &\longmapsto \frac{1}{2}(\xi + \mathrm{i}J\xi). \end{aligned}$$

We thus have a canonical decomposition $\mathbb{C} \otimes T_X = T_X^{1,0} \oplus T_X^{0,1} \simeq T_X \oplus \overline{T_X}$, and by duality a decomposition

$$\mathrm{Hom}_{\mathbb{R}}(T_X^{\mathbb{R}}; \mathbb{C}) \simeq \mathrm{Hom}_{\mathbb{C}}(\mathbb{C} \otimes T_X; \mathbb{C}) \simeq T_X^\star \oplus \overline{T_X^\star}$$

where $T_X^\star$ is the space of $\mathbb{C}$-linear forms and $\overline{T_X^\star}$ the space of conjugate $\mathbb{C}$-linear forms. With these notations, (dx_k, dy_k) is a basis of $\mathrm{Hom}_{\mathbb{R}}(T_{\mathbb{R}}X, \mathbb{C})$, (dz_j) a basis of $T_X^\star$, $(d\overline{z}_j)$ a basis of $\overline{T_X^\star}$, and the differential of a function $f \in C^1(\Omega, \mathbb{C})$ can be written

$$df = \sum_{k=1}^n \frac{\partial f}{\partial x_k}\, dx_k + \frac{\partial f}{\partial y_k}\, dy_k = \sum_{k=1}^n \frac{\partial f}{\partial z_k}\, dz_k + \frac{\partial f}{\partial \overline{z}_k}\, d\overline{z}_k. \tag{3.14}$$

The function f is holomorphic on Ω if and only if df is $\mathbb{C}$-linear, i.e. if and only if f satisfies the *Cauchy-Riemann equations* $\partial f/\partial \overline{z}_k = 0$ on Ω, $1 \leqslant k \leqslant n$. We still denote here by $\mathscr{O}(X)$ the algebra of holomorphic functions on X.

Now, we study the basic rules of complex differential calculus. The complexified exterior algebra $\mathbb{C} \otimes_{\mathbb{R}} \Lambda^\bullet_{\mathbb{R}}(T_X^{\mathbb{R}})^\star = \Lambda^\bullet_{\mathbb{C}}(\mathbb{C} \otimes T_X)^\star$ is given by

$$\Lambda^k(\mathbb{C} \otimes T_X)^\star = \Lambda^k\big(T_X \oplus \overline{T_X}\big)^\star = \bigoplus_{p+q=k} \Lambda^{p,q} T_X^\star, \qquad 0 \leqslant k \leqslant 2n$$

where the exterior products are taken over $\mathbb{C}$, and where the components $\Lambda^{p,q}T_X^\star$ are defined by

$$\Lambda^{p,q}T_X^\star = \Lambda^p T_X^\star \otimes \Lambda^q \overline{T_X^\star}. \tag{3.15}$$

A complex differential form u on X is said to be of *bidegree* or *type* (p,q) if its value at every point lies in the component $\Lambda^{p,q}T_X^\star$; we shall denote by $C^s(\Omega, \Lambda^{p,q}T_X^\star)$ the space of differential forms of bidegree (p,q) and class C^s on any open subset Ω of X. If Ω is a coordinate open set, such a form can be written

$$u(z) = \sum_{|I|=p, |J|=q} u_{I,J}(z)\, dz_I \wedge d\overline{z}_J, \qquad u_{I,J} \in C^s(\Omega, \mathbb{C}).$$

This writing is usually much more convenient than the expression in terms of the real basis $(dx_I \wedge dy_J)_{|I|+|J|=k}$ which is not compatible with the splitting of $\Lambda^k T^\star_{\mathbb{C}}X$ in its (p,q) components. Formula (3.14) shows that the exterior derivative d splits into $d = d' + d''$, where

$$d' : \mathscr{C}^\infty(X, \Lambda^{p,q}T_X^\star) \longrightarrow \mathscr{C}^\infty(X, \Lambda^{p+1,q}T_X^\star),$$
$$d'' : \mathscr{C}^\infty(X, \Lambda^{p,q}T_X^\star) \longrightarrow \mathscr{C}^\infty(X, \Lambda^{p,q+1}T_X^\star),$$

$$d'u = \sum_{I,J} \sum_{1 \leqslant k \leqslant n} \frac{\partial u_{I,J}}{\partial z_k}\, dz_k \wedge dz_I \wedge d\overline{z}_J, \tag{3.16$'$}$$

$$d''u = \sum_{I,J} \sum_{1 \leqslant k \leqslant n.} \frac{\partial u_{I,J}}{\partial \overline{z}_k}\, d\overline{z}_k \wedge dz_I \wedge d\overline{z}_J. \tag{3.16$''$}$$

The identity $d^2 = (d' + d'')^2 = 0$ is equivalent to

$$d'^2 = 0, \qquad d'd'' + d''d' = 0, \qquad d''^2 = 0, \tag{3.17}$$

since these three operators send (p,q)-forms in $(p+2,q)$, $(p+1,q+1)$ and $(p,q+2)$-forms, respectively. In particular, the operator d'' defines for each $p = 0, 1, \ldots, n$ a complex, called the *Dolbeault complex*

$$\mathscr{C}^\infty(X, \Lambda^{p,0}T_X^\star) \xrightarrow{d''} \cdots \longrightarrow \mathscr{C}^\infty(X, \Lambda^{p,q}T_X^\star) \xrightarrow{d''} \mathscr{C}^\infty(X, \Lambda^{p,q+1}T_X^\star)$$

and corresponding *Dolbeault cohomology groups*

$$H^{p,q}(X,\mathbb{C}) = \frac{\operatorname{Ker} d''^{\,p,q}}{\operatorname{Im} d''^{\,p,q-1}}, \tag{3.18}$$

with the convention that the image of d'' is zero for $q = 0$. The cohomology group $H^{p,0}(X,\mathbb{C})$ consists of $(p,0)$-forms $u = \sum_{|I|=p} u_I(z)\, dz_I$ such that $\partial u_I/\partial \overline{z}_k = 0$ for all I, k, i.e. such that all coefficients u_I are holomorphic. Such a form is called a *holomorphic p-form* on X.

Let $F : X_1 \longrightarrow X_2$ be a holomorphic map between complex manifolds. The pull-back $F^\star u$ of a (p,q)-form u of bidegree (p,q) on X_2 is again homogeneous of bidegree (p,q), because the components F_k of F in any coordinate chart are holomorphic, hence $F^\star dz_k = dF_k$ is $\mathbb{C}$-linear. In particular, the equality $dF^\star u = F^\star du$ implies

$$d' F^\star u = F^\star d' u, \qquad d'' F^\star u = F^\star d'' u. \tag{3.19}$$

Note that these commutation relations are no longer true for a non holomorphic change of variable. As in the case of the De Rham cohomology groups, we get a pull-back morphism

$$F^\star : H^{p,q}(X_2,\mathbb{C}) \longrightarrow H^{p,q}(X_1,\mathbb{C}).$$

The rules of complex differential calculus can be easily extended to currents. We use the following notation.

(3.20) Definition. *There are decompositions*

$$\mathscr{D}^k(X,\mathbb{C}) = \bigoplus_{p+q=k} \mathscr{D}^{p,q}(X,\mathbb{C}), \qquad \mathscr{D}'_k(X,\mathbb{C}) = \bigoplus_{p+q=k} \mathscr{D}'_{p,q}(X,\mathbb{C}).$$

The space $\mathscr{D}'_{p,q}(X,\mathbb{C})$ is called the space of currents of bidimension (p,q) and bidegree $(n-p, n-q)$ on X, and is also denoted $\mathscr{D}'^{\,n-p,n-q}(X,\mathbb{C})$.

§ 3.D. Newton and Bochner-Martinelli Kernels

The *Newton kernel* is the elementary solution of the usual Laplace operator $\Delta = \sum \partial^2/\partial x_j^2$ in $\mathbb{R}^m$. We first recall a construction of the Newton kernel.

Let $d\lambda = dx_1 \ldots dx_m$ be the Lebesgue measure on $\mathbb{R}^m$. We denote by $B(a,r)$ the euclidean open ball of center a and radius r in $\mathbb{R}^m$ and by $S(a,r) = \partial B(a,r)$ the corresponding sphere. Finally, we set $\alpha_m = \operatorname{Vol}\big(B(0,1)\big)$ and $\sigma_{m-1} = m\alpha_m$ so that

$$\operatorname{Vol}\big(B(a,r)\big) = \alpha_m r^m, \qquad \operatorname{Area}\big(S(a,r)\big) = \sigma_{m-1} r^{m-1}. \tag{3.21}$$

The second equality follows from the first by derivation. An explicit computation of the integral $\int_{\mathbb{R}^m} e^{-|x|^2} d\lambda(x)$ in polar coordinates shows that $\alpha_m = \pi^{m/2}/(m/2)!$ where $x! = \Gamma(x+1)$ is the Euler Gamma function. The *Newton kernel* is then given by:

$$\begin{cases} N(x) = \dfrac{1}{2\pi} \log|x| & \text{if } m = 2, \\ N(x) = -\dfrac{1}{(m-2)\sigma_{m-1}} |x|^{2-m} & \text{if } m \neq 2. \end{cases} \tag{3.22}$$

The function $N(x)$ is locally integrable on $\mathbb{R}^m$ and satisfies $\Delta N = \delta_0$. When $m = 2$, this follows from Cor. 3.4 and the fact that $\Delta = 4\partial^2/\partial z\partial\overline{z}$. When $m \neq 2$, this can be checked by computing the weak limit

$$\begin{aligned}\lim_{\varepsilon\to 0}\ \Delta(|x|^2+\varepsilon^2)^{1-m/2} &= \lim_{\varepsilon\to 0}\ m(2-m)\varepsilon^2(|x|^2+\varepsilon^2)^{-1-m/2}\\ &= m(2-m)\, I_m\, \delta_0\end{aligned}$$

with $I_m = \int_{\mathbb{R}^m}(|x|^2+1)^{-1-m/2}\,d\lambda(x)$. The last equality is easily seen by performing the change of variable $y = \varepsilon x$ in the integral

$$\int_{\mathbb{R}^m}\varepsilon^2(|x|^2+\varepsilon^2)^{-1-m/2}\,f(x)\,d\lambda(x) = \int_{\mathbb{R}^m}(|y|^2+1)^{-1-m/2}\,f(\varepsilon y)\,d\lambda(y),$$

where f is an arbitrary test function. Using polar coordinates, we find that $I_m = \sigma_{m-1}/m$ and our formula follows.

The *Bochner-Martinelli kernel* is the $(n, n-1)$-differential form on $\mathbb{C}^n$ with L^1_{loc} coefficients defined by

$$\begin{aligned}k_{\mathrm{BM}}(z) &= c_n\sum_{1\leqslant j\leqslant n}(-1)^j\,\frac{\overline{z}_j\,dz_1\wedge\ldots dz_n\wedge d\overline{z}_1\wedge\ldots\widehat{d\overline{z}_j}\ldots\wedge d\overline{z}_n}{|z|^{2n}},\\ c_n &= (-1)^{n(n-1)/2}\frac{(n-1)!}{(2\pi\mathrm{i})^n}.\end{aligned}\tag{3.23}$$

(3.24) Lemma. *$d''k_{\mathrm{BM}} = \delta_0$ on $\mathbb{C}^n$.*

Proof. Since the Lebesgue measure on $\mathbb{C}^n$ is

$$d\lambda(z) = \bigwedge_{1\leqslant j\leqslant n}\frac{\mathrm{i}}{2}dz_j\wedge d\overline{z}_j = \Big(\frac{\mathrm{i}}{2}\Big)^n(-1)^{\frac{n(n-1)}{2}}dz_1\wedge\ldots dz_n\wedge d\overline{z}_1\wedge\ldots d\overline{z}_n,$$

we find

$$\begin{aligned}d''k_{\mathrm{BM}} &= -\frac{(n-1)!}{\pi^n}\sum_{1\leqslant j\leqslant n}\frac{\partial}{\partial\overline{z}_j}\Big(\frac{\overline{z}_j}{|z|^{2n}}\Big)d\lambda(z)\\ &= -\frac{1}{n(n-1)\alpha_{2n}}\sum_{1\leqslant j\leqslant n}\frac{\partial^2}{\partial z_j\partial\overline{z}_j}\Big(\frac{1}{|z|^{2n-2}}\Big)d\lambda(z)\\ &= \Delta N(z)d\lambda(z) = \delta_0.\end{aligned}$$

□

We let $K_{\mathrm{BM}}(z,\zeta)$ be the pull-back of k_{BM} by the map $\pi:\mathbb{C}^n\times\mathbb{C}^n\to\mathbb{C}^n$, $(z,\zeta)\longmapsto z-\zeta$. Then Formula (2.19) implies

$$d''K_{\mathrm{BM}} = \pi^\star\delta_0 = [\Delta],\tag{3.25}$$

where $[\Delta]$ denotes the current of integration on the diagonal $\Delta\subset\mathbb{C}^n\times\mathbb{C}^n$.

(3.26) Koppelman formula. *Let $\Omega \subset \mathbb{C}^n$ be a bounded open set with piecewise C^1 boundary. Then for every (p,q)-form v of class C^1 on $\overline{\Omega}$ we have*

$$v(z) = \int_{\partial\Omega} K_{\mathrm{BM}}^{p,q}(z,\zeta) \wedge v(\zeta) \\ + d''_z \int_{\Omega} K_{\mathrm{BM}}^{p,q-1}(z,\zeta) \wedge v(\zeta) + \int_{\Omega} K_{\mathrm{BM}}^{p,q}(z,\zeta) \wedge d''v(\zeta)$$

on Ω, where $K_{\mathrm{BM}}^{p,q}(z,\zeta)$ denotes the component of $K_{\mathrm{BM}}(z,\zeta)$ of type (p,q) in z and $(n-p, n-q-1)$ in ζ.

Proof. Given $w \in \mathscr{D}^{n-p,n-q}(\Omega)$, we consider the integral

$$\int_{\partial\Omega\times\Omega} K_{\mathrm{BM}}(z,\zeta) \wedge v(\zeta) \wedge w(z).$$

It is well defined since K_{BM} has no singularities on $\partial\Omega \times \operatorname{Supp} v \subset\subset \partial\Omega \times \Omega$. Since $w(z)$ vanishes on $\partial\Omega$ the integral can be extended as well to $\partial(\Omega\times\Omega)$. As $K_{\mathrm{BM}}(z,\zeta)\wedge v(\zeta)\wedge w(z)$ is of total bidegree $(2n, 2n-1)$, its differential d' vanishes. Hence Stokes' formula yields

$$\begin{aligned}\int_{\partial\Omega\times\Omega} K_{\mathrm{BM}}(z,\zeta) \wedge v(\zeta) \wedge w(z) &= \int_{\Omega\times\Omega} d''\big(K_{\mathrm{BM}}(z,\zeta) \wedge v(\zeta) \wedge w(z)\big) \\ &= \int_{\Omega\times\Omega} d''K_{\mathrm{BM}}(z,\zeta) \wedge v(\zeta) \wedge w(z) - K_{\mathrm{BM}}^{p,q}(z,\zeta) \wedge d''v(\zeta) \wedge w(z) \\ &\quad - (-1)^{p+q} \int_{\Omega\times\Omega} K_{\mathrm{BM}}^{p,q-1}(z,\zeta) \wedge v(\zeta) \wedge d''w(z).\end{aligned}$$

By (3.25) we have

$$\int_{\Omega\times\Omega} d''K_{\mathrm{BM}}(z,\zeta) \wedge v(\zeta) \wedge w(z) = \int_{\Omega\times\Omega} [\Delta] \wedge v(\zeta) \wedge w(z) = \int_{\Omega} v(z) \wedge w(z)$$

Denoting $\langle\ ,\ \rangle$ the pairing between currents and test forms on Ω, the above equality is thus equivalent to

$$\begin{aligned}\left\langle \int_{\partial\Omega} K_{\mathrm{BM}}(z,\zeta) \wedge v(\zeta), w(z) \right\rangle &= \left\langle v(z) - \int_{\Omega} K_{\mathrm{BM}}^{p,q}(z,\zeta) \wedge d''v(\zeta), w(z) \right\rangle \\ &\quad - (-1)^{p+q} \left\langle \int_{\Omega} K_{\mathrm{BM}}^{p,q-1}(z,\zeta) \wedge v(\zeta), d''w(z) \right\rangle,\end{aligned}$$

which is itself equivalent to the Koppelman formula by integrating $d''v$ by parts. □

(3.27) Corollary. *Let $v \in {}^s\mathscr{D}^{p,q}(\mathbb{C}^n)$ be a form of class C^s with compact support such that $d''v = 0$, $q \geqslant 1$. Then the $(p,q-1)$-form*

$$u(z) = \int_{\mathbb{C}^n} K_{\mathrm{BM}}^{p,q-1}(z,\zeta) \wedge v(\zeta)$$

is a C^s solution of the equation $d''u = v$. Moreover, if $(p,q) = (0,1)$ and $n \geqslant 2$ then u has compact support, thus the Dolbeault cohomology group with compact support $H_c^{0,1}(\mathbb{C}^n, \mathbb{C})$ vanishes for $n \geqslant 2$.

Proof. Apply the Koppelman formula on a sufficiently large ball $\overline{\Omega} = \overline{B}(0, R)$ containing $\operatorname{Supp} v$. Then the formula immediately gives $d''u = v$. Observe that the coefficients of $K_{\mathrm{BM}}(z, \zeta)$ are $O(|z - \zeta|^{-(2n-1)})$, hence $|u(z)| = O(|z|^{-(2n-1)})$ at infinity. If $q = 1$, then u is holomorphic on $\mathbb{C}^n \smallsetminus \overline{B}(0, R)$. Now, this complement is a union of complex lines when $n \geqslant 2$, hence $u = 0$ on $\mathbb{C}^n \smallsetminus \overline{B}(0, R)$ by Liouville's theorem. □

(3.28) Hartogs extension theorem. *Let Ω be an open set in $\mathbb{C}^n$, $n \geqslant 2$, and let $K \subset \Omega$ be a compact subset such that $\Omega \smallsetminus K$ is connected. Then every holomorphic function $f \in \mathscr{O}(\Omega \smallsetminus K)$ extends into a function $\widetilde{f} \in \mathscr{O}(\Omega)$.*

Proof. Let $\psi \in \mathscr{D}(\Omega)$ be a cut-off function equal to 1 on a neighborhood of K. Set $f_0 = (1 - \psi) f \in \mathscr{C}^\infty(\Omega)$, defined as 0 on K. Then $v = d'' f_0 = -f d''\psi$ can be extended by 0 outside Ω, and can thus be seen as a smooth $(0, 1)$-form with compact support in $\mathbb{C}^n$, such that $d''v = 0$. By Cor. 3.27, there is a smooth function u with compact support in $\mathbb{C}^n$ such that $d''u = v$. Then $\widetilde{f} = f_0 - u \in \mathscr{O}(\Omega)$. Now u is holomorphic outside $\operatorname{Supp} \psi$, so u vanishes on the unbounded component G of $\mathbb{C}^n \smallsetminus \operatorname{Supp} \psi$. The boundary ∂G is contained in $\partial \operatorname{Supp} \psi \subset \Omega \smallsetminus K$, so $\widetilde{f} = (1 - \psi) f - u$ coincides with f on the non empty open set $\Omega \cap G \subset \Omega \smallsetminus K$. Therefore $\widetilde{f} = f$ on the connected open set $\Omega \smallsetminus K$. □

A refined version of the Hartogs extension theorem due to Bochner will be given in Exercise 8.13. It shows that f need only be given as a C^1 function on $\partial\Omega$, satisfying the tangential Cauchy-Riemann equations (a so-called *CR-function*). Then f extends as a holomorphic function $\widetilde{f} \in \mathscr{O}(\Omega) \cap C^0(\overline{\Omega})$, provided that $\partial\Omega$ is connected.

§ 3.E. The Dolbeault-Grothendieck Lemma

We are now in a position to prove the Dolbeault-Grothendieck lemma [Dolbeault 1953], which is the analogue for d'' of the Poincaré lemma. The proof given below makes use of the Bochner-Martinelli kernel. Many other proofs can be given, e.g. by using a reduction to the one dimensional case in combination with the Cauchy formula (3.2), see Exercise 8.5 or [Hörmander 1966].

(3.29) Dolbeault-Grothendieck lemma. *Let Ω be a neighborhood of 0 in $\mathbb{C}^n$ and $v \in {}^s\mathscr{E}^{p,q}(\Omega, \mathbb{C})$, [resp. $v \in {}^s\mathscr{D}'^{p,q}(\Omega, \mathbb{C})$], such that $d''v = 0$, where $1 \leqslant s \leqslant \infty$.*

a) *If $q = 0$, then $v(z) = \sum_{|I|=p} v_I(z)\, dz_I$ is a holomorphic p-form, i.e. a form whose coefficients are holomorphic functions.*

b) *If $q \geqslant 1$, there exists a neighborhood $\omega \subset \Omega$ of 0 and a form u in ${}^s\mathscr{E}^{p,q-1}(\omega, \mathbb{C})$ [resp. a current $u \in {}^s\mathscr{D}'^{p,q-1}(\omega, \mathbb{C})$] such that $d''u = v$ on ω.*

Proof. We assume that Ω is a ball $B(0, r) \subset \mathbb{C}^n$ and take for simplicity $r > 1$ (possibly after a dilation of coordinates). We then set $\omega = B(0, 1)$. Let $\psi \in \mathscr{D}(\Omega)$ be a cut-off function equal to 1 on ω. The Koppelman formula (3.26) applied to the form ψv on Ω gives

$$\psi(z) v(z) = d''_z \int_\Omega K_{\mathrm{BM}}^{p,q-1}(z, \zeta) \wedge \psi(\zeta) v(\zeta) + \int_\Omega K_{\mathrm{BM}}^{p,q}(z, \zeta) \wedge d''\psi(\zeta) \wedge v(\zeta).$$

This formula is valid even when v is a current, because we may regularize v as $v \star \rho_\varepsilon$ and take the limit. We introduce on $\mathbb{C}^n \times \mathbb{C}^n \times \mathbb{C}^n$ the kernel

$$K(z,w,\zeta) = c_n \sum_{j=1}^n \frac{(-1)^j (w_j - \overline{\zeta}_j)}{((z-\zeta)\cdot(w-\overline{\zeta}))^n} \bigwedge_k (dz_k - d\zeta_k) \wedge \bigwedge_{k\neq j} (dw_k - d\overline{\zeta}_k).$$

By construction, $K_{\mathrm{BM}}(z,\zeta)$ is the result of the substitution $w = \overline{z}$ in $K(z,w,\zeta)$, i.e. $K_{\mathrm{BM}} = h^\star K$ where $h(z,\zeta) = (z,\overline{z},\zeta)$. We denote by $K^{p,q}$ the component of K of bidegree $(p,0)$ in z, $(q,0)$ in w and $(n-p, n-q-1)$ in ζ. Then $K^{p,q}_{\mathrm{BM}} = h^\star K^{p,q}$ and we find

$$v = d'' u_0 + g^\star v_1 \quad \text{on } \omega,$$

where $g(z) = (z,\overline{z})$ and

$$u_0(z) = \int_\Omega K^{p,q-1}_{\mathrm{BM}}(z,\zeta) \wedge \psi(\zeta) v(\zeta),$$
$$v_1(z,w) = \int_\Omega K^{p,q}(z,w,\zeta) \wedge d''\psi(\zeta) \wedge v(\zeta).$$

By definition of $K^{p,q}(z,w,\zeta)$, v_1 is holomorphic on the open set

$$U = \big\{(z,w) \in \omega \times \omega\,;\, \forall \zeta \notin \omega,\ \mathrm{Re}(z-\zeta)\cdot(w-\overline{\zeta}) > 0\big\},$$

which contains the "conjugate-diagonal" points $(z,\overline{z})$ as well as the points $(z,0)$ and $(0,w)$ in $\omega \times \omega$. Moreover U clearly has convex slices $(\{z\} \times \mathbb{C}^n) \cap U$ and $(\mathbb{C}^n \times \{w\}) \cap U$. In particular U is starshaped with respect to w, i.e.

$$(z,w) \in U \Longrightarrow (z,tw) \in U, \qquad \forall t \in [0,1].$$

As u_1 is of type $(p,0)$ in z and $(q,0)$ in w, we get $d''_z(g^\star v_1) = g^\star d_w v_1 = 0$, hence $d_w v_1 = 0$. For $q = 0$ we have $K^{p,q-1}_{\mathrm{BM}} = 0$, thus $u_0 = 0$, and v_1 does not depend on w, thus v is holomorphic on ω. For $q \geqslant 1$, we can use the homotopy formula (1.23) with respect to w (considering z as a parameter) to get a holomorphic form $u_1(z,w)$ of type $(p,0)$ in z and $(q-1,0)$ in w, such that $d_w u_1(z,w) = v_1(z,w)$. Then we get $d'' g^\star u_1 = g^\star d_w u_1 = g^\star v_1$, hence

$$v = d''(u_0 + g^\star u_1) \quad \text{on } \omega.$$

Finally, the coefficients of u_0 are obtained as linear combinations of convolutions of the coefficients of ψv with L^1_{loc} functions of the form $\overline{\zeta}_j |\zeta|^{-2n}$. Hence u_0 is of class C^s (resp. is a current of order s), if v is. □

(3.30) Corollary. *The operator d'' is hypoelliptic in bidegree $(p,0)$, i.e. if a current $f \in \mathscr{D}'^{p,0}(X,\mathbb{C})$ satisfies $d''f \in \mathscr{E}^{p,1}(X,\mathbb{C})$, then $f \in \mathscr{E}^{p,0}(X,\mathbb{C})$.*

Proof. The result is local, so we may assume that $X = \Omega$ is a neighborhood of 0 in $\mathbb{C}^n$. The $(p,1)$-form $v = d''f \in \mathscr{E}^{p,1}(X,\mathbb{C})$ satisfies $d''v = 0$, hence there exists $u \in \mathscr{E}^{p,0}(\widetilde{\Omega},\mathbb{C})$ such that $d''u = d''f$. Then $f - u$ is holomorphic and $f = (f-u) + u \in \mathscr{E}^{p,0}(\widetilde{\Omega},\mathbb{C})$. □

§ 4. Subharmonic Functions

A *harmonic* (resp. *subharmonic*) function on an open subset of $\mathbb{R}^m$ is essentially a function (or distribution) u such that $\Delta u = 0$ (resp. $\Delta u \geqslant 0$). A fundamental example of subharmonic function is given by the Newton kernel N, which is actually harmonic on $\mathbb{R}^m \smallsetminus \{0\}$. Subharmonic functions are an essential tool of harmonic analysis and potential theory. Before giving their precise definition and properties, we derive a basic integral formula involving the Green kernel of the Laplace operator on the ball.

§ 4.A. Construction of the Green Kernel

The *Green kernel* $G_\Omega(x,y)$ of a smoothly bounded domain $\Omega \subset\subset \mathbb{R}^m$ is the solution of the following *Dirichlet boundary problem* for the Laplace operator Δ on Ω:

(4.1) Definition. *The Green kernel of a smoothly bounded domain $\Omega \subset\subset \mathbb{R}^m$ is a function $G_\Omega(x,y) : \overline{\Omega} \times \overline{\Omega} \to [-\infty, 0]$ with the following properties:*

a) $G_\Omega(x,y)$ *is* $\mathscr{C}^\infty$ *on* $\overline{\Omega} \times \overline{\Omega} \smallsetminus \mathrm{Diag}_\Omega$ $\quad$ ($\mathrm{Diag}_\Omega =$ *diagonal*) ;

b) $G_\Omega(x,y) = G_\Omega(y,x)$;

c) $G_\Omega(x,y) < 0$ *on* $\Omega \times \Omega$ *and* $G_\Omega(x,y) = 0$ *on* $\partial\Omega \times \Omega$;

d) $\Delta_x G_\Omega(x,y) = \delta_y$ *on* Ω *for every fixed* $y \in \Omega$.

It can be shown that G_Ω always exists and is unique. The uniqueness is an easy consequence of the maximum principle (see Th. 4.14 below). In the case where $\Omega = B(0,r)$ is a ball (the only case we are going to deal with), the existence can be shown through explicit calculations. In fact the Green kernel $G_r(x,y)$ of $B(0,r)$ is

$$G_r(x,y) = N(x-y) - N\Big(\frac{|y|}{r}\Big(x - \frac{r^2}{|y|^2}\,y\Big)\Big), \qquad x,y \in \overline{B}(0,r). \tag{4.2}$$

A substitution of the explicit value of $N(x)$ yields:

$$G_r(x,y) = \frac{1}{4\pi}\log\frac{|x-y|^2}{r^2 - 2\langle x,y\rangle + \frac{1}{r^2}|x|^2\,|y|^2} \qquad \text{if } \ m=2, \quad \text{otherwise}$$

$$G_r(x,y) = \frac{-1}{(m-2)\sigma_{m-1}}\Big(|x-y|^{2-m} - \big(r^2 - 2\langle x,y\rangle + \frac{1}{r^2}|x|^2\,|y|^2\big)^{1-m/2}\Big).$$

(4.3) Theorem. *The above defined function G_r satisfies all four properties* (4.1 a–d) *on $\Omega = B(0,r)$, thus G_r is the Green kernel of $B(0,r)$.*

Proof. The first three properties are immediately verified on the formulas, because

$$r^2 - 2\langle x,y\rangle + \frac{1}{r^2}|x|^2\,|y|^2 = |x-y|^2 + \frac{1}{r^2}\big(r^2 - |x|^2\big)\big(r^2 - |y|^2\big).$$

For property d), observe that $r^2 y/|y|^2 \notin \overline{B}(0,r)$ whenever $y \in B(0,r) \smallsetminus \{0\}$. The second Newton kernel in the right hand side of (4.1) is thus harmonic in x on $B(0,r)$, and

$$\Delta_x G_r(x,y) = \Delta_x N(x-y) = \delta_y \quad \text{on} \quad B(0,r). \qquad \square$$

§ 4.B. Green-Riesz Representation Formula and Dirichlet Problem

§ 4.B.1. *Green-Riesz Formula.* For all smooth functions u, v on a smoothly bounded domain $\Omega \subset\subset \mathbb{R}^m$, we have

$$\int_\Omega (u\,\Delta v - v\,\Delta u)\,d\lambda = \int_{\partial\Omega} \Big(u\,\frac{\partial v}{\partial \nu} - v\,\frac{\partial u}{\partial \nu}\Big)\,d\sigma \tag{4.4}$$

where $\partial/\partial\nu$ is the derivative along the outward normal unit vector ν of $\partial\Omega$ and $d\sigma$ the euclidean area measure. Indeed

$$(-1)^{j-1}\,dx_1 \wedge \ldots \wedge \widehat{dx_j} \wedge \ldots \wedge dx_{m\,\restriction\partial\Omega} = \nu_j\,d\sigma,$$

for the wedge product of $\langle \nu, dx\rangle$ with the left hand side is $\nu_j\,d\lambda$. Therefore

$$\frac{\partial v}{\partial \nu}\,d\sigma = \sum_{j=1}^m \frac{\partial v}{\partial x_j}\,\nu_j\,d\sigma = \sum_{j=1}^m (-1)^{j-1}\frac{\partial v}{\partial x_j}\,dx_1 \wedge \ldots \wedge \widehat{dx_j} \wedge \ldots \wedge dx_m.$$

Formula (4.4) is then an easy consequence of Stokes' theorem. Observe that (4.4) is still valid if v is a distribution with singular support relatively compact in Ω. For $\Omega = B(0,r)$, $u \in C^2\big(\overline{B}(0,r), \mathbb{R}\big)$ and $v(y) = G_r(x,y)$, we get the *Green-Riesz representation formula*:

$$u(x) = \int_{B(0,r)} \Delta u(y)\,G_r(x,y)\,d\lambda(y) + \int_{S(0,r)} u(y)\,P_r(x,y)\,d\sigma(y) \tag{4.5}$$

where $P_r(x,y) = \partial G_r(x,y)/\partial\nu(y)$, $(x,y) \in B(0,r) \times S(0,r)$. The function $P_r(x,y)$ is called the *Poisson kernel*. It is smooth and satisfies $\Delta_x P_r(x,y) = 0$ on $B(0,r)$ by (4.1 d). A simple computation left to the reader yields:

$$P_r(x,y) = \frac{1}{\sigma_{m-1} r}\,\frac{r^2 - |x|^2}{|x-y|^m}. \tag{4.6}$$

Formula (4.5) for $u \equiv 1$ shows that $\int_{S(0,r)} P_r(x,y)\,d\sigma(y) = 1$. When x in $B(0,r)$ tends to $x_0 \in S(0,r)$, we see that $P_r(x,y)$ converges uniformly to 0 on every compact subset of $S(0,r) \smallsetminus \{x_0\}$; it follows that the measure $P_r(x,y)\,d\sigma(y)$ converges weakly to δ_{x_0} on $S(0,r)$.

§ 4.B.2. *Solution of the Dirichlet Problem.* For any bounded measurable function v on $S(a,r)$ we define

$$P_{a,r}[v](x) = \int_{S(a,r)} v(y)\,P_r(x-a, y-a)\,d\sigma(y), \qquad x \in B(a,r). \tag{4.7}$$

If $u \in C^0\big(\overline{B}(a,r), \mathbb{R}\big) \cap C^2\big(B(a,r), \mathbb{R}\big)$ is harmonic, i.e. $\Delta u = 0$ on $B(a,r)$, then (4.5) gives $u = P_{a,r}[u]$ on $B(a,r)$, i.e. the Poisson kernel reproduces harmonic functions. Suppose now that $v \in C^0\big(S(a,r), \mathbb{R}\big)$ is given. Then $P_r(x-a, y-a)\,d\sigma(y)$ converges weakly to δ_{x_0} when x tends to $x_0 \in S(a,r)$, so $P_{a,r}[v](x)$ converges to $v(x_0)$. It follows that the function u defined by

$$\begin{cases} u = P_{a,r}[v] & \text{on } B(a,r), \\ u = v & \text{on } S(a,r) \end{cases}$$

is continuous on $\overline{B}(a,r)$ and harmonic on $B(a,r)$; thus u is the solution of the Dirichlet problem with boundary values v.

§ 4.C. Definition and Basic Properties of Subharmonic Functions

§ 4.C.1. *Definition. Mean Value Inequalities.* If u is a Borel function on $\overline{B}(a,r)$ which is bounded above or below, we consider the mean values of u over the ball or sphere:

$$\mu_B(u\,;a,r) = \frac{1}{\alpha_m r^m}\int_{B(a,r)} u(x)\,d\lambda(x), \tag{4.8}$$

$$\mu_S(u\,;a,r) = \frac{1}{\sigma_{m-1} r^{m-1}}\int_{S(a,r)} u(x)\,d\sigma(x). \tag{4.8$'$}$$

As $d\lambda = dr\,d\sigma$ these mean values are related by

$$\begin{aligned}\mu_B(u\,;a,r) &= \frac{1}{\alpha_m r^m}\int_0^r \sigma_{m-1} t^{m-1}\,\mu_S(u\,;a,t)\,dt\\ &= m\int_0^1 t^{m-1}\,\mu_S(u\,;a,rt)\,dt.\end{aligned} \tag{4.9}$$

Now, apply formula (4.5) with $x=0$. We get $P_r(0,y) = 1/\sigma_{m-1}r^{m-1}$ and $G_r(0,y) = (|y|^{2-m} - r^{2-m})/(2-m)\sigma_{m-1} = -(1/\sigma_{m-1})\int_{|y|}^r t^{1-m}dt$, thus

$$\begin{aligned}\int_{B(0,r)} \Delta u(y)\,G_r(0,y)\,d\lambda(y) &= -\frac{1}{\sigma_{m-1}}\int_0^r \frac{dt}{t^{m-1}}\int_{|y|<t}\Delta u(y)\,d\lambda(y)\\ &= -\frac{1}{m}\int_0^r \mu_B(\Delta u\,;0,t)\,t\,dt\end{aligned}$$

thanks to the Fubini formula. By translating $S(0,r)$ to $S(a,r)$, (4.5) implies the *Gauss formula*

$$\mu_S(u\,;a,r) = u(a) + \frac{1}{m}\int_0^r \mu_B(\Delta u\,;a,t)\,t\,dt. \tag{4.10}$$

Let Ω be an open subset of $\mathbb{R}^m$ and $u \in C^2(\Omega,\mathbb{R})$. If $a\in\Omega$ and $\Delta u(a) > 0$ (resp. $\Delta u(a) < 0$), Formula (4.10) shows that $\mu_S(u\,;a,r) > u(a)$ (resp. $\mu_S(u\,;a,r) < u(a)$) for r small enough. In particular, u is harmonic (i.e. $\Delta u = 0$) if and only if u satisfies the *mean value equality*

$$\mu_S(u\,;a,r) = u(a), \qquad \forall \overline{B}(a,r)\subset\Omega.$$

Now, observe that if (ρ_ε) is a family of radially symmetric smoothing kernels associated with $\rho(x) = \widetilde{\rho}(|x|)$ and if u is a Borel locally bounded function, an easy computation yields

$$\begin{aligned}u\star\rho_\varepsilon(a) &= \int_{B(0,1)} u(a+\varepsilon x)\,\rho(x)\,d\lambda\\ &= \sigma_{m-1}\int_0^1 \mu_S(u\,;a,\varepsilon t)\,\widetilde{\rho}(t)\,t^{m-1}\,dt.\end{aligned} \tag{4.11}$$

Thus, if u is a Borel locally bounded function satisfying the mean value equality on Ω, (4.11) shows that $u \star \rho_\varepsilon = u$ on Ω_ε, in particular u must be smooth. Similarly, if we replace the mean value equality by an inequality, the relevant regularity property to be required for u is just semicontinuity.

(4.12) Theorem and definition. *Let $u : \Omega \longrightarrow [-\infty, +\infty[$ be an upper semicontinuous function. The following various forms of mean value inequalities are equivalent:*

a) $u(x) \leqslant P_{a,r}[u](x), \quad \forall \overline{B}(a,r) \subset \Omega, \quad \forall x \in B(a,r)$;

b) $u(a) \leqslant \mu_S(u\,; a, r), \quad \forall \overline{B}(a,r) \subset \Omega$;

c) $u(a) \leqslant \mu_B(u\,; a, r), \quad \forall \overline{B}(a,r) \subset \Omega$;

d) *for every $a \in \Omega$, there exists a sequence (r_ν) decreasing to 0 such that*

$$u(a) \leqslant \mu_B(u\,; a, r_\nu) \qquad \forall \nu \ ;$$

e) *for every $a \in \Omega$, there exists a sequence (r_ν) decreasing to 0 such that*

$$u(a) \leqslant \mu_S(u\,; a, r_\nu) \qquad \forall \nu.$$

A function u satisfying one of the above properties is said to be subharmonic on Ω. The set of subharmonic functions will be denoted by $\mathrm{Sh}(\Omega)$.

By (4.10) we see that a function $u \in C^2(\Omega, \mathbb{R})$ is subharmonic if and only if $\Delta u \geqslant 0$: in fact $\mu_S(u\,; a, r) < u(a)$ for r small if $\Delta u(a) < 0$. It is also clear on the definitions that every (locally) convex function on Ω is subharmonic.

Proof. We have obvious implications

$$\text{a)} \Longrightarrow \text{b)} \Longrightarrow \text{c)} \Longrightarrow \text{d)} \Longrightarrow \text{e)},$$

the second and last ones by (4.10) and the fact that $\mu_B(u\,; a, r_\nu) \leqslant \mu_S(u\,; a, t)$ for at least one $t \in]0, r_\nu[$. In order to prove e) $\Longrightarrow$ a), we first need a suitable version of the maximum principle.

(4.13) Lemma. *Let $u : \Omega \longrightarrow [-\infty, +\infty[$ be an upper semicontinuous function satisfying property 4.12 e). If u attains its supremum at a point $x_0 \in \Omega$, then u is constant on the connected component of x_0 in Ω.*

Proof. We may assume that Ω is connected. Let

$$W = \{x \in \Omega \ ; \ u(x) < u(x_0)\}.$$

W is open by the upper semicontinuity, and distinct from Ω since $x_0 \notin W$. We want to show that $W = \emptyset$. Otherwise W has a non empty connected component W_0, and W_0 has a boundary point $a \in \Omega$. We have $a \in \Omega \smallsetminus W$, thus $u(a) = u(x_0)$. By assumption 4.12 e), we get $u(a) \leqslant \mu_S(u\,; a, r_\nu)$ for some sequence $r_\nu \to 0$. For r_ν small enough, W_0 intersects $\Omega \smallsetminus \overline{B}(a, r_\nu)$ and $B(a, r_\nu)$; as W_0 is connected, we also have $S(a, r_\nu) \cap W_0 \neq \emptyset$. Since

$u \leqslant u(x_0)$ on the sphere $S(a, r_\nu)$ and $u < u(x_0)$ on its open subset $S(a, r_\nu) \cap W_0$, we get $u(a) \leqslant \mu_S(u\,; a, r) < u(x_0)$, a contradiction. □

(4.14) Maximum principle. *If u is subharmonic in Ω (in the sense that u satisfies the weakest property 4.12 e)), then*

$$\sup_\Omega u = \limsup_{\Omega \ni z \to \partial\Omega \cup \{\infty\}} u(z),$$

and $\sup_K u = \sup_{\partial K} u(z)$ for every compact subset $K \subset \Omega$.

Proof. We have of course $\limsup_{z \to \partial\Omega\cup\{\infty\}} u(z) \leqslant \sup_\Omega u$. If the inequality is strict, this means that the supremum is achieved on some compact subset $L \subset \Omega$. Thus, by the upper semicontinuity, there is $x_0 \in L$ such that $\sup_\Omega u = \sup_L u = u(x_0)$. Lemma 4.13 shows that u is constant on the connected component Ω_0 of x_0 in Ω, hence

$$\sup_\Omega u = u(x_0) = \limsup_{\Omega_0 \ni z \to \partial\Omega_0 \cup \{\infty\}} u(z) \leqslant \limsup_{\Omega \ni z \to \partial\Omega \cup \{\infty\}} u(z),$$

contradiction. The statement involving a compact subset K is obtained by applying the first statement to $\Omega' = K^\circ$. □

Proof of (4.12) e) $\Longrightarrow$ a). Let u be an upper semicontinuous function satisfying 4.12 e) and $\overline{B}(a, r) \subset \Omega$ an arbitrary closed ball. One can find a decreasing sequence of continuous functions $v_k \in C^0\big(S(a, r), \mathbb{R}\big)$ such that $\lim v_k = u$. Set $h_k = P_{a,r}[v_k] \in C^0\big(\overline{B}(a, r), \mathbb{R}\big)$. As h_k is harmonic on $B(a, r)$, the function $u - h_k$ satisfies 4.12 e) on $B(a, r)$. Furthermore $\limsup_{x \to \xi \in S(a,r)} u(x) - h_k(x) \leqslant u(\xi) - v_k(\xi) \leqslant 0$, so $u - h_k \leqslant 0$ on $B(a, r)$ by Th. 4.14. By monotone convergence, we find $u \leqslant P_{a,r}[u]$ on $B(a, r)$ when k tends to $+\infty$. □

§ 4.C.2. *Basic Properties.* Here is a short list of the most basic properties.

(4.15) Theorem. *For any decreasing sequence (u_k) of subharmonic functions, the limit $u = \lim u_k$ is subharmonic.*

Proof. A decreasing limit of upper semicontinuous functions is again upper semicontinuous, and the mean value inequalities 4.12 remain valid for u by Lebesgue's monotone convergence theorem. □

(4.16) Theorem. *Let $u_1, \ldots, u_p \in \mathrm{Sh}(\Omega)$ and $\chi : \mathbb{R}^p \longrightarrow \mathbb{R}$ be a convex function such that $\chi(t_1, \ldots, t_p)$ is non decreasing in each t_j. If χ is extended by continuity into a function $[-\infty, +\infty[^p \longrightarrow [-\infty, +\infty[$, then*

$$\chi(u_1, \ldots, u_p) \in \mathrm{Sh}(\Omega).$$

In particular $u_1 + \cdots + u_p$, $\max\{u_1, \ldots, u_p\}$, $\log(e^{u_1} + \cdots + e^{u_p}) \in \mathrm{Sh}(\Omega)$.

Proof. Every convex function is continuous, hence $\chi(u_1, \ldots, u_p)$ is upper semicontinuous. One can write

$$\chi(t) = \sup_{i \in I} A_i(t)$$

where $A_i(t) = a_1 t_1 + \cdots + a_p t_p + b$ is the family of affine functions that define supporting hyperplanes of the graph of χ. As $\chi(t_1, \ldots, t_p)$ is non-decreasing in each t_j, we have $a_j \geqslant 0$,

thus

$$\sum_{1\leqslant j\leqslant p} a_j u_j(x) + b \leqslant \mu_B\big(\sum a_j u_j + b\,;x,r\big) \leqslant \mu_B\big(\chi(u_1,\ldots,u_p)\,;x,r\big)$$

for every ball $\overline{B}(x,r)\subset\Omega$. If one takes the supremum of this inequality over all the A_i 's, it follows that $\chi(u_1,\ldots,u_p)$ satisfies the mean value inequality 4.12 c). In the last example, the function $\chi(t_1,\ldots,t_p)=\log(e^{t_1}+\cdots+e^{t_p})$ is convex because

$$\sum_{1\leqslant j,k\leqslant p}\frac{\partial^2\chi}{\partial t_j\partial t_k}\,\xi_j\xi_k = e^{-\chi}\sum\xi_j^2\,e^{t_j} - e^{-2\chi}\big(\sum\xi_j\,e^{t_j}\big)^2$$

and $\big(\sum\xi_j\,e^{t_j}\big)^2\leqslant\big(\sum\xi_j^2\,e^{t_j}\big)\,e^{\chi}$ by the Cauchy-Schwarz inequality. □

(4.17) Theorem. *If Ω is connected and $u\in\mathrm{Sh}(\Omega)$, then either $u\equiv-\infty$ or $u\in L^1_{\mathrm{loc}}(\Omega)$.*

Proof. Note that a subharmonic function is always locally bounded above. Let W be the set of points $x\in\Omega$ such that u is integrable in a neighborhood of x. Then W is open by definition and $u>-\infty$ almost everywhere on W. If $x\in\overline{W}$, one can choose $a\in W$ such that $|a-x|<r=\frac{1}{2}d(x,\complement\Omega)$ and $u(a)>-\infty$. Then $B(a,r)$ is a neighborhood of x, $\overline{B}(a,r)\subset\Omega$ and $\mu_B(u\,;a,r)\geqslant u(a)>-\infty$. Therefore $x\in W$, W is also closed. We must have $W=\Omega$ or $W=\emptyset$; in the last case $u\equiv-\infty$ by the mean value inequality. □

(4.18) Theorem. *Let $u\in\mathrm{Sh}(\Omega)$ be such that $u\not\equiv-\infty$ on each connected component of Ω. Then*

a) *$r\longmapsto\mu_S(u\,;a,r)$, $r\longmapsto\mu_B(u\,;a,r)$ are non decreasing functions in the interval $]0,d(a,\complement\Omega)[$, and $\mu_B(u\,;a,r)\leqslant\mu_S(u\,;a,r)$.*

b) *For any family (ρ_ε) of smoothing kernels, $u\star\rho_\varepsilon\in\mathrm{Sh}(\Omega_\varepsilon)\cap\mathscr{C}^\infty(\Omega_\varepsilon,\mathbb{R})$, the family $(u\star\rho_\varepsilon)$ is non decreasing in ε and $\lim_{\varepsilon\to0}u\star\rho_\varepsilon=u$.*

Proof. We first verify statements a) and b) when $u\in C^2(\Omega,\mathbb{R})$. Then $\Delta u\geqslant0$ and $\mu_S(u\,;a,r)$ is non decreasing in virtue of (4.10). By (4.9), we find that $\mu_B(u\,;a,r)$ is also non decreasing and that $\mu_B(u\,;a,r)\leqslant\mu_S(u\,;a,r)$. Furthermore, Formula (4.11) shows that $\varepsilon\longmapsto u\star\rho_\varepsilon(a)$ is non decreasing (provided that ρ_ε is radially symmetric).

In the general case, we first observe that property 4.12 c) is equivalent to the inequality

$$u\leqslant u\star\mu_r\quad\text{ on }\ \Omega_r,\qquad\forall r>0,$$

where μ_r is the probability measure of uniform density on $B(0,r)$. This inequality implies $u\star\rho_\varepsilon\leqslant u\star\rho_\varepsilon\star\mu_r$ on $(\Omega_r)_\varepsilon=\Omega_{r+\varepsilon}$, thus $u\star\rho_\varepsilon\in\mathscr{C}^\infty(\Omega_\varepsilon,\mathbb{R})$ is subharmonic on Ω_ε. It follows that $u\star\rho_\varepsilon\star\rho_\eta$ is non decreasing in η ; by symmetry, it is also non decreasing in ε, and so is $u\star\rho_\varepsilon=\lim_{\eta\to0}u\star\rho_\varepsilon\star\rho_\eta$. We have $u\star\rho_\varepsilon\geqslant u$ by (4.19) and $\limsup_{\varepsilon\to0}u\star\rho_\varepsilon\leqslant u$ by the upper semicontinuity. Hence $\lim_{\varepsilon\to0}u\star\rho_\varepsilon=u$. Property a) for u follows now from its validity for $u\star\rho_\varepsilon$ and from the monotone convergence theorem. □

(4.19) Corollary. *If $u\in\mathrm{Sh}(\Omega)$ is such that $u\not\equiv-\infty$ on each connected component of Ω, then Δu computed in the sense of distribution theory is a positive measure.*

Indeed $\Delta(u \star \rho_\varepsilon) \geqslant 0$ as a function, and $\Delta(u \star \rho_\varepsilon)$ converges weakly to Δu in $\mathscr{D}'(\Omega)$. Corollary 4.19 has a converse, but the correct statement is slightly more involved than for the direct property:

(4.20) Theorem. *If $v \in \mathscr{D}'(\Omega)$ is such that Δv is a positive measure, there exists a unique function $u \in \mathrm{Sh}(\Omega)$ locally integrable such that v is the distribution associated to u.*

We must point out that u need not coincide everywhere with v, even when v is a locally integrable upper semicontinuous function: for example, if v is the characteristic function of a compact subset $K \subset \Omega$ of measure 0, the subharmonic representant of v is $u = 0$.

Proof. Set $v_\varepsilon = v \star \rho_\varepsilon \in \mathscr{C}^\infty(\Omega_\varepsilon, \mathbb{R})$. Then $\Delta v_\varepsilon = (\Delta v) \star \rho_\varepsilon \geqslant 0$, thus $v_\varepsilon \in \mathrm{Sh}(\Omega_\varepsilon)$. Arguments similar to those in the proof of Th. 4.18 show that (v_ε) is non decreasing in ε. Then $u := \lim_{\varepsilon \to 0} v_\varepsilon \in \mathrm{Sh}(\Omega)$ by Th. 4.15. Since v_ε converges weakly to v, the monotone convergence theorem shows that

$$\langle v, f \rangle = \lim_{\varepsilon \to 0} \int_\Omega v_\varepsilon \, f \, d\lambda = \int_\Omega u \, f \, d\lambda, \qquad \forall f \in \mathscr{D}(\Omega), \quad f \geqslant 0,$$

which concludes the existence part. The uniqueness of u is clear from the fact that u must satisfy $u = \lim u \star \rho_\varepsilon = \lim v \star \rho_\varepsilon$. □

The most natural topology on the space $\mathrm{Sh}(\Omega)$ of subharmonic functions is the topology induced by the vector space topology of $L^1_{\mathrm{loc}}(\Omega)$ (Fréchet topology of convergence in L^1 norm on every compact subset of Ω).

(4.21) Proposition. *The convex cone $\mathrm{Sh}(\Omega) \cap L^1_{\mathrm{loc}}(\Omega)$ is closed in $L^1_{\mathrm{loc}}(\Omega)$, and it has the property that every bounded subset is relatively compact.*

Proof. Let (u_j) be a sequence in $\mathrm{Sh}(\Omega) \cap L^1_{\mathrm{loc}}(\Omega)$. If $u_j \to u$ in $L^1_{\mathrm{loc}}(\Omega)$ then $\Delta u_j \to \Delta u$ in the weak topology of distributions, hence $\Delta u \geqslant 0$ and u can be represented by a subharmonic function thanks to Th. 4.20. Now, suppose that $\|u_j\|_{L^1(K)}$ is uniformly bounded for every compact subset K of Ω. Let $\mu_j = \Delta u_j \geqslant 0$. If $\psi \in \mathscr{D}(\Omega)$ is a test function equal to 1 on a neighborhood ω of K and such that $0 \leqslant \psi \leqslant 1$ on Ω, we find

$$\mu_j(K) \leqslant \int_\Omega \psi \, \Delta u_j \, d\lambda = \int_\Omega \Delta\psi \, u_j \, d\lambda \leqslant C \|u_j\|_{L^1(K')},$$

where $K' = \mathrm{Supp}\, \psi$, hence the sequence of measures (μ_j) is uniformly bounded in mass on every compact subset of Ω. By weak compactness, there is a subsequence (μ_{j_ν}) which converges weakly to a positive measure μ on Ω. We claim that $f \star (\psi \mu_{j_\nu})$ converges to $f \star (\psi\mu)$ in $L^1_{\mathrm{loc}}(\mathbb{R}^m)$ for every function $f \in L^1_{\mathrm{loc}}(\mathbb{R}^m)$. In fact, this is clear if $f \in \mathscr{C}^\infty(\mathbb{R}^m)$, and in general we use an approximation of f by a smooth function g together with the estimate

$$\|(f - g) \star (\psi\mu_{j_\nu})\|_{L^1(A)} \leqslant \|(f - g)\|_{L^1(A+K')} \mu_{j_\nu}(K'), \qquad \forall A \subset\subset \mathbb{R}^m$$

to get the conclusion. We apply this when $f = N$ is the Newton kernel. Then $h_j = u_j - N \star (\psi\mu_j)$ is harmonic on ω and bounded in $L^1(\omega)$. As $h_j = h_j \star \rho_\varepsilon$ for any smoothing kernel ρ_ε, we see that all derivatives $D^\alpha h_j = h_j \star (D^\alpha \rho_\varepsilon)$ are in fact uniformly locally bounded

in ω. Hence, after extracting a new subsequence, we may suppose that h_{j_ν} converges uniformly to a limit h on ω. Then $u_{j_\nu} = h_{j_\nu} + N \star (\psi\mu_{j_\nu})$ converges to $u = h + N \star (\psi\mu)$ in $L^1_{\mathrm{loc}}(\omega)$, as desired. $\square$

We conclude this subsection by stating a generalized version of the Green-Riesz formula.

(4.22) Proposition. *Let $u \in \mathrm{Sh}(\Omega) \cap L^1_{\mathrm{loc}}(\Omega)$ and $\overline{B}(0,r) \subset \Omega$.*

a) *The Green-Riesz formula still holds true for such an u, namely, for every $x \in B(0,r)$*

$$u(x) = \int_{B(0,r)} \Delta u(y)\, G_r(x,y)\, d\lambda(y) + \int_{S(0,r)} u(y)\, P_r(x,y)\, d\sigma(y).$$

b) (Harnack inequality)
If $u \geqslant 0$ on $\overline{B}(0,r)$, then for all $x \in B(0,r)$

$$0 \leqslant u(x) \leqslant \int_{S(0,r)} u(y)\, P_r(x,y)\, d\sigma(y) \leqslant \frac{r^{m-2}(r+|x|)}{(r-|x|)^{m-1}}\, \mu_S(u\,;0,r).$$

If $u \leqslant 0$ on $\overline{B}(0,r)$, then for all $x \in B(0,r)$

$$u(x) \leqslant \int_{S(0,r)} u(y)\, P_r(x,y)\, d\sigma(y) \leqslant \frac{r^{m-2}(r-|x|)}{(r+|x|)^{m-1}}\, \mu_S(u\,;0,r) \leqslant 0.$$

Proof. We know that a) holds true if u is of class C^2. In general, we replace u by $u \star \rho_\varepsilon$ and take the limit. We only have to check that

$$\int_{B(0,r)} \mu \star \rho_\varepsilon(y)\, G_r(x,y)\, d\lambda(y) = \lim_{\varepsilon \to 0} \int_{B(0,r)} \mu(y)\, G_r(x,y)\, d\lambda(y)$$

for the positive measure $\mu = \Delta u$. Let us denote by $\widetilde{G}_x(y)$ the function such that

$$\widetilde{G}_x(y) = \begin{cases} G_r(x,y) & \text{if } x \in B(0,r) \\ 0 & \text{if} x \notin B(0,r) \end{cases}$$

Then

$$\begin{aligned} \int_{B(0,r)} \mu \star \rho_\varepsilon(y)\, G_r(x,y)\, d\lambda(y) &= \int_{\mathbb{R}^m} \mu \star \rho_\varepsilon(y)\, \widetilde{G}_x(y)\, d\lambda(y) \\ &= \int_{\mathbb{R}^m} \mu(y)\, \widetilde{G}_x \star \rho_\varepsilon(y)\, d\lambda(y). \end{aligned}$$

However $\widetilde{G}_x$ is continuous on $\mathbb{R}^m \smallsetminus \{x\}$ and subharmonic in a neighborhood of x, hence $\widetilde{G}_x \star \rho_\varepsilon$ converges uniformly to $\widetilde{G}_x$ on every compact subset of $\mathbb{R}^m \smallsetminus \{x\}$, and converges pointwise monotonically in a neighborhood of x. The desired equality follows by the monotone convergence theorem. Finally, b) is a consequence of a), for the integral involving Δu is nonpositive and

$$\frac{1}{\sigma_{m-1} r^{m-1}} \frac{r^{m-2}(r-|x|)}{(r+|x|)^{m-1}} \leqslant P_r(x,y) \leqslant \frac{1}{\sigma_{m-1} r^{m-1}} \frac{r^{m-2}(r+|x|)}{(r-|x|)^{m-1}}$$

by (4.6) combined with the obvious inequality $(r-|x|)^m \leqslant |x-y|^m \leqslant (r+|x|)^m$. $\square$

§ 4.C.3. *Upper Envelopes and Choquet's Lemma.* Let $\Omega \subset \mathbb{R}^n$ and let $(u_\alpha)_{\alpha\in I}$ be a family of upper semicontinuous functions $\Omega \longrightarrow [-\infty, +\infty[$. We assume that (u_α) is locally uniformly bounded above. Then the upper envelope

$$u = \sup u_\alpha$$

need not be upper semicontinuous, so we consider its *upper semicontinuous regularization*:

$$u^\star(z) = \lim_{\varepsilon\to 0} \sup_{B(z,\varepsilon)} u \geqslant u(z).$$

It is easy to check that $u^\star$ is the smallest upper semicontinuous function which is $\geqslant u$. Our goal is to show that $u^\star$ can be computed with a countable subfamily of (u_α). Let $B(z_j, \varepsilon_j)$ be a countable basis of the topology of Ω. For each j, let (z_{jk}) be a sequence of points in $B(z_j, \varepsilon_j)$ such that

$$\sup_k u(z_{jk}) = \sup_{B(z_j,\varepsilon_j)} u,$$

and for each pair (j, k), let $\alpha(j, k, l)$ be a sequence of indices $\alpha \in I$ such that $u(z_{jk}) = \sup_l u_{\alpha(j,k,l)}(z_{jk})$. Set

$$v = \sup_{j,k,l} u_{\alpha(j,k,l)}.$$

Then $v \leqslant u$ and $v^\star \leqslant u^\star$. On the other hand

$$\sup_{B(z_j,\varepsilon_j)} v \geqslant \sup_k v(z_{jk}) \geqslant \sup_{k,l} u_{\alpha(j,k,l)}(z_{jk}) = \sup_k u(z_{jk}) = \sup_{B(z_j,\varepsilon_j)} u.$$

As every ball $B(z, \varepsilon)$ is a union of balls $B(z_j, \varepsilon_j)$, we easily conclude that $v^\star \geqslant u^\star$, hence $v^\star = u^\star$. Therefore:

(4.23) Choquet's lemma. *Every family (u_α) has a countable subfamily $(v_j) = (u_{\alpha(j)})$ such that its upper envelope v satisfies $v \leqslant u \leqslant u^\star = v^\star$.* □

(4.24) Proposition. *If all u_α are subharmonic, the upper regularization $u^\star$ is subharmonic and equal almost everywhere to u.*

Proof. By Choquet's lemma we may assume that (u_α) is countable. Then $u = \sup u_\alpha$ is a Borel function. As each u_α satisfies the mean value inequality on every ball $\overline{B}(z, r) \subset \Omega$, we get

$$u(z) = \sup u_\alpha(z) \leqslant \sup \mu_B(u_\alpha\,;\, z, r) \leqslant \mu_B(u\,;\, z, r).$$

The right-hand side is a continuous function of z, so we infer

$$u^\star(z) \leqslant \mu_B(u\,;\, z, r) \leqslant \mu_B(u^\star\,;\, z, r)$$

and $u^\star$ is subharmonic. By the upper semicontinuity of $u^\star$ and the above inequality we find $u^\star(z) = \lim_{r\to 0} \mu_B(u\,;\, z, r)$, thus $u^\star = u$ almost everywhere by Lebesgue's lemma. □

§ 5. Plurisubharmonic Functions

§ 5.A. Definition and Basic Properties

Plurisubharmonic functions have been introduced independently by [Lelong 1942] and [Oka 1942] for the study of holomorphic convexity. They are the complex counterparts of subharmonic functions.

(5.1) Definition. *A function $u : \Omega \longrightarrow [-\infty, +\infty[$ defined on an open subset $\Omega \subset \mathbb{C}^n$ is said to be plurisubharmonic if*

a) *u is upper semicontinuous* ;

b) *for every complex line $L \subset \mathbb{C}^n$, $u_{\restriction \Omega \cap L}$ is subharmonic on $\Omega \cap L$.*

The set of plurisubharmonic functions on Ω is denoted by $\mathrm{Psh}(\Omega)$.

An equivalent way of stating property b) is: for all $a \in \Omega$, $\xi \in \mathbb{C}^n$, $|\xi| < d(a, \complement\Omega)$, then

$$u(a) \leqslant \frac{1}{2\pi} \int_0^{2\pi} u(a + e^{\mathrm{i}\theta}\, \xi)\, d\theta. \tag{5.2}$$

An integration of (5.2) over $\xi \in S(0, r)$ yields $u(a) \leqslant \mu_S(u\,; a, r)$, therefore

$$\mathrm{Psh}(\Omega) \subset \mathrm{Sh}(\Omega). \tag{5.3}$$

The following results have already been proved for subharmonic functions and are easy to extend to the case of plurisubharmonic functions:

(5.4) Theorem. *For any decreasing sequence of plurisubharmonic functions $u_k \in \mathrm{Psh}(\Omega)$, the limit $u = \lim u_k$ is plurisubharmonic on Ω.*

(5.5) Theorem. *Let $u \in \mathrm{Psh}(\Omega)$ be such that $u \not\equiv -\infty$ on every connected component of Ω. If (ρ_ε) is a family of smoothing kernels, then $u \star \rho_\varepsilon$ is $\mathscr{C}^\infty$ and plurisubharmonic on Ω_ε, the family $(u \star \rho_\varepsilon)$ is non decreasing in ε and $\lim_{\varepsilon \to 0} u \star \rho_\varepsilon = u$.*

(5.6) Theorem. *Let $u_1, \ldots, u_p \in \mathrm{Psh}(\Omega)$ and $\chi : \mathbb{R}^p \longrightarrow \mathbb{R}$ be a convex function such that $\chi(t_1, \ldots, t_p)$ is non decreasing in each t_j. Then $\chi(u_1, \ldots, u_p)$ is plurisubharmonic on Ω. In particular $u_1 + \cdots + u_p$, $\max\{u_1, \ldots, u_p\}$, $\log(e^{u_1} + \cdots + e^{u_p})$ are plurisubharmonic on Ω.*

(5.7) Theorem. *Let $\{u_\alpha\} \subset \mathrm{Psh}(\Omega)$ be locally uniformly bounded from above and $u = \sup u_\alpha$. Then the regularized upper envelope $u^\star$ is plurisubharmonic and is equal to u almost everywhere.*

Proof. By Choquet's lemma, we may assume that (u_α) is countable. Then u is a Borel function which clearly satisfies (5.2), and thus $u \star \rho_\varepsilon$ also satisfies (5.2). Hence $u \star \rho_\varepsilon$ is plurisubharmonic. By Proposition 4.24, $u^\star = u$ almost everywhere and $u^\star$ is subharmonic, so

$$u^\star = \lim u^\star \star \rho_\varepsilon = \lim u \star \rho_\varepsilon$$

is plurisubharmonic. □

If $u \in C^2(\Omega, \mathbb{R})$, the subharmonicity of restrictions of u to complex lines, $\mathbb{C} \ni w \longmapsto u(a + w\xi)$, $a \in \Omega$, $\xi \in \mathbb{C}^n$, is equivalent to

$$\frac{\partial^2}{\partial w \partial \overline{w}} u(a + w\xi) = \sum_{1 \leqslant j,k \leqslant n} \frac{\partial^2 u}{\partial z_j \partial \overline{z}_k}(a + w\xi)\, \xi_j \overline{\xi}_k \geqslant 0.$$

Therefore, u is plurisubharmonic on Ω if and only if $\sum \partial^2 u / \partial z_j \partial \overline{z}_k(a)\, \xi_j \overline{\xi}_k$ is a semipositive hermitian form at every point $a \in \Omega$. This equivalence is still true for arbitrary plurisubharmonic functions, under the following form:

(5.8) Theorem. *If $u \in \mathrm{Psh}(\Omega)$, $u \not\equiv -\infty$ on every connected component of Ω, then for all $\xi \in \mathbb{C}^n$*

$$Hu(\xi) := \sum_{1 \leqslant j,k \leqslant n} \frac{\partial^2 u}{\partial z_j \partial \overline{z}_k}\, \xi_j \overline{\xi}_k \in \mathscr{D}'(\Omega)$$

is a positive measure. Conversely, if $v \in \mathscr{D}'(\Omega)$ is such that $Hv(\xi)$ is a positive measure for every $\xi \in \mathbb{C}^n$, there exists a unique function $u \in \mathrm{Psh}(\Omega)$ locally integrable on Ω such that v is the distribution associated to u.

Proof. If $u \in \mathrm{Psh}(\Omega)$, then $Hu(\xi) = \text{weak } \lim H(u \star \rho_\varepsilon)(\xi) \geqslant 0$. Conversely, $Hv \geqslant 0$ implies $H(v \star \rho_\varepsilon) = (Hv) \star \rho_\varepsilon \geqslant 0$, thus $v \star \rho_\varepsilon \in \mathrm{Psh}(\Omega)$, and also $\Delta v \geqslant 0$, hence $(v \star \rho_\varepsilon)$ is non decreasing in ε and $u = \lim_{\varepsilon \to 0} v \star \rho_\varepsilon \in \mathrm{Psh}(\Omega)$ by Th. 5.4. □

(5.9) Proposition. *The convex cone $\mathrm{Psh}(\Omega) \cap L^1_{\mathrm{loc}}(\Omega)$ is closed in $L^1_{\mathrm{loc}}(\Omega)$, and it has the property that every bounded subset is relatively compact.*

§ 5.B. Relations with Holomorphic Functions

In order to get a better geometric insight, we assume more generally that u is a C^2 function on a complex n-dimensional manifold X. The *complex Hessian* of u at a point $a \in X$ is the hermitian form on T_X defined by

$$Hu_a = \sum_{1 \leqslant j,k \leqslant n} \frac{\partial^2 u}{\partial z_j \partial \overline{z}_k}(a)\, dz_j \otimes d\overline{z}_k. \tag{5.10}$$

If $F : X \longrightarrow Y$ is a holomorphic mapping and if $v \in C^2(Y, \mathbb{R})$, we have $d'd''(v \circ F) = F^\star d'd'' v$. In equivalent notations, a direct calculation gives for all $\xi \in T_{X,a}$

$$H(v \circ F)_a(\xi) = \sum_{j,k,l,m} \frac{\partial^2 v\big(F(a)\big)}{\partial z_l \partial \overline{z}_m} \frac{\partial F_l\big(a\big)}{\partial z_j} \xi_j \overline{\frac{\partial F_m\big(a\big)}{\partial z_k} \xi_k} = Hv_{F(a)}\big(F'(a).\xi\big).$$

In particular Hu_a does not depend on the choice of coordinates $(z_1, \ldots, z_n)$ on X, and $Hv_a \geqslant 0$ on Y implies $H(v \circ F)_a \geqslant 0$ on X. Therefore, the notion of plurisubharmonic function makes sense on any complex manifold.

(5.11) Theorem. *If $F : X \longrightarrow Y$ is a holomorphic map and $v \in \mathrm{Psh}(Y)$, then $v \circ F \in \mathrm{Psh}(X)$.*

Proof. It is enough to prove the result when $X = \Omega_1 \subset \mathbb{C}^n$ and $X = \Omega_2 \subset \mathbb{C}^p$ are open subsets . The conclusion is already known when v is of class C^2, and it can be extended to an arbitrary upper semicontinuous function v by using Th. 5.4 and the fact that $v = \lim v \star \rho_\varepsilon$. $\square$

(5.12) Example. By (3.22) we see that $\log|z|$ is subharmonic on $\mathbb{C}$, thus $\log|f| \in \mathrm{Psh}(X)$ for every holomorphic function $f \in \mathscr{O}(X)$. More generally

$$\log\big(|f_1|^{\alpha_1} + \cdots + |f_q|^{\alpha_q}\big) \in \mathrm{Psh}(X)$$

for every $f_j \in \mathscr{O}(X)$ and $\alpha_j \geqslant 0$ (apply Th. 5.6 with $u_j = \alpha_j \log|f_j|$).

§ 5.C. Convexity Properties

The close analogy of plurisubharmonicity with the concept of convexity strongly suggests that there are deeper connections between these notions. We describe here a few elementary facts illustrating this philosophy. Another interesting connection between plurisubharmonicity and convexity will be seen in § 7.B (Kiselman's minimum principle).

(5.13) Theorem. *If $\Omega = \omega + \mathrm{i}\omega'$ where ω, ω' are open subsets of $\mathbb{R}^n$, and if $u(z)$ is a plurisubharmonic function on Ω that depends only on $x = \mathrm{Re}\, z$, then $\omega \ni x \longmapsto u(x)$ is convex.*

Proof. This is clear when $u \in C^2(\Omega, \mathbb{R})$, for $\partial^2 u/\partial z_j \partial \overline{z}_k = \frac{1}{4}\, \partial^2 u/\partial x_j \partial x_k$. In the general case, write $u = \lim u \star \rho_\varepsilon$ and observe that $u \star \rho_\varepsilon(z)$ depends only on x. $\square$

(5.14) Corollary. *If u is a plurisubharmonic function in the open polydisk $D(a, R) = \prod D(a_j, R_j) \subset \mathbb{C}^n$, then*

$$\begin{aligned}
\mu(u\,;\, r_1, \ldots, r_n) &= \frac{1}{(2\pi)^n} \int_0^{2\pi} u(a_1 + r_1 e^{\mathrm{i}\theta_1}, \ldots, a_n + r_n e^{\mathrm{i}\theta_n})\, d\theta_1 \ldots d\theta_n, \\
m(u\,;\, r_1, \ldots, r_n) &= \sup_{z \in D(a,r)} u(z_1, \ldots, z_n), \qquad r_j < R_j
\end{aligned}$$

are convex functions of $(\log r_1, \ldots, \log r_n)$ that are non decreasing in each variable.

Proof. That μ is non decreasing follows from the subharmonicity of u along every coordinate axis. Now, it is easy to verify that the functions

$$\begin{aligned}
\widetilde{\mu}(z_1, \ldots, z_n) &= \frac{1}{(2\pi)^n} \int_0^{2\pi} u(a_1 + e^{z_1} e^{\mathrm{i}\theta_1}, \ldots, a_n + e^{z_n} e^{\mathrm{i}\theta_n})\, d\theta_1 \ldots d\theta_n, \\
\widetilde{m}(z_1, \ldots, z_n) &= \sup_{|w_j| \leqslant 1} u(a_1 + e^{z_1} w_1, \ldots, a_n + e^{z_n} w_n)
\end{aligned}$$

are upper semicontinuous, satisfy the mean value inequality, and depend only on $\mathrm{Re}\, z_j \in]0, \log R_j[$. Therefore $\widetilde{\mu}$ and $\widetilde{M}$ are convex. Cor. 5.14 follows from the equalities

$$\begin{aligned}
\mu(u\,;\, r_1, \ldots, r_n) &= \widetilde{\mu}(\log r_1, \ldots, \log r_n), \\
m(u\,;\, r_1, \ldots, r_n) &= \widetilde{m}(\log r_1, \ldots, \log r_n). \qquad \square
\end{aligned}$$

§ 5.D. Pluriharmonic Functions

Pluriharmonic functions are the counterpart of harmonic functions in the case of functions of complex variables:

(5.15) Definition. *A function u is said to be pluriharmonic if u and $-u$ are plurisubharmonic.*

A pluriharmonic function is harmonic (in particular smooth) in any $\mathbb{C}$-analytic coordinate system, and is characterized by the condition $Hu = 0$, i.e. $d'd''u = 0$ or

$$\partial^2 u/\partial z_j \partial \overline{z}_k = 0 \quad \text{for all} \quad j, k.$$

If $f \in \mathscr{O}(X)$, it follows that the functions $\operatorname{Re} f$, $\operatorname{Im} f$ are pluriharmonic. Conversely:

(5.16) Theorem. *If the De Rham cohomology group $H^1_{\mathrm{DR}}(X, \mathbb{R})$ is zero, every pluriharmonic function u on X can be written $u = \operatorname{Re} f$ where f is a holomorphic function on X.*

Proof. By hypothesis $H^1_{\mathrm{DR}}(X, \mathbb{R}) = 0$, $u \in \mathscr{C}^\infty(X)$ and $d(d'u) = d''d'u = 0$, hence there exists $g \in \mathscr{C}^\infty(X)$ such that $dg = d'u$. Then dg is of type $(1,0)$, i.e. $g \in \mathscr{O}(X)$ and

$$d(u - 2\operatorname{Re} g) = d(u - g - \overline{g}) = (d'u - dg) + (d''u - d\overline{g}) = 0.$$

Therefore $u = \operatorname{Re}(2g + C)$, where C is a locally constant function. □

§ 5.E. Global Regularization of Plurisubharmonic Functions

We now study a very efficient regularization and patching procedure for continuous plurisubharmonic functions, essentially due to [Richberg 1968]. The main idea is contained in the following lemma:

(5.17) Lemma. *Let $u_\alpha \in \operatorname{Psh}(\Omega_\alpha)$ where $\Omega_\alpha \subset\subset X$ is a locally finite open covering of X. Assume that for every index β*

$$\limsup_{\zeta \to z} u_\beta(\zeta) < \max_{\Omega_\alpha \ni z} \{u_\alpha(z)\}$$

at all points $z \in \partial\Omega_\beta$. Then the function

$$u(z) = \max_{\Omega_\alpha \ni z} \; u_\alpha(z)$$

is plurisubharmonic on X.

Proof. Fix $z_0 \in X$. Then the indices β such that $z_0 \in \partial\Omega_\beta$ or $z_0 \notin \overline{\Omega}_\beta$ do not contribute to the maximum in a neighborhood of z_0. Hence there is a a finite set I of indices α such that $\Omega_\alpha \ni z_0$ and a neighborhood $V \subset \bigcap_{\alpha \in I} \Omega_\alpha$ on which $u(z) = \max_{\alpha \in I} u_\alpha(z)$. Therefore u is plurisubharmonic on V. □

The above patching procedure produces functions which are in general only continuous. When smooth functions are needed, one has to use a regularized max function. Let $\theta \in \mathscr{C}^\infty(\mathbb{R}, \mathbb{R})$ be a nonnegative function with support in $[-1, 1]$ such that $\int_{\mathbb{R}} \theta(h)\, dh = 1$ and $\int_{\mathbb{R}} h\theta(h)\, dh = 0$.

(5.18) Lemma. *For arbitrary $\eta = (\eta_1, \ldots, \eta_p) \in]0, +\infty[^p$, the function*

$$M_\eta(t_1, \ldots, t_p) = \int_{\mathbb{R}^n} \max\{t_1 + h_1, \ldots, t_p + h_p\} \prod_{1 \leqslant j \leqslant n} \theta(h_j/\eta_j)\, dh_1 \ldots dh_p$$

possesses the following properties:

a) *$M_\eta(t_1, \ldots, t_p)$ is non decreasing in all variables, smooth and convex on $\mathbb{R}^n$;*

b) $\max\{t_1, \ldots, t_p\} \leqslant M_\eta(t_1, \ldots, t_p) \leqslant \max\{t_1 + \eta_1, \ldots, t_p + \eta_p\}$;

c) $M_\eta(t_1, \ldots, t_p) = M_{(\eta_1, \ldots, \widehat{\eta_j}, \ldots, \eta_p)}(t_1, \ldots, \widehat{t_j}, \ldots, t_p)$
if $t_j + \eta_j \leqslant \max_{k \neq j}\{t_k - \eta_k\}$;

d) $M_\eta(t_1 + a, \ldots, t_p + a) = M_\eta(t_1, \ldots, t_p) + a, \qquad \forall a \in \mathbb{R}$;

e) *if $u_1, \ldots, u_p$ are plurisubharmonic and satisfy $H(u_j)_z(\xi) \geqslant \gamma_z(\xi)$ where $z \mapsto \gamma_z$ is a continuous hermitian form on T_X, then $u = M_\eta(u_1, \ldots, u_p)$ is plurisubharmonic and satisfies $Hu_z(\xi) \geqslant \gamma_z(\xi)$.*

Proof. The change of variables $h_j \mapsto h_j - t_j$ shows that M_η is smooth. All properties are immediate consequences of the definition, except perhaps e). That $M_\eta(u_1, \ldots, u_p)$ is plurisubharmonic follows from a) and Th. 5.6. Fix a point z_0 and $\varepsilon > 0$. All functions $u'_j(z) = u_j(z) - \gamma_{z_0}(z - z_0) + \varepsilon|z - z_0|^2$ are plurisubharmonic near z_0. It follows that

$$M_\eta(u'_1, \ldots, u'_p) = u - \gamma_{z_0}(z - z_0) + \varepsilon|z - z_0|^2$$

is also plurisubharmonic near z_0. Since $\varepsilon > 0$ was arbitrary, e) follows. □

(5.19) Corollary. *Let $u_\alpha \in \mathscr{C}^\infty(\overline{\Omega}_\alpha) \cap \mathrm{Psh}(\Omega_\alpha)$ where $\Omega_\alpha \subset\subset X$ is a locally finite open covering of X. Assume that $u_\beta(z) < \max\{u_\alpha(z)\}$ at every point $z \in \partial\Omega_\beta$, when α runs over the indices such that $\Omega_\alpha \ni z$. Choose a family (η_α) of positive numbers so small that $u_\beta(z) + \eta_\beta \leqslant \max_{\Omega_\alpha \ni z}\{u_\alpha(z) - \eta_\alpha\}$ for all β and $z \in \partial\Omega_\beta$. Then the function defined by*

$$\widetilde{u}(z) = M_{(\eta_\alpha)}\big(u_\alpha(z)\big) \quad \text{for } \alpha \text{ such that } \ \Omega_\alpha \ni z$$

is smooth and plurisubharmonic on X. □

(5.20) Definition. *A function $u \in \mathrm{Psh}(X)$ is said to be strictly plurisubharmonic if $u \in L^1_{\mathrm{loc}}(X)$ and if for every point $x_0 \in X$ there exists a neighborhood Ω of x_0 and $c > 0$ such that $u(z) - c|z|^2$ is plurisubharmonic on Ω, i.e. $\sum\, (\partial^2 u/\partial z_j \partial \overline{z}_k)\xi_j\overline{\xi}_k \geqslant c|\xi|^2$ (as distributions on Ω) for all $\xi \in \mathbb{C}^n$.*

(5.21) Theorem ([Richberg 1968])**.** *Let $u \in \mathrm{Psh}(X)$ be a continuous function which is strictly plurisubharmonic on an open subset $\Omega \subset X$, with $Hu \geqslant \gamma$ for some continuous positive hermitian form γ on Ω. For any continuous function $\lambda \in C^0(\Omega)$, $\lambda > 0$, there exists a plurisubharmonic function $\widetilde{u}$ in $C^0(X) \cap \mathscr{C}^\infty(\Omega)$ such that $u \leqslant \widetilde{u} \leqslant u + \lambda$ on Ω and $\widetilde{u} = u$ on $X \smallsetminus \Omega$, which is strictly plurisubharmonic on Ω and satisfies $H\widetilde{u} \geqslant (1 - \lambda)\gamma$. In particular, $\widetilde{u}$ can be chosen strictly plurisubharmonic on X if u has the same property.*

Proof. Let (Ω_α) be a locally finite open covering of Ω by relatively compact open balls contained in coordinate patches of X. Choose concentric balls $\Omega''_\alpha \subset \Omega'_\alpha \subset \Omega_\alpha$ of respective radii $r''_\alpha < r'_\alpha < r_\alpha$ and center $z = 0$ in the given coordinates $z = (z_1, \ldots, z_n)$ near $\overline{\Omega}_\alpha$, such that Ω''_α still cover Ω. We set

$$u_\alpha(z) = u \star \rho_{\varepsilon_\alpha}(z) + \delta_\alpha(r'^2_\alpha - |z|^2) \quad \text{on } \overline{\Omega}_\alpha.$$

For $\varepsilon_\alpha < \varepsilon_{\alpha,0}$ and $\delta_\alpha < \delta_{\alpha,0}$ small enough, we have $u_\alpha \leqslant u + \lambda/2$ and $Hu_\alpha \geqslant (1-\lambda)\gamma$ on $\overline{\Omega}_\alpha$. Set

$$\eta_\alpha = \delta_\alpha \min\{r'^2_\alpha - r''^2_\alpha, (r^2_\alpha - r'^2_\alpha)/2\}.$$

Choose first $\delta_\alpha < \delta_{\alpha,0}$ such that $\eta_\alpha < \min_{\overline{\Omega}_\alpha} \lambda/2$, and then $\varepsilon_\alpha < \varepsilon_{\alpha,0}$ so small that $u \leqslant u \star \rho_{\varepsilon_\alpha} < u + \eta_\alpha$ on $\overline{\Omega}_\alpha$. As $\delta_\alpha(r'^2 - |z|^2)$ is $\leqslant -2\eta_\alpha$ on $\partial\Omega_\alpha$ and $> \eta_\alpha$ on $\overline{\Omega}''_\alpha$, we have $u_\alpha < u - \eta_\alpha$ on $\partial\Omega_\alpha$ and $u_\alpha > u + \eta_\alpha$ on $\overline{\Omega}''_\alpha$, so that the condition required in Corollary 5.19 is satisfied. We define

$$\widetilde{u} = \begin{cases} u & \text{on } X \smallsetminus \Omega, \\ M_{(\eta_\alpha)}(u_\alpha) & \text{on } \Omega. \end{cases}$$

By construction, $\widetilde{u}$ is smooth on Ω and satisfies $u \leqslant \widetilde{u} \leqslant u + \lambda$, $Hu \geqslant (1-\lambda)\gamma$ thanks to 5.18 (b,e). In order to see that $\widetilde{u}$ is plurisubharmonic on X, observe that $\widetilde{u}$ is the uniform limit of $\widetilde{u}_\alpha$ with

$$\widetilde{u}_\alpha = \max\left\{u\,,\, M_{(\eta_1 \ldots \eta_\alpha)}(u_1 \ldots u_\alpha)\right\} \quad \text{on} \quad \bigcup_{1 \leqslant \beta \leqslant \alpha} \Omega_\beta$$

and $\widetilde{u}_\alpha = u$ on the complement. □

§ 5.F. Polar and Pluripolar Sets.

Polar and pluripolar sets are sets of $-\infty$ poles of subharmonic and plurisubharmonic functions. Although these functions possess a large amount of flexibility, pluripolar sets have some properties which remind their loose relationship with holomorphic functions.

(5.22) Definition. *A set $A \subset \Omega \subset \mathbb{R}^m$ (resp. $A \subset X$, $\dim_{\mathbb{C}} X = n$) is said to be polar (resp. pluripolar) if for every point $x \in \Omega$ there exist a connected neighborhood W of x and $u \in \mathrm{Sh}(W)$ (resp. $u \in \mathrm{Psh}(W)$), $u \not\equiv -\infty$, such that $A \cap W \subset \{x \in W\ ;\ u(x) = -\infty\}$.*

Theorem 4.17 implies that a polar or pluripolar set is of zero Lebesgue measure. Now, we prove a simple extension theorem.

(5.23) Theorem. *Let $A \subset \Omega$ be a closed polar set and $v \in \mathrm{Sh}(\Omega \smallsetminus A)$ such that v is bounded above in a neighborhood of every point of A. Then v has a unique extension $\widetilde{v} \in \mathrm{Sh}(\Omega)$.*

Proof. The uniqueness is clear because A has zero Lebesgue measure. On the other hand, every point of A has a neighborhood W such that

$$A \cap W \subset \{x \in W\ ;\ u(x) = -\infty\}, \quad u \in \mathrm{Sh}(W), \quad u \not\equiv -\infty.$$

After shrinking W and subtracting a constant to u, we may assume $u \leqslant 0$. Then for every $\varepsilon > 0$ the function $v_\varepsilon = v + \varepsilon u \in \mathrm{Sh}(W \smallsetminus A)$ can be extended as an upper semicontinuous on W by setting $v_\varepsilon = -\infty$ on $A \cap W$. Moreover, v_ε satisfies the mean value inequality $v_\varepsilon(a) \leqslant \mu_S(v_\varepsilon\,; a, r)$ if $a \in W \smallsetminus A$, $r < d(a, A \cup \complement W)$, and also clearly if $a \in A$, $r < d(a, \complement W)$. Therefore $v_\varepsilon \in \mathrm{Sh}(W)$ and $\widetilde{v} = (\sup v_\varepsilon)^\star \in \mathrm{Sh}(W)$. Clearly $\widetilde{v}$ coincides with v on $W \smallsetminus A$. A similar proof gives:

(5.24) Theorem. *Let A be a closed pluripolar set in a complex analytic manifold X. Then every function $v \in \mathrm{Psh}(X \smallsetminus A)$ that is locally bounded above near A extends uniquely into a function $\widetilde{v} \in \mathrm{Psh}(X)$.* □

(5.25) Corollary. *Let $A \subset X$ be a closed pluripolar set. Every holomorphic function $f \in \mathscr{O}(X \smallsetminus A)$ that is locally bounded near A extends to a holomorphic function $\widetilde{f} \in \mathscr{O}(X)$.*

Proof. Apply Th. 5.24 to $\pm \operatorname{Re} f$ and $\pm \operatorname{Im} f$. It follows that $\operatorname{Re} f$ and $\operatorname{Im} f$ have pluriharmonic extensions to X, in particular f extends to $\widetilde{f} \in \mathscr{C}^\infty(X)$. By density of $X \smallsetminus A$, $d''\widetilde{f} = 0$ on X. □

(5.26) Corollary. *Let $A \subset \Omega$ (resp. $A \subset X$) be a closed (pluri)polar set. If Ω (resp. X) is connected, then $\Omega \smallsetminus A$ (resp. $X \smallsetminus A$) is connected.*

Proof. If $\Omega \smallsetminus A$ (resp. $X \smallsetminus A$) is a disjoint union $\Omega_1 \cup \Omega_2$ of non empty open subsets, the function defined by $f \equiv 0$ on Ω_1, $f \equiv 1$ on Ω_2 would have a harmonic (resp. holomorphic) extension through A, a contradiction. □

§ 6. Domains of Holomorphy and Stein Manifolds

§ 6.A. Domains of Holomorphy in $\mathbb{C}^n$. Examples

Loosely speaking, a domain of holomorphy is an open subset Ω in $\mathbb{C}^n$ such that there is no part of $\partial\Omega$ across which all functions $f \in \mathscr{O}(\Omega)$ can be extended. More precisely:

(6.1) Definition. *Let $\Omega \subset \mathbb{C}^n$ be an open subset. Ω is said to be a domain of holomorphy if for every connected open set $U \subset \mathbb{C}^n$ which meets $\partial\Omega$ and every connected component V of $U \cap \Omega$ there exists $f \in \mathscr{O}(\Omega)$ such that $f_{\restriction V}$ has no holomorphic extension to U.*

Under the hypotheses made on U, we have $\emptyset \neq \partial V \cap U \subset \partial\Omega$. In order to show that Ω is a domain of holomorphy, it is thus sufficient to find for every $z_0 \in \partial\Omega$ a function $f \in \mathscr{O}(\Omega)$ which is unbounded near z_0.

(6.2) Examples. Every open subset $\Omega \subset \mathbb{C}$ is a domain of holomorphy (for any $z_0 \in \partial\Omega$, $f(z) = (z - z_0)^{-1}$ cannot be extended at z_0). In $\mathbb{C}^n$, every *convex* open subset is a domain of holomorphy: if $\operatorname{Re}\langle z - z_0, \xi_0\rangle = 0$ is a supporting hyperplane of $\partial\Omega$ at z_0, the function $f(z) = (\langle z - z_0, \xi_0\rangle)^{-1}$ is holomorphic on Ω but cannot be extended at z_0.

(6.3) Hartogs figure. Assume that $n \geqslant 2$. Let $\omega \subset \mathbb{C}^{n-1}$ be a connected open set and $\omega' \subsetneq \omega$ an open subset. Consider the open sets in $\mathbb{C}^n$:

$$\Omega = \big((D(R) \smallsetminus \overline{D}(r)) \times \omega\big) \cup \big(D(R) \times \omega'\big) \qquad \text{(Hartogs figure)},$$

$$\widetilde{\Omega} = D(R) \times \omega \qquad \text{(filled Hartogs figure)}.$$

where $0 \leqslant r < R$ and $D(r) \subset \mathbb{C}$ denotes the open disk of center 0 and radius r in $\mathbb{C}$.

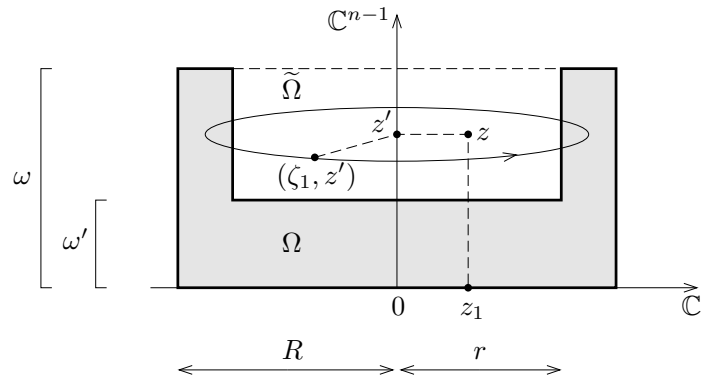

Fig. I.3 Hartogs figure

Then every function $f \in \mathscr{O}(\Omega)$ can be extended to $\widetilde{\Omega} = \omega \times D(R)$ by means of the Cauchy formula:

$$\widetilde{f}(z_1, z') = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta_1|=\rho} \frac{f(\zeta_1, z')}{\zeta_1 - z_1} d\zeta_1, \qquad z \in \widetilde{\Omega}, \quad \max\{|z_1|, r\} < \rho < R.$$

In fact $\widetilde{f} \in \mathscr{O}(D(R) \times \omega)$ and $\widetilde{f} = f$ on $D(R) \times \omega'$, so we must have $\widetilde{f} = f$ on Ω since Ω is connected. It follows that Ω is not a domain of holomorphy. Let us quote two interesting consequences of this example.

(6.4) Corollary (Riemann's extension theorem)**.** *Let X be a complex analytic manifold, and S a closed submanifold of codimension $\geqslant 2$. Then every $f \in \mathscr{O}(X \smallsetminus S)$ extends holomorphically to X.*

Proof. This is a local result. We may choose coordinates $(z_1, \ldots, z_n)$ and a polydisk $D(R)^n$ in the corresponding chart such that $S \cap D(R)^n$ is given by equations $z_1 = \ldots = z_p = 0$, $p = \operatorname{codim} S \geqslant 2$. Then, denoting $\omega = D(R)^{n-1}$ and $\omega' = \omega \smallsetminus \{z_2 = \ldots = z_p = 0\}$, the complement $D(R)^n \smallsetminus S$ can be written as the Hartogs figure

$$D(R)^n \smallsetminus S = \big((D(R) \smallsetminus \{0\}) \times \omega\big) \cup \big(D(R) \times \omega'\big).$$

It follows that f can be extended to $\widetilde{\Omega} = D(R)^n$. □

§ 6.B. Holomorphic Convexity and Pseudoconvexity

Let X be a complex manifold. We first introduce the notion of holomorphic hull of a compact set $K \subset X$. This can be seen somehow as the complex analogue of the notion of (affine) convex hull for a compact set in a real vector space. It is shown that domains of holomorphy in $\mathbb{C}^n$ are characterized a property of holomorphic convexity. Finally, we prove that holomorphic convexity implies pseudoconvexity – a complex analogue of the geometric notion of convexity.

(6.5) Definition. *Let X be a complex manifold and let K be a compact subset of X. Then the holomorphic hull of K in X is defined to be*

$$\widehat{K} = \widehat{K}_{\mathscr{O}(X)} = \big\{z \in X\,;\, |f(z)| \leqslant \sup_K |f|,\ \forall f \in \mathscr{O}(X)\big\}.$$

(6.6) Elementary properties.

a) $\widehat{K}$ is a closed subset of X containing K. Moreover we have

$$\sup_{\widehat{K}} |f| = \sup_K |f|, \qquad \forall f \in \mathscr{O}(X),$$

hence $\widehat{\widehat{K}} = \widehat{K}$.

b) If $h : X \to Y$ is a holomorphic map and $K \subset X$ is a compact set, then $h(\widehat{K}_{\mathscr{O}(X)}) \subset \widehat{h(K)}_{\mathscr{O}(Y)}$. In particular, if $X \subset Y$, then $\widehat{K}_{\mathscr{O}(X)} \subset \widehat{K}_{\mathscr{O}(Y)} \cap X$. This is immediate from the definition.

c) $\widehat{K}$ contains the union of K with all relatively compact connected components of $X \smallsetminus K$ (thus $\widehat{K}$ "fills the holes" of K). In fact, for every connected component U of $X \smallsetminus K$ we have $\partial U \subset \partial K$, hence if $\overline{U}$ is compact the maximum principle yields

$$\sup_{\overline{U}} |f| = \sup_{\partial U} |f| \leqslant \sup_K |f|, \qquad \text{for all } f \in \mathscr{O}(X).$$

d) More generally, suppose that there is a holomorphic map $h : U \to X$ defined on a relatively compact open set U in a complex manifold S, such that h extends as a continuous map $h : \overline{U} \to X$ and $h(\partial U) \subset K$. Then $h(\overline{U}) \subset \widehat{K}$. Indeed, for $f \in \mathscr{O}(X)$, the maximum principle again yields

$$\sup_{\overline{U}} |f \circ h| = \sup_{\partial U} |f \circ h| \leqslant \sup_K |f|.$$

This is especially useful when U is the unit disk in $\mathbb{C}$.

e) Suppose that $X = \Omega \subset \mathbb{C}^n$ is an open set. By taking $f(z) = \exp(A(z))$ where A is an arbitrary affine function, we see that $\widehat{K}_{\mathscr{O}(\Omega)}$ is contained in the intersection of all affine half-spaces containing K. Hence $\widehat{K}_{\mathscr{O}(\Omega)}$ is contained in the affine convex hull $\widehat{K}_{\mathrm{aff}}$. As a consequence $\widehat{K}_{\mathscr{O}(\Omega)}$ is always bounded and $\widehat{K}_{\mathscr{O}(\mathbb{C}^n)}$ is a compact set. However, when Ω is arbitrary, $\widehat{K}_{\mathscr{O}(\Omega)}$ is not always compact; for example, in case $\Omega = \mathbb{C}^n \smallsetminus \{0\}$, $n \geqslant 2$, then $\mathscr{O}(\Omega) = \mathscr{O}(\mathbb{C}^n)$ and the holomorphic hull of $K = S(0,1)$ is the non compact set $\widehat{K} = \overline{B}(0,1) \smallsetminus \{0\}$.

(6.7) Definition. *A complex manifold X is said to be holomorphically convex if the holomorphic hull $\widehat{K}_{\mathscr{O}(X)}$ of every compact set $K \subset X$ is compact.*

(6.8) Remark. *A complex manifold X is holomorphically convex if and only if there is an exhausting sequence of holomorphically compact subsets $K_\nu \subset X$, i.e. compact sets such that*

$$X = \bigcup K_\nu, \qquad \widehat{K}_\nu = K_\nu, \qquad K_\nu^\circ \supset K_{\nu-1}.$$

Indeed, if X is holomorphically convex, we may define K_ν inductively by $K_0 = \emptyset$ and $K_{\nu+1} = (K'_\nu \cup L_\nu)^\wedge_{\mathscr{O}(X)}$, where K'_ν is a neighborhood of K_ν and L_ν a sequence of compact sets of X such that $X = \bigcup L_\nu$. The converse is obvious: if such a sequence (K_ν) exists, then every compact subset $K \subset X$ is contained in some K_ν, hence $\widehat{K} \subset \widehat{K}_\nu = K_\nu$ is compact. □

We now concentrate on domains of holomorphy in $\mathbb{C}^n$. We denote by d and $B(z,r)$ the distance and the open balls associated to an arbitrary norm on $\mathbb{C}^n$, and we set for simplicity $B = B(0,1)$.

(6.9) Proposition. *If Ω is a domain of holomorphy and $K \subset \Omega$ is a compact subset, then $d(\widehat{K}, \complement\Omega) = d(K, \complement\Omega)$ and $\widehat{K}$ is compact.*

Proof. Let $f \in \mathscr{O}(\Omega)$. Given $r < d(K, \complement\Omega)$, we denote by M the supremum of $|f|$ on the compact subset $K + r\overline{B} \subset \Omega$. Then for every $z \in K$ and $\xi \in \overline{B}$, the function

$$\mathbb{C} \ni t \longmapsto f(z + t\xi) = \sum_{k=0}^{+\infty} \frac{1}{k!} D^k f(z)(\xi)^k\, t^k \tag{6.10}$$

is analytic in the disk $|t| < r$ and bounded by M. The Cauchy inequalities imply

$$|D^k f(z)(\xi)^k| \leqslant M k!\, r^{-k}, \quad \forall z \in K, \quad \forall \xi \in \overline{B}.$$

As the left hand side is an analytic fuction of z in Ω, the inequality must also hold for $z \in \widehat{K}$, $\xi \in \overline{B}$. Every $f \in \mathscr{O}(\Omega)$ can thus be extended to any ball $B(z,r)$, $z \in \widehat{K}$, by means of the power series (6.10). Hence $B(z,r)$ must be contained in Ω, and this shows that $d(\widehat{K}, \complement\Omega) \geqslant r$. As $r < d(K, \complement\Omega)$ was arbitrary, we get $d(\widehat{K}, \complement\Omega) \geqslant d(K, \complement\Omega)$ and the converse inequality is clear, so $d(\widehat{K}, \complement\Omega) = d(K, \complement\Omega)$. As $\widehat{K}$ is bounded and closed in Ω, this shows that $\widehat{K}$ is compact. □

(6.11) Theorem. *Let Ω be an open subset of $\mathbb{C}^n$. The following properties are equivalent:*

a) *Ω is a domain of holomorphy;*

b) *Ω is holomorphically convex;*

c) *For every countable subset $\{z_j\}_{j\in\mathbb{N}} \subset \Omega$ without accumulation points in Ω and every sequence of complex numbers (a_j), there exists an interpolation function $F \in \mathscr{O}(\Omega)$ such that $F(z_j) = a_j$.*

d) *There exists a function $F \in \mathscr{O}(\Omega)$ which is unbounded on any neighborhood of any point of $\partial\Omega$.*

Proof. d) $\Longrightarrow$ a) is obvious and a) $\Longrightarrow$ b) is a consequence of Prop. 6.9.

c) $\Longrightarrow$ d). If $\Omega = \mathbb{C}^n$ there is nothing to prove. Otherwise, select a dense sequence (ζ_j) in $\partial\Omega$ and take $z_j \in \Omega$ such that $d(z_j, \zeta_j) < 2^{-j}$. Then the interpolation function $F \in \mathscr{O}(\Omega)$ such that $F(z_j) = j$ satisfies d).

b) $\Longrightarrow$ c). Let $K_\nu \subset \Omega$ be an exhausting sequence of holomorphically convex compact sets as in Remark 6.8. Let $\nu(j)$ be the unique index ν such that $z_j \in K_{\nu(j)+1} \smallsetminus K_{\nu(j)}$. By the definition of a holomorphic hull, we can find a function $g_j \in \mathscr{O}(\Omega)$ such that

$$\sup_{K_{\nu(j)}} |g_j| < |g_j(z_j)|.$$

After multiplying g_j by a constant, we may assume that $g_j(z_j) = 1$. Let $P_j \in \mathbb{C}[z_1, \ldots, z_n]$ be a polynomial equal to 1 at z_j and to 0 at $z_0, z_1, \ldots, z_{j-1}$. We set

$$F = \sum_{j=0}^{+\infty} \lambda_j P_j g_j^{m_j},$$

where $\lambda_j \in \mathbb{C}$ and $m_j \in \mathbb{N}$ are chosen inductively such that

$$\lambda_j = a_j - \sum_{0 \leqslant k < j} \lambda_k P_k(z_j) g_k(z_j)^{m_k},$$
$$|\lambda_j P_j g_j^{m_j}| \leqslant 2^{-j} \quad \text{on} \quad K_{\nu(j)}\ ;$$

once λ_j has been chosen, the second condition holds as soon as m_j is large enough. Since $\{z_j\}$ has no accumulation point in Ω, the sequence $\nu(j)$ tends to $+\infty$, hence the series converges uniformly on compact sets. $\square$

We now show that a holomorphically convex manifold must satisfy some more geometric convexity condition, known as pseudoconvexity, which is most easily described in terms of the existence of plurisubharmonic exhaustion functions.

(6.12) Definition. *A function $\psi : X \longrightarrow [-\infty, +\infty[$ on a topological space X is said to be an exhaustion if all sublevel sets $X_c := \{z \in X\,;\, \psi(z) < c\}$, $c \in \mathbb{R}$, are relatively compact. Equivalently, ψ is an exhaustion if and only if ψ tends to $+\infty$ relatively to the filter of complements $X \smallsetminus K$ of compact subsets of X.*

A function ψ on an open set $\Omega \subset \mathbb{R}^n$ is thus an exhaustion if and only if $\psi(x) \to +\infty$ as $x \to \partial\Omega$ or $x \to \infty$. It is easy to check, cf. Exercise 8.8, that a connected open set $\Omega \subset \mathbb{R}^n$ is convex if and only if Ω has a locally convex exhaustion function. Since plurisubharmonic functions appear as the natural generalization of convex functions in complex analysis, we are led to the following definition.

(6.13) Definition. *Let X be a complex n-dimensional manifold. Then X is said to be*

a) *weakly pseudoconvex if there exists a smooth plurisubharmonic exhaustion function $\psi \in \mathrm{Psh}(X) \cap \mathscr{C}^\infty(X)$;*

b) *strongly pseudoconvex if there exists a smooth strictly plurisubharmonic exhaustion function $\psi \in \mathrm{Psh}(X) \cap \mathscr{C}^\infty(X)$, i.e. $H\psi$ is positive definite at every point.*

(6.14) Theorem. *Every holomorphically convex manifold X is weakly pseudoconvex.*

Proof. Let (K_ν) be an exhausting sequence of holomorphically convex compact sets as in Remark 6.8. For every point $a \in L_\nu := K_{\nu+2} \smallsetminus K_{\nu+1}^\circ$, one can select $g_{\nu,a} \in \mathscr{O}(\Omega)$ such

that $\sup_{K_\nu} |g_{\nu,a}| < 1$ and $|g_{\nu,a}(a)| > 1$. Then $|g_{\nu,a}(z)| > 1$ in a neighborhood of a ; by the Borel-Lebesgue lemma, one can find finitely many functions $(g_{\nu,a})_{a\in I_\nu}$ such that

$$\max_{a\in I_\nu}\big\{|g_{\nu,a}(z)|\big\} > 1 \quad \text{for} \quad z \in L_\nu, \qquad \max_{a\in I_\nu}\big\{|g_{\nu,a}(z)|\big\} < 1 \quad \text{for} \quad z \in K_\nu.$$

For a sufficiently large exponent $p(\nu)$ we get

$$\sum_{a\in I_\nu} |g_{\nu,a}|^{2p(\nu)} \geqslant \nu \quad \text{on} \quad L_\nu, \qquad \sum_{a\in I_\nu} |g_{\nu,a}|^{2p(\nu)} \leqslant 2^{-\nu} \quad \text{on} \quad K_\nu.$$

It follows that the series

$$\psi(z) = \sum_{\nu\in\mathbb{N}} \sum_{a\in I_\nu} |g_{\nu,a}(z)|^{2p(\nu)}$$

converges uniformly to a real analytic function $\psi \in \mathrm{Psh}(X)$ (see Exercise 8.11). By construction $\psi(z) \geqslant \nu$ for $z \in L_\nu$, hence ψ is an exhaustion. □

(6.15) Example. The converse to Theorem 6.14 does not hold. In fact let $X = \mathbb{C}^2/\Gamma$ be the quotient of $\mathbb{C}^2$ by the free abelian group of rank 2 generated by the affine automorphisms

$$g_1(z,w) = (z+1, e^{\mathrm{i}\theta_1}w), \qquad g_2(z,w) = (z+\mathrm{i}, e^{\mathrm{i}\theta_2}w), \qquad \theta_1,\, \theta_2 \in \mathbb{R}.$$

Since Γ acts properly discontinuously on $\mathbb{C}^2$, the quotient has a structure of a complex (non compact) 2-dimensional manifold. The function $w \mapsto |w|^2$ is Γ-invariant, hence it induces a function $\psi((z,w)^\sim) = |w|^2$ on X which is in fact a plurisubharmonic exhaustion function. Therefore X is weakly pseudoconvex. On the other hand, any holomorphic function $f \in \mathscr{O}(X)$ corresponds to a Γ-invariant holomorphic function $\widetilde{f}(z,w)$ on $\mathbb{C}^2$. Then $z \mapsto \widetilde{f}(z,w)$ is bounded for w fixed, because $\widetilde{f}(z,w)$ lies in the image of the compact set $K \times \overline{D}(0,|w|)$, $K =$ unit square in $\mathbb{C}$. By Liouville's theorem, $\widetilde{f}(z,w)$ does not depend on z. Hence functions $f \in \mathscr{O}(X)$ are in one-to-one correspondence with holomorphic functions $\widetilde{f}(w)$ on $\mathbb{C}$ such that $\widetilde{f}(e^{\mathrm{i}\theta_j}w) = \widetilde{f}(w)$. By looking at the Taylor expansion at the origin, we conclude that $\widetilde{f}$ must be a constant if $\theta_1 \notin \mathbb{Q}$ or $\theta_2 \notin \mathbb{Q}$ (if $\theta_1, \theta_2 \in \mathbb{Q}$ and m is the least common denominator of θ_1, θ_2, then $\widetilde{f}$ is a power series of the form $\sum \alpha_k w^{mk}$). From this, it follows easily that X is holomorphically convex if and only if $\theta_1, \theta_2 \in \mathbb{Q}$.

§ 6.C. Stein Manifolds

The class of holomorphically convex manifolds contains two types of manifolds of a rather different nature:

- domains of holomorphy $X = \Omega \subset \mathbb{C}^n$;
- compact complex manifolds.

In the first case we have a lot of holomorphic functions, in fact the functions in $\mathscr{O}(\Omega)$ separate any pair of points of Ω. On the other hand, if X is compact and connected, the sets $\mathrm{Psh}(X)$ and $\mathscr{O}(X)$ consist of constant functions merely (by the maximum principle). It is therefore desirable to introduce a clear distinction between these two subclasses. For

this purpose, [Stein 1951] introduced the class of manifolds which are now called Stein manifolds.

(6.16) Definition. *A complex manifold X is said to be a Stein manifold if*

a) *X is holomorphically convex*;

b) *$\mathscr{O}(X)$ locally separates points in X, i.e. every point $x \in X$ has a neighborhood V such that for any $y \in V \smallsetminus \{x\}$ there exists $f \in \mathscr{O}(X)$ with $f(y) \neq f(x)$.*

The second condition is automatic if $X = \Omega$ is an open subset of $\mathbb{C}^n$. Hence an open set $\Omega \subset \mathbb{C}^n$ is Stein if and only if Ω is a domain of holomorphy.

(6.17) Lemma. *If a complex manifold X satisfies the axiom (6.16 b) of local separation, there exists a smooth nonnegative strictly plurisubharmonic function $u \in \mathrm{Psh}(X)$.*

Proof. Fix $x_0 \in X$. We first show that there exists a smooth nonnegative function $u_0 \in \mathrm{Psh}(X)$ which is strictly plurisubharmonic on a neighborhood of x_0. Let $(z_1, \ldots, z_n)$ be local analytic coordinates centered at x_0, and if necessary, replace z_j by λz_j so that the closed unit ball $\overline{B} = \{\sum |z_j|^2 \leqslant 1\}$ is contained in the neighborhood $V \ni x_0$ on which (6.16 b) holds. Then, for every point $y \in \partial B$, there exists a holomorphic function $f \in \mathscr{O}(X)$ such that $f(y) \neq f(x_0)$. Replacing f with $\lambda(f - f(x_0))$, we can achieve $f(x_0) = 0$ and $|f(y)| > 1$. By compactness of ∂B, we find finitely many functions $f_1, \ldots, f_N \in \mathscr{O}(X)$ such that $v_0 = \sum |f_j|^2$ satisfies $v_0(x_0) = 0$, while $v_0 \geqslant 1$ on ∂B. Now, we set

$$u_0(z) = \begin{cases} v_0(z) & \text{on } X \smallsetminus B, \\ M_\varepsilon\{v_0(z), (|z|^2 + 1)/3\} & \text{on } B. \end{cases}$$

where M_ε are the regularized max functions defined in 5.18. Then u_0 is smooth and plurisubharmonic, coincides with v_0 near ∂B and with $(|z|^2 + 1)/3$ on a neighborhood of x_0. We can cover X by countably many neighborhoods $(V_j)_{j \geqslant 1}$, for which we have a smooth plurisubharmonic functions $u_j \in \mathrm{Psh}(X)$ such that u_j is strictly plurisubharmonic on V_j. Then select a sequence $\varepsilon_j > 0$ converging to 0 so fast that $u = \sum \varepsilon_j u_j \in \mathscr{C}^\infty(X)$. The function u is nonnegative and strictly plurisubharmonic everywhere on X. □

(6.18) Theorem. *Every Stein manifold is strongly pseudoconvex.*

Proof. By Th. 6.14, there is a smooth exhaustion function $\psi \in \mathrm{Psh}(X)$. If $u \geqslant 0$ is strictly plurisubharmonic, then $\psi' = \psi + u$ is a strictly plurisubharmonic exhaustion. □

The converse problem to know whether every strongly pseudoconvex manifold is actually a Stein manifold is known as the *Levi problem*, and was raised by [Levi 1910] in the case of domains $\Omega \subset \mathbb{C}^n$. In that case, the problem has been solved in the affirmative independently by [Oka 1953], [Norguet 1954] and [Bremermann 1954]. The general solution of the Levi problem has been obtained by [Grauert 1958]. Our proof will rely on the theory of L^2 estimates for d'', which will be available only in Chapter VIII.

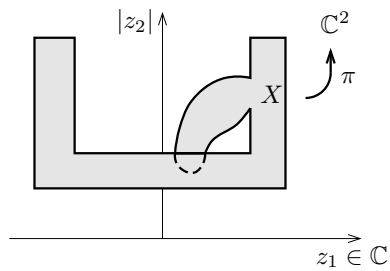

Fig. I-4 Hartogs figure with excrescence

(6.19) Remark. It will be shown later that Stein manifolds always have enough holomorphic functions to separate finitely many points, and one can even interpolate given values of a function and its derivatives of some fixed order at any discrete set of points. In particular, we might have replaced condition (6.16 b) by the stronger requirement that $\mathscr{O}(X)$ separates any pair of points. On the other hand, there are examples of manifolds satisfying the local separation condition (6.16 b), but not global separation. A simple example is obtained by attaching an excrescence inside a Hartogs figure, in such a way that the resulting map $\pi : X \to D = D(0,1)^2$ is not one-to-one (see Figure I-4 above); then $\mathscr{O}(X)$ coincides with $\pi^\star\mathscr{O}(D)$.

§ 6.D. Heredity Properties

Holomorphic convexity and pseudoconvexity are preserved under quite a number of natural constructions. The main heredity properties can be summarized in the following Proposition.

(6.20) Proposition. *Let $\mathscr{C}$ denote the class of holomorphically convex (resp. of Stein, or weakly pseudoconvex, strongly pseudoconvex manifolds).*

a) *If $X, Y \in \mathscr{C}$, then $X \times Y \in \mathscr{C}$.*

b) *If $X \in \mathscr{C}$ and S is a closed complex submanifold of X, then $S \in \mathscr{C}$.*

c) *If $(S_j)_{1\leqslant j\leqslant N}$ is a collection of (not necessarily closed) submanifolds of a complex manifold X such that $S = \bigcap S_j$ is a submanifold of X, and if $S_j \in \mathscr{C}$ for all j, then $S \in \mathscr{C}$.*

d) *If $F : X \to Y$ is a holomorphic map and $S \subset X$, $S' \subset Y$ are (not necessarily closed) submanifolds in the class $\mathscr{C}$, then $S \cap F^{-1}(S')$ is in $\mathscr{C}$, as long as it is a submanifold of X.*

e) *If X is a weakly (resp. strongly) pseudoconvex manifold and u is a smooth plurisubharmonic function on X, then the open set $\Omega = u^{-1}(]-\infty, c[)$ is weakly (resp. strongly) pseudoconvex. In particular the sublevel sets*

$$X_c = \psi^{-1}(]-\infty, c[)$$

of a (strictly) plurisubharmonic exhaustion function are weakly (resp. strongly) pseudoconvex.

Proof. All properties are more or less immediate to check, so we only give the main facts.

a) For $K \subset X$, $L \subset Y$ compact, we have $(K \times L)^{\wedge}_{\mathscr{O}(X\times Y)} = \widehat{K}_{\mathscr{O}(X)} \times \widehat{K}_{\mathscr{O}(Y)}$, and if φ, ψ are plurisubharmonic exhaustions of X, Y, then $\varphi(x) + \psi(y)$ is a plurisubharmonic exhaustion of $X \times Y$.

b) For a compact set $K \subset S$, we have $\widehat{K}_{\mathscr{O}(S)} \subset \widehat{K}_{\mathscr{O}(X)} \cap S$, and if $\psi \in \mathrm{Psh}(X)$ is an exhaustion, then $\psi_{\restriction S} \in \mathrm{Psh}(S)$ is an exhaustion (since S is closed).

c) $\bigcap S_j$ is a closed submanifold in $\prod S_j$ (equal to its intersection with the diagonal of X^N).

d) For a compact set $K \subset S \cap F^{-1}(S')$, we have

$$\widehat{K}_{\mathscr{O}(S\cap F^{-1}(S'))} \subset \widehat{K}_{\mathscr{O}(S)} \cap F^{-1}(\widehat{F(K)}_{\mathscr{O}(S')}),$$

and if φ, ψ are plurisubharmonic exhaustions of S, S', then $\varphi + \psi \circ F$ is a plurisubharmonic exhaustion of $S \cap F^{-1}(S')$.

e) $\varphi(z) := \psi(z) + 1/(c - u(z))$ is a (strictly) plurisubharmonic exhaustion function on Ω. $\square$

§ 7. Pseudoconvex Open Sets in $\mathbb{C}^n$

§ 7.A. Geometric Characterizations of Pseudoconvex Open Sets

We first discuss some characterizations of pseudoconvex open sets in $\mathbb{C}^n$. We will need the following elementary criterion for plurisubharmonicity.

(7.1) Criterion. *Let $v : \Omega \longrightarrow [-\infty, +\infty[$ be an upper semicontinuous function. Then v is plurisubharmonic if and only if for every closed disk $\overline{\Delta} = z_0 + \overline{D}(1)\eta \subset \Omega$ and every polynomial $P \in \mathbb{C}[t]$ such that $v(z_0 + t\eta) \leqslant \mathrm{Re}\, P(t)$ for $|t| = 1$, then $v(z_0) \leqslant \mathrm{Re}\, P(0)$.*

Proof. The condition is necessary because $t \longmapsto v(z_0 + t\eta) - \mathrm{Re}\, P(t)$ is subharmonic in a neighborhood of $\overline{D}(1)$, so it satisfies the maximum principle on $D(1)$ by Th. 4.14. Let us prove now the sufficiency. The upper semicontinuity of v implies $v = \lim_{\nu\to+\infty} v_\nu$ on $\partial\Delta$ where (v_ν) is a strictly decreasing sequence of continuous functions on $\partial\Delta$. As trigonometric polynomials are dense in $C^0(S^1, \mathbb{R})$, we may assume $v_\nu(z_0 + e^{\mathrm{i}\theta}\eta) = \mathrm{Re}\, P_\nu(e^{\mathrm{i}\theta})$, $P_\nu \in \mathbb{C}[t]$. Then $v(z_0 + t\eta) \leqslant \mathrm{Re}\, P_\nu(t)$ for $|t| = 1$, and the hypothesis implies

$$v(z_0) \leqslant \mathrm{Re}\, P_\nu(0) = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{Re}\, P_\nu(e^{\mathrm{i}\theta})\, d\theta = \frac{1}{2\pi}\int_0^{2\pi} v_\nu(z_0 + e^{\mathrm{i}\theta}\eta)\, d\theta.$$

Taking the limit when ν tends to $+\infty$ shows that v satisfies the mean value inequality (5.2). $\square$

For any $z \in \Omega$ and $\xi \in \mathbb{C}^n$, we denote by

$$\delta_\Omega(z, \xi) = \sup\big\{r > 0\ ;\ z + D(r)\,\xi \subset \Omega\big\}$$

the distance from z to $\partial\Omega$ in the complex direction ξ.

(7.2) Theorem. *Let $\Omega \subset \mathbb{C}^n$ be an open subset. The following properties are equivalent:*

a) *Ω is strongly pseudoconvex (according to Def. 6.13* b);

b) *Ω is weakly pseudoconvex*;

c) *Ω has a plurisubharmonic exhaustion function ψ.*

d) *$-\log \delta_\Omega(z,\xi)$ is plurisubharmonic on $\Omega \times \mathbb{C}^n$;*

e) *$-\log d(z, \complement\Omega)$ is plurisubharmonic on Ω.*

If one of these properties hold, Ω is said to be a pseudoconvex open set.

Proof. The implications a) $\Longrightarrow$ b) $\Longrightarrow$ c) are obvious. For the implication c) $\Longrightarrow$ d), we use Criterion 7.1. Consider a disk $\overline{\Delta} = (z_0, \xi_0) + \overline{D}(1)\,(\eta, \alpha)$ in $\Omega \times \mathbb{C}^n$ and a polynomial $P \in \mathbb{C}[t]$ such that

$$-\log \delta_\Omega(z_0 + t\eta, \xi_0 + t\alpha) \leqslant \operatorname{Re} P(t) \quad \text{for} \quad |t| = 1.$$

We have to verify that the inequality also holds when $|t| < 1$. Consider the holomorphic mapping $h : \mathbb{C}^2 \longrightarrow \mathbb{C}^n$ defined by

$$h(t, w) = z_0 + t\eta + we^{-P(t)}(\xi_0 + t\alpha).$$

By hypothesis

$$\begin{aligned} &h\big(\overline{D}(1) \times \{0\}\big) = \mathrm{pr}_1(\overline{\Delta}) \subset \Omega, \\ &h\big(\partial D(1) \times D(1)\big) \subset \Omega \quad (\text{since } |e^{-P}| \leqslant \delta_\Omega \ \text{ on } \partial\Delta), \end{aligned}$$

and the desired conclusion is that $h\big(\overline{D}(1) \times D(1)\big) \subset \Omega$. Let J be the set of radii $r \geqslant 0$ such that $h\big(\overline{D}(1) \times \overline{D}(r)\big) \subset \Omega$. Then J is an open interval $[0, R[$, $R > 0$. If $R < 1$, we get a contradiction as follows. Let $\psi \in \mathrm{Psh}(\Omega)$ be an exhaustion function and

$$K = h\big(\partial D(1) \times \overline{D}(R)\big) \subset\subset \Omega, \qquad c = \sup_K \psi.$$

As $\psi \circ h$ is plurisubharmonic on a neighborhood of $\overline{D}(1) \times D(R)$, the maximum principle applied with respect to t implies

$$\psi \circ h(t, w) \leqslant c \quad \text{on} \quad \overline{D}(1) \times D(R),$$

hence $h\big(\overline{D}(1) \times D(R)\big) \subset \Omega_c \subset\subset \Omega$ and $h\big(\overline{D}(1) \times \overline{D}(R + \varepsilon)\big) \subset \Omega$ for some $\varepsilon > 0$, a contradiction.

d) $\Longrightarrow$ e). The function $-\log d(z, \complement\Omega)$ is continuous on Ω and satisfies the mean value inequality because

$$-\log d(z, \complement\Omega) = \sup_{\xi \in \overline{B}} \big(-\log \delta_\Omega(z, \xi)\big).$$

e) $\Longrightarrow$ a). It is clear that

$$u(z) = |z|^2 + \max\{\log d(z, \complement\Omega)^{-1}, 0\}$$

is a continuous strictly plurisubharmonic exhaustion function. Richberg's theorem 5.21 implies that there exists $\psi \in \mathscr{C}^\infty(\Omega)$ strictly plurisubharmonic such that $u \leqslant \psi \leqslant u + 1$. Then ψ is the required exhaustion function. $\square$

(7.3) Proposition.

a) *Let $\Omega \subset \mathbb{C}^n$ and $\Omega' \subset \mathbb{C}^p$ be pseudoconvex. Then $\Omega \times \Omega'$ is pseudoconvex. For every holomorphic map $F : \Omega \to \mathbb{C}^p$ the inverse image $F^{-1}(\Omega')$ is pseudoconvex.*

b) *If $(\Omega_\alpha)_{\alpha \in I}$ is a family of pseudoconvex open subsets of $\mathbb{C}^n$, the interior of the intersection $\Omega = \big(\bigcap_{\alpha \in I} \Omega_\alpha\big)^\circ$ is pseudoconvex.*

c) *If $(\Omega_j)_{j \in \mathbb{N}}$ is a non decreasing sequence of pseudoconvex open subsets of $\mathbb{C}^n$, then $\Omega = \bigcup_{j \in \mathbb{N}} \Omega_j$ is pseudoconvex.*

Proof. a) Let φ, ψ be smooth plurisubharmonic exhaustions of Ω, Ω'. Then $(z, w) \longmapsto \varphi(z) + \psi(w)$ is an exhaustion of $\Omega \times \Omega'$ and $z \longmapsto \varphi(z) + \psi(F(z))$ is an exhaustion of $F^{-1}(\Omega')$.

b) We have $-\log d(z, \complement\Omega) = \sup_{\alpha \in I} -\log d(z, \complement\Omega_\alpha)$, so this function is plurisubharmonic.

c) The limit $-\log d(z, \complement\Omega) = \lim\!\downarrow_{j \to +\infty} -\log d(z, \complement\Omega_j)$ is plurisubharmonic, hence Ω is pseudoconvex. This result cannot be generalized to strongly pseudoconvex manifolds: J.E. Fornæss in [Fornaess 1976] has constructed an increasing sequence of 2-dimensional Stein (even affine algebraic) manifolds X_ν whose union is not Stein; see Exercise 8.16. $\square$

(7.4) Examples. a) An *analytic polyhedron* in $\mathbb{C}^n$ is an open subset of the form

$$P = \{z \in \mathbb{C}^n \,;\, |f_j(z)| < 1,\ 1 \leqslant j \leqslant N\}$$

where $(f_j)_{1 \leqslant j \leqslant N}$ is a family of analytic functions on $\mathbb{C}^n$. By 7.3 a), every analytic polyhedron is pseudoconvex.

b) Let $\omega \subset \mathbb{C}^{n-1}$ be pseudoconvex and let $u : \omega \longrightarrow [-\infty, +\infty[$ be an upper semicontinuous function. Then the *Hartogs domain*

$$\Omega = \big\{(z_1, z') \in \mathbb{C} \times \omega \,;\, \log|z_1| + u(z') < 0\big\}$$

is pseudoconvex if and and only if u is plurisubharmonic. To see that the plurisubharmonicity of u is necessary, observe that

$$u(z') = -\log \delta_\Omega\big((0, z'), (1, 0)\big).$$

Conversely, assume that u is plurisubharmonic and continuous. If ψ is a plurisubharmonic exhaustion of ω, then

$$\psi(z') + \big|\log|z_1| + u(z')\big|^{-1}$$

is an exhaustion of Ω. This is no longer true if u is not continuous, but in this case we may apply Property 7.3 c) to conclude that

$$\Omega_\varepsilon = \big\{(z_1, z')\,;\, d(z', \complement\omega) > \varepsilon,\ \log|z_1| + u \star \rho_\varepsilon(z') < 0\big\}, \qquad \Omega = \bigcup \Omega_\varepsilon$$

are pseudoconvex.

c) An open set $\Omega \subset \mathbb{C}^n$ is called a *tube* of base ω if $\Omega = \omega + \mathrm{i}\mathbb{R}^n$ for some open subset $\omega \subset \mathbb{R}^n$. Then of course $-\log d(z, \complement\Omega) = -\log(x, \complement\omega)$ depends only on the real part $x = \operatorname{Re} z$. By Th. 5.13, this function is plurisubharmonic if and only if it is locally convex in x. Therefore Ω if pseudoconvex if and only if every connected component of ω is convex.

d) An open set $\Omega \subset \mathbb{C}^n$ is called a *Reinhardt domain* if $(e^{\mathrm{i}\theta_1} z_1, \ldots, e^{\mathrm{i}\theta_n} z_n)$ is in Ω for every $z = (z_1, \ldots, z_n) \in \Omega$ and $\theta_1, \ldots, \theta_n \in \mathbb{R}^n$. For such a domain, we consider the *logarithmic indicatrix*

$$\omega^\star = \Omega^\star \cap \mathbb{R}^n \quad \text{with} \quad \Omega^\star = \{\zeta \in \mathbb{C}^n\,;\, (e^{\zeta_1}, \ldots, e^{\zeta_n}) \in \Omega\}.$$

It is clear that $\Omega^\star$ is a tube of base $\omega^\star$. Therefore every connected component of $\omega^\star$ must be convex if Ω is pseudoconvex. The converse is not true: $\Omega = \mathbb{C}^n \smallsetminus \{0\}$ is not pseudoconvex for $n \geqslant 2$ although $\omega^\star = \mathbb{R}^n$ is convex. However, the Reinhardt open set

$$\Omega^\bullet = \big\{(z_1, \ldots, z_n) \in (\mathbb{C} \smallsetminus \{0\})^n\,;\, (\log|z_1|, \ldots, \log|z_n|) \in \omega^\star\big\} \subset \Omega$$

is easily seen to be pseudoconvex if $\omega^\star$ is convex: if χ is a convex exhaustion of $\omega^\star$, then $\psi(z) = \chi(\log|z_1|, \ldots, \log|z_n|)$ is a plurisubharmonic exhaustion of $\Omega^\bullet$. Similarly, if $\omega^\star$ is convex and such that $x \in \omega^\star \Longrightarrow y \in \omega^\star$ for $y_j \leqslant x_j$, we can take χ increasing in all variables and tending to $+\infty$ on $\partial\omega^\star$, hence the set

$$\widetilde{\Omega} = \big\{(z_1, \ldots, z_n) \in \mathbb{C}^n\,;\, |z_j| \leqslant e^{x_j} \ \text{ for some } \ x \in \omega^\star\big\}$$

is a pseudoconvex Reinhardt open set containing 0. □

§ 7.B. Kiselman's Minimum Principle

We already know that a maximum of plurisubharmonic functions is plurisubharmonic. However, if v is a plurisubharmonic function on $X \times \mathbb{C}^n$, the partial minimum function on X defined by $u(\zeta) = \inf_{z \in \Omega} v(\zeta, z)$ need not be plurisubharmonic. A simple counterexample in $\mathbb{C} \times \mathbb{C}$ is given by

$$v(\zeta, z) = |z|^2 + 2\operatorname{Re}(z\zeta) = |z + \overline{\zeta}|^2 - |\zeta|^2, \qquad u(\zeta) = -|\zeta|^2.$$

It follows that the image $F(\Omega)$ of a pseudoconvex open set Ω by a holomorphic map F need not be pseudoconvex. In fact, if

$$\Omega = \{(t, \zeta, z) \in \mathbb{C}^3\,;\, \log|t| + v(\zeta, z) < 0\}$$

and if $\Omega' \subset \mathbb{C}^2$ is the image of Ω by the projection map $(t, \zeta, z) \longmapsto (t, \zeta)$, then $\Omega' = \{(t, \zeta) \in \mathbb{C}^2\,;\, \log|t| + u(\zeta) < 0\}$ is not pseudoconvex. However, the minimum property holds true when $v(\zeta, z)$ depends only on $\operatorname{Re} z$:

(7.5) Theorem ([Kiselman 1978]). *Let $\Omega \subset \mathbb{C}^p \times \mathbb{C}^n$ be a pseudoconvex open set such that each slice*

$$\Omega_\zeta = \{z \in \mathbb{C}^n\,;\,(\zeta, z) \in \Omega\}, \qquad \zeta \in \mathbb{C}^p,$$

is a convex tube $\omega_\zeta + \mathrm{i}\mathbb{R}^n$, $\omega_\zeta \subset \mathbb{R}^n$. For every plurisubharmonic function $v(\zeta, z)$ on Ω that does not depend on $\operatorname{Im} z$, the function

$$u(\zeta) = \inf_{z \in \Omega_\zeta} v(\zeta, z)$$

is plurisubharmonic or locally $\equiv -\infty$ on $\Omega' = \mathrm{pr}_{\mathbb{C}^n}(\Omega)$.

Proof. The hypothesis implies that $v(\zeta, z)$ is convex in $x = \operatorname{Re} z$. In addition, we first assume that v is smooth, plurisubharmonic in (ζ, z), strictly convex in x and

$$\lim_{x \to \{\infty\} \cup \partial\omega_\zeta} v(\zeta, x) = +\infty$$

for every $\zeta \in \Omega'$. Then $x \longmapsto v(\zeta, x)$ has a unique minimum point $x = g(\zeta)$, solution of the equations $\partial v/\partial x_j(x, \zeta) = 0$. As the matrix $(\partial^2 v/\partial x_j \partial x_k)$ is positive definite, the implicit function theorem shows that g is smooth. Now, if $\mathbb{C} \ni w \longmapsto \zeta_0 + wa$, $a \in \mathbb{C}^n$, $|w| \leqslant 1$ is a complex disk Δ contained in Ω, there exists a holomorphic function f on the unit disk, smooth up to the boundary, whose real part solves the Dirichlet problem

$$\operatorname{Re} f(e^{\mathrm{i}\theta}) = g(\zeta_0 + e^{\mathrm{i}\theta} a).$$

Since $v(\zeta_0 + wa, f(w))$ is subharmonic in w, we get the mean value inequality

$$v(\zeta_0, f(0)) \leqslant \frac{1}{2\pi} \int_0^{2\pi} v\big(\zeta_0 + e^{\mathrm{i}\theta} a, f(e^{\mathrm{i}\theta})\big) d\theta = \frac{1}{2\pi} \int_{\partial\Delta} v(\zeta, g(\zeta)) d\theta.$$

The last equality holds because $\operatorname{Re} f = g$ on $\partial\Delta$ and $v(\zeta, z) = v(\zeta, \operatorname{Re} z)$ by hypothesis. As $u(\zeta_0) \leqslant v(\zeta_0, f(0))$ and $u(\zeta) = v(\zeta, g(\zeta))$, we see that u satisfies the mean value inequality, thus u is plurisubharmonic.

Now, this result can be extended to arbitrary functions v as follows: let $\psi(\zeta, z) \geqslant 0$ be a continuous plurisubharmonic function on Ω which is independent of $\operatorname{Im} z$ and is an exhaustion of $\Omega \cap (\mathbb{C}^p \times \mathbb{R}^n)$, e.g.

$$\psi(\zeta, z) = \max\{|\zeta|^2 + |\operatorname{Re} z|^2, -\log \delta_\Omega(\zeta, z)\}.$$

There is slowly increasing sequence $C_j \to +\infty$ such that each function $\psi_j = (C_j - \psi \star \rho_{1/j})^{-1}$ is an "exhaustion" of a pseudoconvex open set $\Omega_j \subset\subset \Omega$ whose slices are convex tubes and such that $d(\Omega_j, \complement\Omega) > 2/j$. Then

$$v_j(\zeta, z) = v \star \rho_{1/j}(\zeta, z) + \frac{1}{j} |\operatorname{Re} z|^2 + \psi_j(\zeta, z)$$

is a decreasing sequence of plurisubharmonic functions on Ω_j satisfying our previous conditions. As $v = \lim v_j$, we see that $u = \lim u_j$ is plurisubharmonic. □

(7.6) Corollary. *Let $\Omega \subset \mathbb{C}^p \times \mathbb{C}^n$ be a pseudoconvex open set such that all slices Ω_ζ, $\zeta \in \mathbb{C}^p$, are convex tubes in $\mathbb{C}^n$. Then the projection Ω' of Ω on $\mathbb{C}^p$ is pseudoconvex.*

Proof. Take $v \in \mathrm{Psh}(\Omega)$ equal to the function ψ defined in the proof of Th. 7.5. Then u is a plurisubharmonic exhaustion of Ω'. □

§ 7.C. Levi Form of the Boundary

For an arbitrary domain in $\mathbb{C}^n$, we first show that pseudoconvexity is a local property of the boundary.

(7.7) Theorem. *Let $\Omega \subset \mathbb{C}^n$ be an open subset such that every point $z_0 \in \partial\Omega$ has a neighborhood V such that $\Omega \cap V$ is pseudoconvex. Then Ω is pseudoconvex.*

Proof. As $d(z, \complement\Omega)$ coincides with $d\big(z, \complement(\Omega \cap V)\big)$ in a neighborhood of z_0, we see that there exists a neighborhood U of $\partial\Omega$ such that $-\log d(z, \complement\Omega)$ is plurisubharmonic on $\Omega \cap U$. Choose a convex increasing function χ such that

$$\chi(r) > \sup_{(\Omega \smallsetminus U) \cap \overline{B}(0,r)} -\log d(z, \complement\Omega), \qquad \forall r \geqslant 0.$$

Then the function

$$\psi(z) = \max\big\{\chi(|z|), -\log d(z, \complement\Omega)\big\}$$

coincides with $\chi(|z|)$ in a neighborhood of $\Omega \smallsetminus U$. Therefore $\psi \in \mathrm{Psh}(\Omega)$, and ψ is clearly an exhaustion. $\square$

Now, we give a geometric characterization of the pseudoconvexity property when $\partial\Omega$ is of class C^2. Let $\rho \in C^2(\overline{\Omega})$ be a *defining function* of Ω, i.e. a function such that

$$\rho < 0 \ \text{ on } \ \Omega, \qquad \rho = 0 \ \text{ and } \ d\rho \neq 0 \ \text{ on } \ \partial\Omega. \tag{7.8}$$

The *holomorphic tangent space* to $\partial\Omega$ is by definition the largest complex subspace which is contained in the tangent space $T_{\partial\Omega}$ to the boundary:

$${}^hT_{\partial\Omega} = T_{\partial\Omega} \cap JT_{\partial\Omega}. \tag{7.9}$$

It is easy to see that ${}^hT_{\partial\Omega,z}$ is the complex hyperplane of vectors $\xi \in \mathbb{C}^n$ such that

$$d'\rho(z) \cdot \xi = \sum_{1 \leqslant j \leqslant n} \frac{\partial\rho}{\partial z_j}\, \xi_j = 0.$$

The *Levi form* on ${}^hT_{\partial\Omega}$ is defined at every point $z \in \partial\Omega$ by

$$L_{\partial\Omega,z}(\xi) = \frac{1}{|\nabla\rho(z)|} \sum_{j,k} \frac{\partial^2\rho}{\partial z_j \partial \overline{z}_k}\, \xi_j \overline{\xi}_k, \qquad \xi \in {}^hT_{\partial\Omega,z}. \tag{7.10}$$

The Levi form does not depend on the particular choice of ρ, as can be seen from the following intrinsic computation of $L_{\partial\Omega}$ (we still denote by $L_{\partial\Omega}$ the associated sesquilinear form).

(7.11) Lemma. *Let ξ, η be C^1 vector fields on $\partial\Omega$ with values in ${}^hT_{\partial\Omega}$. Then*

$$\langle [\xi, \eta], J\nu \rangle = 4\,\mathrm{Im}\, L_{\partial\Omega}(\xi, \eta)$$

where ν is the outward normal unit vector to $\partial\Omega$, $[\ ,\ ]$ the Lie bracket of vector fields and $\langle\ ,\ \rangle$ the hermitian inner product.

Proof. Extend first ξ, η as vector fields in a neighborhood of $\partial\Omega$ and set

$$\xi' = \sum \xi_j \frac{\partial}{\partial z_j} = \frac{1}{2}(\xi - \mathrm{i}J\xi), \qquad \eta'' = \sum \overline{\eta}_k \frac{\partial}{\partial \overline{z}_k} = \frac{1}{2}(\eta + \mathrm{i}J\eta).$$

As $\xi, J\xi, \eta, J\eta$ are tangent to $\partial\Omega$, we get on $\partial\Omega$:

$$0 = \xi'.(\eta''.\rho) + \eta''.(\xi'.\rho) = \sum_{1\leqslant j,k\leqslant n} 2\frac{\partial^2\rho}{\partial z_j \partial \overline{z}_k}\,\xi_j\overline{\eta}_k + \xi_j \frac{\partial\overline{\eta}_k}{\partial z_j}\frac{\partial\rho}{\partial\overline{z}_k} + \overline{\eta}_k \frac{\partial\xi_j}{\partial\overline{z}_k}\frac{\partial\rho}{\partial z_j}.$$

Since $[\xi,\eta]$ is also tangent to $\partial\Omega$, we have $\mathrm{Re}\langle[\xi,\eta],\nu\rangle = 0$, hence $\langle J[\xi,\eta],\nu\rangle$ is real and

$$\langle[\xi,\eta], J\nu\rangle = -\langle J[\xi,\eta],\nu\rangle = -\frac{1}{|\nabla\rho|}\big(J[\xi,\eta].\rho\big) = -\frac{2}{|\nabla\rho|}\,\mathrm{Re}\,\big(J[\xi',\eta''].\rho\big)$$

because $J[\xi',\eta'] = i[\xi',\eta']$ and its conjugate $J[\xi'',\eta'']$ are tangent to $\partial\Omega$. We find now

$$J[\xi',\eta''] = -\mathrm{i}\sum \xi_j \frac{\partial\overline{\eta}_k}{\partial z_j}\frac{\partial}{\partial\overline{z}_k} + \overline{\eta}_k\frac{\partial\xi_j}{\partial\overline{z}_k}\frac{\partial}{\partial\overline{z}_j},$$

$$\mathrm{Re}\,\big(J[\xi',\eta''].\rho\big) = \mathrm{Im}\sum \xi_j\frac{\partial\overline{\eta}_k}{\partial z_j}\frac{\partial\rho}{\partial\overline{z}_k} + \overline{\eta}_k\frac{\partial\xi_j}{\partial\overline{z}_k}\frac{\partial\rho}{\partial z_j} = -2\,\mathrm{Im}\sum\frac{\partial^2\rho}{\partial z_j\partial\overline{z}_k}\,\xi_j\overline{\eta}_k,$$

$$\langle[\xi,\eta], J\nu\rangle = \frac{4}{|\nabla\rho|}\,\mathrm{Im}\sum\frac{\partial^2\rho}{\partial z_j\partial\overline{z}_k}\,\xi_j\overline{\eta}_k = 4\,\mathrm{Im}\,L_{\partial\Omega}(\xi,\eta). \quad \square$$

(7.12) Theorem. *An open subset $\Omega\subset\mathbb{C}^n$ with C^2 boundary is pseudoconvex if and only if the Levi form $L_{\partial\Omega}$ is semipositive at every point of $\partial\Omega$.*

Proof. Set $\delta(z) = d(z, \complement\Omega)$, $z\in\overline{\Omega}$. Then $\rho = -\delta$ is C^2 near $\partial\Omega$ and satisfies (7.8). If Ω is pseudoconvex, the plurisubharmonicity of $-\log(-\rho)$ means that for all $z\in\Omega$ near $\partial\Omega$ and all $\xi\in\mathbb{C}^n$ one has

$$\sum_{1\leqslant j,k\leqslant n}\Big(\frac{1}{|\rho|}\,\frac{\partial^2\rho}{\partial z_j\partial\overline{z}_k} + \frac{1}{\rho^2}\,\frac{\partial\rho}{\partial z_j}\,\frac{\partial\rho}{\partial\overline{z}_k}\Big)\xi_j\overline{\xi}_k \geqslant 0.$$

Hence $\sum(\partial^2\rho/\partial z_j\partial\overline{z}_k)\xi_j\overline{\xi}_k \geqslant 0$ if $\sum(\partial\rho/\partial z_j)\xi_j = 0$, and an easy argument shows that this is also true at the limit on $\partial\Omega$.

Conversely, if Ω is not pseudoconvex, Th. 7.2 and 7.7 show that $-\log\delta$ is not plurisubharmonic in any neighborhood of $\partial\Omega$. Hence there exists $\xi\in\mathbb{C}^n$ such that

$$c = \Big(\frac{\partial^2}{\partial t\partial\overline{t}}\,\log\delta(z+t\xi)\Big)_{|t=0} > 0$$

for some z in the neighborhood of $\partial\Omega$ where $\delta\in C^2$. By Taylor's formula, we have

$$\log\delta(z+t\xi) = \log\delta(z) + \mathrm{Re}(at+bt^2) + c|t|^2 + \mathrm{o}(|t|^2)$$

with $a,b\in\mathbb{C}$. Now, choose $z_0\in\partial\Omega$ such that $\delta(z) = |z-z_0|$ and set

$$h(t) = z + t\xi + e^{at+bt^2}(z_0 - z), \qquad t\in\mathbb{C}.$$

Then we get $h(0) = z_0$ and

$$\begin{aligned}\delta(h(t)) &\geqslant \delta(z + t\xi) - \delta(z)\left|e^{at+bt^2}\right| \\ &\geqslant \delta(z)\left|e^{at+bt^2}\right|\left(e^{c|t|^2/2} - 1\right) \geqslant \delta(z)\, c|t|^2/3\end{aligned}$$

when $|t|$ is sufficiently small. Since $\delta(h(0)) = \delta(z_0) = 0$, we obtain at $t = 0$:

$$\frac{\partial}{\partial t}\,\delta(h(t)) = \sum \frac{\partial \delta}{\partial z_j}(z_0)\, h'_j(0) = 0,$$

$$\frac{\partial^2}{\partial t \partial \bar{t}}\,\delta(h(t)) = \sum \frac{\partial^2 \delta}{\partial z_j \partial \bar{z}_k}(z_0)\, h'_j(0)\overline{h'_k(0)} > 0,$$

hence $h'(0) \in {}^hT_{\partial\Omega, z_0}$ and $L_{\partial\Omega, z_0}(h'(0)) < 0$. □

(7.13) Definition. *The boundary $\partial\Omega$ is said to be weakly (resp. strongly) pseudoconvex if $L_{\partial\Omega}$ is semipositive (resp. positive definite) on $\partial\Omega$. The boundary is said to be Levi flat if $L_{\partial\Omega} \equiv 0$.*

(7.14) Remark. Lemma 7.11 shows that $\partial\Omega$ is Levi flat if and only if the subbundle ${}^hT_{\partial\Omega} \subset T_{\partial\Omega}$ is integrable (i.e. stable under the Lie bracket). Assume that $\partial\Omega$ is of class $\mathscr{C}^k$, $k \geqslant 2$. Then ${}^hT_{\partial\Omega}$ is of class C^{k-1}. By Frobenius' theorem, the integrability condition implies that ${}^hT_{\partial\Omega}$ is the tangent bundle to a $\mathscr{C}^k$ foliation of $\partial\Omega$ whose leaves have real dimension $2n - 2$. But the leaves themselves must be complex analytic since ${}^hT_{\partial\Omega}$ is a complex vector space (cf. Lemma 7.15 below). Therefore $\partial\Omega$ is Levi flat if and only if it is foliated by complex analytic hypersurfaces.

(7.15) Lemma. *Let Y be a C^1-submanifold of a complex analytic manifold X. If the tangent space $T_{Y,x}$ is a complex subspace of $T_{X,x}$ at every point $x \in Y$, then Y is complex analytic.*

Proof. Let $x_0 \in Y$. Select holomorphic coordinates $(z_1, \ldots, z_n)$ on X centered at x_0 such that T_{Y,x_0} is spanned by $\partial/\partial z_1, \ldots, \partial/\partial z_p$. Then there exists a neighborhood $U = U' \times U''$ of x_0 such that $Y \cap U$ is a graph

$$z'' = h(z'), \qquad z' = (z_1, \ldots, z_p) \in U', \quad z'' = (z_{p+1}, \ldots, z_n)$$

with $h \in C^1(U')$ and $dh(0) = 0$. The differential of h at z' is the composite of the projection of $\mathbb{C}^p \times \{0\}$ on $T_{Y,(z',h(z'))}$ along $\{0\} \times \mathbb{C}^{n-p}$ and of the second projection $\mathbb{C}^n \to \mathbb{C}^{n-p}$. Hence $dh(z')$ is $\mathbb{C}$-linear at every point and h is holomorphic. □

§ 8. Exercises

§ 8.1. Let $\Omega \subset \mathbb{C}^n$ be an open set such that

$$z \in \Omega, \quad \lambda \in \mathbb{C}, \quad |\lambda| \leqslant 1 \Longrightarrow \lambda z \in \Omega.$$

Show that Ω is a union of polydisks of center 0 (with arbitrary linear changes of coordinates) and infer that the space of polynomials $\mathbb{C}[z_1, \ldots, z_n]$ is dense in $\mathscr{O}(\Omega)$ for the topology of uniform convergence on compact subsets and in $\mathscr{O}(\Omega) \cap C^0(\overline{\Omega})$ for the topology of uniform convergence on $\overline{\Omega}$.
Hint: consider the Taylor expansion of a function $f \in \mathscr{O}(\Omega)$ at the origin, writing it as a series of homogeneous polynomials. To deal with the case of $\overline{\Omega}$, first apply a dilation to f.

§ 8.2. Let $B \subset \mathbb{C}^n$ be the unit euclidean ball, $S = \partial B$ and $f \in \mathscr{O}(B) \cap C^0(\overline{B})$. Our goal is to check the following Cauchy formula:

$$f(w) = \frac{1}{\sigma_{2n-1}} \int_S \frac{f(z)}{(1 - \langle w, z\rangle)^n}\, d\sigma(z).$$

a) By means of a unitary transformation and Exercise 8.1, reduce the question to the case when $w = (w_1, 0, \ldots, 0)$ and $f(z)$ is a monomial z^α.

b) Show that the integral $\int_B z^\alpha \overline{z}_1^k \, d\lambda(z)$ vanishes unless $\alpha = (k, 0, \ldots, 0)$. Compute the value of the remaining integral by the Fubini theorem, as well as the integrals $\int_S z^\alpha \overline{z}_1^k \, d\sigma(z)$.

c) Prove the formula by a suitable power series expansion.

§ 8.3. A current $T \in \mathscr{D}'_p(M)$ is said to be *normal* if both T and dT are of order zero, i.e. have measure coefficients.

a) If T is normal and has support contained in a C^1 submanifold $Y \subset M$, show that there exists a normal current Θ on Y such that $T = j_\star \Theta$, where $j : Y \longrightarrow M$ is the inclusion.
Hint: if $x_1 = \ldots = x_q = 0$ are equations of Y in a coordinate system $(x_1, \ldots, x_n)$, observe that $x_j T = x_j dT = 0$ for $1 \leqslant j \leqslant q$ and infer that $dx_1 \wedge \ldots \wedge dx_q$ can be factorized in all terms of T.

b) What happens if $p > \dim Y$?

c) Are a) and b) valid when the normality assumption is dropped ?

§ 8.4. Let $T = \sum_{1 \leqslant j \leqslant n} T_j d\overline{z}_j$ be a closed current of bidegree $(0,1)$ with compact support in $\mathbb{C}^n$ such that $d''T = 0$.

a) Show that the partial convolution $S = (1/\pi z_1) \star_1 T_1$ is a solution of the equation $d''S = T$.

b) Let $K = \operatorname{Supp} T$. If $n \geqslant 2$, show that S has support in the compact set $\widetilde{K}$ equal to the union of K and of all bounded components of $\mathbb{C}^n \smallsetminus K$.
Hint: observe that S is holomorphic on $\mathbb{C}^n \smallsetminus K$ and that S vanishes for $|z_2| + \ldots + |z_n|$ large.

§ 8.5. Alternative proof of the Dolbeault-Grothendieck lemma. Let $v = \sum_{|J|=q} v_J d\overline{z}_J$, $q \geqslant 1$, be a smooth form of bidegree $(0,q)$ on a polydisk $\Omega = D(0,R) \subset \mathbb{C}^n$, such that $d''v = 0$, and let $\omega = D(0,r) \subset\subset \omega$. Let k be the smallest integer such that the monomials $d\overline{z}_J$ appearing in v only involve $d\overline{z}_1, \ldots, d\overline{z}_k$. Prove by induction on k that the equation $d''u = v$ can be solved on ω.
Hint: set $v = f \wedge d\overline{z}_k + g$ where f, g only involve $d\overline{z}_1, \ldots, d\overline{z}_{k-1}$. Then consider $v - d''F$ where

$$F = \sum_{|J|=q-1} F_J d\overline{z}_J, \qquad F_J(z) = (\psi(z_k) f_J) \star_k \Big(\frac{1}{\pi z_k}\Big),$$

where $\star_k$ denotes the partial convolution with respect to z_k, $\psi(z_k)$ is a cut-off function equal to 1 on $D(0, r_k + \varepsilon)$ and $f = \sum_{|J|=q-1} f_J d\overline{z}_J$.

§ 8.6. Construct locally bounded non continuous subharmonic functions on $\mathbb{C}$.
Hint: consider e^u where $u(z) = \sum_{j \geqslant 1} 2^{-j} \log|z - 1/j|$.

§ 8.7. Let ω be an open subset of $\mathbb{R}^n$, $n \geqslant 2$, and u a subharmonic function which is not locally $-\infty$.

a) For every open set $\omega \subset\subset \Omega$, show that there is a positive measure μ with support in $\overline{\omega}$ and a harmonic function h on ω such that $u = N \star \mu + h$ on ω.

b) Use this representation to prove the following properties: $u \in L^p_{\mathrm{loc}}$ for all $p < n/(n-2)$ and $\partial u / \partial x_j \in L^p_{\mathrm{loc}}$ for all $p < n/(n-1)$.

§ 8.8. Show that a connected open set $\Omega \subset \mathbb{R}^n$ is convex if and only if Ω has a locally convex exhaustion function φ.
Hint: to show the sufficiency, take a path $\gamma : [0,1] \to \Omega$ joining two arbitrary points $a, b \in \Omega$ and consider the restriction of φ to $[a, \gamma(t_0)] \cap \Omega$ where t_0 is the supremum of all t such that $[a, \gamma(u)] \subset \Omega$ for $u \in [0,t]$.

§ 8.9. Let $r_1, r_2 \in]1, +\infty[$. Consider the compact set

$$K = \{|z_1| \leqslant r_1\,,\ |z_2| \leqslant 1\} \cup \{|z_1| \leqslant 1\,,\ |z_2| \leqslant r_2\} \subset \mathbb{C}^2.$$

Show that the holomorphic hull of K in $\mathbb{C}^2$ is

$$\widehat{K} = \{|z_1| \leqslant r_1\,,\ |z_2| \leqslant r_2\,,\ |z_1|^{1/\log r_1}|z_2|^{1/\log r_2} \leqslant e\}.$$

Hint: to show that $\widehat{K}$ is contained in this set, consider all holomorphic monomials $f(z_1, z_2) = z_1^{\alpha_1} z_2^{\alpha_2}$. To show the converse inclusion, apply the maximum principle to the domain $|z_1| \leqslant r_1$, $|z_2| \leqslant r_2$ on suitably chosen Riemann surfaces $z_1^{\alpha_1} z_2^{\alpha_2} = \lambda$.

§ 8.10. Compute the rank of the Levi form of the ellipsoid $|z_1|^2 + |z_3|^4 + |z_3|^6 < 1$ at every point of the boundary.

§ 8.11. Let X be a complex manifold and let $u(z) = \sum_{j\in\mathbb{N}} |f_j|^2$, $f_j \in \mathscr{O}(X)$, be a series converging uniformly on every compact subset of X. Prove that the limit is real analytic and that the series remains uniformly convergent by taking derivatives term by term.
Hint: since the problem is local, take $X = B(0, r)$, a ball in $\mathbb{C}^n$. Let $g_j(z) = \overline{g_j(\overline{z})}$ be the conjugate function of f_j and let $U(z, w) = \sum_{j\in\mathbb{N}} f_j(z)g_j(w)$ on $X \times X$. Using the Cauchy-Schwarz inequality, show that this series of holomorphic functions is uniformly convergent on every compact subset of $X \times X$.

§ 8.12. Let $\Omega \subset \mathbb{C}^n$ be a bounded open set with C^2 boundary.

a) Let $a \in \partial\Omega$ be a given point. Let e_n be the outward normal vector to $T_{\partial\Omega,a}$, $(e_1, \ldots, e_{n-1})$ an orthonormal basis of ${}^hT_a(\partial\Omega)$ in which the Levi form is diagonal and $(z_1, \ldots, z_n)$ the associated linear coordinates centered at a. Show that there is a neighborhood V of a such that $\partial\Omega \cap V$ is the graph $\operatorname{Re} z_n = -\varphi(z_1, \ldots, z_{n-1}, \operatorname{Im} z_n)$ of a function φ such that $\varphi(z) = O(|z|^2)$ and the matrix $\partial^2\varphi/\partial z_j\partial\overline{z}_k(0)$, $1 \leqslant j, k \leqslant n-1$ is diagonal.

b) Show that there exist local analytic coordinates $w_1 = z_1, \ldots, w_{n-1} = z_{n-1}$, $w_n = z_n + \sum c_{jk} z_j z_k$ on a neighborhood V' of $a = 0$ such that

$$\Omega \cap V' = V' \cap \{\operatorname{Re} w_n + \sum_{1\leqslant j\leqslant n} \lambda_j |w_j|^2 + \mathrm{o}(|w|^2) < 0\}, \qquad \lambda_j \in \mathbb{R}$$

and that λ_n can be assigned to any given value by a suitable choice of the coordinates.
Hint: Consider the Taylor expansion of order 2 of the defining function $\rho(z) = (\operatorname{Re} z_n + \varphi(z))(1 + \operatorname{Re} \sum c_j z_j)$ where $c_j \in \mathbb{C}$ are chosen in a suitable way.

c) Prove that $\partial\Omega$ is strongly pseudoconvex at a if and only if there is a neighborhood U of a and a biholomorphism Φ of U onto some open set of $\mathbb{C}^n$ such that $\Phi(\Omega \cap U)$ is strongly convex.

d) Assume that the Levi form of $\partial\Omega$ is not semipositive. Show that all holomorphic functions $f \in \mathscr{O}(\Omega)$ extend to some (fixed) neighborhood of a.
Hint: assume for example $\lambda_1 < 0$. For $\varepsilon > 0$ small, show that Ω contains the Hartogs figure

$$\{\varepsilon/2 < |w_1| < \varepsilon\} \times \{|w_j| < \varepsilon^2\}_{1<j<n} \times \{|w_n| < \varepsilon^{3/2}\,,\ \operatorname{Re} w_n < \varepsilon^3\} \ \cup$$
$$\{|w_1| < \varepsilon\} \times \{|w_j| < \varepsilon^2\}_{1<j<n} \times \{|w_n| < \varepsilon^{3/2}\,,\ \operatorname{Re} w_n < -\varepsilon^2\}.$$

§ 8.13. Let $\Omega \subset \mathbb{C}^n$ be a bounded open set with C^2 boundary and $\rho \in C^2(\Omega, \mathbb{R})$ such that $\rho < 0$ on Ω, $\rho = 0$ and $d\rho \neq 0$ on $\partial\Omega$. Let $f \in C^1(\partial\Omega, \mathbb{C})$ be a function satisfying the *tangential Cauchy-Riemann equations*

$$\xi'' \cdot f = 0, \qquad \forall \xi \in {}^hT_{\partial\Omega}, \qquad \xi'' = \frac{1}{2}(\xi + \mathrm{i}J\xi).$$

a) Let f_0 be a C^1 extension of f to $\overline{\Omega}$. Show that $d''f_0 \wedge d''\rho = 0$ on $\partial\Omega$ and infer that $v = \mathbb{1}_\Omega d''f_0$ is a d''-closed current on $\mathbb{C}^n$.

b) Show that the solution u of $d''u = v$ provided by Cor. 3.27 is continuous and that f admits an extension $\widetilde{f} \in \mathscr{O}(\Omega) \cap C^0(\overline{\Omega})$ if $\partial\Omega$ is connected.

§ 8.14. Let $\Omega \subset \mathbb{C}^n$ be a bounded pseudoconvex domain with C^2 boundary and let $\delta(z) = d(z, \complement\Omega)$ be the euclidean distance to the boundary.

a) Use the plurisubharmonicity of $-\log\delta$ to prove the following fact: for every $\varepsilon > 0$ there is a constant $C_\varepsilon > 0$ such that

$$\frac{-H\delta_z(\xi)}{\delta(z)} + \varepsilon\frac{|d'\delta_z.\xi|^2}{|\delta(z)|^2} + C_\varepsilon|\xi|^2 \geqslant 0$$

for $\xi \in \mathbb{C}^n$ and z near $\partial\Omega$.

b) Set $\psi(z) = -\log\delta(z) + K|z|^2$. Show that for K large and α small the function

$$\rho(z) = -\exp\big(-\alpha\psi(z)\big) = -\big(e^{-K|z|^2}\delta(z)\big)^\alpha$$

is plurisubharmonic.

c) Prove the existence of a plurisubharmonic exhaustion function $u : \Omega \to [-1, 0[$ of class C^2 such that $|u(z)|$ has the same order of magnitude as $\delta(z)^\alpha$ when z tends to $\partial\Omega$.
Hint: consult [Diederich-Fornaess 1977].

§ 8.15. Let $\Omega = \omega + \mathrm{i}\mathbb{R}^n$ be a connected tube in $\mathbb{C}^n$ of base ω.

a) Assume first that $n = 2$. Let $T \subset \mathbb{R}^2$ be the triangle $x_1 \geqslant 0$, $x_2 \geqslant 0$, $x_1 + x_2 \leqslant 1$, and assume that the two edges $[0, 1] \times \{0\}$ and $\{0\} \times [0, 1]$ are contained in ω. Show that every holomorphic function $f \in \mathscr{O}(\Omega)$ extends to a neighborhood of $T + \mathrm{i}\mathbb{R}^2$.
Hint: let $\pi : \mathbb{C}^2 \longrightarrow \mathbb{R}^2$ be the projection on the real part and M_ε the intersection of $\pi^{-1}((1+\varepsilon)T)$ with the Riemann surface $z_1 + z_2 - \frac{\varepsilon}{2}(z_1^2 + z_2^2) = 1$ (a non degenerate affine conic). Show that M_ε is compact and that

$$\pi(\partial M_\varepsilon) \subset ([0, 1+\varepsilon] \times \{0\}) \cup (\{0\} \times [0, 1+\varepsilon]) \subset \omega,$$
$$\pi([0, 1] \cdot M_\varepsilon) \supset T$$

for ε small. Use the Cauchy formula along ∂M_ε (in some parametrization of the conic) to obtain an extension of f to $[0, 1] \cdot M_\varepsilon + \mathrm{i}\mathbb{R}^n$.

b) In general, show that every $f \in \mathscr{O}(\Omega)$ extends to the convex hull $\widehat{\Omega}$.
Hint: given $a, b \in \omega$, consider a polygonal line joining a and b and apply a) inductively to obtain an extension along $[a, b] + \mathrm{i}\mathbb{R}^n$.

§ 8.16. For each integer $\nu \geqslant 1$, consider the algebraic variety

$$X_\nu = \big\{(z, w, t) \in \mathbb{C}^3\,;\, wt = p_\nu(z)\big\}, \qquad p_\nu(z) = \prod_{1 \leqslant k \leqslant \nu}(z - 1/k),$$

and the map $j_\nu : X_\nu \to X_{\nu+1}$ such that

$$j_\nu(z, w, t) = \Big(z,\, w,\, t\Big(z - \frac{1}{\nu+1}\Big)\Big).$$

a) Show that X_ν is a Stein manifold, and that j_ν is an embedding of X_ν onto an open subset of $X_{\nu+1}$.

b) Define $X = \lim(X_\nu, j_\nu)$, and let $\pi_\nu : X_\nu \to \mathbb{C}^2$ be the projection to the first two coordinates. Since $\pi_{\nu+1} \circ j_\nu = \pi_\nu$, there exists a holomorphic map $\pi : X \to \mathbb{C}^2$, $\pi = \lim \pi_\nu$. Show that

$$\mathbb{C}^2 \smallsetminus \pi(X) = \big\{(z, 0) \in \mathbb{C}^2\,;\, z \neq 1/\nu,\ \forall\nu \in \mathbb{N},\ \nu \geqslant 1\big\},$$

and especially, that $(0, 0) \notin \pi(X)$.

c) Consider the compact set

$$K = \pi^{-1}\big(\{(z, w) \in \mathbb{C}^2\,;\, |z| \leqslant 1,\ |w| = 1\}\big).$$

By looking at points of the forms $(1/\nu, w, 0)$, $|w| = 1$, show that $\pi^{-1}(1/\nu, 1/\nu) \in \widehat{K}_{\mathscr{O}(X)}$. Conclude from this that X is not holomorphically convex (this example is due to [Fornaess 1976]).

§ 8.17. Let X be a complex manifold, and let $\pi : \widetilde{X} \to X$ be a holomorphic unramified covering of X (X and $\widetilde{X}$ are assumed to be connected).

a) Let g be a complete riemannian metric on X, and let $\widetilde{d}$ be the geodesic distance on $\widetilde{X}$ associated to $\widetilde{g} = \pi^\star g$ (see VIII-2.3 for definitions). Show that $\widetilde{g}$ is complete and that $\delta_0(x) := \widetilde{d}(x, x_0)$ is a continuous exhaustion function on $\widetilde{X}$, for any given point $x_0 \in \widetilde{X}$.

b) Let (U_α) be a locally finite covering of X by open balls contained in coordinate open sets, such that all intersections $U_\alpha \cap U_\beta$ are diffeomorphic to convex open sets (see Lemma IV-6.9). Let θ_α be a partition of unity subordinate to the covering (U_α), and let $\delta_{\varepsilon_\alpha}$ be the convolution of δ_0 with a regularizing kernel $\rho_{\varepsilon_\alpha}$ on each piece of $\pi^{-1}(U_\alpha)$ which is mapped biholomorphically onto U_α. Finally, set $\delta = \sum(\theta_\alpha \circ \pi)\delta_{\varepsilon_\alpha}$. Show that if (ε_α) is a collection of sufficiently small positive numbers, then δ is a smooth exhaustion function on $\widetilde{X}$.

c) Using the fact that δ_0 is 1-Lipschitz with respect to $\widetilde{d}$, show that derivatives $\partial^{|\nu|}\delta(x)/\partial x^\nu$ of a given order with respect to coordinates in U_α are uniformly bounded in all components of $\pi^{-1}(U_\alpha)$, at least when x lies in the compact subset $\operatorname{Supp}\theta_\alpha$. Conclude from this that there exists a positive hermitian form with continuous coefficients on X such that $H\delta \geqslant -\pi^\star\gamma$ on $\widetilde{X}$.

d) If X is strongly pseudoconvex, show that $\widetilde{X}$ is also strongly pseudoconvex.
Hint: let ψ be a smooth strictly plurisubharmonic exhaustion function on X. Show that there exists a smooth convex increasing function $\chi : \mathbb{R} \to \mathbb{R}$ such that $\delta + \chi \circ \psi$ is strictly plurisubharmonic.

Chapter II

Coherent Sheaves and Analytic Spaces

The chapter starts with rather general and abstract concepts concerning sheaves and ringed spaces. Introduced in the decade 1950-1960 by Leray, Cartan, Serre and Grothendieck, sheaves and ringed spaces have since been recognized as the adequate tools to handle algebraic varieties and analytic spaces in a unified framework. We then concentrate ourselves on the theory of complex analytic functions. The second section is devoted to a proof of the Weierstrass preparation theorem, which is nothing but a division algorithm for holomorphic functions. It is used to derive algebraic properties of the ring $\mathcal{O}_n$ of germs of holomorphic functions in $\mathbb{C}^n$. Coherent analytic sheaves are then introduced and the fundamental coherence theorem of Oka is proved. Basic properties of analytic sets are investigated in detail: local parametrization theorem, Hilbert's Nullstellensatz, coherence of the ideal sheaf of an analytic set, analyticity of the singular set. The formalism of complex spaces is then developed and gives a natural setting for the proof of more global properties (decomposition into global irreducible components, maximum principle). After a few definitions concerning cycles, divisors and meromorphic functions, we investigate the important notion of normal space and establish the Oka normalization theorem. Next, the Remmert-Stein extension theorem and the Remmert proper mapping theorem on images of analytic sets are proved by means of semi-continuity results on the rank of morphisms. As an application, we give a proof of Chow's theorem asserting that every analytic subset of $\mathbb{P}^n$ is algebraic. Finally, the concept of analytic scheme with nilpotent elements is introduced as a generalization of complex spaces, and we discuss the concepts of bimeromorphic maps, modifications and blowing-up.

§ 1. Presheaves and Sheaves

§ 1.A. Main Definitions.

Sheaves have become a very important tool in analytic or algebraic geometry as well as in algebraic topology. They are especially useful when one wants to relate global properties of an object to its local properties (the latter being usually easier to establish). We first introduce the axioms of presheaves and sheaves in full generality and give some basic examples.

(1.1) Definition. *Let X be a topological space. A presheaf $\mathcal{A}$ on X consists of the following data:*

a) *a collection of non empty sets $\mathcal{A}(U)$ associated with every open set $U \subset X$,*

b) *a collection of maps $\rho_{U,V} : \mathcal{A}(V) \longrightarrow \mathcal{A}(U)$ defined whenever $U \subset V$ and satisfying the transitivity property*

c) $\rho_{U,V} \circ \rho_{V,W} = \rho_{U,W}$ *for* $U \subset V \subset W$, $\qquad \rho_{U,U} = \mathrm{Id}_U$ *for every* U.

The set $\mathscr{A}(U)$ *is called the set of sections of the presheaf* $\mathscr{A}$ *over* U.

Most often, the presheaf $\mathscr{A}$ is supposed to carry an additional algebraic structure. For instance:

(1.2) Definition. *A presheaf* $\mathscr{A}$ *is said to be a presheaf of abelian groups* (*resp. rings, R-modules, algebras*) *if all sets* $\mathscr{A}(U)$ *are abelian groups* (*resp. rings, R-modules, algebras*) *and if the maps* $\rho_{U,V}$ *are morphisms of these algebraic structures. In this case, we always assume that* $\mathscr{A}(\emptyset) = \{0\}$.

(1.3) Example. If we assign to each open set $U \subset X$ the set $\mathscr{C}(U)$ of all real valued continuous functions on U and let $\rho_{U,V}$ be the obvious restriction morphism $\mathscr{C}(V) \to \mathscr{C}(U)$, then $\mathscr{C}$ is a presheaf of rings on X. Similarly if X is differentiable (resp. complex analytic) manifold, there are well defined presheaves of rings $\mathscr{C}^k$ of functions of class $\mathscr{C}^k$ (resp. $\mathscr{O}$) of holomorphic functions) on X. Because of these examples, the maps $\rho_{U,V}$ in Def. 1.1 are often viewed intuitively as "restriction homomorphisms", although the sets $\mathscr{A}(U)$ are not necessarily sets of functions defined over U. For the simplicity of notation we often just write $\rho_{U,V}(f) = f_{\restriction U}$ whenever $f \in \mathscr{A}(V)$, $V \supset U$. $\square$

For the above presheaves $\mathscr{C}$, $\mathscr{C}^k$, $\mathscr{O}$, the properties of functions under consideration are purely local. As a consequence, these presheaves satisfy the following additional *gluing axioms*, where (U_α) and $U = \bigcup U_\alpha$ are arbitrary open subsets of X:

(1.4′) If $F_\alpha \in \mathscr{A}(U_\alpha)$ are such that $\rho_{U_\alpha \cap U_\beta, U_\alpha}(F_\alpha) = \rho_{U_\alpha \cap U_\beta, U_\beta}(F_\beta)$ for all α, β, there exists $F \in \mathscr{A}(U)$ such that $\rho_{U_\alpha, U}(F) = F_\alpha$;

(1.4″) If $F, G \in \mathscr{A}(U)$ and $\rho_{U_\alpha, U}(F) = \rho_{U_\alpha, U}(G)$ for all α, then $F = G$;

in other words, local sections over the sets U_α can be glued together if they coincide in the intersections and the resulting section on U is uniquely defined. Not all presheaves satisfy (1.4′) and (1.4″):

(1.5) Example. Let E be an arbitrary set with a distinguished element 0 (e.g. an abelian group, a R-module, ...). The *constant presheaf* E_X on X is defined to be $E_X(U) = E$ for all $\emptyset \neq U \subset X$ and $E_X(\emptyset) = \{0\}$, with restriction maps $\rho_{U,V} = \mathrm{Id}_E$ if $\emptyset \neq U \subset V$ and $\rho_{U,V} = 0$ if $U = \emptyset$. Then axiom (1.4′) is not satisfied if U is the union of two disjoint open sets U_1, U_2 and E contains a non zero element.

(1.6) Definition. *A presheaf* $\mathscr{A}$ *is said to be a sheaf if it satisfies the gluing axioms* (1.4′) *and* (1.4″).

If $\mathscr{A}$, $\mathscr{B}$ are presheaves of abelian groups (or of some other algebraic structure) on the same space X, a presheaf morphism $\varphi : \mathscr{A} \to \mathscr{B}$ is a collection of morphisms $\varphi_U : \mathscr{A}(U) \to \mathscr{B}(U)$ commuting with the restriction morphisms, i.e. such that for each pair $U \subset V$ there

is a commutative diagram

$$\begin{array}{ccc} \mathscr{A}(V) & \xrightarrow{\varphi_V} & \mathscr{B}(V) \\ \rho^{\mathscr{A}}_{U,V}\downarrow & & \downarrow\rho^{\mathscr{B}}_{U,V} \\ \mathscr{A}(U) & \xrightarrow{\varphi_U} & \mathscr{B}(U). \end{array}$$

We say that $\mathscr{A}$ is a subpresheaf of $\mathscr{B}$ in the case where $\varphi_U : \mathscr{A}(U) \subset \mathscr{B}(U)$ is the inclusion morphism; the commutation property then means that $\rho^{\mathscr{B}}_{U,V}(\mathscr{A}(V)) \subset \mathscr{A}(U)$ for all U, V, and that $\rho^{\mathscr{A}}_{U,V}$ coincides with $\rho^{\mathscr{B}}_{U,V}$ on $\mathscr{A}(V)$. If $\mathscr{A}$ is a subpresheaf of a presheaf $\mathscr{B}$ of abelian groups, there is a presheaf quotient $\mathscr{C} = \mathscr{B}/\mathscr{A}$ defined by $\mathscr{C}(U) = \mathscr{B}(U)/\mathscr{A}(U)$. In a similar way, one defines the presheaf kernel (resp. presheaf image, presheaf cokernel) of a presheaf morphism $\varphi : \mathscr{A} \to \mathscr{B}$ to be the presheaves

$$U \mapsto \operatorname{Ker}\varphi_U, \qquad U \mapsto \operatorname{Im}\varphi_U, \qquad U \mapsto \operatorname{Coker}\varphi_U.$$

The direct sum $\mathscr{A} \oplus \mathscr{B}$ of presheaves of abelian groups $\mathscr{A}$, $\mathscr{B}$ is the presheaf $U \mapsto \mathscr{A}(U) \oplus \mathscr{B}(U)$, the tensor product $\mathscr{A} \otimes \mathscr{B}$ of presheaves of R-modules is $U \mapsto \mathscr{A}(U) \otimes_R \mathscr{B}(U)$, etc ...

(1.7) Remark. The reader should take care of the fact that the presheaf quotient of a sheaf by a subsheaf is not necessarily a sheaf. To give a specific example, let $X = S^1$ be the unit circle in $\mathbb{R}^2$, let $\mathscr{C}$ be the sheaf of continuous complex valued functions and $\mathscr{Z}$ the subsheaf of integral valued continuous functions (i.e. locally constant functions to $\mathbb{Z}$). The exponential map

$$\varphi = \exp(2\pi\mathrm{i}\bullet) : \mathscr{C} \longrightarrow \mathscr{C}^\star$$

is a morphism from $\mathscr{C}$ to the sheaf $\mathscr{C}^\star$ of invertible continuous functions, and the kernel of φ is precisely $\mathscr{Z}$. However φ_U is surjective for all $U \neq X$ but maps $\mathscr{C}(X)$ onto the multiplicative subgroup of continuous functions of $\mathscr{C}^\star(X)$ of degree 0. Therefore the quotient presheaf $\mathscr{C}/\mathscr{Z}$ is not isomorphic with $\mathscr{C}^\star$, although their groups of sections are the same for all $U \neq X$. Since $\mathscr{C}^\star$ is a sheaf, we see that $\mathscr{C}/\mathscr{Z}$ does not satisfy property (1.4′). □

In order to overcome the difficulty appearing in Remark 1.7, it is necessary to introduce a suitable process by which we can produce a sheaf from a presheaf. For this, it is convenient to introduce a slightly modified viewpoint for sheaves.

(1.8) Definition. *If $\mathscr{A}$ is a presheaf, we define the set $\widetilde{\mathscr{A}}_x$ of germs of $\mathscr{A}$ at a point $x \in X$ to be the abstract inductive limit*

$$\widetilde{\mathscr{A}}_x = \varinjlim_{U \ni x} \big(\mathscr{A}(U), \rho_{U,V}\big).$$

More explicitely, $\widetilde{\mathscr{A}}_x$ is the set of equivalence classes of elements in the disjoint union $\coprod_{U \ni x} \mathscr{A}(U)$ taken over all open neighborhoods U of x, with two elements $F_1 \in \mathscr{A}(U_1)$, $F_2 \in \mathscr{A}(U_2)$ being equivalent, $F_1 \sim F_2$, if and only if there is a neighborhood $V \subset U_1, U_2$ such that $F_{1\restriction V} = F_{2\restriction V}$, i.e., $\rho_{VU_1}(F_1) = \rho_{VU_2}(F_2)$. The germ of an element $F \in \mathscr{A}(U)$ at a point $x \in U$ will be denoted by F_x.

Let $\mathcal{A}$ be an arbitrary presheaf. The disjoint union $\widetilde{\mathcal{A}} = \coprod_{x\in X} \widetilde{\mathcal{A}}_x$ can be equipped with a natural topology as follows: for every $F \in \mathcal{A}(U)$, we set

$$\Omega_{F,U} = \big\{F_x \ ; \ x \in U\big\}$$

and choose the $\Omega_{F,U}$ to be a basis of the topology of $\widetilde{\mathcal{A}}$; note that this family is stable by intersection: $\Omega_{F,U} \cap \Omega_{G,V} = \Omega_{H,W}$ where W is the (open) set of points $x \in U \cap V$ at which $F_x = G_x$ and $H = \rho_{W,U}(F)$. The obvious projection map $\pi : \widetilde{\mathcal{A}} \to X$ which sends $\widetilde{\mathcal{A}}_x$ to $\{x\}$ is then a local homeomorphism (it is actually a homeomorphism from $\Omega_{F,U}$ onto U). This leads in a natural way to the following definition:

(1.9) Definition. *Let X and $\mathcal{S}$ be topological spaces (not necessarily Hausdorff), and let $\pi : \mathcal{S} \longrightarrow X$ be a mapping such that*

a) *π maps $\mathcal{S}$ onto X ;*

b) *π is a local homeomorphism, that is, every point in $\mathcal{S}$ has an open neighborhood which is mapped homeomorphically by π onto an open subset of X.*

Then $\mathcal{S}$ is called a sheaf-space on X and π is called the projection of $\mathcal{S}$ on X. If $x \in X$, then $\mathcal{S}_x = \pi^{-1}(x)$ is called the stalk of $\mathcal{S}$ at x.

If Y is a subset of X, we denote by $\Gamma(Y, \mathcal{S})$ the set of sections of $\mathcal{S}$ on Y, i.e. the set of continuous functions $F : Y \to \mathcal{S}$ such that $\pi \circ F = \mathrm{Id}_Y$. It is clear that the presheaf defined by the collection of sets $\mathcal{S}'(U) := \Gamma(U, \mathcal{S})$ for all open sets $U \subset X$ together with the restriction maps $\rho_{U,V}$ satisfies axioms (1.4′) and (1.4″), hence $\mathcal{S}'$ is a sheaf. The set of germs of $\mathcal{S}'$ at x is in one-to-one correspondence with the stalk $\mathcal{S}_x = \pi^{-1}(x)$, thanks to the local homeomorphism assumption 1.9 b). This shows that one can associate in a natural way a sheaf $\mathcal{S}'$ to every sheaf-space $\mathcal{S}$, and that the sheaf-space $(\mathcal{S}')^\sim$ can be considered to be identical to the original sheaf-space $\mathcal{S}$. Since the assignment $\mathcal{S} \mapsto \mathcal{S}'$ from sheaf-spaces to sheaves is an equivalence of categories, we will usually omit the prime sign in the notation of $\mathcal{S}'$ and thus use the same symbols for a sheaf-space and its associated sheaf of sections; in a corresponding way, we write $\Gamma(U, \mathcal{S}) = \mathcal{S}(U)$ when U is an open set.

Conversely, given a presheaf $\mathcal{A}$ on X, we have an associated sheaf-space $\widetilde{\mathcal{A}}$ and an obvious presheaf morphism

$$\mathcal{A}(U) \longrightarrow \widetilde{\mathcal{A}}'(U) = \Gamma(U, \widetilde{\mathcal{A}}), \qquad F \longmapsto \widetilde{F} = (U \ni x \mapsto F_x). \tag{1.10}$$

This morphism is clearly injective if and only if $\mathcal{A}$ satisfies axiom (1.4″), and it is not difficult to see that (1.4′) and (1.4″) together imply surjectivity. Therefore $\mathcal{A} \to \widetilde{\mathcal{A}}'$ is an isomorphism if and only if $\mathcal{A}$ is a sheaf. According to the equivalence of categories between sheaves and sheaf-spaces mentioned above, we will use from now on the same symbol $\widetilde{\mathcal{A}}$ for the sheaf-space and its associated sheaf $\widetilde{\mathcal{A}}'$; one says that $\widetilde{\mathcal{A}}$ is the *sheaf associated with the presheaf* $\mathcal{A}$. If $\mathcal{A}$ itself is a sheaf, we will again identify $\widetilde{\mathcal{A}}$ and $\mathcal{A}$, but we will of course keep the notational difference for a presheaf $\mathcal{A}$ which is not a sheaf.

(1.11) Example. The sheaf associated to the constant presheaf of stalk E over X is the sheaf of locally constant functions $X \to E$. This sheaf will be denoted merely by E_X

or E if there is no risk of confusion with the corresponding presheaf. In Remark 1.7, we have $\mathcal{Z} = \mathbb{Z}_X$ and the sheaf $(\mathscr{C}/\mathbb{Z}_X)^\sim$ associated with the quotient presheaf $\mathscr{C}/\mathbb{Z}_X$ is isomorphic to $\mathscr{C}^\star$ via the exponential map. $\square$

In the sequel, we usually work in the category of sheaves rather than in the category of presheaves themselves. For instance, the quotient $\mathscr{B}/\mathscr{A}$ of a sheaf $\mathscr{B}$ by a subsheaf $\mathscr{A}$ generally refers to the sheaf associated with the quotient presheaf: its stalks are equal to $\mathscr{B}_x/\mathscr{A}_x$, but a section G of $\mathscr{B}/\mathscr{A}$ over an open set U need not necessarily come from a global section of $\mathscr{B}(U)$; what can be only said is that there is a covering (U_α) of U and local sections $F_\alpha \in \mathscr{B}(U_\alpha)$ representing $G_{\restriction U_\alpha}$ such that $(F_\beta - F_\alpha)_{\restriction U_\alpha \cap U_\beta}$ belongs to $\mathscr{A}(U_\alpha \cap U_\beta)$. A sheaf morphism $\varphi : \mathscr{A} \to \mathscr{B}$ is said to be injective (resp. surjective) if the germ morphism $\varphi_x : \mathscr{A}_x \to \mathscr{B}_x$ is injective (resp. surjective) for every $x \in X$. Let us note again that a surjective sheaf morphism φ does not necessarily give rise to surjective morphisms $\varphi_U : \mathscr{A}(U) \to \mathscr{B}(U)$.

§ 1.B. Direct and Inverse Images of Sheaves

Let X, Y be topological spaces and let $f : X \to Y$ be a continuous map. If $\mathscr{A}$ is a presheaf on X, the *direct image* $f_\star\mathscr{A}$ is the presheaf on Y defined by

$$f_\star\mathscr{A}(U) = \mathscr{A}\big(f^{-1}(U)\big) \tag{1.12}$$

for all open sets $U \subset Y$. When $\mathscr{A}$ is a sheaf, it is clear that $f_\star\mathscr{A}$ also satisfies axioms (1.4′) and (1.4″), thus $f_\star\mathscr{A}$ is a sheaf. Its stalks are given by

$$(f_\star\mathscr{A})_y = \varinjlim_{V \ni y} \mathscr{A}\big(f^{-1}(V)\big) \tag{1.13}$$

where V runs over all open neighborhoods of $y \in Y$.

Now, let $\mathscr{B}$ be a sheaf on Y, viewed as a sheaf-space with projection map $\pi : \mathscr{B} \to Y$. We define the *inverse image* $f^{-1}\mathscr{B}$ by

$$f^{-1}\mathscr{B} = \mathscr{B} \times_Y X = \big\{(s, x) \in \mathscr{B} \times X\,;\ \pi(s) = f(x)\big\} \tag{1.14}$$

with the topology induced by the product topology on $\mathscr{B} \times X$. It is then easy to see that the projection $\pi' = \mathrm{pr}_2 : f^{-1}\mathscr{B} \to X$ is a local homeomorphism, therefore $f^{-1}\mathscr{B}$ is a sheaf on X. By construction, the stalks of $f^{-1}\mathscr{B}$ are

$$(f^{-1}\mathscr{B})_x = \mathscr{B}_{f(x)}, \tag{1.15}$$

and the sections $\sigma \in f^{-1}\mathscr{B}(U)$ can be considered as continuous mappings $s : U \to \mathscr{B}$ such that $\pi \circ \sigma = f$. In particular, any section $s \in \mathscr{B}(V)$ on an open set $V \subset Y$ has a *pull-back*

$$f^\star s = s \circ f \quad \in \quad f^{-1}\mathscr{B}\big(f^{-1}(V)\big). \tag{1.16}$$

There are always natural sheaf morphisms

$$f^{-1}f_\star\mathscr{A} \longrightarrow \mathscr{A}, \qquad \mathscr{B} \longrightarrow f_\star f^{-1}\mathscr{B} \tag{1.17}$$

defined as follows. A germ in $(f^{-1}f_\star\mathscr{A})_x = (f_\star\mathscr{A})_{f(x)}$ is defined by a local section $s \in (f_\star\mathscr{A})(V) = \mathscr{A}(f^{-1}(V))$ for some neighborhood V of $f(x)$; this section can be mapped to

the germ $s_x \in \mathscr{A}_x$. In the opposite direction, the pull-back $f^\star s$ of a section $s \in \mathscr{B}(V)$ can be seen by (1.16) as a section of $f_\star f^{-1}\mathscr{B}(V)$. It is not difficult to see that these natural morphisms are not isomorphisms in general. For instance, if f is a finite covering map with q sheets and if we take $\mathscr{A} = E_X$, $\mathscr{B} = E_Y$ to be constant sheaves, then $f_\star E_X \simeq E_Y^q$ and $f^{-1}E_Y = E_X$, thus $f^{-1}f_\star E_X \simeq E_X^q$ and $f_\star f^{-1}E_Y \simeq E_Y^q$.

§ 1.C. Ringed Spaces

Many natural geometric structures considered in analytic or algebraic geometry can be described in a convenient way as topological spaces equipped with a suitable "structure sheaf" which, most often, is a sheaf of commutative rings. For instance, a lot of properties of $\mathscr{C}^k$ differentiable (resp. real analytic, complex analytic) manifolds can be described in terms of their sheaf of rings $\mathscr{C}_X^k$ of differentiable functions (resp. $\mathscr{C}_X^\omega$ of real analytic functions, $\mathscr{O}_X$ of holomorphic functions). We first recall a few standard definitions concerning rings, referring to textbooks on algebra for more details (see e.g. [Lang 1965]).

(1.18) Some definitions and conventions about rings. *All our rings R are supposed implicitly to have a unit element 1_R (if $R = \{0\}$, we agree that $1_R = 0_R$), and a ring morphism $R \to R'$ is supposed to map 1_R to $1_{R'}$. In the subsequent definitions, we assume that all rings under consideration are commutative.*

a) *An ideal $I \subset R$ is said to be prime if $xy \in I$ implies $x \in I$ or $y \in I$, i.e., if the quotient ring R/I is entire.*

b) *An ideal $I \subset R$ is said to be maximal if $I \neq R$ and there are no ideals J such that $I \subsetneq J \subsetneq R$ (equivalently, if the quotient ring R/I is a field).*

c) *The ring R is said to be a local ring if R has a unique maximal ideal $\mathfrak{m}$ (equivalently, if R has an ideal $\mathfrak{m}$ such that all elements of $R \smallsetminus \mathfrak{m}$ are invertible). Its residual field is defined to be the quotient field $R/\mathfrak{m}$.*

d) *The ring R is said to be Noetherian if every ideal $I \subset R$ is finitely generated (equivalently, if every increasing sequence of ideals $I_1 \subset I_2 \subset \ldots$ is stationary).*

e) *The radical $\sqrt{I}$ of an ideal I is the set of all elements $x \in R$ such that some power x^m, $m \in \mathbb{N}^\star$, lies in in I. Then $\sqrt{I}$ is again an ideal of R.*

f) *The nilradical $N(R) = \sqrt{\{0\}}$ is the ideal of nilpotent elements of R. The ring R is said to be reduced if $N(R) = \{0\}$. Otherwise, its reduction is defined to be the reduced ring $R/N(R)$.*

We now introduce the general notion of a ringed space.

(1.19) Definition. *A ringed space is a pair $(X, \mathscr{R}_X)$ consisting of a topological space X and of a sheaf of rings $\mathscr{R}_X$ on X, called the structure sheaf. A morphism*

$$F : (X, \mathscr{R}_X) \to (Y, \mathscr{R}_Y)$$

of ringed spaces is a pair $(f, F^\star)$ where $f : X \to Y$ is a continuous map and

$$F^\star \ : \ f^{-1}\mathscr{R}_Y \to \mathscr{R}_X, \qquad F_x^\star \ : \ \mathscr{R}_{Y,f(x)} \to \mathscr{R}_{X,x}$$

a homomorphism of sheaves of rings on X, called the comorphism of F.

If $F : (X, \mathscr{R}_X) \to (Y, \mathscr{R}_Y)$ and $G : (Y, \mathscr{R}_Y) \to (Z, \mathscr{R}_Z)$ are morphisms of ringed spaces, the composite $G \circ F$ is the pair consisting of the map $g \circ f : X \to Z$ and of the comorphism $(G \circ F)^\star = F^\star \circ f^{-1} G^\star$:

$$F^\star \circ f^{-1} G^\star \colon f^{-1} g^{-1} \mathscr{R}_Z \xrightarrow{f^{-1} G^\star} f^{-1} \mathscr{R}_Y \xrightarrow{F^\star} \mathscr{R}_X, \tag{1.20}$$

$$F_x^\star \circ G_{f(x)}^\star \colon \mathscr{R}_{Z, g \circ f(x)} \longrightarrow \mathscr{R}_{Y, f(x)} \longrightarrow \mathscr{R}_{X,x}.$$

We say of course that F is an isomorphism of ringed spaces if there exists G such that $G \circ F = \mathrm{Id}_X$ and $F \circ G = \mathrm{Id}_Y$.

If $(X, \mathscr{R}_X)$ is a ringed space, the nilradical of $\mathscr{R}_X$ defines an ideal subsheaf $\mathscr{N}_X$ of $\mathscr{R}_X$, and the identity map $\mathrm{Id}_X : X \to X$ together with the ring homomorphism $\mathscr{R}_X \to \mathscr{R}_X / \mathscr{N}_X$ defines a ringed space morphism

$$(X, \mathscr{R}_X / \mathscr{N}_X) \to (X, \mathscr{R}_X) \tag{1.21}$$

called the *reduction morphism*. Quite often, the letter X by itself is used to denote the ringed space $(X, \mathscr{R}_X)$; we then denote by $X_{\mathrm{red}} = (X, \mathscr{R}_X / \mathscr{N}_X)$ its reduction. The ringed space X is said to be *reduced* if $\mathscr{N}_X = 0$, in which case the reduction morphism $X_{\mathrm{red}} \to X$ is an isomorphism. In all examples considered later on in this book, the structure sheaf $\mathscr{R}_X$ will be a sheaf of *local rings* over some field k. The relevant definition is as follows.

(1.22) Definition.

a) *A local ringed space is a ringed space $(X, \mathscr{R}_X)$ such that all stalks $\mathscr{R}_{X,x}$ are local rings. The maximal ideal of $\mathscr{R}_{X,x}$ will be denoted by $\mathfrak{m}_{X,x}$. A morphism $F = (f, F^\star) : (X, \mathscr{R}_X) \to (Y, \mathscr{R}_Y)$ of local ringed spaces is a morphism of ringed spaces such that $F_x^\star(\mathfrak{m}_{Y,f(x)}) \subset \mathfrak{m}_{X,x}$ at any point $x \in X$ (i.e., $F_x^\star$ is a "local" homomorphism of rings).*

b) *A local ringed space over a field k is a local ringed space $(X, \mathscr{R}_X)$ such that all rings $\mathscr{R}_{X,x}$ are local k-algebras with residual field $\mathscr{R}_{X,x}/\mathfrak{m}_{X,x} \simeq k$. A morphism F between such spaces is supposed to have its comorphism defined by local k-homomorphisms $F_x^\star : \mathscr{R}_{Y,f(x)} \to \mathscr{R}_{X,x}$.*

If $(X, \mathscr{R}_X)$ is a local ringed space over k, we can associate to each section $s \in \mathscr{R}_X(U)$ a function

$$\overline{s} : U \to k, \qquad x \mapsto \overline{s}(x) \in k = \mathscr{R}_{X,x}/\mathfrak{m}_{X,x},$$

and we get a sheaf morphism $\mathscr{R}_X \to \overline{\mathscr{R}}_X$ onto a subsheaf of rings $\overline{\mathscr{R}}_X$ of the sheaf of functions from X to k. We clearly have a factorization

$$\mathscr{R}_X \to \mathscr{R}_X / \mathscr{N}_X \to \overline{\mathscr{R}}_X,$$

and thus a corresponding factorization of ringed space morphisms (with Id_X as the underlying set theoretic map)

$$X_{\text{st-red}} \to X_{\mathrm{red}} \to X$$

where $X_{\text{st-red}} = (X, \overline{R}_X)$ is called the *strong reduction* of $(X, \mathscr{R}_X)$. It is easy to see that $X_{\text{st-red}}$ is actually a reduced local ringed space over k. We say that X is strongly reduced if

$\mathscr{R}_X \to \overline{\mathscr{R}}_X$ is an isomorphism, that is, if $\mathscr{R}_X$ can be identified with a subsheaf of the sheaf of functions $X \to k$ (in our applications to the theory of algebraic or analytic schemes, the concepts of reduction and strong reduction will actually be the same ; in general, these notions differ). It is important to observe that reduction (resp. strong reduction) is a functorial process:
if $F = (f, F^\star) : (X, \mathscr{R}_X) \to (Y, \mathscr{R}_Y)$ is a morphism of ringed spaces (resp. of local ringed spaces over k), there are natural reductions

$$F_{\text{red}} = (f, F^\star_{\text{red}}) : X_{\text{red}} \to Y_{\text{red}}, \quad F^\star_{\text{red}} : \mathscr{R}_{Y,f(x)}/\mathscr{N}_{Y,f(x)} \to \mathscr{R}_{X,x}/\mathscr{N}_{X,x},$$
$$F_{\text{st-red}} = (f, f^\star) : X_{\text{st-red}} \to Y_{\text{st-red}}, \quad f^\star : \overline{\mathscr{R}}_{Y,f(x)} \to \overline{\mathscr{R}}_{X,x}, \quad \overline{s} \mapsto \overline{s} \circ f$$

where $f^\star$ is the usual pull-back comorphism associated with f. Therefore, if $(X, \mathscr{R}_X)$ and $(Y, \mathscr{R}_Y)$ are strongly reduced, the morphism F is completely determined by the underlying set-theoretic map f. Our first basic examples of (strongly reduced) ringed spaces are the various types of manifolds already defined in Chapter I. The language of ringed spaces provides an equivalent but more elegant and more intrinsic definition.

(1.23) Definition. *Let X be a Hausdorff separable topological space. One can define the category of $\mathscr{C}^k$, $k \in \mathbb{N} \cup \{\infty, \omega\}$, differentiable manifolds (resp. complex analytic manifolds) to be the category of reduced local ringed spaces $(X, \mathscr{R}_X)$ over $\mathbb{R}$ (resp. over $\mathbb{C}$), such that every point $x \in X$ has a neighborhood U on which the restriction $(U, \mathscr{R}_{X\restriction U})$ is isomorphic to a ringed space $(\Omega, \mathscr{C}^k_\Omega)$ where $\Omega \subset \mathbb{R}^n$ is an open set and $\mathscr{C}^k_\Omega$ is the sheaf of $\mathscr{C}^k$ differentiable functions (resp. $(\Omega, \mathscr{O}_\Omega)$, where $\Omega \subset \mathbb{C}^n$ is an open subset, and $\mathscr{O}_\Omega$ is the sheaf of holomorphic functions on Ω).*

We say that the ringed spaces $(\Omega, \mathscr{C}^k_\Omega)$ (resp. $(\Omega, \mathscr{O}_\Omega)$) are the *models* of the category of differentiable (resp. complex analytic) manifolds, and that a general object $(X, \mathscr{R}_X)$ in the category is *locally isomorphic* to one of the given model spaces. It is easy to see that the corresponding ringed spaces morphisms are nothing but the usual concepts of differentiable and holomorphic maps.

§ 1.D. Algebraic Varieties over a Field

As a second illustration of the notion of ringed space, we present here a brief introduction to the formalism of algebraic varieties, referring to [Hartshorne 1977] or [EGA 1967] for a much more detailed exposition. Our hope is that the reader who already has some background of analytic or algebraic geometry will find some hints of the strong interconnections between both theories. Beginners are invited to skip this section and proceed directly to the theory of complex analytic sheaves in § 2. All rings or algebras occurring in this section are supposed to be commutative rings with unit.

§ 1.D.1. *Affine Algebraic Sets.* Let k be an algebraically closed field of any characteristic. An *affine algebraic set* is a subset $X \subset k^N$ of the affine space k^N defined by an arbitrary collection $S \subset k[T_1, \dots, T_N]$ of polynomials, that is,

$$X = V(S) = \big\{(z_1, \dots, z_N) \in k^N \,;\, P(z_1, \dots, z_N) = 0,\ \forall P \in S\big\}.$$

Of course, if $J \subset k[T_1, \dots, T_N]$ is the ideal generated by S, then $V(S) = V(J)$. As $k[T_1, \dots, T_N]$ is Noetherian, J is generated by finitely many elements $(P_1, \dots, P_m)$, thus

$X = V(\{P_1, \dots, P_m\})$ is always defined by finitely many equations. Conversely, for any subset $Y \subset k^N$, we consider the ideal $I(Y)$ of $k[T_1, \dots, T_N]$, defined by

$$I(Y) = \big\{P \in k[T_1, \dots, T_N]\,;\, P(z) = 0,\ \forall z \in Y\big\}.$$

Of course, if $Y \subset k^N$ is an algebraic set, we have $V(I(Y)) = Y$. In the opposite direction, we have the following fundamental result.

(1.24) Hilbert's Nullstellensatz (see [Lang 1965]). *If $J \subset k[T_1, \dots, T_N]$ is an ideal, then $I(V(J)) = \sqrt{J}$.*

If $X = V(J) \subset k^N$ is an affine algebraic set, we define the (reduced) ring $\mathscr{O}(X)$ of algebraic functions on X to be the set of all functions $X \to k$ which are restrictions of polynomials, i.e.,

$$\mathscr{O}(X) = k[T_1, \dots, T_N]/I(X) = k[T_1, \dots, T_N]/\sqrt{J}. \tag{1.25}$$

This is clearly a reduced k-algebra. An (algebraic) morphism of affine algebraic sets $X = V(J) \subset k^N$, $Y = V(J') \subset k^{N'}$ is a map $f : Y \to X$ which is the restriction of a polynomial map $k^{N'} to k^N$. We then get a k-algebra homomomorphism

$$f^\star : \mathscr{O}(X) \to \mathscr{O}(Y), \qquad s \mapsto s \circ f,$$

called the *comorphism* of f. In this way, we have defined a contravariant functor

$$X \mapsto \mathscr{O}(X), \qquad f \mapsto f^\star \tag{1.26}$$

from the category of affine algebraic sets to the category of finitely generated reduced k-algebras.

We are going to show the existence of a natural functor going in the opposite direction. In fact, let us start with an arbitrary finitely generated algebra A (not necessarily reduced at this moment). For any choice of generators $(g_1, \dots, g_N)$ of A we get a surjective morphism of the polynomial ring $k[T_1, \dots, T_N]$ onto A,

$$k[T_1, \dots, T_N] \to A, \qquad T_j \mapsto g_j,$$

and thus $A \simeq k[T_1, \dots, T_N]/J$ with the ideal J being the kernel of this morphism. It is well-known that every maximal ideal $\mathfrak{m}$ of A has codimension 1 in A (see [Lang 1965]), so that $\mathfrak{m}$ gives rise to a k-algebra homomorphism $A \to A/\mathfrak{m} = k$. We thus get a bijection

$$\mathrm{Hom}_{\mathrm{alg}}(A, k) \to \mathrm{Spm}(A), \qquad u \mapsto \mathrm{Ker}\, u$$

between the set of k-algebra homomorphisms and the set $\mathrm{Spm}(A)$ of maximal ideals of A. In fact, if $A = k[T_1, \dots, T_N]/J$, an element $\varphi \in \mathrm{Hom}_{\mathrm{alg}}(A, k)$ is completely determined by the values $z_j = \varphi(T_j \bmod J)$, and the corresponding algebra homomorphism $k[T_1, \dots, T_N] \to k$, $P \mapsto P(z_1, \dots, z_N)$ can be factorized mod J if and only if $z = (z_1, \dots, z_N) \in k^N$ satisfies the equations

$$P(z_1, \dots, z_N) = 0, \qquad \forall P \in J.$$

We infer from this that

$$\mathrm{Spm}(A) \simeq V(J) = \big\{(z_1, \dots, z_N) \in k^N \,;\, P(z_1, \dots, z_N) = 0, \forall P \in J\big\}$$

can be identified with the *affine algebraic set* $V(J) \subset k^N$. If we are given an algebra homomorphism $\Phi : A \to B$ of finitely generated k-algebras we get a corresponding map $\mathrm{Spm}(\Phi) : \mathrm{Spm}(B) \to \mathrm{Spm}(A)$ described either as

$$\begin{aligned} &\mathrm{Spm}(B) \to \mathrm{Spm}(A), \quad \mathfrak{m} \mapsto \Phi^{-1}(\mathfrak{m}) \quad \text{or} \\ &\mathrm{Hom}_{\mathrm{alg}}(B, k) \to \mathrm{Hom}_{\mathrm{alg}}(A, k), \quad v \mapsto v \circ \Phi. \end{aligned}$$

If $B = k[T'_1, \dots, T'_{N'}]/J'$ and $\mathrm{Spm}(B) = V(J') \subset k^{N'}$, it is easy to see that $\mathrm{Spm}(\Phi) : \mathrm{Spm}(B) \to \mathrm{Spm}(A)$ is the restriction of the polynomial map

$$f : k^{N'} \to k^N, \qquad w \mapsto f(w) = (P_1(w), \dots, P_N(w)),$$

where $P_j \in k[T'_1, \dots, T'_{N'}]$ are polynomials such that $P_j = \Phi(T_j) \bmod J'$ in B. We have in this way defined a contravariant functor

$$A \mapsto \mathrm{Spm}(A), \qquad \Phi \mapsto \mathrm{Spm}(\Phi) \tag{1.27}$$

from the category of finitely generated k-algebras to the category of affine algebraic sets.

Since $A = k[T_1, \dots, T_N]/J$ and its reduction $A/N(A) = k[T_1, \dots, T_N]/\sqrt{J}$ give rise to the same algebraic set

$$V(J) = \mathrm{Spm}(A) = \mathrm{Spm}(A/N(A)) = V(\sqrt{J}),$$

we see that the category of affine algebraic sets is actually equivalent to the subcategory of *reduced* finitely generated k-algebras.

(1.28) Example. The simplest example of an affine algebraic set is the affine space

$$k^N = \mathrm{Spm}(k[T_1, \dots, T_N]),$$

in particular $\mathrm{Spm}(k) = k^0$ is just one point. We agree that $\mathrm{Spm}(\{0\}) = \emptyset$ (observe that $V(J) = \emptyset$ when J is the unit ideal in $k[T_1, \dots, T_N]$).

§ 1.D.2. *Zariski Topology and Affine Algebraic Schemes.* Let A be a finitely generated algebra and $X = \mathrm{Spm}(A)$. To each ideal $\mathfrak{a} \subset A$ we associate the zero variety $V(\mathfrak{a}) \subset X$ which consists of all elements $\mathfrak{m} \in X = \mathrm{Spm}(A)$ such that $\mathfrak{m} \supset \mathfrak{a}$; if

$$A \simeq k[T_1, \dots, T_N]/J \quad \text{and} \quad X \simeq V(J) \subset k^N,$$

then $V(\mathfrak{a})$ can be identified with the zero variety $V(J_{\mathfrak{a}}) \subset X$ of the inverse image $J_{\mathfrak{a}}$ of $\mathfrak{a}$ in $k[T_1, \dots, T_N]$. For any family $(\mathfrak{a}_\alpha)$ of ideals in A we have

$$V(\sum \mathfrak{a}_\alpha) = \bigcap V(\mathfrak{a}_\alpha), \qquad V(\mathfrak{a}_1) \cup V(\mathfrak{a}_2) = V(\mathfrak{a}_1\mathfrak{a}_2),$$

hence there exists a unique topology on X such that the closed sets consist precisely of all algebraic subsets $(V(\mathfrak{a}))_{\mathfrak{a} \subset A}$ of X. This topology is called the Zariski topology. The

Zariski topology is almost never Hausdorff (for example, if $X = k$ is the affine line, the open sets are $\emptyset$ and complements of finite sets, thus any two nonempty open sets have nonempty intersection). However, X is a *Noetherian space*, that is, a topological space in which every decreasing sequence of closed sets is stationary; an equivalent definition is to require that every open set is quasi-compact (from any open covering of an open set, one can extract a finite covering).

We now come to the concept of affine open subsets. For $s \in A$, the open set $D(s) = X \smallsetminus V(s)$ can be given the structure of an affine algebraic variety. In fact, if $A = k[T_1, \dots, T_N]/J$ and s is represented by a polynomial in $k[T_1, \dots, T_N]$, the localized ring $A[1/s]$ can be written as $A[1/s] = k[T_1, \dots, T_N, T_{N+1}]/J_s$ where $J_s = J[T_{N+1}] + (sT_{N+1} - 1)$, thus

$$V(J_s) = \{(z, w) \in V(J) \times k\,;\, s(z)\, w = 1\} \simeq V(I) \smallsetminus s^{-1}(0)$$

and $D(s)$ can be identified with $\mathrm{Spm}(A[1/s])$. We have $D(s_1) \cap D(s_2) = D(s_1 s_2)$, and the sets $(D(s))_{s \in A}$ are easily seen to be a basis of the Zariski topology on X. The open sets $D(s)$ are called *affine open sets*. Since the open sets $D(s)$ containing a given point $x \in X$ form a basis of neighborhoods, one can define a sheaf space $\mathscr{O}_X$ such that the ring of germs $\mathscr{O}_{X,x}$ is the inductive limit

$$\mathscr{O}_{X,x} = \varinjlim_{D(s) \ni x} A[1/s] = \{\text{fractions } p/q\,;\, p, q \in A,\, q(x) \neq 0\}.$$

This is a local ring with maximal ideal

$$\mathfrak{m}_{X,x} = \{p/q\,;\, p, q \in A,\, p(x) = 0,\, q(x) \neq 0\},$$

and residual field $\mathscr{O}_{X,x}/\mathfrak{m}_{X,x} = k$. In this way, we get a ringed space $(X, \mathscr{O}_X)$ over k. It is easy to see that $\Gamma(X, \mathscr{O}_X)$ coincides with the finitely generated k-algebra A. In fact, from the definition of $\mathscr{O}_X$, a global section is obtained by gluing together local sections p_j/s_j on affine open sets $D(s_j)$ with $\bigcup D(s_j) = X$, $1 \leqslant j \leqslant m$. This means that the ideal $\mathfrak{a} = (s_1, \dots, s_m) \subset A$ has an empty zero variety $V(\mathfrak{a})$, thus $\mathfrak{a} = A$ and there are elements $u_j \in A$ with $\sum u_j s_j = 1$. The compatibility condition $p_j/s_j = p_k/s_k$ implies that these elements are induced by

$$\sum u_j p_j / \sum u_j s_j = \sum u_j p_j \in A,$$

as desired. More generally, since the open sets $D(s)$ are affine, we get

$$\Gamma(D(s), \mathscr{O}_X) = A[1/s].$$

It is easy to see that the ringed space $(X, \mathscr{O}_X)$ is reduced if and only if A itself is reduced; in this case, X is even strongly reduced as Hilbert's Nullstellensatz shows. Otherwise, the reduction X_{red} can obtained from the reduced algebra $A_{\mathrm{red}} = A/N(A)$.

Ringed spaces $(X, \mathscr{O}_X)$ as above are called *affine algebraic schemes* over k (although substantially different from the usual definition, our definition can be shown to be equivalent in this special situation; compare with [Hartshorne 1977]). The category of affine algebraic schemes is equivalent to the category of finitely generated k-algebras (with the arrows reversed).

§ 1.D.3. *Algebraic Schemes.* Algebraic schemes over k are defined to be ringed spaces over k which are locally isomorphic to affine algebraic schemes, modulo an ad hoc separation condition.

(1.29) Definition. *An algebraic scheme over k is a local ringed space $(X, \mathscr{O}_X)$ over k such that*

a) *X has a finite covering by open sets U_α such that $(U_\alpha, \mathscr{O}_{X\restriction U_\alpha})$ is isomorphic as a ringed space to an affine algebraic scheme $(\mathrm{Spm}(A_\alpha), \mathscr{O}_{\mathrm{Spm}(A_\alpha)})$.*

b) *X satisfies the algebraic separation axiom, namely the diagonal Δ_X of $X \times X$ is closed for the Zariski topology.*

A morphism of algebraic schemes is just a morphism of the underlying local ringed spaces. An (abstract) algebraic variety is the same as a reduced algebraic scheme.

In the above definition, some words of explanation are needed for b), since the product $X \times Y$ of algebraic schemes over k *is not* the ringed space theoretic product, i.e., the product topological space equipped with the structure sheaf $\mathrm{pr}_1^\star \mathscr{O}_X \otimes_k \mathrm{pr}_2^\star \mathscr{O}_Y$. Instead, we define the product of two affine algebraic schemes $X = \mathrm{Spm}(A)$ and $Y = \mathrm{Spm}(B)$ to be $X \times Y = \mathrm{Spm}(A \otimes_k B)$, equipped with the Zariski topology and the structural sheaf associated with $A \otimes_k B$. Notice that the Zariski topology on $X \times Y$ *is not the product topology* of the Zariski topologies on X, Y, as the example $k^2 = k \times k$ shows; also, the rational function $1/(1 - z_1 - z_2) \in \mathscr{O}_{k^2,(0,0)}$ is not in $\mathscr{O}_{k,0} \otimes_k \mathscr{O}_{k,0}$. In general, if X, Y are written as $X = \bigcup U_\alpha$ and $Y = \bigcup V_\beta$ with affine open sets U_α, V_β, we define $X \times Y$ to be the union of all open affine charts $U_\alpha \times V_\beta$ with their associated structure sheaves of affine algebraic varieties, the open sets of $X \times Y$ being all unions of open sets in the various charts $U_\alpha \times V_\beta$. The separation axiom b) is introduced for the sake of excluding pathological examples such as an affine line $k \amalg \{0'\}$ with the origin changed into a double point.

§ 1.D.4. *Subschemes.* If $(X, \mathscr{O}_X)$ is an affine algebraic scheme and $A = \Gamma(X, \mathscr{O}_X)$ is the associated algebra, we say that $(Y, \mathscr{O}_Y)$ is a *subscheme* of $(X, \mathscr{O}_X)$ if there is an ideal $\mathfrak{a}$ of A such that $Y \hookrightarrow X$ is the morphism defined by the algebra morphism $A \to A/\mathfrak{a}$ as its comorphism. As $\mathrm{Spm}(A/\mathfrak{a}) \to \mathrm{Spm}(A)$ has for image the set $V(\mathfrak{a})$ of maximal ideals $\mathfrak{m}$ of A containing $\mathfrak{a}$, we see that $Y = V(\mathfrak{a})$ as a set; let us introduce the ideal subsheaf $\mathscr{J} = \mathfrak{a}\mathscr{O}_X \subset \mathscr{O}_X$. Since the structural sheaf $\mathscr{O}_Y$ is obtained by taken localizations $A/\mathfrak{a}[1/s]$, it is easy to see that $\mathscr{O}_Y$ coincides with the quotient sheaf $\mathscr{O}_X/\mathscr{J}$ restricted to Y. Since $\mathfrak{a}$ has finitely many generators, the ideal sheaf $\mathscr{J}$ is locally finitely generated (see § 2 below). This leads to the following definition.

(1.30) Definition. *If $(X, \mathscr{O}_X)$ is an algebraic scheme, a (closed) subscheme is an algebraic scheme $(Y, \mathscr{O}_Y)$ such that Y is a Zariski closed subset of X, and there is a locally finitely generated ideal subsheaf $\mathscr{J} \subset \mathscr{O}_X$ such that $Y = V(\mathscr{J})$ and $\mathscr{O}_Y = (\mathscr{O}_X/\mathscr{J})_{\restriction Y}$.*

If $(Y, \mathscr{O}_Y)$, $(Z, \mathscr{O}_Z)$ are subschemes of $(X, \mathscr{O}_X)$ defined by ideal subsheaves $\mathscr{J}$, $\mathscr{J}' \subset \mathscr{O}_X$, there are corresponding subschemes $Y \cap Z$ and $Y \cup Z$ defined as ringed spaces

$$(Y \cap Z, \mathscr{O}_X/(\mathscr{J} + \mathscr{J}')), \qquad (Y \cup Z, \mathscr{O}_X/\mathscr{J}\mathscr{J}').$$

§ **1.D.5.** *Projective Algebraic Varieties.* A very important subcategory of the category of algebraic varieties is provided by *projective algebraic varieties.* Let $\mathbb{P}_k^N$ be the projective N-space, that is, the set $k^{N+1} \smallsetminus \{0\}/k^\star$ of equivalence classes of $(N+1)$-tuples $(z_0, \dots, z_N) \in k^{N+1} \smallsetminus \{0\}$ under the equivalence relation given by $(z_0, \dots, z_N) \sim \lambda(z_0, \dots, z_N)$, $\lambda \in k^\star$. The corresponding element of $\mathbb{P}_k^N$ will be denoted $[z_0 : z_1 : \dots : z_N]$. It is clear that $\mathbb{P}_N^k$ can be covered by the $(N+1)$ affine charts U_α, $0 \leqslant \alpha \leqslant N$, such that

$$U_\alpha = \big\{[z_0 : z_1 : \dots : z_N] \in \mathbb{P}_k^N \;\; z_\alpha \neq 0\big\}.$$

The set U_α can be identified with the affine N-space k^N by the map

$$U_\alpha \to k^N, \qquad [z_0 : z_1 : \dots : z_N] \mapsto \Big(\frac{z_0}{z_\alpha}, \frac{z_1}{z_\alpha}, \dots, \frac{z_{\alpha-1}}{z_\alpha}, \frac{z_{\alpha+1}}{z_\alpha}, \dots, \frac{z_N}{z_\alpha}\Big).$$

With this identification, $\mathscr{O}(U_\alpha)$ is the algebra of homogeneous rational functions of degree 0 in $z_0, \dots, z_N$ which have just a power of z_α in their denominator. It is easy to see that the structure sheaves $\mathscr{O}_{U_\alpha}$ and $\mathscr{O}_{U_\beta}$ coincide in the intersections $U_\alpha \cap U_\beta$; they can be glued together to define an algebraic variety structure $(\mathbb{P}_k^N, \mathscr{O}_{\mathbb{P}^N})$, such that $\mathscr{O}_{\mathbb{P}^N,[z]}$ consists of all homogeneous rational functions p/q of degree 0 (i.e., $\deg p = \deg q$), such that $q(z) \neq 0$.

(1.31) Definition. *An algebraic scheme or variety $(X, \mathscr{O}_X)$ is said to be projective if it is isomorphic to a closed subscheme of some projective space $(\mathbb{P}_k^N, \mathscr{O}_{\mathbb{P}^N})$.*

We now indicate a standard way of constructing projective schemes. Let S be a collection of homogeneous polynomials $P_j \in k[z_0, \dots, z_N]$, of degree $d_j \in \mathbb{N}$. We define an associated *projective algebraic set*

$$\widetilde{V}(S) = \big\{[z_0 : \dots : z_N] \in \mathbb{P}_k^N \,;\, P(z) = 0,\ \forall P \in S\big\}.$$

Let J be the *homogeneous ideal* of $k[z_0, \dots, z_N]$ generated by S (recall that an ideal J is said to be homogeneous if $J = \bigoplus J_m$ is the direct sum of its homogeneous components, or equivalently, if J is generated by homogeneous elements). We have an associated graded algebra

$$B = k[z_0, \dots, z_N]/J = \bigoplus B_m, \qquad B_m = k[z_0, \dots, z_N]_m/J_m$$

such that B is generated by B_1 and B_m is a finite dimensional vector space over k for each k. This is enough to construct the desired scheme structure on $\widetilde{V}(J) := \bigcap \widetilde{V}(J_m)$, as we see in the next subsection.

§ **1.D.6.** *Projective Scheme Associated with a Graded Algebra.* Let us start with a reduced graded k-algebra

$$B = \bigoplus_{m \in \mathbb{N}} B_m$$

such that B is generated by B_0 and B_1 as an algebra, and B_0, B_1 are finite dimensional vector spaces over k (it then follows that B is finitely generated and that all B_m are finite dimensional vector spaces). Given $s \in B_m$, $m > 0$, we define a k-algebra A_s to be the ring of all fractions of homogeneous degree 0 with a power of s as their denominator, i.e.,

$$A_s = \big\{p/s^d \,;\, p \in B_{dm},\ d \in \mathbb{N}\big\}. \tag{1.32}$$

Since A_s is generated by $\frac{1}{s}B_1^m$ over B_0, A_s is a finitely generated algebra. We define $U_s = \mathrm{Spm}(A_s)$ to be the associated affine algebraic variety. For $s \in B_m$ and $s' \in B_{m'}$, we clearly have algebra homomorphisms

$$A_s \to A_{ss'}, \qquad A_{s'} \to A_{ss'},$$

since $A_{ss'}$ is the algebra of all 0-homogeneous fractions with powers of s and s' in the denominator. As $A_{ss'}$ is the same as the localized ring $A_s[s^{m'}/s'^m]$, we see that $U_{ss'}$ can be identified with an affine open set in U_s, and we thus get canonical injections

$$U_{ss'} \hookrightarrow U_s, \qquad U_{ss'} \hookrightarrow U_{s'}.$$

(1.33) Definition. *If $B = \bigoplus_{m\in\mathbb{N}} B_m$ is a reduced graded algebra generated by its finite dimensional vector subspaces B_0 and B_1, we associate an algebraic scheme $(X, \mathscr{O}_X) = \mathrm{Proj}(B)$ as follows. To each finitely generated algebra $A_s = \{p/s^d\,;\, p \in B_{dm},\, d \in \mathbb{N}\}$ we associate an affine algebraic variety $U_s = \mathrm{Spm}(A_s)$. We let X be the union of all open charts U_s with the identifications $U_s \cap U_{s'} = U_{ss'}$; then the collection (U_s) is a basis of the topology of X, and $\mathscr{O}_X$ is the unique sheaf of local k-algebras such that $\Gamma(U_s, \mathscr{O}_X) = A_s$ for each U_s.*

The following proposition shows that only finitely many open charts are actually needed to describe X (as required in Def. 1.29 a)).

(1.34) Lemma. *If $s_0, \ldots, s_N$ is a basis of B_1, then* $\mathrm{Proj}(B) = \bigcup_{0\leqslant j\leqslant N} U_{s_j}$.

Proof. In fact, if $x \in X$ is contained in a chart U_s for some $s \in B_m$, then $U_s = \mathrm{Spm}(A_s) \neq \emptyset$, and therefore $A_s \neq \{0\}$. As A_s is generated by $\frac{1}{s}B_1^m$, we can find a fraction $f = s_{j_1} \ldots s_{j_m}/s$ representing an element $f \in \mathscr{O}(U_s)$ such that $f(x) \neq 0$. Then $x \in U_s \smallsetminus f^{-1}(0)$, and $U_s \smallsetminus f^{-1}(0) = \mathrm{Spm}(A_s[1/f]) = U_s \cap U_{s_{j_1}} \cap \ldots \cap U_{s_{j_m}}$. In particular $x \in U_{s_{j_1}}$. $\square$

(1.35) Example. One can consider the *projective space* $\mathbb{P}_k^N$ to be the algebraic scheme

$$\mathbb{P}_k^N = \mathrm{Proj}(k[T_0, \ldots, T_N]).$$

The Proj construction is functorial in the following sense: if we have a graded homomorphism $\Phi : B \to B'$ (i.e. an algebra homomorphism such that $\Phi(B_m) \subset B'_m$, then there are corresponding morphisms $A_s \to A'_{\Phi(s)}$, $U'_{\Phi(s)} \to U_s$, and we thus find a scheme morphism

$$F : \mathrm{Proj}(B') \to \mathrm{Proj}(B).$$

Also, since $p/s^d = ps^l/s^{d+l}$, the algebras A_s depend only on components B_m of large degree, and we have $A_s = A_{s^l}$. It follows easily that there is a canonical isomorphism

$$\mathrm{Proj}(B) \simeq \mathrm{Proj}\Big(\bigoplus_m B_{lm}\Big).$$

Similarly, we may if we wish change a finite number of components B_m without affecting $\mathrm{Proj}(B)$. In particular, we may alway assume that $B_0 = k\,1_B$. By selecting

finitely many generators $g_0, \ldots, g_N$ in B_1, we then find a surjective graded homomorphism $k[T_0, \ldots, T_N] \to B$, thus $B \simeq k[T_0, \ldots, T_N]/J$ for some graded ideal $J \subset B$. The algebra homomorphism $k[T_0, \ldots, T_N] \to B$ therefore yields a scheme embedding $\mathrm{Proj}(B) \to \mathbb{P}^N$ onto $V(J)$.

We will not pursue further the study of algebraic varieties from this point of view ; in fact we are mostly interested in the case $k = \mathbb{C}$, and algebraic varieties over $\mathbb{C}$ are a special case of the more general concept of complex analytic space.

§ 2. The Local Ring of Germs of Analytic Functions

§ 2.A. The Weierstrass Preparation Theorem

Our first goal is to establish a basic factorization and division theorem for analytic functions of several variables, which is essentially due to Weierstrass. We follow here a simple proof given by C.L. Siegel, based on a clever use of the Cauchy formula. Let g be a holomorphic function defined on a neighborhood of 0 in $\mathbb{C}^n$, $g \not\equiv 0$. There exists a dense set of vectors $v \in \mathbb{C}^n \smallsetminus \{0\}$ such that the function $\mathbb{C} \ni t \longmapsto g(tv)$ is not identically zero. In fact the Taylor series of g at the origin can be written

$$g(tv) = \sum_{k=0}^{+\infty} \frac{1}{k!}\, t^k\, g^{(k)}(v)$$

where $g^{(k)}$ is a homogeneous polynomial of degree k on $\mathbb{C}^n$ and $g^{(k_0)} \not\equiv 0$ for some index k_0. Thus it suffices to select v such that $g^{(k_0)}(v) \neq 0$. After a change of coordinates, we may assume that $v = (0, \ldots, 0, 1)$. Let s be the vanishing order of $z_n \longmapsto g(0, \ldots, 0, z_n)$ at $z_n = 0$. There exists $r_n > 0$ such that $g(0, \ldots, 0, z_n) \neq 0$ when $0 < |z_n| \leqslant r_n$. By continuity of g and compactness of the circle $|z_n| = r_n$, there exists $r' > 0$ and $\varepsilon > 0$ such that

$$g(z', z_n) \neq 0 \quad \text{for } z' \in \mathbb{C}^{n-1}, \quad |z'| \leqslant r', \quad r_n - \varepsilon \leqslant |z_n| \leqslant r_n + \varepsilon.$$

For every integer $k \in \mathbb{N}$, let us consider the integral

$$S_k(z') = \frac{1}{2\pi \mathrm{i}} \int_{|z_n| = r_n} \frac{1}{g(z', z_n)} \frac{\partial g}{\partial z_n}(z', z_n)\, z_n^k\, dz_n.$$

Then S_k is holomorphic in a neighborhood of $|z'| \leqslant r'$. Rouché's theorem shows that $S_0(z')$ is the number of roots z_n of $g(z', z_n) = 0$ in the disk $|z_n| < r_n$, thus by continuity $S_0(z')$ must be a constant s. Let us denote by $w_1(z'), \ldots, w_s(z')$ these roots, counted with multiplicity. By definition of r_n, we have $w_1(0) = \ldots = w_s(0) = 0$, and by the choice of r', ε we have $|w_j(z')| < r_n - \varepsilon$ for $|z'| \leqslant r'$. The Cauchy residue formula yields

$$S_k(z') = \sum_{j=1}^{s} w_j(z')^k.$$

Newton's formula shows that the elementary symmetric function $c_k(z')$ of degree k in $w_1(z'), \ldots, w_s(z')$ is a polynomial in $S_1(z'), \ldots, S_k(z')$. Hence $c_k(z')$ is holomorphic in a

neighborhood of $|z'| \leqslant r'$. Let us set

$$P(z', z_n) = z_n^s - c_1(z')z_n^{s-1} + \cdots + (-1)^s c_s(z') = \prod_{j=1}^{s} \big(z_n - w_j(z')\big).$$

For $|z'| \leqslant r'$, the quotient $f = g/P$ (resp. $f = P/g$) is holomorphic in z_n on the disk $|z_n| < r_n + \varepsilon$, because g and P have the same zeros with the same multiplicities, and $f(z', z_n)$ is holomorphic in z' for $r_n - \varepsilon \leqslant |z_n| \leqslant r_n + \varepsilon$. Therefore

$$f(z', z_n) = \frac{1}{2\pi \mathrm{i}} \int_{|w_n| = r_n + \varepsilon} \frac{f(z', w_n)\, dw_n}{w_n - z_n}$$

is holomorphic in z on a neighborhood of the closed polydisk $\overline{\Delta}(r', r_n) = \{|z'| \leqslant r'\} \times \{|z_n| \leqslant r_n\}$. Thus g/P is invertible and we obtain:

(2.1) Weierstrass preparation theorem. *Let g be holomorphic on a neighborhood of 0 in $\mathbb{C}^n$, such that $g(0, z_n)/z_n^s$ has a not zero finite limit at $z_n = 0$. With the above choice of r' and r_n, one can write $g(z) = u(z)P(z', z_n)$ where u is an invertible holomorphic function in a neighborhood of the polydisk $\overline{\Delta}(r', r_n)$, and P is a Weierstrass polynomial in z_n, that is, a polynomial of the form*

$$P(z', z_n) = z_n^s + a_1(z')z_n^{s-1} + \cdots + a_s(z'), \qquad a_k(0) = 0,$$

with holomorphic coefficients $a_k(z')$ on a neighborhood of $|z'| \leqslant r'$ in $\mathbb{C}^{n-1}$.

(2.2) Remark. If g vanishes at order m at 0 and $v \in \mathbb{C}^n \smallsetminus \{0\}$ is selected such that $g^{(m)}(v) \neq 0$, then $s = m$ and P must also vanish at order m at 0. In that case, the coefficients $a_k(z')$ are such that $a_k(z') = O(|z'|^k)$, $1 \leqslant k \leqslant s$.

(2.3) Weierstrass division theorem. *Every bounded holomorphic function f on $\Delta = \Delta(r', r_n)$ can be represented in the form*

$$f(z) = g(z)q(z) + R(z', z_n), \tag{2.4}$$

where q and R are analytic in Δ, $R(z', z_n)$ is a polynomial of degree $\leqslant s - 1$ in z_n, and

$$\sup_{\Delta} |q| \leqslant C \sup_{\Delta} |f|, \qquad \sup_{\Delta} |R| \leqslant C \sup_{\Delta} |f| \tag{2.5}$$

for some constant $C \geqslant 0$ independent of f. The representation (2.4) is unique.

Proof (Siegel). It is sufficient to prove the result when $g(z) = P(z', z_n)$ is a Weierstrass polynomial.

Let us first prove the uniqueness. If $f = Pq_1 + R_1 = Pq_2 + R_2$, then

$$P(q_2 - q_1) + (R_2 - R_1) = 0.$$

It follows that the s roots z_n of $P(z', \bullet) = 0$ are zeros of $R_2 - R_1$. Since $\deg_{z_n}(R_2 - R_1) \leqslant s - 1$, we must have $R_2 - R_1 \equiv 0$, thus $q_2 - q_1 \equiv 0$.

In order to prove the existence of (q, R), we set

$$q(z', z_n) = \lim_{\varepsilon \to 0+} \frac{1}{2\pi \mathrm{i}} \int_{|w_n| = r_n - \varepsilon} \frac{f(z', w_n)}{P(z', w_n)(w_n - z_n)} \, dw_n, \qquad z \in \Delta \ ;$$

observe that the integral does not depend on ε when $\varepsilon < r_n - |z_n|$ is small enough. Then q is holomorphic on Δ. The function $R = f - Pq$ is also holomorphic on Δ and

$$R(z) = \lim_{\varepsilon \to 0+} \frac{1}{2\pi \mathrm{i}} \int_{|w_n| = r_n - \varepsilon} \frac{f(z', w_n)}{P(z', w_n)} \Big[\frac{P(z', w_n) - P(z', z_n)}{(w_n - z_n)} \Big] \, dw_n.$$

The expression in brackets has the form

$$\big[(w_n^s - z_n^s) + \sum_{j=1}^{s} a_j(z')(w_n^{s-j} - z_n^{s-j}) \big] / (w_n - z_n)$$

hence is a polynomial in z_n of degree $\leqslant s - 1$ with coefficients that are holomorphic functions of z'. Thus we have the asserted decomposition $f = Pq + R$ and

$$\sup_{\Delta} |R| \leqslant C_1 \sup_{\Delta} |f|$$

where C_1 depends on bounds for the $a_j(z')$ and on $\mu = \min |P(z', z_n)|$ on the compact set $\{|z'| \leqslant r'\} \times \{|z_n| = r_n\}$. By the maximum principle applied to $q = (f - R)/P$ on each disk $\{z'\} \times \{|z_n| < r_n - \varepsilon\}$, we easily get

$$\sup_{\Delta} |q| \leqslant \mu^{-1}(1 + C_1) \sup_{\Delta} |f|. \qquad \square$$

§ 2.B. Algebraic Properties of the Ring $\mathscr{O}_n$

We give here important applications of the Weierstrass preparation theorem to the study of the ring of germs of holomorphic functions in $\mathbb{C}^n$.

(2.6) Notation. *We let $\mathscr{O}_n$ be the ring of germs of holomorphic functions on $\mathbb{C}^n$ at 0. Alternatively, $\mathscr{O}_n$ can be identified with the ring $\mathbb{C}\{z_1, \ldots, z_n\}$ of convergent power series in $z_1, \ldots, z_n$.*

(2.7) Theorem. *The ring $\mathscr{O}_n$ is Noetherian, i.e. every ideal $\mathscr{I}$ of $\mathscr{O}_n$ is finitely generated.*

Proof. By induction on n. For $n = 1$, $\mathscr{O}_n$ is principal: every ideal $\mathscr{I} \neq \{0\}$ is generated by z^s, where s is the minimum of the vanishing orders at 0 of the non zero elements of $\mathscr{I}$. Let $n \geqslant 2$ and $\mathscr{I} \subset \mathscr{O}_n$, $\mathscr{I} \neq \{0\}$. After a change of variables, we may assume that $\mathscr{I}$ contains a Weierstrass polynomial $P(z', z_n)$. For every $f \in \mathscr{I}$, the Weierstrass division theorem yields

$$f(z) = P(z', z_n) q(z) + R(z', z_n), \qquad R(z', z_n) = \sum_{k=0}^{s-1} c_k(z') \, z_n^k,$$

and we have $R \in \mathscr{I}$. Let us consider the set $\mathscr{M}$ of coefficients $(c_0, \ldots, c_{s-1})$ in $\mathscr{O}_{n-1}^{\oplus s}$ corresponding to the polynomials $R(z', z_n)$ which belong to $\mathscr{I}$. Then $\mathscr{M}$ is a $\mathscr{O}_{n-1}$-submodule

of $\mathcal{O}_{n-1}^{\oplus s}$. By the induction hypothesis $\mathcal{O}_{n-1}$ is Noetherian; furthermore, every submodule of a finitely generated module over a Noetherian ring is finitely generated ([Lang 1965], Chapter VI). Therefore $\mathcal{M}$ is finitely generated, and $\mathcal{I}$ is generated by P and by polynomials $R_1, \dots, R_N$ associated with a finite set of generators of $\mathcal{M}$. □

Before going further, we need two lemmas which relate the algebraic properties of $\mathcal{O}_n$ to those of the polynomial ring $\mathcal{O}_{n-1}[z_n]$.

(2.8) Lemma. *Let $P, F \in \mathcal{O}_{n-1}[z_n]$ where P is a Weierstrass polynomial. If P divides F in $\mathcal{O}_n$, then P divides F in $\mathcal{O}_{n-1}[z_n]$.*

Proof. Assume that $F(z', z_n) = P(z', z_n)h(z)$, $h \in \mathcal{O}_n$. The standard division algorithm of F by P in $\mathcal{O}_{n-1}[z_n]$ yields

$$F = PQ + R, \qquad Q, R \in \mathcal{O}_{n-1}[z_n], \quad \deg R < \deg P.$$

The uniqueness part of Th. 2.3 implies $h(z) = Q(z', z_n)$ and $R \equiv 0$. □

(2.9) Lemma. *Let $P(z', z_n)$ be a Weierstrass polynomial.*

a) *If $P = P_1 \dots P_N$ with $P_j \in \mathcal{O}_{n-1}[z_n]$, then, up to invertible elements of $\mathcal{O}_{n-1}$, all P_j are Weierstrass polynomials.*

b) *$P(z', z_n)$ is irreducible in $\mathcal{O}_n$ if and only if it is irreducible in $\mathcal{O}_{n-1}[z_n]$.*

Proof. a) Assume that $P = P_1 \dots P_N$ with polynomials $P_j \in \mathcal{O}_{n-1}[z_n]$ of respective degrees s_j, $\sum_{1 \leqslant j \leqslant N} s_j = s$. The product of the leading coefficients of $P_1, \dots, P_N$ in $\mathcal{O}_{n-1}$ is equal to 1; after normalizing these polynomials, we may assume that $P_1, \dots, P_N$ are unitary and $s_j > 0$ for all j. Then

$$P(0, z_n) = z_n^s = P_1(0, z_n) \dots P_N(0, z_n),$$

hence $P_j(0, z_n) = z_n^{s_j}$ and therefore P_j is a Weierstrass polynomial.

b) Set $s = \deg P$ and $P(0, z_n) = z_n^s$. Assume that P is reducible in $\mathcal{O}_n$, with $P(z', z_n) = g_1(z)g_2(z)$ for non invertible elements $g_1, g_2 \in \mathcal{O}_n$. Then $g_1(0, z_n)$ and $g_2(0, z_n)$ have vanishing orders $s_1, s_2 > 0$ with $s_1 + s_2 = s$, and

$$g_j = u_j P_j, \qquad \deg P_j = s_j, \quad j = 1, 2,$$

where P_j is a Weierstrass polynomial and $u_j \in \mathcal{O}_n$ is invertible. Therefore $P_1 P_2 = uP$ for an invertible germ $u \in \mathcal{O}_n$. Lemma 2.8 shows that P divides $P_1 P_2$ in $\mathcal{O}_{n-1}[z_n]$; since P_1, P_2 are unitary and $s = s_1 + s_2$, we get $P = P_1 P_2$, hence P is reducible in $\mathcal{O}_{n-1}[z_n]$. The converse implication is obvious from a). □

(2.10) Theorem. *$\mathcal{O}_n$ is a factorial ring, i.e. $\mathcal{O}_n$ is entire and:*

a) *every non zero germ $f \in \mathcal{O}_n$ admits a factorization $f = f_1 \dots f_N$ in irreducible elements;*

b) *the factorization is unique up to invertible elements.*

Proof. The existence part a) follows from Lemma 2.9 if we take f to be a Weierstrass polynomial and $f = f_1 \dots f_N$ be a decomposition of maximal length N into polynomials of positive degree. In order to prove the uniqueness, it is sufficient to verify the following statement:

b′) *If g is an irreducible element that divides a product $f_1 f_2$, then g divides either f_1 or f_2.*

By Th. 2.1, we may assume that f_1, f_2, g are Weierstrass polynomials in z_n. Then g is irreducible and divides $f_1 f_2$ in $\mathscr{O}_{n-1}[z_n]$ thanks to Lemmas 2.8 and 2.9 b). By induction on n, we may assume that $\mathscr{O}_{n-1}$ is factorial. The standard Gauss lemma ([Lang 1965], Chapter V) says that the polynomial ring $A[T]$ is factorial if the ring A is factorial. Hence $\mathscr{O}_{n-1}[z_n]$ is factorial by induction and thus g must divide f_1 or f_2 in $\mathscr{O}_{n-1}[z_n]$. □

(2.11) Theorem. *If $f, g \in \mathscr{O}_n$ are relatively prime, then the germs f_z, g_z at every point $z \in \mathbb{C}^n$ near 0 are again relatively prime.*

Proof. One may assume that $f = P$, $g = Q$ are Weierstrass polynomials. Let us recall that unitary polynomials $P, Q \in \mathscr{A}[X]$ ($\mathscr{A}$ = a factorial ring) are relatively prime if and only if their resultant $R \in \mathscr{A}$ is non zero. Then the resultant $R(z') \in \mathscr{O}_{n-1}$ of $P(z', z_n)$ and $Q(z', z_n)$ has a non zero germ at 0. Therefore the germ $R_{z'}$ at points $z' \in \mathbb{C}^{n-1}$ near 0 is also non zero. □

§ 3. Coherent Sheaves

§ 3.A. Locally Free Sheaves and Vector Bundles

Before introducing the more general notion of a coherent sheaf, we discuss the notion of locally free sheaves over a sheaf a ring. All rings occurring in the sequel are supposed to be commutative with unit (the non commutative case is also of considerable interest, e.g. in view of the theory of $\mathscr{D}$-modules, but this subject is beyond the scope of the present book).

(3.1) Definition. *Let $\mathscr{A}$ be a sheaf of rings on a topological space X and let $\mathscr{S}$ a sheaf of modules over $\mathscr{A}$ (or briefly a $\mathscr{A}$-module). Then $\mathscr{S}$ is said to be locally free of rank r over $\mathscr{A}$, if $\mathscr{S}$ is locally isomorphic to $\mathscr{A}^{\oplus r}$ on a neighborhood of every point, i.e. for every $x_0 \in X$ one can find a neighborhood Ω and sections $F_1, \dots, F_r \in \mathscr{S}(\Omega)$ such that the sheaf homomorphism*

$$F : \mathscr{A}^{\oplus r}_{\restriction\Omega} \longrightarrow \mathscr{S}_{\restriction\Omega}, \qquad \mathscr{A}^{\oplus r}_x \ni (w_1, \dots, w_r) \longmapsto \sum_{1 \leqslant j \leqslant r} w_j F_{j,x} \in \mathscr{S}_x$$

is an isomorphism.

By definition, if $\mathscr{S}$ is locally free, there is a covering $(U_\alpha)_{\alpha \in I}$ by open sets on which $\mathscr{S}$ admits free generators $F^1_\alpha, \dots, F^r_\alpha \in \mathscr{S}(U_\alpha)$. Because the generators can be uniquely expressed in terms of any other system of independent generators, there is for each pair (α, β) a $r \times r$ matrix

$$G_{\alpha\beta} = (G^{jk}_{\alpha\beta})_{1 \leqslant j,k \leqslant r}, \qquad G^{jk}_{\alpha\beta} \in \mathscr{A}(U_\alpha \cap U_\beta),$$

such that

$$F_\beta^k = \sum_{1\leqslant j\leqslant r} F_\alpha^j G_{\alpha\beta}^{jk} \qquad \text{on} \quad U_\alpha \cap U_\beta.$$

In other words, we have a commutative diagram

$$\begin{array}{ccc} \mathscr{A}^{\oplus r}_{\restriction U_\alpha\cap U_\beta} & \xrightarrow{F_\alpha} & \mathscr{S}_{\restriction U_\alpha\cap U_\beta} \\ G_{\alpha\beta}\Big\uparrow & & \Big\| \\ \mathscr{A}^{\oplus r}_{\restriction U_\alpha\cap U_\beta} & \xrightarrow[F_\beta]{} & \mathscr{S}_{\restriction U_\alpha\cap U_\beta} \end{array}$$

It follows easily from the equality $G_{\alpha\beta} = F_\alpha^{-1} \circ F_\beta$ that the *transition matrices* $G_{\alpha\beta}$ are invertible matrices satisfying the transition relation

$$G_{\alpha\gamma} = G_{\alpha\beta} G_{\beta\gamma} \qquad \text{on} \quad U_\alpha \cap U_\beta \cap U_\gamma \tag{3.2}$$

for all indices $\alpha, \beta, \gamma \in I$. In particular $G_{\alpha\alpha} = \mathrm{Id}$ on U_α and $G_{\alpha\beta}^{-1} = G_{\beta\alpha}$ on $U_\alpha \cap U_\beta$.

Conversely, if we are given a system of invertible $r \times r$ matrices $G_{\alpha\beta}$ with coefficients in $\mathscr{A}(U_\alpha \cap U_\beta)$ satisfying the transition relation (3.2), we can define a locally free sheaf $\mathscr{S}$ of rank r over $\mathscr{A}$ by taking $\mathscr{S} \simeq \mathscr{A}^{\oplus r}$ over each U_α, the identification over $U_\alpha \cap U_\beta$ being given by the isomorphism $G_{\alpha\beta}$. A section H of $\mathscr{S}$ over an open set $\Omega \subset X$ can just be seen as a collection of sections $H_\alpha = (H_\alpha^1, \ldots, H_\alpha^r)$ of $\mathscr{A}^{\oplus r}(\Omega \cap U_\alpha)$ satisfying the transition relations $H_\alpha = G_{\alpha\beta} H_\beta$ over $\Omega \cap U_\alpha \cap U_\beta$.

The notion of locally free sheaf is closely related to another essential notion of differential geometry, namely the notion of vector bundle (resp. topological, differentiable, holomorphic ..., vector bundle). To describe the relation between these notions, we assume that the sheaf of rings $\mathscr{A}$ is a subsheaf of the sheaf $\mathscr{C}_{\mathbb{K}}$ of continous functions on X with values in the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$, containing the sheaf of locally constant functions $X \to \mathbb{K}$. Then, for each $x \in X$, there is an evaluation map

$$\mathscr{A}_x \to \mathbb{K}, \qquad w \mapsto w(x)$$

whose kernel is a maximal ideal $\mathfrak{m}_x$ of $\mathscr{A}_x$, and $\mathscr{A}_x/\mathfrak{m}_x = \mathbb{K}$. Let $\mathscr{S}$ be a locally free sheaf of rank r over $\mathscr{A}$. To each $x \in X$, we can associate a $\mathbb{K}$-vector space $E_x = \mathscr{S}_x/\mathfrak{m}_x\mathscr{S}_x$: since $\mathscr{S}_x \simeq \mathscr{A}_x^{\oplus r}$, we have $E_x \simeq (\mathscr{A}_x/\mathfrak{m}_x)^{\oplus r} = \mathbb{K}^r$. The set $E = \coprod_{x\in X} E_x$ is equipped with a natural projection

$$\pi : E \to X, \qquad \xi \in E_x \mapsto \pi(\xi) := x,$$

and the fibers $E_x = \pi^{-1}(x)$ have a structure of r-dimensional $\mathbb{K}$-vector space: such a structure E is called a $\mathbb{K}$-*vector bundle of rank* r over X. Every section $s \in \mathscr{S}(U)$ gives rise to a *section* of E over U by setting $s(x) = s_x \bmod \mathfrak{m}_x$. We obtain a function (still denoted by the same symbol) $s : U \to E$ such that $s(x) \in E_x$ for every $x \in U$, i.e. $\pi \circ s = \mathrm{Id}_U$. It is clear that $\mathscr{S}(U)$ can be considered as a $\mathscr{A}(U)$-submodule of the $\mathbb{K}$-vector space of functions $U \to E$ mapping a point $x \in U$ to an element in the fiber E_x. Thus we get a subsheaf of the sheaf of E-valued sections, which is in a natural way a $\mathscr{A}$-module isomorphic to $\mathscr{S}$. This subsheaf will be denoted by $\mathscr{A}(E)$ and will be called the *sheaf of* $\mathscr{A}$*-sections* of E. If we are given a $\mathbb{K}$-vector bundle E over X and a subsheaf $\mathscr{S} = \mathscr{A}(E)$ of

the sheaf of all sections of E which is in a natural way a locally free $\mathcal{A}$-module of rank r, we say that E (or more precisely the pair $(E, \mathcal{A}(E))$) is a $\mathcal{A}$-vector bundle of rank r over X.

(3.3) Example. In case $\mathcal{A} = \mathscr{C}_{X,\mathbb{K}}$ is the sheaf of all $\mathbb{K}$-valued continuous functions on X, we say that E is a *topological* vector bundle over X. When X is a manifold and $\mathcal{A} = \mathscr{C}^p_{X,\mathbb{K}}$, we say that E is a *C^p-differentiable* vector bundle; finally, when X is complex analytic and $\mathcal{A} = \mathscr{O}_X$, we say that E is a *holomorphic* vector bundle.

Let us introduce still a little more notation. Since $\mathcal{A}(E)$ is a locally free sheaf of rank r over any open set U_α in a suitable covering of X, a choice of generators $(F^1_\alpha, \dots, F^r_\alpha)$ for $\mathcal{A}(E)_{\restriction U_\alpha}$ yields corresponding generators $(e^1_\alpha(x), \dots, e^r_\alpha(x))$ of the fibers E_x over $\mathbb{K}$. Such a system of generators is called a *$\mathcal{A}$-admissible frame* of E over U_α. There is a corresponding isomorphism

$$\theta_\alpha : E_{\restriction U_\alpha} := \pi^{-1}(U_\alpha) \longrightarrow U_\alpha \times \mathbb{K}^r \tag{3.4}$$

which to each $\xi \in E_x$ associates the pair $(x, (\xi^1_\alpha, \dots, \xi^r_\alpha)) \in U_\alpha \times \mathbb{K}^r$ composed of x and of the components $(\xi^j_\alpha)_{1 \leqslant j \leqslant r}$ of ξ in the basis $(e^1_\alpha(x), \dots, e^r_\alpha(x))$ of E_x. The bundle E is said to be *trivial* if it is of the form $X \times \mathbb{K}^r$, which is the same as saying that $\mathcal{A}(E) = \mathcal{A}^{\oplus r}$. For this reason, the isomorphisms θ_α are called *trivializations* of E over U_α. The corresponding *transition automorphisms* are

$$\begin{aligned} &\theta_{\alpha\beta} := \theta_\alpha \circ \theta_\beta^{-1} : (U_\alpha \cap U_\beta) \times \mathbb{K}^r \longrightarrow (U_\alpha \cap U_\beta) \times \mathbb{K}^r, \\ &\theta_{\alpha\beta}(x, \xi) = (x, g_{\alpha\beta}(x) \cdot \xi), \qquad (x, \xi) \in (U_\alpha \cap U_\beta) \times \mathbb{K}^r, \end{aligned}$$

where $(g_{\alpha\beta}) \in \mathrm{GL}_r(\mathcal{A})(U_\alpha \cap U_\beta)$ are the transition matrices already described (except that they are just seen as matrices with coefficients in $\mathbb{K}$ rather than with coefficients in a sheaf). Conversely, if we are given a collection of matrices $g_{\alpha\beta} = (g^{jk}_{\alpha\beta}) \in \mathrm{GL}_r(\mathcal{A})(U_\alpha \cap U_\beta)$ satisfying the transition relation

$$g_{\alpha\gamma} = g_{\alpha\beta} g_{\beta\gamma} \qquad \text{on} \quad U_\alpha \cap U_\beta \cap U_\gamma,$$

we can define a $\mathcal{A}$-vector bundle

$$E = \Big(\coprod_{\alpha \in I} U_\alpha \times \mathbb{K}^r \Big) / \sim$$

by gluing the charts $U_\alpha \times \mathbb{K}^r$ via the identification $(x_\alpha, \xi_\alpha) \sim (x_\beta, \xi_\beta)$ if and only if $x_\alpha = x_\beta = x \in U_\alpha \cap U_\beta$ and $\xi_\alpha = g_{\alpha\beta}(x) \cdot \xi_\beta$.

(3.5) Example. When X is a real differentiable manifold, an interesting example of real vector bundle is the *tangent bundle* T_X ; if $\tau_\alpha : U_\alpha \to \mathbb{R}^n$ is a collection of coordinate charts on X, then $\theta_\alpha = \pi \times d\tau_\alpha : T_{X \restriction U_\alpha} \to U_\alpha \times \mathbb{R}^m$ define trivializations of T_X and the transition matrices are given by $g_{\alpha\beta}(x) = d\tau_{\alpha\beta}(x^\beta)$ where $\tau_{\alpha\beta} = \tau_\alpha \circ \tau_\beta^{-1}$ and $x^\beta = \tau_\beta(x)$. The dual $T^\star_X$ of T_X is called the *cotangent bundle* of X. If X is complex analytic, then T_X has the structure of a holomorphic vector bundle.

We now briefly discuss the concept of sheaf and bundle morphisms. If $\mathcal{S}$ and $\mathcal{S}'$ are sheaves of $\mathcal{A}$-modules over a topological space X, then by a morphism $\varphi : \mathcal{S} \to \mathcal{S}'$ we just

mean a $\mathscr{A}$-linear sheaf morphism. If $\mathscr{S} = \mathscr{A}(E)$ and $\mathscr{S}' = \mathscr{A}(E')$ are locally free sheaves, this is the same as a $\mathscr{A}$-linear bundle morphism, that is, a fiber preserving $\mathbb{K}$-linear morphism $\varphi(x) : E_x \to E'_x$ such that the matrix representing φ in any local $\mathscr{A}$-admissible frames of E and E' has coefficients in $\mathscr{A}$.

(3.6) Proposition. *Suppose that $\mathscr{A}$ is a sheaf of local rings, i.e. that a section of $\mathscr{A}$ is invertible in $\mathscr{A}$ if and only if it never takes the zero value in $\mathbb{K}$. Let $\varphi : \mathscr{S} \to \mathscr{S}'$ be a $\mathscr{A}$-morphism of locally free $\mathscr{A}$-modules of rank r, r'. If the rank of the $r' \times r$ matrix $\varphi(x) \in M_{r'r}(\mathbb{K})$ is constant for all $x \in X$, then* $\operatorname{Ker}\varphi$ *and* $\operatorname{Im}\varphi$ *are locally free subsheaves of $\mathscr{S}$, $\mathscr{S}'$ respectively, and* $\operatorname{Coker}\varphi = \mathscr{S}'/\operatorname{Im}\varphi$ *is locally free.*

Proof. This is just a consequence of elementary linear algebra, once we know that non zero determinants with coefficients in $\mathscr{A}$ can be inverted. □

Note that all three sheaves $\mathscr{C}_{X,\mathbb{K}}$, $\mathscr{C}^p_{X,\mathbb{K}}$, $\mathscr{O}_X$ are sheaves of local rings, so Prop. 3.6 applies to these cases. However, even if we work in the holomorphic category ($\mathscr{A} = \mathscr{O}_X$), a difficulty immediately appears that the kernel or cokernel of an arbitrary morphism of locally free sheaves is in general not locally free.

(3.7) Examples.

a) Take $X = \mathbb{C}$, let $\mathscr{S} = \mathscr{S}' = \mathscr{O}$ be the trivial sheaf, and let $\varphi : \mathscr{O} \to \mathscr{O}$ be the morphism $u(z) \mapsto z\,u(z)$. It is immediately seen that φ is injective as a sheaf morphism ($\mathscr{O}$ being an entire ring), and that $\operatorname{Coker}\varphi$ is the *skyscraper sheaf* $\mathbb{C}_0$ of stalk $\mathbb{C}$ at $z = 0$, having zero stalks at all other points $z \neq 0$. Thus $\operatorname{Coker}\varphi$ is not a locally free sheaf, although φ is everywhere injective (note however that the corresponding morphism $\varphi : E \to E'$, $(z, \xi) \mapsto (z, z\xi)$ of trivial rank 1 vector bundles $E = E' = \mathbb{C} \times \mathbb{C}$ is *not injective* on the zero fiber E_0).

b) Take $X = \mathbb{C}^3$, $\mathscr{S} = \mathscr{O}^{\oplus 3}$, $\mathscr{S}' = \mathscr{O}$ and

$$\varphi : \mathscr{O}^{\oplus 3} \to \mathscr{O}, \qquad (u_1, u_2, u_3) \mapsto \sum_{1 \leqslant j \leqslant 3} z_j u_j(z_1, z_2, z_3).$$

Since φ yields a surjective bundle morphism on $\mathbb{C}^3 \smallsetminus \{0\}$, one easily sees that $\operatorname{Ker}\varphi$ is locally free of rank 2 over $\mathbb{C}^3 \smallsetminus \{0\}$. However, by looking at the Taylor expansion of the u_j's at 0, it is not difficult to check that $\operatorname{Ker}\varphi$ is the $\mathscr{O}$-submodule of $\mathscr{O}^{\oplus 3}$ generated by the three sections $(-z_2, z_1, 0)$, $(-z_3, 0, z_1)$ and $(0, z_3, -z_2)$, and that any two of these three sections cannot generate the 0-stalk $(\operatorname{Ker}\varphi)_0$. Hence $\operatorname{Ker}\varphi$ is not locally free.

Since the category of locally free $\mathscr{O}$-modules is not stable by taking kernels or cokernels, one is led to introduce a more general category which will be stable under these operations. This leads to the notion of *coherent sheaves*.

§ 3.B. Notion of Coherence

The notion of coherence again deals with sheaves of modules over a sheaf of rings. It is a semi-local property which says roughly that the sheaf of modules locally has a finite presentation in terms of generators and relations. We describe here some general properties of this notion, before concentrating ourselves on the case of coherent $\mathscr{O}_X$-modules.

(3.8) Definition. *Let $\mathscr{A}$ be a sheaf of rings on a topological space X and $\mathscr{S}$ a sheaf of modules over $\mathscr{A}$ (or briefly a $\mathscr{A}$-module). Then $\mathscr{S}$ is said to be locally finitely generated if for every point $x_0 \in X$ one can find a neighborhood Ω and sections $F_1, \ldots, F_q \in \mathscr{S}(\Omega)$ such that for every $x \in \Omega$ the stalk $\mathscr{S}_x$ is generated by the germs $F_{1,x}, \ldots, F_{q,x}$ as an $\mathscr{A}_x$-module.*

(3.9) Lemma. *Let $\mathscr{S}$ be a locally finitely generated sheaf of $\mathscr{A}$-modules on X and $G_1, \ldots, G_N$ sections in $\mathscr{S}(U)$ such that $G_{1,x_0}, \ldots, G_{N,x_0}$ generate $\mathscr{S}_{x_0}$ at $x_0 \in U$. Then $G_{1,x}, \ldots, G_{N,x}$ generate $\mathscr{S}_x$ for x near x_0.*

Proof. Take $F_1, \ldots, F_q$ as in Def. 3.8. As $G_1, \ldots, G_N$ generate $\mathscr{S}_{x_0}$, one can find a neighborhood $\Omega' \subset \Omega$ of x_0 and $H_{jk} \in \mathscr{A}(\Omega')$ such that $F_j = \sum H_{jk} G_k$ on Ω'. Thus $G_{1,x}, \ldots, G_{N,x}$ generate $\mathscr{S}_x$ for all $x \in \Omega'$. $\square$

§ 3.2.1. *Definition of Coherent Sheaves.* If U is an open subset of X, we denote by $\mathscr{S}_{\restriction U}$ the restriction of $\mathscr{S}$ to U, i.e. the union of all stalks $\mathscr{S}_x$ for $x \in U$. If $F_1, \ldots, F_q \in \mathscr{S}(U)$, the kernel of the sheaf homomorphism $F : \mathscr{A}_{\restriction U}^{\oplus q} \longrightarrow \mathscr{S}_{\restriction U}$ defined by

$$\mathscr{A}_x^{\oplus q} \ni (g^1, \ldots, g^q) \longmapsto \sum_{1 \leqslant j \leqslant q} g^j F_{j,x} \in \mathscr{S}_x, \qquad x \in U \tag{3.10}$$

is a subsheaf $\mathscr{R}(F_1, \ldots, F_q)$ of $\mathscr{A}_{\restriction U}^{\oplus q}$, called the *sheaf of relations* between $F_1, \ldots, F_q$.

(3.11) Definition. *A sheaf $\mathscr{S}$ of $\mathscr{A}$-modules on X is said to be coherent if:*

a) *$\mathscr{S}$ is locally finitely generated* ;

b) *for any open subset U of X and any $F_1, \ldots, F_q \in \mathscr{S}(U)$, the sheaf of relations $\mathscr{R}(F_1, \ldots, F_q)$ is locally finitely generated.*

Assumption a) means that every point $x \in X$ has a neighborhood Ω such that there is a surjective sheaf morphism $F : \mathscr{A}_{\restriction \Omega}^{\oplus q} \longrightarrow \mathscr{S}_{\restriction \Omega}$, and assumption b) implies that the kernel of F is locally finitely generated. Thus, after shrinking Ω, we see that $\mathscr{S}$ admits over Ω a finite presentation under the form of an exact sequence

$$\mathscr{A}_{\restriction \Omega}^{\oplus p} \xrightarrow{G} \mathscr{A}_{\restriction \Omega}^{\oplus q} \xrightarrow{F} \mathscr{S}_{\restriction \Omega} \longrightarrow 0, \tag{3.12}$$

where G is given by a $q \times p$ matrix (G_{jk}) of sections of $\mathscr{A}(\Omega)$ whose columns (G_{j1}), ..., (G_{jp}) are generators of $\mathscr{R}(F_1, \ldots, F_q)$.

It is clear that every locally finitely generated subsheaf of a coherent sheaf is coherent. From this we easily infer:

(3.13) Theorem. *Let $\varphi : \mathscr{F} \longrightarrow \mathscr{G}$ be a $\mathscr{A}$-morphism of coherent sheaves. Then $\operatorname{Im} \varphi$ and $\ker \varphi$ are coherent.*

Proof. Clearly $\operatorname{Im} \varphi$ is a locally finitely generated subsheaf of $\mathscr{G}$, so it is coherent. Let $x_0 \in X$, let $F_1, \ldots, F_q \in \mathscr{F}(\Omega)$ be generators of $\mathscr{F}$ on a neighborhood Ω of x_0, and $G_1, \ldots, G_r \in \mathscr{A}(\Omega')^{\oplus q}$ be generators of $\mathscr{R}\big(\varphi(F_1), \ldots, \varphi(F_q)\big)$ on a neighborhood $\Omega' \subset \Omega$ of x_0. Then $\ker \varphi$ is generated over Ω' by the sections

$$H_j = \sum_{k=1}^{q} G_j^k F_k \in \mathscr{F}(\Omega'), \qquad 1 \leqslant j \leqslant r. \qquad \square$$

(3.14) Theorem. *Let $0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{S} \longrightarrow \mathcal{G} \longrightarrow 0$ be an exact sequence of $\mathcal{A}$-modules. If two of the sheaves $\mathcal{F}, \mathcal{S}, \mathcal{G}$ are coherent, then all three are coherent.*

Proof. If $\mathcal{S}$ and $\mathcal{G}$ are coherent, then $\mathcal{F} = \ker(\mathcal{S} \to \mathcal{G})$ is coherent by Th. 3.13. If $\mathcal{S}$ and $\mathcal{F}$ are coherent, then $\mathcal{G}$ is locally finitely generated; to prove the coherence, let $G_1, \ldots, G_q \in \mathcal{G}(U)$ and $x_0 \in U$. Then there is a neighborhood Ω of x_0 and sections $\tilde{G}_1, \ldots, \tilde{G}_q \in \mathcal{S}(\Omega)$ which are mapped to $G_1, \ldots, G_q$ on Ω. After shrinking Ω, we may assume also that $\mathcal{F}_{\restriction\Omega}$ is generated by sections $F_1, \ldots, F_p \in \mathcal{F}(\Omega)$. Then $\mathcal{R}(G_1, \ldots, G_q)$ is the projection on the last q-components of $\mathcal{R}(F_1, \ldots, F_p, \tilde{G}_1, \ldots, \tilde{G}_q) \subset \mathcal{A}^{p+q}$, which is finitely generated near x_0 by the coherence of $\mathcal{S}$. Hence $\mathcal{R}(G_1, \ldots, G_q)$ is finitely generated near x_0 and $\mathcal{G}$ is coherent.

Finally, assume that $\mathcal{F}$ and $\mathcal{G}$ are coherent. Let $x_0 \in X$ be any point, let $F_1, \ldots, F_p \in \mathcal{F}(\Omega)$ and $G_1, \ldots, G_q \in \mathcal{G}(\Omega)$ be generators of $\mathcal{F}$, $\mathcal{G}$ on a neighborhood Ω of x_0. There is a neighborhood Ω' of x_0 such that $G_1, \ldots, G_q$ admit liftings $\tilde{G}_1, \ldots, \tilde{G}_q \in \mathcal{S}(\Omega')$. Then $(F_1, \ldots, F_q, \tilde{G}_1, \ldots, \tilde{G}_q)$ generate $\mathcal{S}_{\restriction\Omega'}$, so $\mathcal{S}$ is locally finitely generated. Now, let $S_1, \ldots, S_q$ be arbitrary sections in $\mathcal{S}(U)$ and $\overline{S}_1, \ldots, \overline{S}_q$ their images in $\mathcal{G}(U)$. For any $x_0 \in U$, the sheaf of relations $\mathcal{R}(\overline{S}_1, \ldots, \overline{S}_q)$ is generated by sections $P_1, \ldots, P_s \in \mathcal{A}(\Omega)^{\oplus q}$ on a small neighborhood Ω of x_0. Set $P_j = (P_j^k)_{1 \leqslant k \leqslant q}$. Then $H_j = P_j^1 S_1 + \ldots + P_j^q S_q$, $1 \leqslant j \leqslant s$, are mapped to 0 in $\mathcal{G}$ so they can be seen as sections of $\mathcal{F}$. The coherence of $\mathcal{F}$ shows that $\mathcal{R}(H_1, \ldots, H_s)$ has generators $Q_1, \ldots, Q_t \in \mathcal{A}(\Omega')^s$ on a small neighborhood $\Omega' \subset \Omega$ of x_0. Then $\mathcal{R}(S_1, \ldots, S_q)$ is generated over Ω' by $R_j = \sum Q_j^k P_k \in \mathcal{A}(\Omega')$, $1 \leqslant j \leqslant t$, and $\mathcal{S}$ is coherent. □

(3.15) Corollary. *If $\mathcal{F}$ and $\mathcal{G}$ are coherent subsheaves of a coherent analytic sheaf $\mathcal{S}$, the intersection $\mathcal{F} \cap \mathcal{G}$ is a coherent sheaf.*

Proof. Indeed, the intersection sheaf $\mathcal{F} \cap \mathcal{G}$ is the kernel of the composite morphism $\mathcal{F} \hookrightarrow \mathcal{S} \longrightarrow \mathcal{S}/\mathcal{G}$, and $\mathcal{S}/\mathcal{G}$ is coherent. □

§ 3.2.2. *Coherent Sheaf of Rings.* A sheaf of rings $\mathcal{A}$ is said to be coherent if it is coherent as a module over itself. By Def. 3.11, this means that for any open set $U \subset X$ and any sections $F_j \in \mathcal{A}(U)$, the sheaf of relations $\mathcal{R}(F_1, \ldots, F_q)$ is finitely generated. The above results then imply that all free modules $\mathcal{A}^{\oplus p}$ are coherent. As a consequence:

(3.16) Theorem. *If $\mathcal{A}$ is a coherent sheaf of rings, any locally finitely generated subsheaf of $\mathcal{A}^{\oplus p}$ is coherent. In particular, if $\mathcal{S}$ is a coherent $\mathcal{A}$-module and $F_1, \ldots, F_q \in \mathcal{S}(U)$, the sheaf of relations $\mathcal{R}(F_1, \ldots, F_q) \subset \mathcal{A}^{\oplus q}$ is also coherent.*

Let $\mathcal{S}$ be a coherent sheaf of modules over a coherent sheaf of ring $\mathcal{A}$. By an iteration of construction (3.12), we see that for every integer $m \geqslant 0$ and every point $x \in X$ there is a neighborhood Ω of x on which there is an exact sequence of sheaves

$$\mathcal{A}_{\restriction\Omega}^{\oplus p_m} \xrightarrow{F_m} \mathcal{A}_{\restriction\Omega}^{\oplus p_{m-1}} \longrightarrow \cdots \longrightarrow \mathcal{A}_{\restriction\Omega}^{\oplus p_1} \xrightarrow{F_1} \mathcal{A}_{\restriction\Omega}^{\oplus p_0} \xrightarrow{F_0} \mathcal{S}_{\restriction\Omega} \longrightarrow 0, \tag{3.17}$$

where F_j is given by a $p_{j-1} \times p_j$ matrix of sections in $\mathcal{A}(\Omega)$.

§ 3.C. Analytic Sheaves and the Oka Theorem

Many properties of holomorphic functions which will be considered in this book can be expressed in terms of sheaves. Among them, analytic sheaves play a central role. The

Oka theorem [Oka 1950] asserting the coherence of the sheaf of holomorphic functions can be seen as a far-reaching deepening of the noetherian property seen in § 2. The theory of analytic sheaves could not be presented without it.

(3.18) Definition. *Let M be a n-dimensional complex analytic manifold and let $\mathscr{O}_M$ be the sheaf of germs of analytic functions on M. An analytic sheaf over M is by definition a sheaf $\mathscr{S}$ of modules over $\mathscr{O}_M$.*

(3.19) Coherence theorem of Oka. *The sheaf of rings $\mathscr{O}_M$ is coherent for any complex manifold M.*

Let $F_1, \ldots, F_q \in \mathscr{O}(U)$. Since $\mathscr{O}_{M,x}$ is Noetherian, we already know that every stalk $\mathscr{R}(F_1, \ldots, F_q)_x \subset \mathscr{O}_{M,x}^{\oplus q}$ is finitely generated, but the important new fact expressed by the theorem is that the sheaf of relations is locally finitely generated, namely that the "same" generators can be chosen to generate each stalk in a neighborhood of a given point.

Proof. By induction on $n = \dim_{\mathbb{C}} M$. For $n = 0$, the stalks $\mathscr{O}_{M,x}$ are equal to $\mathbb{C}$ and the result is trivial. Assume now that $n \geqslant 1$ and that the result has already been proved in dimension $n-1$. Let U be an open set of M and $F_1, \ldots, F_q \in \mathscr{O}_M(U)$. To show that $\mathscr{R}(F_1, \ldots, F_q)$ is locally finitely generated, we may assume that $U = \Delta = \Delta' \times \Delta_n$ is a polydisk in $\mathbb{C}^n$ centered at $x_0 = 0$; after a change of coordinates and multiplication of $F_1, \ldots, F_q$ by invertible functions, we may also suppose that $F_1, \ldots, F_q$ are Weierstrass polynomials in z_n with coefficients in $\mathscr{O}(\Delta')$. We need a lemma.

(3.20) Lemma. *If $x = (x', x_n) \in \Delta$, the $\mathscr{O}_{\Delta,x}$-module $\mathscr{R}(F_1, \ldots, F_q)_x$ is generated by those of its elements whose components are germs of analytic polynomials in $\mathscr{O}_{\Delta',x'}[z_n]$ with a degree in z_n at most equal to μ, the maximum of the degrees of $F_1, \ldots, F_q$.*

Proof. Assume for example that F_q is of the maximum degree μ. By the Weierstrass preparation Th. 2.1 and Lemma 2.9 applied at x, we can write $F_{q,x} = f'f''$ where $f', f'' \in \mathscr{O}_{\Delta',x'}[z_n]$, f' is a Weierstrass polynomial in $z_n - x_n$ and $f''(x) \neq 0$. Let μ' and μ'' denote the degrees of f' and f'' with respect to z_n, so $\mu' + \mu'' = \mu$. Now, take $(g^1, \ldots, g^q) \in \mathscr{R}(F_1, \ldots, F_q)_x$. The Weierstrass division theorem gives

$$g^j = F_{q,x} t^j + r^j, \qquad j = 1, \ldots, q-1,$$

where $t^j \in \mathscr{O}_{\Delta,x}$ and $r^j \in \mathscr{O}_{\Delta',x'}[z_n]$ is a polynomial of degree $< \mu'$. For $j = q$, define $r^q = g^q + \sum_{1 \leqslant j \leqslant q-1} t^j F_{j,x}$. We can write

$$(g^1, \ldots, g^q) = \sum_{1 \leqslant j \leqslant q} t^j (0, \ldots, F_q, \ldots, 0, -F_j)_x + (r^1, \ldots, r^q) \tag{3.21}$$

where F_q is in the j-th position in the q-tuples of the summation. Since these q-tuples are in $\mathscr{R}(F_1, \ldots, F_q)_x$, we have $(r^1, \ldots, r^q) \in \mathscr{R}(F_1, \ldots, F_q)_x$, thus

$$\sum_{1 \leqslant j \leqslant q-1} F_{j,x} r^j + f'f''r^q = 0.$$

As the sum is a polynomial in z_n of degree $< \mu + \mu'$, it follows from Lemma 2.9 that $f''r^q$ is a polynomial in z_n of degree $< \mu$. Now we have

$$(r^1, \ldots, r^q) = 1/f''(f''r^1, \ldots, f''r^q)$$

where $f''r^j$ is of degree $< \mu' + \mu'' = \mu$. In combination with (3.21) this proves the lemma. □

Proof of Theorem 3.19 (end). If $g = (g^1, \dots, g^q)$ is one of the polynomials of $\mathscr{R}(F_1, \dots, F_q)_x$ described in Lemma 3.20, we can write

$$g^j = \sum_{0 \leqslant k \leqslant \mu} u^{jk} z_n^k, \qquad u^{jk} \in \mathscr{O}_{\Delta', x'}.$$

The condition for $(g^1, \dots, g^q)$ to belong to $\mathscr{R}(F_1, \dots, F_q)_x$ therefore consists of $2\mu + 1$ linear conditions for the germ $u = (u^{jk})$ with coefficients in $\mathscr{O}(\Delta')$. By the induction hypothesis, $\mathscr{O}_{\Delta'}$ is coherent and Th. 3.16 shows that the corresponding modules of relations are generated over $\mathscr{O}_{\Delta', x'}$, for x' in a neighborhood Ω' of 0, by finitely many $(q \times \mu)$-tuples $U_1, \dots, U_N \in \mathscr{O}(\Omega')^{q\mu}$. By Lemma 3.20, $\mathscr{R}(F_1, \dots, F_q)_x$ is generated at every point $x \in \Omega = \Omega' \times \Delta_n$ by the germs of the corresponding polynomials

$$G_l(z) = \Big(\sum_{0 \leqslant k \leqslant \mu} U_l^{jk}(z') z_n^k \Big)_{1 \leqslant j \leqslant q}, \qquad z \in \Omega, \quad 1 \leqslant l \leqslant N. \qquad \square$$

(3.22) Strong Noetherian property. *Let $\mathscr{F}$ be a coherent analytic sheaf on a complex manifold M and let $\mathscr{F}_1 \subset \mathscr{F}_2 \subset \dots$ be an increasing sequence of coherent subsheaves of $\mathscr{F}$. Then the sequence $(\mathscr{F}_k)$ is stationary on every compact subset of M.*

Proof. Since $\mathscr{F}$ is locally a quotient of a free module $\mathscr{O}_M^{\oplus q}$, we can pull back the sequence to $\mathscr{O}_M^{\oplus q}$ and thus suppose $\mathscr{F} = \mathscr{O}_M$ (by easy reductions similar to those in the proof of Th. 3.14). Suppose M connected and $\mathscr{F}_{k_0} \neq \{0\}$ for some index k_0 (otherwise, there is nothing to prove). By the analytic continuation theorem, we easily see that $\mathscr{F}_{k_0, x} \neq \{0\}$ for every $x \in M$. We can thus find a non zero Weierstrass polynomial $P \in \mathscr{F}_{k_0}(V)$, $\deg_{z_n} P(z', z_n) = \mu$, in a coordinate neighborhood $V = \Delta' \times \Delta_n$ of any point $x \in M$. A division by P shows that for $k \geqslant k_0$ and $x \in V$, all stalks $\mathscr{F}_{k,x}$ are generated by P_x and by polynomials of degree $< \mu$ in z_n with coefficients in $\mathscr{O}_{\Delta', x'}$. Therefore, we can apply induction on n to the coherent $\mathscr{O}_{\Delta'}$-module

$$\mathscr{F}' = \mathscr{F} \cap \big\{ Q \in \mathscr{O}_{\Delta'}[z_n] \,;\, \deg Q \leqslant \mu \big\}$$

and its increasing sequence of coherent subsheaves $\mathscr{F}'_k = \mathscr{F}_k \cap \mathscr{F}'$. □

§ 4. Complex Analytic Sets. Local Properties

§ 4.A. Definition. Irreducible Components

A complex analytic set is a set which can be defined locally by finitely many holomorphic equations; such a set has in general singular points, because no assumption is made on the differentials of the equations. We are interested both in the description of the singularities and in the study of algebraic properties of holomorphic functions on analytic sets. For a more detailed study than ours, we refer to H. Cartan's seminar [Cartan 1950], to the books of [Gunning-Rossi 1965], [Narasimhan 1966] or the recent book by [Grauert-Remmert 1984].

(4.1) Definition. *Let M be a complex analytic manifold. A subset $A \subset M$ is said to be an analytic subset of M if A is closed and if for every point $x_0 \in A$ there exist a neighborhood U of x_0 and holomorphic functions $g_1, \ldots, g_n$ in $\mathscr{O}(U)$ such that*

$$A \cap U = \{z \in U \ ; \ g_1(z) = \ldots = g_N(z) = 0\}.$$

Then $g_1, \ldots, g_N$ are said to be (local) equations of A in U.

It is easy to see that a finite union or intersection of analytic sets is analytic: if (g'_j), (g''_k) are equations of A', A'' in the open set U, then the family of all products $(g'_j g''_k)$ and the family $(g'_j) \cup (g''_k)$ define equations of $A' \cup A''$ and $A' \cap A''$ respectively.

(4.2) Remark. Assume that M is connected. The analytic continuation theorem shows that either $A = M$ or A has no interior point. In the latter case, each piece $A \cap U = g^{-1}(0)$ is the set of points where the function $\log|g|^2 = \log(|g_1|^2 + \cdots + |g_N|^2) \in \mathrm{Psh}(U)$ takes the value $-\infty$, hence A is pluripolar. In particular $M \smallsetminus A$ is connected and every function $f \in \mathscr{O}(M \smallsetminus A)$ that is locally bounded near A can be extended to a function $\tilde{f} \in \mathscr{O}(M)$. $\square$

We focus now our attention on local properties of analytic sets. By definition, a germ of set at a point $x \in M$ is an equivalence class of elements in the power set $\mathscr{P}(M)$, with $A \sim B$ if there is an open neighborhood V of x such that $A \cap V = B \cap V$. The germ of a subset $A \subset M$ at x will be denoted by (A, x). We most often consider the case when $A \subset M$ is a analytic set in a neighborhood U of x, and in this case we denote by $\mathscr{I}_{A,x}$ the ideal of germs $f \in \mathscr{O}_{M,x}$ which vanish on (A, x). Conversely, if $\mathscr{J} = (g_1, \ldots, g_N)$ is an ideal of $\mathscr{O}_{M,x}$, we denote by $\big(V(\mathscr{J}), x\big)$ the germ at x of the zero variety $V(\mathscr{J}) = \{z \in U \ ; \ g_1(z) = \ldots = g_N(z) = 0\}$, where U is a neighborhood of x such that $g_j \in \mathscr{O}(U)$. It is easy to check that the germ $(V(cJ), x)$ does not depend on the choice of generators. Moreover, it is clear that

(4.3′) for every ideal $\mathscr{J}$ in the ring $\mathscr{O}_{M,x}$, $\quad \mathscr{I}_{V(\mathscr{J}),x} \supset \mathscr{J}$,

(4.3″) for every germ of analytic set (A, x), $\quad \big(V(\mathscr{I}_{A,x}), x\big) = (A, x)$.

(4.4) Definition. *A germ (A, x) is said to be irreducible if it has no decomposition $(A, x) = (A_1, x) \cup (A_2, x)$ with analytic sets $(A_j, x) \neq (A, x)$, $j = 1, 2$.*

(4.5) Proposition. *A germ (A, x) is irreducible if and only if $\mathscr{I}_{A,x}$ is a prime ideal of the ring $\mathscr{O}_{M,x}$.*

Proof. Let us recall that an ideal $\mathscr{J}$ is said to be *prime* if $fg \in \mathscr{J}$ implies $f \in \mathscr{J}$ or $g \in \mathscr{J}$. Assume that (A, x) is irreducible and that $fg \in \mathscr{I}_{A,x}$. As we can write $(A, x) = (A_1, x) \cup (A_2, x)$ with $A_1 = A \cap f^{-1}(0)$ and $A_2 = A \cap g^{-1}(0)$, we must have for example $(A_1, x) = (A, x)$; thus $f \in \mathscr{I}_{A,x}$ and $\mathscr{I}_{A,x}$ is prime. Conversely, if $(A, x) = (A_1, x) \cup (A_2, x)$ with $(A_j, x) \neq (A, x)$, there exist $f \in \mathscr{I}_{A_1,x}$, $g \in \mathscr{I}_{A_2,x}$ such that $f, g \notin \mathscr{I}_{A,x}$. However $fg \in \mathscr{I}_{A,x}$, thus $\mathscr{I}_{A,x}$ is not prime. $\square$

(4.6) Theorem. *Every decreasing sequence of germs of analytic sets (A_k, x) is stationary.*

Proof. In fact, the corresponding sequence of ideals $\mathscr{J}_k = \mathscr{I}_{A_k,x}$ is increasing, thus $\mathscr{J}_k = \mathscr{J}_{k_0}$ for $k \geqslant k_0$ large enough by the Noetherian property of $\mathscr{O}_{M,x}$. Hence $(A_k, x) =$

$\big(V(\mathcal{J}_k), x\big)$ is constant for $k \geqslant k_0$. This result has the following straightforward consequence: $\square$

(4.7) Theorem. *Every analytic germ (A, x) has a finite decomposition*

$$(A, x) = \bigcup_{1 \leqslant k \leqslant N} (A_k, x)$$

where the germs (A_j, x) are irreducible and $(A_j, x) \not\subset (A_k, x)$ for $j \neq k$. The decomposition is unique apart from the ordering.

Proof. If (A, x) can be split in several components, we split repeatedly each component as long as one of them is reducible. The process must stop by Th. 4.6, whence the existence. For the uniqueness, assume that $(A, x) = \bigcup (A'_l, x)$, $1 \leqslant l \leqslant N'$, is another decomposition. Since $(A_k, x) = \bigcup_l (A_k \cap A'_l, x)$, we must have $(A_k, x) = (A_k \cap A'_l, x)$ for some $l = l(k)$, i.e. $(A_k, x) \subset (A'_{l(k)}, x)$, and likewise $(A'_{l(k)}, x) \subset (A_j, x)$ for some j. Hence $j = k$ and $(A'_{l(k)}, x) = (A_k, x)$. $\square$

§ 4.B. Local Structure of a Germ of Analytic Set

We are going to describe the local structure of a germ, both from the holomorphic and topological points of view. By the above decomposition theorem, we may restrict ourselves to the case of irreducible germs Let $\mathcal{J}$ be a prime ideal of $\mathcal{O}_n = \mathcal{O}_{\mathbb{C}^n,0}$ and let $A = V(\mathcal{J})$ be its zero variety. We set $\mathcal{J}_k = \mathcal{J} \cap \mathbb{C}\{z_1, \ldots, z_k\}$ for each $k = 0, 1, \ldots, n$.

(4.8) Proposition. *There exist an integer d, a basis $(e_1, \ldots, e_n)$ of $\mathbb{C}^n$ and associated coordinates $(z_1, \ldots, z_n)$ with the following properties: $\mathcal{J}_d = \{0\}$ and for every integer $k = d+1, \ldots, n$ there is a Weierstrass polynomial $P_k \in \mathcal{J}_k$ of the form*

$$P_k(z', z_k) = z_k^{s_k} + \sum_{1 \leqslant j \leqslant s_k} a_{j,k}(z')\, z_k^{s_k - j}, \qquad a_{j,k}(z') \in \mathcal{O}_{k-1}, \tag{4.9}$$

where $a_{j,k}(z') = O(|z'|^j)$. Moreover, the basis $(e_1, \ldots, e_n)$ can be chosen arbitrarily close to any preassigned basis $(e_1^0, \ldots, e_n^0)$.

Proof. By induction on n. If $\mathcal{J} = \mathcal{J}_n = \{0\}$, then $d = n$ and there is nothing to prove. Otherwise, select a non zero element $g_n \in \mathcal{J}$ and a vector e_n such that $\mathbb{C} \ni w \longmapsto g_n(we_n)$ has minimum vanishing order s_n. This choice excludes at most the algebraic set $g_n^{(s_n)}(v) = 0$, so e_n can be taken arbitrarily close to e_n^0. Let $(\tilde{z}_1, \ldots, \tilde{z}_{n-1}, z_n)$ be the coordinates associated to the basis $(e_1^0, \ldots, e_{n-1}^0, e_n)$. After multiplication by an invertible element, we may assume that g_n is a Weierstrass polynomial

$$P_n(\tilde{z}, z_n) = z_n^{s_n} + \sum_{1 \leqslant j \leqslant s_n} a_{j,n}(\tilde{z})\, z_n^{s_n - j}, \qquad a_{j,n} \in \mathcal{O}_{n-1},$$

and $a_{j,n}(\tilde{z}) = O(|\tilde{z}|^j)$ by Remark 2.2. If $\mathcal{J}_{n-1} = \mathcal{J} \cap \mathbb{C}\{\tilde{z}\} = \{0\}$ then $d = n-1$ and the construction is finished. Otherwise we apply the induction hypothesis to the ideal $\mathcal{J}_{n-1} \subset \mathcal{O}_{n-1}$ in order to find a new basis $(e_1, \ldots, e_{n-1})$ of $\mathrm{Vect}(e_1^0, \ldots, e_{n-1}^0)$, associated coordinates $(z_1, \ldots, z_{n-1})$ and Weierstrass polynomials $P_k \in \mathcal{J}_k$, $d+1 \leqslant k \leqslant n-1$, as stated in the lemma. $\square$

(4.10) Lemma. *If $w \in \mathbb{C}$ is a root of $w^d + a_1 w^{d-1} + \cdots + a_d = 0$, $a_j \in \mathbb{C}$, then $|w| \leqslant 2 \max |a_j|^{1/j}$.*

Proof. Otherwise $|w| > 2|a_j|^{1/j}$ for all $j = 1, \ldots, d$ and the given equation $-1 = a_1/w + \cdots + a_d/w^d$ implies $1 \leqslant 2^{-1} + \cdots + 2^{-d}$, a contradiction. □

(4.11) Corollary. *Set $z' = (z_1, \ldots, z_d)$, $z'' = (z_{d+1}, \ldots, z_n)$, and let Δ' in $\mathbb{C}^d$, Δ'' in $\mathbb{C}^{n-d}$ be polydisks of center 0 and radii $r', r'' > 0$. Then the germ $(A, 0)$ is contained in a cone $|z''| \leqslant C|z'|$, $C =$ constant, and the restriction of the projection map $\mathbb{C}^n \longrightarrow \mathbb{C}^d$, $(z', z'') \longmapsto z'$:*

$$\pi : A \cap (\Delta' \times \Delta'') \longrightarrow \Delta'$$

is proper if r'' is small enough and $r' \leqslant r''/C$.

Proof. The polynomials $P_k(z_1, \ldots, z_{k-1}\,;\, z_k)$ vanish on $(A, 0)$. By Lemma 4.10 and (4.9), every point $z \in A$ sufficiently close to 0 satisfies

$$|z_k| \leqslant C_k(|z_1| + \cdots + |z_{k-1}|), \qquad d+1 \leqslant k \leqslant n,$$

thus $|z''| \leqslant C|z'|$ and the Corollary follows. □

Since $\mathcal{J}_d = \{0\}$, we have an injective ring morphism

$$\mathcal{O}_d = \mathbb{C}\{z_1, \ldots, z_d\} \hookrightarrow \mathcal{O}_n/\mathcal{J}. \tag{4.12}$$

(4.13) Proposition. *$\mathcal{O}_n/\mathcal{J}$ is a finite integral extension of $\mathcal{O}_d$.*

Proof. Let $f \in \mathcal{O}_n$. A division by P_n yields $f = P_n q_n + R_n$ with a remainder $R_n \in \mathcal{O}_{n-1}[z_n]$, $\deg_{z_n} R_n < s_n$. Further divisions of the coefficients of R_n by P_{n-1}, P_{n-2} etc ... yield

$$R_{k+1} = P_k q_k + R_k, \qquad R_k \in \mathcal{O}_k[z_{k+1}, \ldots, z_n],$$

where $\deg_{z_j} R_k < s_j$ for $j > k$. Hence

$$f = R_d + \sum_{d+1 \leqslant k \leqslant n} P_k q_k = R_d \mod (P_{d+1}, \ldots, P_n) \subset \mathcal{J} \tag{4.14}$$

and $\mathcal{O}_n/\mathcal{J}$ is finitely generated as an $\mathcal{O}_d$-module by the family of monomials $z_{d+1}^{\alpha_{d+1}} \ldots z_n^{\alpha_n}$ with $\alpha_j < s_j$. □

As $\mathcal{J}$ is prime, $\mathcal{O}_n/\mathcal{J}$ is an entire ring. We denote by $\tilde{f}$ the class of any germ $f \in \mathcal{O}_n$ in $\mathcal{O}_n/\mathcal{J}$, by $\mathcal{M}_A$ and $\mathcal{M}_d$ the quotient fields of $\mathcal{O}_n/\mathcal{J}$ and $\mathcal{O}_d$ respectively. Then $\mathcal{M}_A = \mathcal{M}_d[\tilde{z}_{d+1}, \ldots, \tilde{z}_n]$ is a finite algebraic extension of $\mathcal{M}_d$. Let $q = [\mathcal{M}_A : \mathcal{M}_d]$ be its degree and let $\sigma_1, \ldots, \sigma_q$ be the embeddings of $\mathcal{M}_A$ over $\mathcal{M}_d$ in an algebraic closure $\overline{\mathcal{M}}_A$. Let us recall that a factorial ring is integrally closed in its quotient field ([Lang 1965], Chapter IX). Hence every element of $\mathcal{M}_d$ which is integral over $\mathcal{O}_d$ lies in fact in $\mathcal{O}_d$. By the primitive element theorem, there exists a linear form $u(z'') = c_{d+1} z_{d+1} + \cdots + c_n z_n$, $c_k \in \mathbb{C}$, such that $\mathcal{M}_A = \mathcal{M}_d[\tilde{u}]$; in fact, u is of degree q if and only if $\sigma_1 \tilde{u}, \ldots, \sigma_q \tilde{u}$ are all distinct, and this excludes at most a finite number of vector subspaces in the space $\mathbb{C}^{n-d}$ of coefficients $(c_{d+1}, \ldots, c_n)$. As $\tilde{u} \in \mathcal{O}_n/\mathcal{J}$ is integral over the integrally closed ring $\mathcal{O}_d$, the unitary irreducible polynomial W_u of $\tilde{u}$ over $\mathcal{M}_d$ has coefficients in $\mathcal{O}_d$:

$$W_u(z'\,;T) = T^q + \sum_{1 \leqslant j \leqslant q} a_j(z_1, \ldots, z_d)\, T^{q-j}, \qquad a_j \in \mathcal{O}_d.$$

W_u must be a Weierstrass polynomial, otherwise there would exist a factorization $W_u = W'Q$ in $\mathscr{O}_d[T]$ with a Weierstrass polynomial W' of degree $\deg W' < q = \deg \tilde{u}$ and $Q(0) \neq 0$, hence $W'(\tilde{u}) = 0$, a contradiction. In the same way, we see that $\tilde{z}_{d+1}, \ldots, \tilde{z}_n$ have irreducible equations $W_k(z'\,;\,\tilde{z}_k) = 0$ where $W_k \in \mathscr{O}_d[T]$ is a Weierstrass polynomial of degree $= \deg \tilde{z}_k \leqslant q$, $d+1 \leqslant k \leqslant n$.

(4.15) Lemma. *Let $\delta(z') \in \mathscr{O}_d$ be the discriminant of $W_u(z'\,;T)$. For every element g of $\mathscr{M}_A$ which is integral over $\mathscr{O}_d$ (or equivalently over $\mathscr{O}_n/\mathscr{J}$) we have $\delta g \in \mathscr{O}_d[\tilde{u}]$.*

Proof. We have $\delta(z') = \prod_{j<k}(\sigma_k\tilde{u} - \sigma_j\tilde{u})^2 \not\equiv 0$, and $g \in \mathscr{M}_A = \mathscr{M}_d[\tilde{u}]$ can be written

$$g = \sum_{0\leqslant j\leqslant q-1} b_j\,\tilde{u}^j, \qquad b_j \in \mathscr{M}_d,$$

where $b_0, \ldots, b_{d-1}$ are the solutions of the linear system $\sigma_k g = \sum b_j(\sigma_k\tilde{u})^j$; the determinant (of Van der Monde type) is $\delta^{1/2}$. It follows that $\delta b_j \in \mathscr{M}_d$ are polynomials in $\sigma_k g$ and $\sigma_k\tilde{u}$, thus δb_j is integral over $\mathscr{O}_d$. As $\mathscr{O}_d$ is integrally closed, we must have $\delta b_j \in \mathscr{O}_d$, hence $\delta g \in \mathscr{O}_d[\tilde{u}]$. □

In particular, there exist unique polynomials $B_{d+1}, \ldots, B_n \in \mathscr{O}_d[T]$ with $\deg B_k \leqslant q-1$, such that

$$\delta(z')z_k = B_k(z'\,;u(z'')) \pmod{\mathscr{J}}. \tag{4.16}$$

Then we have

$$\delta(z')^q W_k\big(z'\,;B_k(z'\,;T)/\delta(z')\big) \in \text{ideal } \; W_u(z'\,;T)\,\mathscr{O}_d[T]\ ; \tag{4.17}$$

indeed, the left-hand side is a polynomial in $\mathscr{O}_d[T]$ and admits $T = \tilde{u}$ as a root in $\mathscr{O}_n/\mathscr{J}$ since $B_k(z'\,;\tilde{u})/\delta(z') = \tilde{z}_k$ and $W_k(z'\,;\tilde{z}_k) = 0$.

(4.18) Lemma. *Consider the ideal*

$$\mathscr{G} = \big(W_u(z'\,;u(z'))\,,\,\delta(z')z_k - B_k(z'\,;u(z''))\big) \subset \mathscr{J}$$

and set $m = \max\{q, (n-d)(q-1)\}$. For every germ $f \in \mathscr{O}_n$, there exists a unique polynomial $R \in \mathscr{O}_d[T]$, $\deg_T R \leqslant q-1$, such that

$$\delta(z')^m f(z) = R(z'\,;u(z'')) \pmod{\mathscr{G}}.$$

Moreover $f \in \mathscr{J}$ implies $R = 0$, hence $\delta^m\mathscr{J} \subset \mathscr{G}$.

Proof. By (4.17) and a substitution of z_k, we find $\delta(z')^q W_k(z'\,;z_k) \in \mathscr{G}$. The analogue of formula (4.14) with W_k in place of P_k yields

$$f = R_d + \sum_{d+1\leqslant k\leqslant n} W_k q_k, \qquad R_d \in \mathscr{O}_d[z_{d+1}, \ldots, z_n],$$

with $\deg_{z_k} R_d < \deg W_k \leqslant q$, thus $\delta^m f = \delta^m R_d \bmod \mathscr{G}$. We may therefore replace f by R_d and assume that $f \in \mathscr{O}_d[z_{d+1}, \ldots, z_n]$ is a polynomial of total degree $\leqslant (n-d)(q-1) \leqslant m$. A substitution of z_k by $B_k(z'\,;u(z''))/\delta(z')$ yields $G \in \mathscr{O}_d[T]$ such that

$$\delta(z')^m f(z) = G(z'\,;u(z'')) \mod \big(\delta(z')z_k - B_k(z'\,;u(z''))\big).$$

Finally, a division $G = W_u Q + R$ gives the required polynomial $R \in \mathcal{O}_d[T]$. The last statement is clear: if $f \in \mathcal{J}$ satisfies $\delta^m(z')f(z) = R(z\,;u(z''))$ mod $\mathcal{G}$ for $\deg_T R < q$, then $R(z'\,;\tilde{u}) = 0$, and as $\tilde{u} \in \mathcal{O}_n/\mathcal{J}$ is of degree q, we must have $R = 0$. The uniqueness of R is proved similarly. $\square$

(4.19) Local parametrization theorem. *Let $\mathcal{J}$ be a prime ideal of $\mathcal{O}_n$ and let $A = V(\mathcal{J})$. Assume that the coordinates*

$$(z'\,;z'') = (z_1, \ldots, z_d\,; z_{d+1}, \ldots, z_n)$$

are chosen as above. Then the ring $\mathcal{O}_n/\mathcal{J}$ is a finite integral extension of $\mathcal{O}_d$; let q be the degree of the extension and let $\delta(z') \in \mathcal{O}_d$ be the discriminant of the irreducible polynomial of a primitive element $u(z'') = \sum_{k>d} c_k z_k$. If Δ', Δ'' are polydisks of sufficiently small radii r', r'' and if $r' \leqslant r''/C$ with C large, the projection map $\pi : A \cap (\Delta' \times \Delta'') \longrightarrow \Delta'$ is a ramified covering with q sheets, whose ramification locus is contained in $S = \{z' \in \Delta'; \delta(z') = 0\}$. This means that:

a) *the open subset $A_S = A \cap \big((\Delta' \smallsetminus S) \times \Delta''\big)$ is a smooth d-dimensional manifold, dense in $A \cap (\Delta' \times \Delta'')$;*

b) *$\pi : A_S \longrightarrow \Delta' \smallsetminus S$ is a covering ;*

c) *the fibers $\pi^{-1}(z')$ have exactly q elements if $z' \notin S$ and at most q if $z' \in S$.*

Moreover, A_S is a connected covering of $\Delta' \smallsetminus S$, and $A \cap (\Delta' \times \Delta'')$ is contained in a cone $|z''| \leqslant C|z'|$ (see Fig. 1).

Proof. After a linear change in the coordinates $z_{d+1}, \ldots, z_n$, we may assume $u(z'') = z_{d+1}$, so $W_u = W_{d+1}$ and $B_{d+1}(z'\,;T) = \delta(z')T$. By Lemma 4.18, we have

$$\mathcal{G} = \big(W_{d+1}(z', z_{d+1})\,,\, \delta(z')z_k - B_k(z', z_{d+1})\big)_{k \geqslant d+2} \subset \mathcal{J}, \qquad \delta^m \mathcal{J} \subset \mathcal{G}.$$

We can thus find a polydisk $\Delta = \Delta' \times \Delta''$ of sufficiently small radii r', r'' such that $V(\mathcal{J}) \subset V(\mathcal{G}) \subset V(\delta^m \mathcal{J})$ in Δ. As $V(\mathcal{J}) = A$ and $V(\delta) \cap \Delta = S \times \Delta''$, this implies

$$A \cap \Delta \subset V(\mathcal{G}) \cap \Delta \subset (A \cap \Delta) \cup (S \times \Delta'').$$

In particular, the set $A_S = A \cap \big((\Delta' \smallsetminus S) \times \Delta''\big)$ lying above $\Delta' \smallsetminus S$ coincides with $V(\mathcal{G}) \cap \big((\Delta' \smallsetminus S) \times \Delta''\big)$, which is the set of points $z \in \Delta$ parametrized by the equations

$$\begin{cases} \delta(z') \neq 0, \quad W_{d+1}(z', z_{d+1}) = 0, \\ z_k = B_k(z', z_{d+1})/\delta(z'), \quad d+2 \leqslant k \leqslant n. \end{cases} \tag{4.20}$$

As $\delta(z')$ is the resultant of W_{d+1} and $\partial W_{d+1}/\partial T$, we have

$$\partial W_{d+1}/\partial T(z', z_{d+1}) \neq 0 \quad \text{ on } \; A_S.$$

The implicit function theorem shows that z_{d+1} is locally a holomorphic function of z' on A_S, and the same is true for $z_k = B_k(z', z_{d+1})/\delta(z')$, $k \geqslant d+2$. Hence A_S is a smooth

manifold, and for $r' \leqslant r''/C$ small, the projection map $\pi : A_S \longrightarrow \Delta' \smallsetminus S$ is a proper local diffeomorphism; by (4.20) the fibers $\pi^{-1}(z')$ have at most q points corresponding to some of the q roots w of $W_{d+1}(z'\,;w) = 0$. Since $\Delta' \smallsetminus S$ is connected (Remark 4.2), either $A_S = \emptyset$ or the map π is a covering of constant sheet number $q' \leqslant q$. However, if w is a root of $W_{d+1}(z', w) = 0$ with $z' \in \Delta' \smallsetminus S$ and if we set $z_{d+1} = w$, $z_k = B_k(z', w)/\delta(z')$, $k \geqslant d+2$, relation (4.17) shows that $W_k(z', z_k) = 0$, in particular $|z_k| = O(|z'|^{1/q})$ by Lemma 4.10. For z' small enough, the q points $z = (z', z'')$ defined in this way lie in Δ, thus $q' = q$. Property b) and the first parts of a) and c) follow. Now, we need the following lemma.

(4.21) Lemma. *If $\mathcal{J} \subset \mathcal{O}_n$ is prime and $A = V(\mathcal{J})$, then $\mathcal{I}_{A,0} = \mathcal{J}$.*

I.t is obvious that $\mathcal{I}_{A,0} \supset \mathcal{J}$. Now, for any $f \in \mathcal{I}_{A,0}$, Prop. 4.13 implies that $\tilde{f}$ satisfies in $\mathcal{O}_n/\mathcal{I}$ an irreducible equation

$$f^r + b_1(z')\, f^{r-1} + \cdots + b_r(z') = 0 \quad (\mathrm{mod}\ \mathcal{J}).$$

Then $b_r(z')$ vanishes on $(A, 0)$ and the first part of c) gives $b_r = 0$ on $\Delta' \smallsetminus S$. Hence $\tilde{b}_r = 0$ and the irreducibility of the equation of $\tilde{f}$ implies $r = 1$, so $f \in \mathcal{J}$, as desired. □

Proof of Theorem 4.19 (end). It only remains to prove that A_S is connected and dense in $A \cap \Delta$ and that the fibers $\pi^{-1}(z')$, $z' \in S$, have at most q elements. Let $A_{S,1}, \ldots, A_{S,N}$ be the connected components of A_S. Then $\pi : A_{S,j} \longrightarrow \Delta' \smallsetminus S$ is a covering with q_j sheets, $q_1 + \cdots + q_N = q$. For every point $\zeta' \in \Delta' \smallsetminus S$, there exists a neighborhood U of ζ' such that $A_{S,j} \cap \pi^{-1}(U)$ is a disjoint union of graphs $z'' = g_{j,k}(z')$ of analytic functions $g_{j,k} \in \mathcal{O}(U)$, $1 \leqslant k \leqslant q_j$. If $\lambda(z'')$ is an arbitrary linear form in $z_{d+1}, \ldots, z_n$ and $z' \in \Delta' \smallsetminus S$, we set

$$P_{\lambda,j}(z'\,;T) = \prod_{\{z''\,;\,(z',z'') \in A_{S,j}\}} \big(T - \lambda(z'')\big) = \prod_{1 \leqslant k \leqslant k_j} \big(T - \lambda \circ g_{j,k}(z')\big).$$

This defines a polynomial in T with bounded analytic coefficients on $\Delta' \smallsetminus S$. These coefficients have analytic extensions to Δ' (Remark 4.2), thus $P_{\lambda,j} \in \mathcal{O}(\Delta')[T]$. By construction, $P_{\lambda,j}\big(z'\,;\lambda(z'')\big)$ vanishes identically on $A_{S,j}$. Set

$$P_\lambda = \prod_{1 \leqslant j \leqslant N} P_{\lambda,j}, \qquad f(z) = \delta(z')\, P_\lambda\big(z'\,;\lambda(z'')\big)\ ;$$

f vanishes on $A_{S,1} \cup \ldots \cup A_{S,N} \cup (S \times \Delta'') \supset A \cap \Delta$. Lemma 4.21 shows that $\mathcal{I}_{A,0}$ is prime; as $\delta \notin \mathcal{I}_{A,0}$, we get $P_{\lambda,j}\big(z'\,;\lambda(z'')\big) \in \mathcal{I}_{A,0}$ for some j. This is a contradiction if $N \geqslant 2$ and if λ is chosen in such a way that λ separates the q points z''_ν in each fiber $\pi^{-1}(z'_\nu)$, for a sequence $z'_\nu \to 0$ in $\Delta' \smallsetminus S$. Hence $N = 1$, A_S is connected, and for every $\lambda \in (\mathbb{C}^{n-d})^\star$ we have $P_\lambda\big(z', \lambda(z'') \in \mathcal{I}_{(A,0)}$. By construction $P_\lambda\big(z', \lambda(z'')\big)$ vanishes on A_S, so it vanishes on $\overline{A}_S$; hence, for every $z' \in S$, the fiber $\overline{A}_S \cap \pi^{-1}(z')$ has at most q elements, otherwise selecting λ which separates $q+1$ of these points would yield $q+1$ roots $\lambda(z'')$ of $P_\lambda(z'\,;T)$, a contradiction. Assume now that A_S is not dense in $A \cap \Delta$ for arbitrarily small polydisks Δ. Then there exists a sequence $A \ni z_\nu = (z'_\nu, z''_\nu) \to 0$ such that $z'_\nu \in S$ and z''_ν is not in $F_\nu := \mathrm{pr}''\big(\overline{A}_S \cap \pi^{-1}(z'_\nu)\big)$. The continuity of the roots of the polynomial $P_\lambda(z'\,;T)$ as $\Delta' \smallsetminus S \ni z' \to z'_\nu$ implies that the set of roots of $P_\lambda(z'_\nu\,;T)$ is $\lambda(F_\nu)$. Select λ such that

$\lambda(z''_\nu) \notin \lambda(F_\nu)$ for all ν. Then $P_\lambda\big(z'_\nu\,;\lambda(z''_\nu)\big) \neq 0$ for every ν and $P_\lambda\big(z'\,;\lambda(z'')\big) \notin \mathcal{I}_{A,0}$, a contradiction. $\square$

At this point, it should be observed that many of the above statements completely fail in the case of real analytic sets. In $\mathbb{R}^2$, for example, the prime ideal $\mathcal{J} = (x^5 + y^4)$ defines an irreducible germ of curve $(A, 0)$ and there is an injective integral extension of rings $\mathbb{R}\{x\} \hookrightarrow \mathbb{R}\{x, y\}/\mathcal{J}$ of degree 4; however, the projection of $(A, 0)$ on the first factor, $(x, y) \mapsto x$, has not a constant sheet number near 0, and this number is not related to the degree of the extension. Also, the prime ideal $\mathcal{J} = (x^2+y^2)$ has an associated variety $V(\mathcal{J})$ reduced to $\{0\}$, hence $\mathcal{I}_{A,0} = (x, y)$ is strictly larger than $\mathcal{J}$, in contrast with Lemma 4.21.

Let us return to the complex situation, which is much better behaved. The result obtained in Lemma 4.21 can then be extended to non prime ideals and we get the following important result:

(4.22) Hilbert's Nullstellensatz. *For every ideal $\mathcal{J} \subset \mathcal{O}_n$*

$$\mathcal{I}_{V(\mathcal{J}),0} = \sqrt{\mathcal{J}},$$

where $\sqrt{\mathcal{J}}$ is the radical of $\mathcal{J}$, i.e. the set of germs $f \in \mathcal{O}_n$ such that some power f^k lies in $\mathcal{J}$.

Proof. Set $B = V(\mathcal{J})$. If $f^k \in \mathcal{J}$, then f^k vanishes on $(B, 0)$ and $f \in \mathcal{I}_{B,0}$. Thus $\sqrt{\mathcal{J}} \subset \mathcal{I}_{B,0}$. Conversely, it is well known that $\sqrt{\mathcal{J}}$ is the intersection of all prime ideals $\mathcal{P} \supset \mathcal{J}$ ([Lang 1965], Chapter VI). For such an ideal $(B, 0) = \big(V(\mathcal{J}), 0\big) \supset \big(V(\mathcal{P}), 0\big)$, thus $\mathcal{I}_{B,0} \subset \mathcal{I}_{V(\mathcal{P}),0} = \mathcal{P}$ in view of Lemma 4.21. Therefore $\mathcal{I}_{B,0} \subset \bigcap_{\mathcal{P}\supset\mathcal{J}} \mathcal{P} = \sqrt{\mathcal{J}}$ and the Theorem is proved. $\square$

In other words, if a germ $(B, 0)$ is defined by an arbitrary ideal $\mathcal{J} \subset \mathcal{O}_n$ and if $f \in \mathcal{O}_n$ vanishes on $(B, 0)$, then some power f^k lies in $\mathcal{J}$.

§ 4.C. Regular and Singular Points. Dimension.

The above powerful results enable us to investigate the structure of singularities of an analytic set. We first give a few definitions.

(4.23) Definition. *Let $A \subset M$ be an analytic set and $x \in A$. We say that $x \in A$ is a regular point of A if $A \cap \Omega$ is a $\mathbb{C}$-analytic submanifold of Ω for some neighborhood Ω of x. Otherwise x is said to be singular. The corresponding subsets of A will be denoted respectively A_{reg} and A_{sing}.*

It is clear from the definition that A_{reg} is an open subset of A (thus A_{sing} is closed), and that the connected components of A_{reg} are $\mathbb{C}$-analytic submanifolds of M (non necessarily closed).

(4.24) Proposition. *If (A, x) is irreducible, there exist arbitrarily small neighborhoods Ω of x such that $A_{\mathrm{reg}} \cap \Omega$ is dense and connected in $A \cap \Omega$.*

Proof. Take $\Omega = \Delta$ as in Th. 4.19. Then $A_S \subset A_{\mathrm{reg}} \cap \Omega \subset A \cap \Omega$, where A_S is connected and dense in $A \cap \Omega$; hence $A_{\mathrm{reg}} \cap \Omega$ has the same properties. $\square$

(4.25) Definition. *The dimension of an irreducible germ of analytic set (A,x) is defined by* $\dim(A,x)=\dim(A_{\mathrm{reg}},x)$. *If (A,x) has several irreducible components (A_l,x), we set*

$$\dim(A,x)=\max\{\dim(A_l,x)\},\qquad \operatorname{codim}(A,x)=n-\dim(A,x).$$

(4.26) Proposition. *Let $(B,x)\subset(A,x)$ be germs of analytic sets. If (A,x) is irreducible and $(B,x)\neq(A,x)$, then $\dim(B,x)<\dim(A,x)$ and $B\cap\Omega$ has empty interior in $A\cap\Omega$ for all sufficiently small neighborhoods Ω of x.*

Proof. We may assume $x=0$, $(A,0)\subset(\mathbb{C}^n,0)$ and $(B,0)$ irreducible. Then $\mathcal{I}_{A,0}\subset\mathcal{I}_{B,0}$ are prime ideals. When we choose suitable coordinates for the ramified coverings, we may at each step select vectors $e_n,e_{n-1},\dots$ that work simultaneously for A and B. If $\dim B=\dim A$, the process stops for both at the same time, i.e. we get ramified coverings

$$\pi: A\cap(\Delta'\times\Delta'')\longrightarrow\Delta',\qquad \pi: B\cap(\Delta'\times\Delta'')\longrightarrow\Delta'$$

with ramification loci S_A,S_B. Then $B\cap\big((\Delta'\smallsetminus(S_A\cup S_B))\times\Delta''\big)$ is an open subset of the manifold $A_S=A\cap\big((\Delta'\smallsetminus S_A)\times\Delta''\big)$, therefore $B\cap A_S$ is an analytic subset of A_S with non empty interior. The same conclusion would hold if $B\cap\Delta$ had non empty interior in $A\cap\Delta$. As A_S is connected, we get $B\cap A_S=A_S$, and as $B\cap\Delta$ is closed in Δ we infer $B\cap\Delta\supset\overline{A}_S=A\cap\Delta$, hence $(B,0)=(A,0)$, in contradiction with the hypothesis. □

(4.27) Example: parametrization of curves. Suppose that $(A,0)$ is an irreducible germ of curve ($\dim(A,0)=1$). If the disk $\Delta'\subset\mathbb{C}$ is chosen so small that $S=\{0\}$, then A_S is a connected covering of $\Delta'\smallsetminus\{0\}$ with q sheets. Hence, there exists a covering isomorphism between π and the standard covering

$$\mathbb{C}\supset\Delta(r)\smallsetminus\{0\}\longrightarrow\Delta(r^q)\smallsetminus\{0\},\qquad t\longmapsto t^q,\qquad r^q=\text{radius of }\ \Delta',$$

i.e. a map $\gamma:\Delta(r)\smallsetminus\{0\}\longrightarrow A_S$ such that $\pi\circ\gamma(t)=t^q$. This map extends into a bijective holomorphic map $\gamma:\Delta(r)\longrightarrow A\cap\Delta$ with $\gamma(0)=0$. This means that every irreducible germ of curve can be parametrized by a bijective holomorphic map defined on a disk in $\mathbb{C}$ (see also Exercise 11.8).

§ 4.D. Coherence of Ideal Sheaves

Let A be an analytic set in a complex manifold M. The *sheaf of ideals* $\mathcal{I}_A$ is the subsheaf of $\mathcal{O}_M$ consisting of germs of holomorphic functions on M which vanish on A. Its stalks are the ideals $\mathcal{I}_{A,x}$ already considered; note that $\mathcal{I}_{A,x}=\mathcal{O}_{M,x}$ if $x\notin A$. If $x\in A$, we let $\mathcal{O}_{A,x}$ be the ring of germs of functions on (A,x) which can be extended as germs of holomorphic functions on (M,x). By definition, there is a surjective morphism $\mathcal{O}_{M,x}\longrightarrow\mathcal{O}_{A,x}$ whose kernel is $\mathcal{I}_{A,x}$, thus

$$\mathcal{O}_{A,x}=\mathcal{O}_{M,x}/\mathcal{I}_{A,x},\qquad\forall x\in A,\tag{4.28}$$

i.e. $\mathcal{O}_A=(\mathcal{O}_M/\mathcal{I}_A)_{\restriction A}$. Since $\mathcal{I}_{A,x}=\mathcal{O}_{M,x}$ for $x\notin A$, the quotient sheaf $\mathcal{O}_M/\mathcal{I}_A$ is zero on $M\smallsetminus A$.

(4.29) Theorem ([Cartan 1950])**.** *For any analytic set $A\subset M$, the sheaf of ideals $\mathcal{I}_A$ is a coherent analytic sheaf.*

Proof. It is sufficient to prove the result when A is an analytic subset in a neighborhood of 0 in $\mathbb{C}^n$. If $(A,0)$ is not irreducible, there exists a neighborhood Ω such that $A\cap\Omega = A_1\cup\ldots\cup A_N$ where A_k are analytic sets in Ω and $(A_k,0)$ is irreducible. We have $\mathscr{I}_{A\cap\Omega} = \bigcap \mathscr{I}_{A_k}$, so by Cor. 3.15 we may assume that $(A,0)$ is irreducible. Then we can choose coordinates z', z'', polydisks Δ',Δ'' and a primitive element $u(z'') = c_{d+1}z_{d+1}+\cdots+c_nz_n$ such that Th. 4.19 is valid. Since $\delta(z') = \prod_{j<k}(\sigma_k\tilde{u}-\sigma_j\tilde{u})^2$, we see that $\delta(z')$ is in fact a polynomial in the c_j's with coefficients in $\mathscr{O}_d$. The same is true for the coefficients of the polynomials $W_u(z'\,;T)$ and $B_k(z'\,;T)$ which can be expressed in terms of the elementary symmetric functions of the $\sigma_k\tilde{u}$'s. We suppose that Δ' is chosen small enough in order that all coefficients of these $\mathscr{O}_d[c_{d+1},\ldots,c_n]$ polynomials are in $\mathscr{O}(\Delta')$. Let $\delta_\alpha\in\mathscr{O}(\Delta')$ be some non zero coefficient appearing in $\delta^m = \sum\delta_\alpha c^\alpha$. Also, let $G_1,\ldots,G_N\in\mathscr{O}(\Delta')[z'']$ be the coefficients of all monomials c^α appearing in the expansion of the functions $W_u(z'\,;u(z''))$ or $\delta(z')z_k - B_k(z'\,;u(z''))$. Clearly, $G_1,\ldots,G_N$ vanish on $A\cap\Delta$. We contend that

$$\mathscr{I}_{A,x} = \big\{f\in\mathscr{O}_{M,x}\ ;\ \delta_\alpha f\in(G_{1,x},\ldots,G_{N,x})\big\}. \tag{4.30}$$

This implies that the sheaf $\mathscr{I}_A$ is the projection on the first factor of the sheaf of relations $\mathscr{R}(\delta_\alpha,G_1,\ldots,G_N)\subset\mathscr{O}_\Delta^{N+1}$, which is coherent by the Oka theorem; Theorem 4.29 then follows.

We first prove that the inclusion $\mathscr{I}_{A,x}\supset\{\ldots\}$ holds in (4.30). In fact, if $\delta_\alpha f\in(G_{1,x},\ldots,G_{N,x})$, then f vanishes on $A\smallsetminus\{\delta_\alpha=0\}$ in some neighborhood of x. Since $(A\cap\Delta)\smallsetminus\{\delta_\alpha=0\}$ is dense in $A\cap\Delta$, we conclude that $f\in\mathscr{I}_{A,x}$.

To prove the other inclusion $\mathscr{I}_{A,x}\subset\{\ldots\}$, we repeat the proof of Lemma 4.18 with a few modifications. Let $x\in\Delta$ be a given point. At x, the irreducible polynomials $W_u(z'\,;T)$ and $W_k(z'\,;T)$ of $\tilde{u}$ and $\tilde{z}_k$ in $\mathscr{O}_{M,0}/\mathscr{I}_{A,0}$ split into

$$\begin{aligned} W_u(z'\,;T) &= W_{u,x}\big(z'\,;T-u(x'')\big)\,Q_{u,x}\big(z'\,;T-u(x'')\big),\\ W_k(z'\,;T) &= W_{k,x}(z'\,;T-x_k)\,Q_{k,x}(z'\,;T-x_k), \end{aligned}$$

where $W_{u,x}(z'\,;T)$ and $W_{k,x}(z'\,;T)$ are Weierstrass polynomials in T and $Q_{u,x}(x',0)\neq 0$, $Q_{k,x}(x',0)\neq 0$. For all $z'\in\Delta'$, the roots of $W_u(z'\,;T)$ are the values $u(z'')$ at all points $z\in A\cap\pi^{-1}(z')$. As A is closed, any point $z\in A\cap\pi^{-1}(z')$ with z' near x' has to be in a small neighborhood of one of the points $y\in A\cap\pi^{-1}(x')$. Choose $c_{d+1},\ldots,c_n$ such that the linear form $u(z'')$ separates all points in the fiber $A\cap\pi^{-1}(x')$. Then, for a root $u(z'')$ of $W_{u,x}\big(z'\,;T-u(x'')\big)$, the point z must be in a neighborhood of $y=x$, otherwise $u(z'')$ would be near $u(y'')\neq u(x'')$ and the Weierstrass polynomial $W_{u,x}(z'\,;T)$ would have a root away from 0, in contradiction with (4.10). Conversely, if $z\in A\cap\pi^{-1}(z')$ is near x, then $Q_{u,x}\big(z'\,;u(z'')-u(x'')\big)\neq 0$ and $u(z'')$ is a root of $W_{u,x}\big(z'\,;T-u(x'')\big)$. From this, we infer that every polynomial $P(z'\,;T)\in\mathscr{O}_{\Delta',x'}[T]$ such that $P\big(z'\,;u(z'')\big)=0$ on (A,x) is a multiple of $W_{u,x}\big(z'\,;T-u(x'')\big)$, because the roots of the latter polynomial are simple for z' in the dense set $(\Delta'\smallsetminus S,x)$. In particular $\deg P<\deg W_{u,x}$ implies $P=0$ and

$$\delta(z')^q W_{k,x}\big(z'\,;B_k(z'\,;u(z''))/\delta(z')-x_k\big)$$

is a multiple of $W_{u,x}\big(z'\,;T-u(x'')\big)$. If we replace W_u, W_k by $W_{u,x}$, $W_{k,x}$ respectively, the proof of Lemma 4.18 shows that for every $f\in\mathscr{O}_{M,x}$ there is a polynomial $R\in\mathscr{O}_{\Delta',x'}[T]$

of degree $\deg R < \deg W_{u,x}$ such that

$$\begin{aligned}&\delta(z')^m f(z) = R\big(z'\,;u(z'')\big) \quad \text{modulo the ideal}\\ &\big(\ W_{u,x}\big(z'\,;u(z'')-u(x'')\big),\ \delta(z')z_k - B_k\big(z'\,;u(z'')\big)\ \big),\end{aligned}$$

and $f \in \mathcal{I}_{A,x}$ implies $R = 0$. Since $W_{u,x}$ differs from W_u only by an invertible element in $\mathcal{O}_{M,x}$, we conclude that

$$\Big(\sum \delta_\alpha c^\alpha\Big)\mathcal{I}_{A,x} = \delta^m \mathcal{I}_{A,x} \subset (G_{1,x},\ldots,G_{N,x}).$$

This is true for a dense open set of coefficients $c_{d+1},\ldots,c_n$, therefore by expressing the coefficients δ_α through interpolation of $\sum \delta_\alpha c^\alpha$ at suitable points c we infer

$$\delta_\alpha \mathcal{I}_{A,x} \subset (G_{1,x},\ldots,G_{N,x}) \quad \text{for all } \alpha. \qquad \square$$

(4.31) Theorem. *A_{sing} is an analytic subset of A.*

Proof. The statement is local. Assume first that $(A,0)$ is an irreducible germ in $\mathbb{C}^n$. Let $g_1,\ldots,g_N$ be generators of the sheaf $\mathcal{I}_A$ on a neighborhood Ω of 0. Set $d = \dim A$. In a neighborhood of every point $x \in A_{\text{reg}} \cap \Omega$, A can be defined by holomorphic equations $u_1(z) = \ldots = u_{n-d}(z) = 0$ such that $du_1,\ldots,du_{n-d}$ are linearly independant. As $u_1,\ldots,u_{n-d}$ are generated by $g_1,\ldots,g_N$, one can extract a subfamily $g_{j_1},\ldots,g_{j_{n-d}}$ that has at least one non zero Jacobian determinant of rank $n-d$ at x. Therefore $A_{\text{sing}} \cap \Omega$ is defined by the equations

$$\det\Big(\frac{\partial g_j}{\partial z_k}\Big)_{\substack{j\in J\\ k\in K}} = 0, \qquad J \subset \{1,\ldots,N\},\ \ K \subset \{1,\ldots,n\},\ \ |J| = |K| = n-d.$$

Assume now that $(A,0) = \bigcup (A_l,0)$ with $(A_l,0)$ irreducible. The germ of an analytic set at a regular point is irreducible, thus every point which belongs simultaneously to at least two components is singular. Hence

$$(A_{\text{sing}},0) = \bigcup (A_{l,\text{sing}},0) \cup \bigcup_{k\neq l} (A_k \cap A_l,0),$$

and A_{sing} is analytic. $\square$

Now, we give a characterization of regular points in terms of a simple algebraic property of the ring $\mathcal{O}_{A,x}$.

(4.32) Proposition. *Let (A,x) be a germ of analytic set of dimension d and let $\mathfrak{m}_{A,x} \subset \mathcal{O}_{A,x}$ be the maximal ideal of functions that vanish at x. Then $\mathfrak{m}_{A,x}$ cannot have less than d generators and $\mathfrak{m}_{A,x}$ has d generators if and only if x is a regular point.*

Proof. If $A \subset \mathbb{C}^n$ is a d-dimensional submanifold in a neighborhood of x, there are local coordinates centered at x such that A is given by the equations $z_{d+1} = \ldots = z_n$ near $z = 0$. Then $\mathcal{O}_{A,x} \simeq \mathcal{O}_d$ and $\mathfrak{m}_{A,x}$ is generated by $z_1,\ldots,z_d$. Conversely, assume that $\mathfrak{m}_{A,x}$ has s generators $g_1(z),\ldots,g_s(z)$ in $\mathcal{O}_{A,x} = \mathcal{O}_{\mathbb{C}^n,x}/\mathcal{I}_{A,x}$. Letting $x = 0$ for simplicity, we can write

$$z_j = \sum_{1\leqslant k\leqslant s} u_{jk}(z)g_k(z) + f_j(z), \qquad u_{jk} \in \mathcal{O}_n,\ \ f_j \in \mathcal{I}_{A,0},\ \ 1 \leqslant j \leqslant n.$$

Then we find $dz_j = \sum c_{jk}(0)dg_k(0) + df_j(0)$, so that the rank of the system of differentials $\big(df_j(0)\big)_{1\leqslant j\leqslant n}$ is at least equal to $n-s$. Assume for example that $df_1(0),\ldots,df_{n-s}(0)$ are linearly independent. By the implicit function theorem, the equations $f_1(z) = \ldots = f_{n-s}(z) = 0$ define a germ of submanifold of dimension s containing $(A,0)$, thus $s \geqslant d$ and $(A,0)$ equals this submanifold if $s = d$. □

(4.33) Corollary. *Let $A \subset M$ be an analytic set of pure dimension d and let $B \subset A$ be an analytic subset of codimension $\geqslant p$ in A. Then, as an $\mathscr{O}_{A,x}$-module, the ideal $\mathscr{I}_{B,x}$ cannot be generated by less than p generators at any point $x \in B$, and by less than $p+1$ generators at any point $x \in B_{\mathrm{reg}} \cap A_{\mathrm{sing}}$.*

Proof. Suppose that $\mathscr{I}_{B,x}$ admits s-generators $(g_1,\ldots,g_s)$ at x. By coherence of $\mathscr{I}_B$ these germs also generate $\mathscr{I}_B$ in a neighborhood of x, so we may assume that x is a regular point of B. Then there are local coordinates $(z_1,\ldots,z_n)$ on M centered at x such that (B,x) is defined by $z_{k+1} = \ldots = z_n = 0$, where $k = \dim(B,x)$. Then the maximal ideal $\mathfrak{m}_{B,x} = \mathfrak{m}_{A,x}/\mathscr{I}_{B,x}$ is generated by $z_1,\ldots,z_k$, so that $\mathfrak{m}_{A,x}$ is generated by $(z_1,\ldots,z_k,g_1,\ldots,g_s)$. By Prop. 4.32, we get $k+s \geqslant d$, thus $s \geqslant d-k \geqslant p$, and we have strict inequalities when $x \in A_{\mathrm{sing}}$. □

§ 5. Complex Spaces

Much in the same way a manifold is constructed by piecing together open patches isomorphic to open sets in a vector space, a complex space is obtained by gluing together open patches isomorphic to analytic subsets. The general concept of analytic morphism (or holomorphic map between analytic sets) is first needed.

§ 5.A. Morphisms and Comorphisms

Let $A \subset \Omega \subset \mathbb{C}^n$ and $B \subset \Omega' \subset \mathbb{C}^p$ be analytic sets. A morphism from A to B is by definition a map $F : A \longrightarrow B$ such that for every $x \in A$ there is a neighborhood U of x and a holomorphic map $\tilde{F} : U \longrightarrow \mathbb{C}^p$ such that $\tilde{F}_{\restriction A\cap U} = F_{\restriction A\cap U}$. Equivalently, such a morphism can be defined as a continuous map $F : A \longrightarrow B$ such that for all $x \in A$ and $g \in \mathscr{O}_{B,F(x)}$ we have $g \circ F \in \mathscr{O}_{A,x}$. The induced ring morphism

$$F_x^\star \ : \ \mathscr{O}_{B,F(x)} \ni g \longmapsto g \circ F \in \mathscr{O}_{A,x} \tag{5.1}$$

is called the *comorphism* of F at point x.

§ 5.B. Definition of Complex Spaces

(5.2) Definition. *A complex space X is a locally compact Hausdorff space, countable at infinity, together with a sheaf $\mathscr{O}_X$ of continuous functions on X, such that there exists an open covering (U_λ) of X and for each λ a homeomorphism $F_\lambda : U_\lambda \longrightarrow A_\lambda$ onto an analytic set $A_\lambda \subset \Omega_\lambda \subset \mathbb{C}^{n_\lambda}$ such that the comorphism $F_\lambda^\star : \mathscr{O}_{A_\lambda} \longrightarrow \mathscr{O}_{X\restriction U_\lambda}$ is an isomorphism of sheaves of rings. $\mathscr{O}_X$ is called the structure sheaf of X.*

By definition a complex space X is locally isomorphic to an analytic set, so the concepts of holomorphic function on X, of analytic subset, of analytic morphism, etc ... are

meaningful. If X is a complex space, Th. 4.31 implies that X_{sing} is an analytic subset of X.

(5.3) Theorem and definition. *For every complex space X, the set X_{reg} is a dense open subset of X, and consists of a disjoint union of connected complex manifolds X'_α. Let X_α be the closure of X'_α in X. Then (X_α) is a locally finite family of analytic subsets of X, and $X = \bigcup X_\alpha$. The sets X_α are called the global irreducible components of X.*

Observe that the germ at a given point of a global irreducible component can be reducible, as shows the example of the cubic curve $\Gamma : y^2 = x^2(1+x)$; the germ $(\Gamma, 0)$ has two analytic branches $y = \pm x\sqrt{1+x}$, however $\Gamma \smallsetminus \{0\}$ is easily seen to be a connected smooth Riemann surface (the real points of γ corresponding to $-1 \leqslant x \leqslant 0$ form a path connecting the two branches). This example shows that the notion of *global irreducible component* is quite different from the notion of local irreducible component introduced in (4.4).

Proof. By definition of X_{reg}, the connected components X'_α are (disjoint) complex manifolds. Let us show that the germ of $X_\alpha = \overline{X}'_\alpha$ at any point $x \in X$ is analytic. We may assume that (X, x) is a germ of analytic set A in an open subset of $\mathbb{C}^n$. Let (A_l, x), $1 \leqslant l \leqslant N$, be the irreducible components of this germ and U a neighborhood of x such that $X \cap U = \bigcup A_l \cap U$. Let $\Omega_l \subset U$ be a neighborhood of x such that $A_{l,\text{reg}} \cap \Omega_l$ is connected and dense in $A_l \cap \Omega_l$ (Prop. 4.24). Then $A'_l := X_{\text{reg}} \cap A_l \cap \Omega_l$ equals $(A_{l,\text{reg}} \cap \Omega_l) \smallsetminus \bigcup_{k \neq l} A_{l,\text{reg}} \cap \Omega_l \cap A_k$. However, $A_{l,\text{reg}} \cap \Omega_l \cap A_k$ is an analytic subset of $A_{l,\text{reg}} \cap \Omega_l$, distinct from $A_{l,\text{reg}} \cap \Omega_l$, otherwise $A_{l,\text{reg}} \cap \Omega_l$ would be contained in A_k, thus $(A_l, x) \subset (A_k, x)$ by density. Remark 4.2 implies that A'_l is connected and dense in $A_{l,\text{reg}} \cap \Omega_l$, hence in $A_l \cap \Omega_l$. Set $\Omega = \bigcap \Omega_l$ and let $(X_\alpha)_{\alpha \in J}$ be the family of global components which meet Ω (i.e. such that $X'_\alpha \cap \Omega \neq \emptyset$). As $X_{\text{reg}} \cap \Omega = \bigcup A'_l \cap \Omega$, each X'_α, $\alpha \in J$, meets at least one set A'_l, and as $A'_l \subset X_{\text{reg}}$ is connected, we have in fact $A'_l \subset X'_\alpha$. It follows that there exists a partition $(L_\alpha)_{\alpha \in J}$ of $\{1, \dots, N\}$ such that $X'_\alpha \cap \Omega = \bigcup_{l \in L_\alpha} A'_l \cap \Omega$, $\alpha \in J$. Hence J is finite, $\operatorname{card} J \leqslant N$, and

$$X_\alpha \cap \Omega = \overline{X}'_\alpha \cap \Omega = \bigcup_{l \in L_\alpha} \overline{A}'_l \cap \Omega = \bigcup_{l \in L_\alpha} A_l \cap \Omega$$

is analytic for all $\alpha \in J$. □

(5.4) Corollary. *If A, B are analytic subsets in a complex space X, then the closure $\overline{A \smallsetminus B}$ is an analytic subset, consisting of the union of all global irreducible components A_λ of A which are not contained in B.*

Proof. Let $C = \bigcup A_\lambda$ be the union of these components. Since (A_λ) is locally finite, C is analytic. Clearly $A \smallsetminus B = C \smallsetminus B = \bigcup A_\lambda \smallsetminus B$. The regular part A'_λ of each A_λ is a connected manifold and $A'_\lambda \cap B$ is a proper analytic subset (otherwise $A'_\lambda \subset B$ would imply $A_\lambda \subset B$). Thus $A'_\lambda \smallsetminus (A'_\lambda \cap B)$ is dense in A'_λ which is dense in A_λ, so $\overline{A \smallsetminus B} = \bigcup A_\lambda = C$. □

(5.5) Theorem. *For any family (A_λ) of analytic sets in a complex space X, the intersection $A = \bigcap A_\lambda$ is an analytic subset of X. Moreover, the intersection is stationary on every compact subset of X.*

Proof. It is sufficient to prove the last statement, namely that every point $x \in X$ has a neighborhood Ω such that $A \cap \Omega$ is already obtained as a finite intersection. However, since $\mathscr{O}_{X,x}$ is Noetherian, the family of germs of finite intersections has a minimum element (B, x), $B = \bigcap A_{\lambda_j}$, $1 \leqslant j \leqslant N$. Let $\tilde{B}$ be the union of the global irreducible components B_α of B which contain the point x ; clearly $(B, x) = (\tilde{B}, x)$. For any set A_λ in the family, the minimality of B implies $(B, x) \subset (A_\lambda, x)$. Let B'_α be the regular part of any global irreducible component B_α of $\tilde{B}$. Then $B'_\alpha \cap A_\lambda$ is a closed analytic subset of B'_α containing a non empty open subset (the intersection of B'_α with some neighborhood of x), so we must have $B'_\alpha \cap A_\lambda = B'_\alpha$. Hence $B_\alpha = \overline{B}'_\alpha \subset A_\lambda$ for all $B_\alpha \subset \tilde{B}$ and all A_λ, thus $\tilde{B} \subset A = \bigcap A_\lambda$. We infer

$$(B, x) = (\tilde{B}, x) \subset (A, x) \subset (B, x),$$

and the proof is complete. □

As a consequence of these general results, it is not difficult to show that a complex space always admits a (locally finite) stratification such that the strata are smooth manifolds.

(5.6) Proposition. *Let X be a complex space. Then there is a locally stationary increasing sequence of analytic subsets $Y_k \subset X$, $k \in \mathbb{N}$, such that Y_0 is a discrete set and such that $Y_k \smallsetminus Y_{k-1}$ is a smooth k-dimensional complex manifold for $k \geqslant 1$. Such a sequence is called a stratification of X, and the sets $Y_k \smallsetminus Y_{k-1}$ are called the strata (the strata may of course be empty for some indices $k < \dim X$).*

Proof. Let $\mathscr{F}$ be the family of irreducible analytic subsets $Z \subset X$ which can be obtained through a finite sequence of steps of the following types:

a) Z is an irreducible component of X ;

b) Z is an irreducible component of the singular set Z'_{sing} of some member $Z' \in \mathscr{F}$;

c) Z is an irreducible component of some finite intersection of sets $Z_j \in \mathscr{F}$.

Since X has locally finite dimension and since steps b) or c) decrease the dimension of our sets Z, it is clear that $\mathscr{F}$ is a locally finite family of analytic sets in X. Let Y_k be the union of all sets $Z \in \mathscr{F}$ of dimension $\leqslant k$. It is easily seen that $\bigcup Y_k = X$ and that the irreducible components of $(Y_k)_{\text{sing}}$ are contained in Y_{k-1} (these components are either intersections of components $Z_j \subset Y_k$ or parts of the singular set of some component $Z \subset Y_k$, so there are in either case obtained by step b) or c) above). Hence $Y_k \smallsetminus Y_{k-1}$ is a smooth manifold and it is of course k-dimensional, because the components of Y_k of dimension $< k$ are also contained in Y_{k-1} by definition.

(5.7) Theorem. *Let X be an irreducible complex space. Then every non constant holomorphic function f on X defines an open map $f : X \longrightarrow \mathbb{C}$.*

Proof. We show that the image $f(\Omega)$ of any neighborhood Ω of $x \in X$ contains a neighborhood of $f(x)$. Let (X_l, x) be an irreducible component of the germ (X, x) (embedded in $\mathbb{C}^n$) and $\Delta = \Delta' \times \Delta'' \subset \Omega$ a polydisk such that the projection $\pi : X_l \cap \Delta \longrightarrow \Delta'$ is a ramified covering. The function f is non constant on the dense open manifold X_{reg}, so we may select a complex line $L \subset \Delta'$ through 0, not contained in the ramification locus of π,

such that f is non constant on the one dimensional germ $\pi^{-1}(L)$. Therefore we can find a germ of curve

$$(\mathbb{C}, 0) \ni t \longmapsto \gamma(t) \in (X, x)$$

such that $f \circ \gamma$ is non constant. This implies that the image of every neighborhood of $0 \in \mathbb{C}$ by $f \circ \gamma$ already contains a neighborhood of $f(x)$. □

(5.8) Corollary. *If X is a compact irreducible analytic space, then every holomorphic function $f \in \mathcal{O}(X)$ is constant.*

In fact, if $f \in \mathcal{O}(X)$ was non constant, $f(X)$ would be compact and also open in $\mathbb{C}$ by Th. 5.7, a contradiction. This result implies immediately the following consequence.

(5.9) Theorem. *Let X be a complex space such that the global holomorphic functions in $\mathcal{O}(X)$ separate the points of X. Then every compact analytic subset A of X is finite.*

Proof. A has a finite number of irreducible components A_λ which are also compact. Every function $f \in \mathcal{O}(X)$ is constant on A_λ, so A_λ must be reduced to a single point. □

§ 5.C. Coherent Sheaves over Complex Spaces

Let X be a complex space and $\mathcal{O}_X$ its structure sheaf. Locally, X can be identified to an analytic set A in an open set $\Omega \subset \mathbb{C}^n$, and we have $\mathcal{O}_X = \mathcal{O}_\Omega / \mathcal{I}_A$. Thus $\mathcal{O}_X$ is coherent over the sheaf of rings $\mathcal{O}_\Omega$. It follows immediately that $\mathcal{O}_X$ is coherent over itself. Let $\mathcal{S}$ be a $\mathcal{O}_X$-module. If $\tilde{\mathcal{S}}$ denotes the extension of $\mathcal{S}_{\restriction A}$ to Ω obtained by setting $\tilde{\mathcal{S}}_x = 0$ for $x \in \Omega \smallsetminus A$, then $\tilde{\mathcal{S}}$ is a $\mathcal{O}_\Omega$-module, and it is easily seen that $\mathcal{S}_{\restriction A}$ is coherent over $\mathcal{O}_{X \restriction A}$ if and only if $\tilde{\mathcal{S}}$ is coherent over $\mathcal{O}_\Omega$. If Y is an analytic subset of X, then Y is locally given by an analytic subset B of A and the sheaf of ideals of Y in $\mathcal{O}_X$ is the quotient $\mathcal{I}_Y = \mathcal{I}_B / \mathcal{I}_A$; hence $\mathcal{I}_Y$ is coherent. Let us mention the following important property of supports.

(5.10) Theorem. *If $\mathcal{S}$ is a coherent $\mathcal{O}_X$-module, the support of $\mathcal{S}$, defined as* $\operatorname{Supp} \mathcal{S} = \{x \in X \,;\, \mathcal{S}_x \neq 0\}$ *is an analytic subset of X.*

Proof. The result is local, thus after extending $\mathcal{S}$ by 0, we may as well assume that X is an open subset $\Omega \subset \mathbb{C}^n$. By (3.12), there is an exact sequence of sheaves

$$\mathcal{O}_U^{\oplus p} \xrightarrow{G} \mathcal{O}_U^{\oplus q} \xrightarrow{F} \mathcal{S}_{\restriction U} \longrightarrow 0$$

in a neighborhood U of any point. If $G : \mathcal{O}_x^{\oplus p} \longrightarrow \mathcal{O}_x^{\oplus q}$ is surjective it is clear that the linear map $G(x) : \mathbb{C}^p \longrightarrow \mathbb{C}^q$ must be surjective; conversely, if $G(x)$ is surjective, there is a q-dimensional subspace $E \subset \mathbb{C}^p$ on which the restriction of $G(x)$ is a bijection onto $\mathbb{C}^q$; then $G_{\restriction E} : \mathcal{O}_U \otimes_{\mathbb{C}} E \longrightarrow \mathcal{O}_U^{\oplus q}$ is bijective near x and G is surjective. The support of $\mathcal{S}_{\restriction U}$ is thus equal to the set of points $x \in U$ such that all minors of $G(x)$ of order q vanish. □

§ 6. Analytic Cycles and Meromorphic Functions

§ 6.A. Complete Intersections

Our goal is to study in more details the dimension of a subspace given by a set of equations. The following proposition is our starting point.

(6.1) Proposition. *Let X be a complex space of pure dimension p and A an analytic subset of X with* $\operatorname{codim}_X A \geqslant 2$. *Then every function $f \in \mathscr{O}(X \smallsetminus A)$ is locally bounded near A.*

Proof. The statement is local on X, so we may assume that X is an irreducible germ of analytic set in $(\mathbb{C}^n, 0)$. Let $(A_k, 0)$ be the irreducible components of $(A, 0)$. By a reasoning analogous to that of Prop. 4.26, we can choose coordinates $(z_1, \ldots, z_n)$ on $\mathbb{C}^n$ such that all projections

$$\begin{aligned} \pi &: z \longmapsto (z_1, \ldots, z_p), \quad p = \dim X, \\ \pi_k &: z \longmapsto (z_1, \ldots, z_{p_k}), \quad p_k = \dim A_k, \end{aligned}$$

define ramified coverings $\pi : X \cap \Delta \longrightarrow \Delta'$, $\pi_k : A_k \cap \Delta \longrightarrow \Delta'_k$. By construction $\pi(A_k) \subset \Delta'$ is contained in the set B_k defined by some Weierstrass polynomials in the variables $z_{p_k+1}, \ldots, z_p$ and $\operatorname{codim}_{\Delta'} B_k = p - p_k \geqslant 2$. Let S be the ramification locus of π and $B = \bigcup B_k$. We have $\pi(A \cap \Delta) \subset B$. For $z' \in \Delta' \smallsetminus (S \cup B)$, we let

$$\sigma_k(z') = \text{elementary symmetric function of degree } k \text{ in } f(z', z''_\alpha),$$

where (z', z''_α) are the q points of X projecting on z'. Then σ_k is holomorphic on $\Delta' \smallsetminus (S \cup B)$ and locally bounded near every point of $S \smallsetminus B$, thus σ_k extends holomorphically to $\Delta' \smallsetminus B$ by Remark 4.2. Since $\operatorname{codim} B \geqslant 2$, σ_k extends to Δ' by Cor. I.5.25. Now, f satisfies $f^q - \sigma_1 f^{q-1} + \ldots + (-1)^q \sigma_q = 0$, thus f is locally bounded on $X \cap \Delta$. □

(6.2) Theorem. *Let X be an irreducible complex space and $f \in \mathscr{O}(X)$, $f \not\equiv 0$. Then $f^{-1}(0)$ is empty or of pure dimension* $\dim X - 1$.

Proof. Let $A = f^{-1}(0)$. By Prop. 4.26, we know that $\dim A \leqslant \dim X - 1$. If A had an irreducible branch A_j of dimension $\leqslant \dim X - 2$, then in virtue of Prop. 6.1 the function $1/f$ would be bounded in a neighborhood of $A_j \smallsetminus \bigcup_{k \neq j} A_k$, a contradiction. □

(6.3) Corollary. *If $f_1, \ldots, f_p$ are holomorphic functions on an irreducible complex space X, then all irreducible components of $f_1^{-1}(0) \cap \ldots \cap f_p^{-1}(0)$ have codimension $\geqslant p$.* □

(6.4) Definition. *Let X be a complex space of pure dimension n and A an analytic subset of X of pure dimension. Then A is said to be a local (set theoretic) complete intersection in X if every point of A has a neighborhood Ω such that*

$$A \cap \Omega = \{x \in \Omega\,;\, f_1(x) = \ldots = f_p(x) = 0\}$$

with exactly $p = \operatorname{codim} A$ functions $f_j \in \mathscr{O}(\Omega)$.

(6.5) Remark. As a converse to Th. 6.2, one may ask whether every hypersurface A in X is locally defined by a single equation $f = 0$. In general the answer is negative. A simple counterexample for $\dim X = 3$ is obtained with the singular quadric $X = \{z_1 z_2 + z_3 z_4 = 0\} \subset \mathbb{C}^4$ and the plane $A = \{z_1 = z_3 = 0\} \subset X$. Then A cannot be defined by a single equation $f = 0$ near the origin, otherwise the plane $B = \{z_2 = z_4 = 0\}$ would be such that

$$f^{-1}(0) \cap B = A \cap B = \{0\},$$

in contradiction with Th. 6.2 (also, by Exercise 11.11, we would get the inequality $\operatorname{codim}_X A \cap B \leqslant 2$). However, the answer is positive when X is a manifold:

(6.6) Theorem. *Let M be a complex manifold with $\dim_{\mathbb{C}} M = n$, let (A, x) be an analytic germ of pure dimension $n-1$ and let A_j, $1 \leqslant j \leqslant N$, be its irreducible components.*

a) *The ideal of (A, x) is a principal ideal $\mathcal{I}_{A,x} = (g)$ where g is a product of irreducible germs g_j such that $\mathcal{I}_{A_j,x} = (g_j)$.*

b) *For every $f \in \mathcal{O}_{M,x}$ such that $f^{-1}(0) \subset (A, x)$, there is a unique decomposition $f = u g_1^{m_1} \dots g_N^{m_N}$ where u is an invertible germ and m_j is the order of vanishing of f at any point $z \in A_{j,\mathrm{reg}} \smallsetminus \bigcup_{k \neq j} A_k$.*

Proof. a) In a suitable local coordinate system centered at x, the projection $\pi : \mathbb{C}^n \longrightarrow \mathbb{C}^{n-1}$ realizes all A_j as ramified coverings

$$\pi : A_j \cap \Delta \longrightarrow \Delta' \subset \mathbb{C}^{n-1}, \qquad \text{ramification locus} = S_j \subset \Delta'.$$

The function

$$g_j(z', z_n) = \prod_{w \in A_j \cap \pi^{-1}(z')} (z_n - w_n), \qquad z' \in \Delta' \smallsetminus S_j$$

extends into a holomorphic function in $\mathcal{O}_{\Delta'}[z_n]$ and is irreducible at x. Set $g = \prod g_j \in \mathcal{I}_{A,x}$. For any $f \in \mathcal{I}_{A,x}$, the Weierstrass division theorem yields $f = gQ + R$ with $R \in \mathcal{O}_{n-1}[z_n]$ and $\deg R < \deg g$. As $R(z', z_n)$ vanishes when z_n is equal to w_n for each point $w \in A \cap \pi^{-1}(z')$, R has exactly $\deg g$ roots when $z' \in \Delta' \smallsetminus \big(\bigcup S_j \cup \bigcup \pi(A_j \cap A_k)\big)$, so $R = 0$. Hence $\mathcal{I}_{A,x} = (g)$ and similarly $\mathcal{I}_{A_j,x} = (g_j)$. Since $\mathcal{I}_{A_j}$ is coherent, g_j is also a generator of $\mathcal{I}_{A_j,z}$ for z near x and we infer that g_j has order 1 at any regular point $z \in A_{j,\mathrm{reg}}$.

b) As $\mathcal{O}_{M,x}$ is factorial, any $f \in \mathcal{O}_{M,x}$ can be written $f = u\, g_1^{m_1} \dots g_N^{m_N}$ where u is either invertible or a product of irreducible elements distinct from the g_j's. In the latter case the hypersurface $u^{-1}(0)$ cannot be contained in (A, x), otherwise it would be a union of some of the components A_j and u would be divisible by some g_j. This proves b). $\square$

(6.7) Definition. *Let X be an complex space of pure dimension n.*

a) *An analytic q-cycle Z on X is a formal linear combination $\sum \lambda_j A_j$ where (A_j) is a locally finite family of irreducible analytic sets of dimension q in X and $\lambda_j \in \mathbb{Z}$. The support of Z is $|Z| = \bigcup_{\lambda_j \neq 0} A_j$. The group of all q-cycles on X is denoted $\mathrm{Cycl}^q(X)$. Effective q-cycles are elements of the subset $\mathrm{Cycl}^q_+(X)$ of cycles such that all coefficients λ_j are $\geqslant 0$; rational, real cycles are cycles with coefficients $\lambda_j \in \mathbb{Q}$, $\mathbb{R}$.*

b) *An analytic $(n-1)$-cycle is called a (Weil) divisor, and we set*

$$\mathrm{Div}(X) = \mathrm{Cycl}^{n-1}(X).$$

c) *Assume that $\dim X_{\mathrm{sing}} \leqslant n-2$. If $f \in \mathcal{O}(X)$ does not vanish identically on any irreducible component of X, we associate to f a divisor*

$$\mathrm{div}(f) = \sum m_j A_j \in \mathrm{Div}_+(X)$$

in the following way: the components A_j are the irreducible components of $f^{-1}(0)$ and the coefficient m_j is the vanishing order of f at every regular point in $X_{\mathrm{reg}} \cap A_{j,\mathrm{reg}} \smallsetminus \bigcup_{k\neq j} A_k$. It is clear that we have

$$\mathrm{div}(fg) = \mathrm{div}(f) + \mathrm{div}(g).$$

d) *A Cartier divisor is a divisor $D = \sum \lambda_j A_j$ that is equal locally to a $\mathbb{Z}$-linear combination of divisors of the form* $\mathrm{div}(f)$.

It is easy to check that the collection of abelian groups $\mathrm{Cycl}^q(U)$ over all open sets $U \subset X$, together with the obvious restriction morphisms, satisfies axioms (1.4) of sheaves; observe however that the restriction of an irreducible component A_j to a smaller open set may subdivide in several components. Hence we obtain sheaves of abelian groups Cycl^q and $\mathrm{Div} = \mathrm{Cycl}^{n-1}$ on X. The stalk Cycl^q_x is the free abelian group generated by the set of irreducible germs of q-dimensional analytic sets at the point x. These sheaves carry a natural partial ordering determined by the subsheaf of positive elements Cycl^q_+. We define the sup and inf of two analytic cycles $Z = \sum \lambda_j A_j$, $Z' = \sum \mu_j A_j$ by

$$\sup\{Z, Z'\} = \sum \sup\{\lambda_j, \mu_j\}\, A_j, \quad \inf\{Z, Z'\} = \sum \inf\{\lambda_j, \mu_j\}\, A_j\,; \tag{6.8}$$

it is clear that these operations are compatible with restrictions, i.e. they are defined as sheaf operations.

(6.9) Remark. When X is a manifold, Th. 6.6 shows that every effective $\mathbb{Z}$-divisor is locally the divisor of a holomorphic function; thus, for manifolds, the concepts of Weil and Cartier divisors coincide. This is not always the case in general: in Example 6.5, one can show that A is not a Cartier divisor.

§ 6.B. Divisors and Meromorphic Functions

Let X be a complex space. For $x \in X$, let $\mathscr{M}_{X,x}$ be the ring of quotients of $\mathscr{O}_{X,x}$, i.e. the set of formal quotients g/h, $g, h \in \mathscr{O}_{X,x}$, where h is not a zero divisor in $\mathscr{O}_{X,x}$, with the identification $g/h = g'/h'$ if $gh' = g'h$. We consider the disjoint union

$$\mathscr{M}_X = \coprod_{x\in X} \mathscr{M}_{X,x} \tag{6.10}$$

with the topology in which the open sets open sets are unions of sets of the type $\{G_x/H_x\,;\, x \in V\} \subset \mathscr{M}_X$ where V is open in X and $G, H \in \mathscr{O}_X(V)$. Then $\mathscr{M}_X$ is a sheaf over X, and the sections of $\mathscr{M}_X$ over an open set U are called *meromorphic functions* on U. By definition, these sections can be represented locally as quotients of holomorphic functions, but there need not exist such a global representation on U.

A point $x \in X$ is called a *pole* of a meromorphic function f on X if $f_x \notin \mathscr{O}_{X,x}$. Clearly, the set P_f of poles of f is a closed subset of X with empty interior: if $f = g/h$ on U, then $h \not\equiv 0$ on any irreducible component and $P_f \cap U \subset h^{-1}(0)$. For $x \notin P_f$, one can speak of *the value* $f(x)$. If the restriction of f to $X_{\mathrm{reg}} \smallsetminus P_f$ does not vanish identically on any irreducible component of (X, x), then $1/f$ is a meromorphic function

in a neighborhood of x ; the set of poles of $1/f$ will be denoted Z_f and called the *zero set* of f. If f vanishes on some connected open subset of $X_{\text{reg}} \smallsetminus P_f$, then f vanishes identically (outside P_f) on the global irreducible component X_α containing this set; we agree that these components X_α are contained in Z_f. For every point x in the complement of $Z_f \cap P_f$, we have either $f_x \in \mathscr{O}_{X,x}$ or $(1/f)_x \in \mathscr{O}_{X,x}$, thus f defines a holomorphic map $X \smallsetminus (Z_f \cap P_f) \longrightarrow \mathbb{C} \cup \{\infty\} = \mathbb{P}^1$ with values in the projective line. In general, no value (finite or infinite) can be assigned to f at a point $x \in Z_f \cap P_f$, as shows the example of the function $f(z) = z_2/z_1$ in $\mathbb{C}^2$. The set $Z_f \cap P_f$ is called the *indeterminacy set* of f.

(6.11) Theorem. *For every meromorphic function f on X, the sets P_f, Z_f and the indeterminacy set $Z_f \cap P_f$ are analytic subsets.*

Proof. Let $\mathscr{J}_x$ be the ideal of germs $u \in \mathscr{O}_{X,x}$ such that $uf_x \in \mathscr{O}_{X,x}$. Let us write $f = g/h$ on a small open set U. Then $\mathscr{J}_{\restriction U}$ appears as the projection on the first factor of the sheaf of relations $\mathscr{R}(g,h) \subset \mathscr{O}_U \times \mathscr{O}_U$, so $\mathscr{J}$ is a coherent sheaf of ideals. Now

$$P_f = \big\{x \in X\,;\, \mathscr{J}_x = \mathscr{O}_{X,x}\big\} = \operatorname{Supp} \mathscr{O}_X/\mathscr{J},$$

thus P_f is analytic by Th. 5.10. Similarly, the projection of $\mathscr{R}(g,h)$ on the second factor defines a sheaf of ideals $\mathscr{J}'$ such that $Z_f = \operatorname{Supp} \mathscr{O}_X/\mathscr{J}'$. □

When X has pure dimension n and $\dim X_{\text{sing}} \leqslant n-2$, Def. 6.7 c) can be extended to meromorphic functions: if $f = g/h$ locally, we set

$$\operatorname{div}(f) = \operatorname{div}(g) - \operatorname{div}(h). \tag{6.12}$$

By 6.7 c), we immediately see that this definition does not depend on the choice of the local representant g/h. Furthermore, *Cartier divisors* are precisely those divisors which are associated locally to meromorphic functions.

Assume from now on that M is a connected n-dimensional complex manifold. Then, for every point $x \in M$, the ring $\mathscr{O}_{M,x} \simeq \mathscr{O}_n$ is factorial. This property makes the study of meromorphic functions much easier.

(6.13) Theorem. *Let f be a non zero meromorphic function on a manifold M, with $\dim_{\mathbb{C}} M = n$. Then the sets Z_f, P_f are purely $(n-1)$-dimensional, and the indeterminacy set $Z_f \cap P_f$ is purely $(n-2)$-dimensional.*

Proof. For every point $a \in M$, the germ f_a can be written g_a/h_a where $g_a, h_a \in \mathscr{O}_{M,a}$ are relatively prime holomorphic germs. By Th. 2.11, the germs g_x, h_x are still relatively prime for x in a neighborhood U of a. Thus the ideal $\mathscr{J}$ associated to f coincides with (h) on U, and we have

$$P_f \cap U = \operatorname{Supp} \mathscr{O}_U/(h) = h^{-1}(0), \qquad Z_f \cap U = g^{-1}(0).$$

Th. 6.2 implies our contentions: if g_λ and h_μ are the irreducible components of g, h, then $Z_f \cap P_f = \bigcup g_\lambda^{-1}(0) \cap h_\mu^{-1}(0)$ is $(n-2)$-dimensional. As we will see in the next section, Th. 6.13 does not hold on an arbitrary complex space. □

Let (A_j), resp. (B_j), be the global irreducible components of Z_f, resp. P_f. In a neighborhood V_j of the $(n-1)$-dimensional analytic set

$$A'_j = A_j \smallsetminus \big(P_f \cup \bigcup_{k \neq j} A_k\big)$$

f is holomorphic and $V \cap f^{-1}(0) = A'_j$. As $A'_{j,\mathrm{reg}}$ is connected, we must have $\mathrm{div}(f_{\restriction V_j}) = m_j A'_j$ for some constant multiplicity m_j equal to the vanishing order of f along $A'_{j,\mathrm{reg}}$. Similarly, $1/f$ is holomorphic in a neighborhood W_j of

$$B'_j = B_j \smallsetminus \big(Z_f \cup \bigcup_{k \neq j} B_k\big)$$

and we have $\mathrm{div}(f_{\restriction V}) = -p_j B'_j$ where p_j is the vanishing order of $1/f$ along $B'_{j,\mathrm{reg}}$. At a point $x \in M$ the germs $A_{j,x}$ and $B_{j,x}$ may subdivide in irreducible local components $A_{j,\lambda,x}$ and $B_{j,\lambda,x}$. If $g_{j,\lambda}$ and $h_{j,\lambda}$ are local generators of the corresponding ideals, we may a priori write

$$f_x = u\, g/h \quad \text{where} \quad g = \prod g_{j,\lambda}^{m_{j,\lambda}}, \quad h = \prod h_{j,\lambda}^{p_{j,\lambda}}$$

and where u is invertible. Then necessarily $m_{j,\lambda} = m_j$ and $p_{j,\lambda} = p_j$ for all λ, and we see that the global divisor of f on M is

$$\mathrm{div}(f) = \sum m_j A_j - \sum p_j B_j. \tag{6.14}$$

Let us denote by $\mathscr{M}^\star$ the multiplicative sheaf of germs of non zero meromorphic functions, and by $\mathscr{O}^\star$ the sheaf of germs of invertible holomorphic functions. Then we have an exact sequence of sheaves

$$1 \longrightarrow \mathscr{O}^\star \longrightarrow \mathscr{M}^\star \xrightarrow{\mathrm{div}} \mathrm{Div} \longrightarrow 0. \tag{6.15}$$

Indeed, the surjectivity of div is a consequence of Th. 6.6. Moreover, any meromorphic function that has a positive divisor must be holomorphic by the fact that $\mathscr{O}_n$ is factorial. Hence a meromorphic function f with $\mathrm{div}(f) = 0$ is an invertible holomorphic function.

§ 7. Normal Spaces and Normalization

§ 7.A. Weakly Holomorphic Functions

The goal of this section is to show that the singularities of X can be studied by enlarging the structure sheaf $\mathscr{O}_X$ into a sheaf $\tilde{\mathscr{O}}_X$ of so-called weakly holomorphic functions.

(7.1) Definition. *Let X be a complex space. A weakly holomorphic function f on X is a holomorphic function on X_{reg} such that every point of X_{sing} has a neighborhood V for which f is bounded on $X_{\mathrm{reg}} \cap V$. We denote by $\tilde{\mathscr{O}}_{X,x}$ the ring of germs of weakly holomorphic functions over neighborhoods of x and $\tilde{\mathscr{O}}_X$ the associated sheaf.*

Clearly, $\tilde{\mathscr{O}}_{X,x}$ is a ring containing $\mathscr{O}_{X,x}$. If (X_j, x) are the irreducible components of (X, x), there is a fundamental system of neighborhoods V of x such that $X_{\mathrm{reg}} \cap V$ is a disjoint union of connected open sets

$$X_{j,\mathrm{reg}} \cap V \smallsetminus \bigcup_{k \neq j} X_k \cap X_{j,\mathrm{reg}} \cap V$$

which are dense in $X_{j,\mathrm{reg}} \cap V$. Therefore any bounded holomorphic function on $X_{\mathrm{reg}} \cap V$ extends to each component $X_{j,\mathrm{reg}} \cap V$ and we see that

$$\tilde{\mathscr{O}}_{X,x} = \bigoplus \tilde{\mathscr{O}}_{X_j,x}.$$

The first important fact is that weakly holomorphic functions are always meromorphic and possess "universal denominators".

(7.2) Theorem. *For every point $x \in X$, there is a neighborhood V of x and $h \in \mathscr{O}_X(V)$ such that $h^{-1}(0)$ is nowhere dense in V and $h_y\tilde{\mathscr{O}}_{X,y} \subset \mathscr{O}_{X,y}$ for all $y \in V$; such a function h is called a universal denominator on V. In particular $\tilde{\mathscr{O}}_X$ is contained in the ring $\mathscr{M}_X$ of meromorphic functions.*

Proof. First assume that (X, x) is irreducible and that we have a ramified covering $\pi : X \cap \Delta \longrightarrow \Delta'$ with ramification locus S. We claim that the discriminant $\delta(z')$ of a primitive element $u(z'') = c_{d+1}z_{d+1} + \cdots + c_n z_n$ is a universal denominator on $X \cap \Delta$. To see this, we imitate the proof of Lemma 4.15. Let $f \in \tilde{\mathscr{O}}_{X,y}$, $y \in X \cap \Delta$. Then we solve the equation

$$f(z) = \sum_{0 \leqslant j \leqslant q} b_j(z')u(z'')^j$$

in a neighborhood of y. For $z' \in \Delta' \smallsetminus S$, let us denote by (z', z''_α), $1 \leqslant \alpha \leqslant q$, the points in the fiber $X \cap \pi^{-1}(z')$. Among these, only q' are close to y, where q' is the sum of the sheet numbers of the irreducible components of (X, y) by the projection π. The other points (z', z''_α), say $q' < \alpha \leqslant q$, are in neighborhoods of the points of $\pi^{-1}(y') \smallsetminus \{y\}$. We take $\big(b_j(z')\big)$ to be the solution of the linear system

$$\sum_{0 \leqslant j \leqslant q} b_j(z')u(z''_\alpha)^j = \begin{cases} f(z', z''_\alpha) & \text{for } \ 1 \leqslant \alpha \leqslant q', \\ 0 & \text{for } \ q' < \alpha \leqslant n. \end{cases}$$

The solutions $b_j(z')$ are holomorphic on $\Delta' \smallsetminus S$ near y'. Since the determinant is $\delta(z')^{1/2}$, we see that δb_j is bounded, thus $\delta b_j \in \mathscr{O}_{\Delta',y'}$ and $\delta_y f \in \mathscr{O}_{X,y}$.

Now, assume that $(X, x) \subset (\mathbb{C}^n, 0)$ has irreducible components (X_j, x). We can find for each j a neighborhood Ω_j of 0 in $\mathbb{C}^n$ and a function $\delta_j \in \mathscr{O}_n(\Omega_j)$ which is a universal denominator on $X_j \cap \Omega_j$. After adding to δ_j a function which is identically zero on (X_j, x) and non zero on (X_k, x), $k \neq j$, we may assume that $\delta_j^{-1}(0) \cap X_k \cap \Omega$ is nowhere dense in $X_k \cap \Omega$ for all j and k and some small $\Omega \subset \bigcap \Omega_j$. Then $\delta = \prod \delta_j$ is a universal denominator on each component $X_j \cap \Omega$. For some possibly smaller Ω, select a function $v_j \in \mathscr{O}_n(\Omega)$ such that v_j vanishes identically on $\bigcup_{k \neq j} X_k \cap \Omega$ and $v_j^{-1}(0)$ is nowhere dense in $X_j \cap \Omega$, and set $h = \delta \sum v_k$. For any germ $f \in \tilde{\mathscr{O}}_{X,y}$, $y \in X \cap \Omega$, there is a germ $g_j \in \mathscr{O}_{\Omega,y}$ with $\delta f = g_j$ on (X_j, y). We have $h = \delta v_j$ on $X_j \cap \Omega$, so $h^{-1}(0)$ is nowhere dense in $X \cap \Omega$ and

$$hf = v_j \delta f = v_j g_j = \sum v_k g_k \quad \text{on each} \ \ (X_j, y).$$

Since $\sum v_k g_k \in \mathscr{O}_{\Omega,y}$, we get $h\tilde{\mathscr{O}}_{X,y} \subset \mathscr{O}_{X,y}$. □

(7.3) Theorem. *If (X, x) is irreducible, $\tilde{\mathscr{O}}_{X,x}$ is the integral closure of $\mathscr{O}_{X,x}$ in its quotient field $\mathscr{M}_{X,x}$. Moreover, every germ $f \in \tilde{\mathscr{O}}_{X,x}$ admits a limit*

$$\lim_{X_{\mathrm{reg}} \ni z \to x} f(z).$$

Observe that $\mathscr{O}_{X,x}$ is an entire ring, so the ring of quotients $\mathscr{M}_{X,x}$ is actually a field. A simple illustration of the theorem is obtained with the irreducible germ of curve $X : z_1^3 = z_2^2$ in $(\mathbb{C}^2, 0)$. Then X can be parametrized by $z_1 = t^2$, $z_2 = t^3$, $t \in \mathbb{C}$, and $\mathscr{O}_{X,0} = \mathbb{C}\{z_1, z_2\}/(z_1^3 - z_2^2) = \mathbb{C}\{t^2, t^3\}$ consists of all convergent series $\sum a_n t^n$ with $a_1 = 0$. The function $z_2/z_1 = t$ is weakly holomorphic on X and satisfies the integral equation $t^2 - z_1 = 0$. Here we have $\tilde{\mathscr{O}}_{X,0} = \mathbb{C}\{t\}$.

Proof. a) Let $f = g/h$ be an element in $\mathscr{M}_{X,x}$ satisfying an integral equation

$$f^m + a_1 f^{m-1} + \ldots + a_m = 0, \qquad a_k \in \mathscr{O}_{X,x}.$$

Set $A = h^{-1}(0)$. Then f is holomorphic on $X \smallsetminus A$ near x, and Lemma 4.10 shows that f is bounded on a neighborhood of x. By Remark 4.2, f can be extended as a holomorphic function on X_{reg} in a neighborhood of x, thus $f \in \tilde{\mathscr{O}}_{X,x}$.

b) Let $f \in \tilde{\mathscr{O}}_{X,x}$ and let $\pi : X \cap \Delta \longrightarrow \Delta'$ be a ramified covering in a neighborhood of x, with ramification locus S. As in the proof of Th. 6.1, f satisfies an equation

$$f^q - \sigma_1 f^{q-1} + \cdots + (-1)^q \sigma_q = 0, \qquad \sigma_k \in \mathscr{O}(\Delta') \ ;$$

indeed the elementary symmetric functions $\sigma_k(z')$ are holomorphic on $\Delta' \smallsetminus S$ and bounded, so they extend holomorphically to Δ'. Hence $\tilde{\mathscr{O}}_{X,x}$ is integral over $\mathscr{O}_{X,x}$ and we already know that $\tilde{\mathscr{O}}_{X,x} \subset \mathscr{M}_{X,x}$.

c) Finally, the cluster set $\bigcap_{V \ni x} \overline{f(X_{\mathrm{reg}} \cap V)}$ is connected, because there is a fundamental system of neighborhoods V of x such that $X_{\mathrm{reg}} \cap V$ is connected, and any intersection of a decreasing sequence of compact connected sets is connected. However the limit set is contained in the finite set of roots of equation b) at point $x' \in \Delta'$, so it must be reduced to one element. $\square$

§ 7.B. Normal Spaces

Normal spaces are spaces for which all weakly holomorphic functions are actually holomorphic. These spaces will be seen later to have "simpler" singularities than general analytic spaces.

(7.4) Definition. *A complex space X is said to be normal at a point x if (X, x) is irreducible and $\tilde{\mathscr{O}}_{X,x} = \mathscr{O}_{X,x}$, that is, $\mathscr{O}_{X,x}$ is integrally closed in its field of quotients. The set of normal (resp. non-normal) points will be denoted X_{norm} (resp. $X_{\text{n-n}}$). The space X itself is said to be normal if X is normal at every point.*

Observe that any regular point x is normal: in fact $\mathscr{O}_{X,x} \simeq \mathscr{O}_n$ is then factorial, hence integrally closed. Therefore $X_{\text{n-n}} \subset X_{\mathrm{sing}}$.

(7.5) Theorem. *The non-normal set $X_{\text{n-n}}$ is an analytic subset of X. In particular, X_{norm} is open in X.*

Proof. We give here a beautifully simple proof due to [Grauert and Remmert 1984]. Let h be a universal denominator on a neighborhood V of a given point and let $\mathscr{I} = \sqrt{h\mathscr{O}_X}$ be the sheaf of ideals of $h^{-1}(0)$ by Hilbert's Nullstellensatz. Finally, let $\mathscr{F} = \hom_{\mathscr{O}}(\mathscr{I}, \mathscr{I})$ be the sheaf of $\mathscr{O}_X$-endomorphisms of $\mathscr{I}$. Since $\mathscr{I}$ is coherent, so is $\mathscr{F}$ (cf. Exercise 11.3).

Clearly, the homotheties of $\mathcal{I}$ give an injection $\mathcal{O}_X \subset \mathcal{F}$ over V. We claim that there is a natural injection $\mathcal{F} \subset \tilde{\mathcal{O}}_X$. In fact, any endomorphism of $\mathcal{I}$ yields by restriction a homomorphism $h\mathcal{O}_X \longrightarrow \mathcal{O}_X$, and by $\mathcal{O}_X$-linearity such a homomorphism is obtained by multiplication by an element in $h^{-1}\mathcal{O}_X$. Thus $\mathcal{F} \subset h^{-1}\mathcal{O}_X \subset \mathcal{M}_X$. Since each stalk $\mathcal{I}_x$ is a finite $\mathcal{O}_{X,x}$-module containing non-zero divisors, it follows that that any meromorphic germ f such that $f\mathcal{I}_x \subset \mathcal{I}_x$ is integral over $\mathcal{O}_{X,x}$ ([Lang 1965], Chapter IX, § 1), hence $\mathcal{F}_x \subset \tilde{\mathcal{O}}_{X,x}$. Thus we have inclusions $\mathcal{O}_X \subset \mathcal{F} \subset \tilde{\mathcal{O}}_X$. Now, we assert that

$$X_{\text{n-n}} \cap V = \{x \in V\,;\, \mathcal{F}_x \neq \mathcal{O}_{X,x}\} = \mathcal{F}/\mathcal{O}_X.$$

This will imply the theorem by 5.10. To prove the equality, we first observe that $\mathcal{F}_x \neq \mathcal{O}_{X,x}$ implies $\tilde{\mathcal{O}}_{X,x} \neq \mathcal{O}_{X,x}$, thus $x \in X_{\text{n-n}}$. Conversely, assume that x is non normal, that is, $\tilde{\mathcal{O}}_{X,x} \neq \mathcal{O}_{X,x}$. Let k be the smallest integer such that $\mathcal{I}_x^k\tilde{\mathcal{O}}_{X,x} \subset \mathcal{O}_{X,x}$; such an integer exists since $\mathcal{I}_x^l\tilde{\mathcal{O}}_{X,x} \subset h\tilde{\mathcal{O}}_{X,x} \subset \mathcal{O}_{X,x}$ for l large. Then there is an element $w \in \mathcal{I}_x^{k-1}\tilde{\mathcal{O}}_{X,x}$ such that $w \notin \mathcal{O}_{X,x}$. We have $w\mathcal{I}_x \subset \mathcal{O}_{X,x}$; moreover, as w is locally bounded near X_{sing}, any germ wg in $w\mathcal{I}_x$ satisfies $\lim w(z)g(z) = 0$ when $z \in X_{\text{reg}}$ tends to a point of the zero variety $h^{-1}(0)$ of $\mathcal{I}_x$. Hence $w\mathcal{I}_x \subset \mathcal{I}_x$, i.e. $w \in \mathcal{F}_x$, but $w \notin \mathcal{O}_{X,x}$, so $\mathcal{F}_x \neq \mathcal{O}_{X,x}$. □

(7.6) Theorem. *If $x \in X$ is a normal point, then (X_{sing}, x) has codimension at least 2 in (X, x).*

Proof. We suppose that $\Sigma = X_{\text{sing}}$ has codimension 1 in a neighborhood of x and try to get a contradiction. By restriction to a smaller neighborhood, we may assume that X itself is normal and irreducible (since X_{norm} is open), $\dim X = n$, that Σ has pure dimension $n-1$ and that the ideal sheaf $\mathcal{I}_\Sigma$ has global generators $(g_1, \ldots, g_k)$. Then $\Sigma \subset \bigcup g_j^{-1}(0)$; both sets have pure dimension $n-1$ and thus singular sets of dimension $\leqslant n-2$. Hence there is a point $a \in \Sigma$ that is regular on Σ and on $\bigcup g_j^{-1}(0)$, in particular there is a neighborhood V of a such that $g_1^{-1}(0) \cap V = \ldots = g_k^{-1}(0) \cap V = \Sigma \cap V$ is a smooth $(n-1)$-dimensional manifold. Since $\operatorname{codim}_X \Sigma = 1$ and a is a singular point of X, $\mathcal{I}_{\Sigma,a}$ cannot have less than 2 generators in $\mathcal{O}_{X,a}$ by Cor. 4.33. Take $(g_1, \ldots, g_l)$, $l \geqslant 2$, to be a minimal subset of generators. Then $f = g_2/g_1$ cannot belong to $\mathcal{O}_{X,a}$, but f is holomorphic on $V \smallsetminus \Sigma$. We may assume that there is a sequence $a_\nu \in V \smallsetminus \Sigma$ converging to a such that $f(a_\nu)$ remains bounded (otherwise reverse g_1 and g_2 and pass to a subsequence). Since $g_1^{-1}(0) \cap V = \Sigma \cap V$, Hilbert's Nullstellensatz gives an integer m such that $\mathcal{I}_{\Sigma,a}^m \subset g_1\mathcal{O}_{X,a}$, hence $f_a\mathcal{I}_{\Sigma,a}^m \subset \mathcal{O}_{X,a}$. We take m to be the smallest integer such that the latter inclusion holds. Then there is a product $g^\alpha = g_1^{\alpha_1} \ldots g_l^{\alpha_l}$ with $|\alpha| = m-1$ such that $fg^\alpha \notin \mathcal{O}_{X,a}$ but $fg^\alpha g_j \in \mathcal{O}_{X,a}$ for each j. Since the sequence $f(a_\nu)$ is bounded we conclude that $fg^\alpha g_j$ vanishes at a. The zero set of this function has dimension $n-1$ and is contained in $\bigcup g_k^{-1}(0) \cap V = \Sigma \cap V$ so it must contain the germ (Σ, a). Hence $fg^\alpha g_j \in \mathcal{I}_{\Sigma,a}$ and $fg^\alpha\mathcal{I}_{\Sigma,a} \subset \mathcal{I}_{\Sigma,a}$. As $\mathcal{I}_{\Sigma,a}$ is a finitely generated $\mathcal{O}_{X,a}$-module, this implies $fg^\alpha \in \tilde{\mathcal{O}}_{X,a} = \mathcal{O}_{X,a}$, a contradiction. □

(7.7) Corollary. *A complex curve is normal if and only if it is regular.*

(7.8) Corollary. *Let X be a normal complex space and Y an analytic subset of X such that $\dim(Y, x) \leqslant \dim(X, x) - 2$ for any $x \in X$. Then any holomorphic function on $X \smallsetminus Y$ can be extended to a holomorphic function on X.*

Proof. By Cor. I.5.25, every holomorphic function f on $X_{\text{reg}} \smallsetminus Y$ extends to X_{reg}. Since $\operatorname{codim} X_{\text{sing}} \geqslant 2$, Th. 6.1 shows that f is locally bounded near X_{sing}. Therefore f extends to X by definition of a normal space. □

§ 7.C. The Oka Normalization Theorem

The important normalization theorem of [Oka 1950] shows that $\tilde{\mathscr{O}}_X$ can be used to define the structure sheaf of a new analytic space $\tilde{X}$ which is normal and is obtained by "simplifying" the singular set of X. More precisely:

(7.9) Definition. *Let X be a complex space. A normalization (Y, π) of X is a normal complex space Y together with a holomorphic map $\pi : Y \longrightarrow X$ such that the following conditions are satisfied.*

a) *$\pi : Y \longrightarrow X$ is proper and has finite fibers;*

b) *if Σ is the set of singular points of X and $A = \pi^{-1}(\Sigma)$, then $Y \smallsetminus A$ is dense in Y and $\pi : Y \smallsetminus A \longrightarrow X \smallsetminus \Sigma = X_{\text{reg}}$ is an analytic isomorphism.*

It follows from b) that $Y \smallsetminus A \subset Y_{\text{reg}}$. Thus Y is obtained from X by a suitable "modification" of its singular points. Observe that Y_{reg} may be larger than $Y \smallsetminus A$, as is the case in the following two examples.

(7.10) Examples. a) Let $X = \mathbb{C} \times \{0\} \cup \{0\} \times \mathbb{C}$ be the complex curve $z_1 z_2 = 0$ in $\mathbb{C}^2$. Then the normalization of X is the disjoint union $Y = \mathbb{C} \times \{1, 2\}$ of two copies of $\mathbb{C}$, with the map $\pi(t_1) = (t_1, 0)$, $\pi(t_2) = (0, t_2)$. The set $A = \pi^{-1}(0, 0)$ consists of exactly two points.

b) The cubic curve $X : z_1^3 = z_2^2$ is normalized by the map $\pi : \mathbb{C} \longrightarrow X$, $t \longmapsto (t^2, t^3)$. Here π is a homeomorphism but π^{-1} is not analytic at $(0, 0)$. □

We first show that the normalization is essentially unique up to isomorphism and postpone the proof of its existence for a while.

(7.11) Lemma. *If (Y_1, π_1) and (Y_2, π_2) are normalizations of X, there is a unique analytic isomorphism $\varphi : Y_1 \longrightarrow Y_2$ such that $\pi_1 = \pi_2 \circ \varphi$.*

Proof. Let Σ be the set of singular points of X and $A_j = \pi_j^{-1}(\Sigma)$, $j = 1, 2$. Let $\varphi' : Y_1 \smallsetminus A_1 \longrightarrow Y_2 \smallsetminus A_2$ be the analytic isomorphism $\pi_2^{-1} \circ \pi_1$. We assert that φ' can be extended into a map $\varphi : Y_1 \longrightarrow Y_2$. In fact, let $a \in A_1$ and $s = \pi_1(a) \in \Sigma$. Then $\pi_2^{-1}(s)$ consists of a finite set of points $y_j \in Y_2$. Take disjoint neighborhoods U_j of y_j such that U_j is an analytic subset in an open set $\Omega_j \subset\subset \mathbb{C}^N$. Since π_2 is proper, there is a neighborhood V of s in X such that $\pi_2^{-1}(V) \subset \bigcup U_j$ and by continuity of π_1 a neighborhood W of a such that $\pi_1(W) \subset V$. Then $\varphi' = \pi_2^{-1} \circ \pi_1$ maps $W \smallsetminus A_1$ into $\bigcup U_j$ and can be seen as a bounded holomorphic map into $\mathbb{C}^N$ through the embeddings $U_j \subset \Omega_j \subset\subset \mathbb{C}^N$. Since Y_1 is normal, φ' extends to W, and the extension takes values in $\bigcup \overline{U}_j$ which is contained in Y_2 (shrink U_j if necessary). Thus φ' extends into a map $\varphi : Y_1 \longrightarrow Y_2$ and similarly φ'^{-1} extends into a map $\psi : Y_2 \longrightarrow Y_1$. By density of $Y_j \smallsetminus A_j$, we have $\psi \circ \varphi = \mathrm{Id}_{Y_1}$, $\varphi \circ \psi = \mathrm{Id}_{Y_2}$. □

(7.12) Oka normalization theorem. *Let X be any complex space. Then X has a normalization (Y, π).*

Proof. Because of the previous lemma, it suffices to prove that any point $x \in X$ has a neighborhood U such that U admits a normalization; all these local normalizations will then glue together. Hence we may suppose that X is an analytic set in an open set of $\mathbb{C}^n$. Moreover, if (X, x) splits into irreducible components (X_j, x) and if (Y_j, π_j) is a normalization of $X_j \cap U$, then the disjoint union $Y = \coprod Y_j$ with $\pi = \coprod \pi_j$ is easily seen to be a normalization of $X \cap U$. We may therefore assume that (X, x) is irreducible. Let h be a universal denominator in a neighborhood of x. Then $\tilde{\mathscr{O}}_{X,x}$ is isomorphic to its image $h\tilde{\mathscr{O}}_{X,x} \subset \mathscr{O}_{X,x}$, so it is a finitely generated $\mathscr{O}_{X,x}$-module. Let $(f_1, \dots, f_m)$ be a finite set of generators of $\mathscr{O}_{X,x}$. After shrinking X again, we may assume the following two points:

- X is an analytic set in an open set $\Omega \subset \mathbb{C}^n$, (X, x) is irreducible and X_{reg} is connected;
- f_j is holomorphic in X_{reg}, can be written $f_j = g_j/h$ on X with g_j, h in $\mathscr{O}_n(\Omega)$ and satisfies an integral equation $P_j(z\,;\, f_j(z)) = 0$ where $P_j(z\,;\, T)$ is a unitary polynomial with holomorphic coefficients on X.

Set $X' = X \smallsetminus h^{-1}(0)$. Consider the holomorphic map

$$F : X_{\mathrm{reg}} \longrightarrow \Omega \times \mathbb{C}^m, \qquad z \longmapsto \big(z, f_1(z), \dots, f_m(z)\big)$$

and the image $Y' = F(X')$. We claim that the closure Y of Y' in $\Omega \times \mathbb{C}^m$ is an analytic set. In fact, the set

$$Z = \big\{(z, w) \in \Omega \times \mathbb{C}^m\,;\, z \in X\,,\ h(z)w_j = g_j(z)\big\}$$

is analytic and $Y' = Z \smallsetminus \{h(z) = 0\}$, so we may apply Cor. 5.4. Observe that Y' is contained in the set defined by $P_j(z\,; w_j) = 0$, thus so is its closure Y. The first projection $\Omega \times \mathbb{C}^m \longrightarrow \Omega$ gives a holomorphic map $\pi : Y \longrightarrow X$ such that $\pi \circ F = \mathrm{Id}$ on X', hence also on X_{reg}. If $\Sigma = X_{\mathrm{sing}}$ and $A = \pi^{-1}(\Sigma)$, the restriction $\pi : Y \smallsetminus A \longrightarrow X \smallsetminus \Sigma = X_{\mathrm{reg}}$ is thus an analytic isomorphism and F is its inverse. Since (X, x) is irreducible, each f_j has a limit ℓ_j at x by Th. 7.3 and the fiber $\pi^{-1}(x)$ is reduced to the single point $y = (x, \ell)$. The other fibers $\pi^{-1}(z)$ are finite because they are contained in the finite set of roots of the equations $P_j(z\,; w_j) = 0$. The same argument easily shows that π is proper (use Lemma 4.10).

Next, we show that Y is normal at the point $y = \pi^{-1}(x)$. In fact, for any bounded holomorphic function u on (Y_{reg}, y) the function $u \circ F$ is bounded and holomorphic on (X_{reg}, x). Hence $u \circ F \in \tilde{\mathscr{O}}_{X,x} = \mathscr{O}_{X,x}[f_1, \dots, f_m]$ and we can write $u \circ F(z) = Q(z\,;\, f_1(z), \dots, f_m(z)) = Q \circ F(z)$ where $Q(z\,;\, w) = \sum a_\alpha(z) w^\alpha$ is a polynomial in w with coefficients in $\mathscr{O}_{X,x}$. Thus u coincides with Q on (Y_{reg}, y), and as Q is holomorphic on $(X, x) \times \mathbb{C}^m \supset (Y, y)$, we conclude that $u \in \mathscr{O}_{Y,y}$. Therefore $\tilde{\mathscr{O}}_{Y,y} = \mathscr{O}_{Y,y}$.

Finally, by Th. 7.5, there is a neighborhood $V \subset Y$ of y such that every point of V is normal. As π is proper, we can find a neighborhood U of x with $\pi^{-1}(U) \subset V$. Then $\pi : \pi^{-1}(U) \longrightarrow U$ is the required normalization in a neighborhood of x. $\square$

The proof of Th. 7.12 shows that the fiber $\pi^{-1}(x)$ has exactly one point y_j for each irreducible component (X_j, x) of (X, x). As a one-to-one proper map is a homeomorphism, we get in particular:

(7.13) Corollary. *If X is a locally irreducible complex space, the normalization $\pi : Y \longrightarrow X$ is a homeomorphism.* $\square$

(7.14) Remark. In general, for any open set $U \subset X$, we have an isomorphism

$$\pi^\star : \tilde{\mathscr{O}}_X(U) \xrightarrow{\simeq} \mathscr{O}_Y\big(\pi^{-1}(U)\big), \tag{7.15}$$

whose inverse is given by the comorphism of $\pi^{-1} : X_{\text{reg}} \longrightarrow Y$; note that $\tilde{\mathscr{O}}_Y(U) = \mathscr{O}_Y(U)$ since Y is normal. Taking the direct limit over all neighborhoods U of a given point $x \in X$, we get an isomorphism

$$\pi^\star : \tilde{\mathscr{O}}_{X,x} \longrightarrow \bigoplus_{y_j \in \pi^{-1}(x)} \mathscr{O}_{Y,y_j}. \tag{7.15$'$}$$

In other words, $\tilde{\mathscr{O}}_X$ is isomorphic to the direct image sheaf $\pi_\star \mathscr{O}_Y$, see (1.12). We will prove later on the deep fact that the direct image of a coherent sheaf by a proper holomorphic map is always coherent ([Grauert 1960], see IX.5.1). Hence $\tilde{\mathscr{O}}_X = \pi_\star \mathscr{O}_Y$ is a coherent sheaf over $\mathscr{O}_X$.

§ 8. Holomorphic Mappings and Extension Theorems

§ 8.A. Rank of a Holomorphic Mapping

Our goal here is to introduce the general concept of the rank of a holomorphic map and to relate the rank to the dimension of the fibers. As in the smooth case, the rank is shown to satisfy semi-continuity properties.

(8.1) Lemma. *Let $F : X \longrightarrow Y$ be a holomorphic map from a complex space X to a complex space Y.*

a) *If F is finite, i.e. proper with finite fibers, then* $\dim X \leqslant \dim Y$.

b) *If F is finite and surjective, then* $\dim X = \dim Y$.

Proof. a) Let $x \in X$, (X_j, x) an irreducible component and $m = \dim(X_j, x)$. If (Y_k, y) are the irreducible components of Y at $y = F(x)$, then (X_j, x) is contained in $\bigcup F^{-1}(Y_k)$, hence (X_j, x) is contained in one of the sets $F^{-1}(Y_k)$. If $p = \dim(Y_k, y)$, there is a ramified covering π from some neighborhood of y in Y_k onto a polydisk in $\Delta' \subset \mathbb{C}^p$. Replacing X by some neighborhood of x in X_j and F by the finite map $\pi \circ F_{\restriction X_j} : X_j \longrightarrow \Delta'$, we may suppose that $Y = \Delta'$ and that X is irreducible, $\dim X = m$. Let $r = \operatorname{rank} dF_{x_0}$ be the maximum of the rank of the differential of F on X_{reg}. Then $r \leqslant \min\{m, p\}$ and the rank of dF is constant equal to r on a neighborhood U of x_0. The constant rank theorem implies that the fibers $F^{-1}(y) \cap U$ are $(m-r)$-dimensional submanifolds, hence $m - r = 0$ and $m = r \leqslant p$.

b) We only have to show that $\dim X \geqslant \dim Y$. Fix a regular point $y \in Y$ of maximal dimension. By taking the restriction $F : F^{-1}(U) \longrightarrow U$ to a small neighborhood U of y, we may assume that Y is an open subset of $\mathbb{C}^p$. If $\dim X < \dim Y$, then X is a union of

analytic manifolds of dimension $< \dim Y$ and Sard's theorem implies that $F(X)$ has zero Lebesgue measure in Y, a contradiction. $\square$

(8.2) Proposition. *For any holomorphic map $F : X \longrightarrow Y$, the fiber dimension $\dim\big(F^{-1}(F(x)), x\big)$ is an upper semi-continuous function of x.*

Proof. Without loss of generality, we may suppose that X is an analytic set in $\Omega \subset \mathbb{C}^n$, that $F(X)$ is contained in a small neighborhood of $F(x)$ in Y which is embedded in $\mathbb{C}^N$, and that $x = 0$, $F(x) = 0$. Set $A = F^{-1}(0)$ and $s = \dim(A, 0)$. We can find a linear form ξ_1 on $\mathbb{C}^n$ such that $\dim(A \cap \xi_1^{-1}(0), 0) = s - 1$; in fact we need only select a point $x_j \neq 0$ on each irreducible component $(A_j, 0)$ of $(A, 0)$ and take $\xi_1(x_j) \neq 0$. By induction, we can find linearly independent forms $\xi_1, \ldots, \xi_s$ on $\mathbb{C}^n$ such that

$$\dim\big(A \cap \xi_1^{-1}(0) \cap \ldots \cap \xi_j^{-1}(0), 0\big) = s - j$$

for all $j = 1, \ldots, s$; in particular 0 is an isolated point in the intersection when $j = s$. After a change of coordinates, we may suppose that $\xi_j(z) = z_j$. Fix r'' so small that the ball $\overline{B}'' \subset \mathbb{C}^{n-s}$ of center 0 and radius r'' satisfies $A \cap (\{0\} \times \overline{B}'') = \{0\}$. Then A is disjoint from the compact set $\{0\} \times \partial B''$, so there exists a small ball $B' \subset \mathbb{C}^s$ of center 0 such that $A \cap (\overline{B}' \times \partial B'') = \emptyset$, i.e. F does not vanish on the compact set $K = X \cap (\overline{B}' \times \partial B'')$. Set $\varepsilon = \min_K |F|$. Then for $|y| < \varepsilon$ the fiber $F^{-1}(y)$ does not intersect $\overline{B}' \times \partial B''$. This implies that the projection map $\pi : F^{-1}(y) \cap (B' \times B'') \longrightarrow B'$ is proper. The fibers of π are then compact analytic subsets of B'', so they are finite by 5.9. Lemma 8.1 a) implies

$$\dim F^{-1}(y) \cap (B' \times B'') \leqslant \dim B' = s = \dim(A, 0) = \dim(F^{-1}(0), 0). \qquad \square$$

Let X be a pure dimensional complex space and $F : X \longrightarrow Y$ a holomorphic map. For any point $x \in X$, we define the *rank of F at x* by

$$\rho_F(x) = \dim(X, x) - \dim\big(F^{-1}(F(x)), x\big). \tag{8.3}$$

By the above proposition, ρ_F is a lower semi-continuous function on X. In particular, if ρ_F is maximum at some point x_0, it must be constant in a neighborhood of x_0. The maximum $\overline{\rho}(F) = \max_X \rho_F$ is thus attained on X_{reg} or on any dense open subset $X' \subset X_{\mathrm{reg}}$. If X is not pure dimensional, we define $\overline{\rho}(F) = \max_\alpha \overline{\rho}(F_{\restriction X_\alpha})$ where (X_α) are the irreducible components of X. For a map $F : X \longrightarrow \mathbb{C}^N$, the constant rank theorem implies that $\overline{\rho}(F)$ is equal to the maximum of the rank of the jacobian matrix dF at points of X_{reg} (or of X').

(8.4) Proposition. *If $F : X \longrightarrow Y$ is a holomorphic map and Z an analytic subset of X, then $\overline{\rho}(F_{\restriction Z}) \leqslant \overline{\rho}(F)$.*

Proof. Since each irreducible component of Z is contained in an irreducible component of X, we may assume X irreducible. Let $\pi : \tilde{X} \longrightarrow X$ be the normalization of X and $\tilde{Z} = \pi^{-1}(Z)$. Since π is finite and surjective, the fiber of $F \circ \pi$ at point x has the same dimension than the fiber of F at $\pi(x)$ by Lemma 8.1 b). Therefore $\overline{\rho}(F \circ \pi) = \overline{\rho}(F)$ and $\overline{\rho}(F \circ \pi_{\restriction \tilde{Z}}) = \overline{\rho}(F_{\restriction Z})$, so we may assume X normal. By induction on $\dim X$, we may also suppose that Z has pure codimension 1 in X (every point of Z has a neighborhood $V \subset X$

such that $Z \cap V$ is contained in a pure one codimensional analytic subset of V). But then $Z_{\mathrm{reg}} \cap X_{\mathrm{reg}}$ is dense in Z_{reg} because $\operatorname{codim} X_{\mathrm{sing}} \geqslant 2$. Thus we are reduced to the case when X is a manifold and Z a submanifold, and this case is clear if we consider the rank of the jacobian matrix. □

(8.5) Theorem. *Let $F : X \longrightarrow Y$ be a holomorphic map. If Y is pure dimensional and $\overline{\rho}(F) < \dim Y$, then $F(X)$ has empty interior in Y.*

Proof. Taking the restriction of F to $F^{-1}(Y_{\mathrm{reg}})$, we may assume that Y is a manifold. Since X is a countable union of compact sets, so is $F(X)$, and Baire's theorem shows that the result is local for X. By Prop. 8.4 and an induction on $\dim X$, $F(X_{\mathrm{sing}})$ has empty interior in Y. The set $Z \subset X_{\mathrm{reg}}$ of points where the jacobian matrix of F has rank $< \overline{\rho}(F)$ is an analytic subset hence, by induction again, $F(Z)$ has empty interior. The constant rank theorem finally shows that every point $x \in X_{\mathrm{reg}} \smallsetminus Z$ has a neighborhood V such that $F(V)$ is a submanifold of dimension $\overline{\rho}(F)$ in Y, thus $F(V)$ has empty interior and Baire's theorem completes the proof. □

(8.6) Corollary. *Let $F : X \longrightarrow Y$ be a surjective holomorphic map. Then $\dim Y = \overline{\rho}(F)$.*

Proof. By the remark before Prop. 8.4, there is a regular point $x_0 \in X$ such that the jacobian matrix of F has rank $\overline{\rho}(F)$. Hence, by the constant rank theorem $\dim Y \geqslant \overline{\rho}(F)$. Conversely, let Y_α be an irreducible component of Y of dimension equal to $\dim Y$, and $Z = F^{-1}(Y_\alpha) \subset X$. Then $F(Z) = Y_\alpha$ and Th. 8.5 implies $\overline{\rho}(F) \geqslant \overline{\rho}(F_{\restriction Z}) \geqslant \dim Y_\alpha$. □

§ 8.B. Remmert and Remmert-Stein Theorems

We are now ready to prove two important results: the extension theorem for analytic subsets due to [Remmert and Stein 1953] and the theorem of [Remmert 1956, 1957] which asserts that the image of a complex space under a proper holomorphic map is an analytic set. These will be obtained by a simultaneous induction on the dimension.

(8.7) Remmert-Stein theorem. *Let X be a complex space, A an analytic subset of X and Z an analytic subset of $X \smallsetminus A$. Suppose that there is an integer $p \geqslant 0$ such that $\dim A \leqslant p$, while $\dim(Z, x) > p$ for all $x \in Z$. Then the closure $\overline{Z}$ of Z in X is an analytic subset.*

(8.8) Remmert's proper mapping theorem. *Let $F : X \longrightarrow Y$ be a proper holomorphic map. Then $F(X)$ is an analytic subset of Y.*

Proof. We let (8.7_m) denote statement (8.7) for $\dim Z \leqslant m$ and (8.8_m) denote statement (8.8) for $\dim X \leqslant m$. We proceed by induction on m in two steps:

Step 1. (8.7_m) and (8.8_{m-1}) imply (8.8_m).
Step 2. (8.8_{m-1}) implies (8.7_m). As (8.8_m) is obvious for $m = 0$, our statements will then be valid for all m, i.e. for all complex spaces of bounded dimension. However, Th. 8.7 is local on X and Th. 8.8 is local on Y, so the general case is immediately reduced to the finite dimensional case. *Proof of step 1.* The analyticity of $F(X)$ is a local question in Y.

Since $F : F^{-1}(U) \longrightarrow U$ is proper for any open set $U \subset Y$ and $F^{-1}(U) \subset\subset X$ if $U \subset\subset Y$, we may suppose that Y is embedded in an open set $\Omega \subset \mathbb{C}^n$ and that X only has finitely

many irreducible components X_α. Then we have $F(X) = \bigcup F(X_\alpha)$ and we are reduced to the case when X is irreducible, $\dim X = m$ and $Y = \Omega$.

First assume that X is a manifold and that the rank of dF is constant. The constant rank theorem implies that every point in X has a neighborhood V such that $F(V)$ is a closed submanifold in a neighborhood W of $F(x)$ in Y. For any point $y \in Y$, the fiber $F^{-1}(y)$ can be covered by finitely many neighborhoods V_j of points $x_j \in F^{-1}(y)$ such that $F(V_j)$ is a closed submanifold in a neighborhood W_j of y. Then there is a neighborhood of y $W \subset \bigcap W_j$ such that $F^{-1}(W) \subset \bigcup V_j$, so $F(X) \cap W = \bigcup F(V_j) \cap W$ is a finite union of closed submanifolds in W and $F(X)$ is analytic in Y.

Now suppose that X is a manifold, set $r = \overline{\rho}(F)$ and let $Z \subset X$ be the analytic subset of points x where the rank of dF_x is $< r$. Since $\dim Z < m = \dim X$, the hypothesis (8.8_{m-1}) shows that $F(Z)$ is analytic. We have $\dim F(Z) = \overline{\rho}(F_{\restriction Z}) < r$. If $F(Z) = F(X)$, then $F(X)$ is analytic. Otherwise $A = F^{-1}\big(F(Z)\big)$ is a proper analytic subset of X, dF has constant rank on $X \smallsetminus A \subset X \smallsetminus Z$ and the morphism $F : X \smallsetminus A \longrightarrow Y \smallsetminus F(Z)$ is proper. Hence the image $F(X \smallsetminus A)$ is analytic in $Y \smallsetminus F(Z)$. Since $\dim F(X \smallsetminus A) = r \leqslant m$ and $\dim F(Z) < r$, hypothesis (8.7_m) implies that $F(X) = \overline{F(X \smallsetminus A)}$ is analytic in Y. When X is not a manifold, we apply the same reasoning with $Z = X_{\text{sing}}$ in order to be reduced to the case of $F : X \smallsetminus A \longrightarrow Y \smallsetminus F(Z)$ where $X \smallsetminus A$ is a manifold. $\square$

Proof of step 2. Since Th. 8.7 is local on X, we may suppose that X is an open set $\Omega \subset \mathbb{C}^n$. Then we use induction on p to reduce the situation to the case when A is a p-dimensional submanifold (if this case is taken for granted, the closure of Z in $\Omega \smallsetminus A_{\text{sing}}$ is analytic and we conclude by the induction hypothesis). By a local analytic change of coordinates, we may assume that $0 \in A$ and that $A = \Omega \cap L$ where L is a vector subspace of $\mathbb{C}^n$ of dimension p. By writing $Z = \bigcup_{p<s\leqslant m} Z_s$ where Z_s is an analytic subset of $\Omega \smallsetminus Y$ of pure dimension s, we may suppose that Z has pure dimension s, $p < s \leqslant m$. We are going to show that $\overline{Z}$ is analytic in a neighborhood of 0.

Let ξ_1 be a linear form on $\mathbb{C}^n$ which is not identically zero on L nor on any irreducible component of Z (just pick a point x_ν on each component and take $\xi_1(x_\nu) \neq 0$ for all ν). Then $\dim L \cap \xi_1^{-1}(0) = p - 1$ and the analytic set $Z \cap \xi_1^{-1}(0)$ has pure dimension $s - 1$. By induction, there exist linearly independent forms $\xi_1, \ldots, \xi_s$ such that

$$\begin{aligned} &\dim L \cap \xi_1^{-1}(0) \cap \ldots \cap \xi_j^{-1}(0) = p - j, \quad 1 \leqslant j \leqslant p, \\ &\dim Z \cap \xi_1^{-1}(0) \cap \ldots \cap \xi_j^{-1}(0) = s - j, \quad 1 \leqslant j \leqslant s. \end{aligned} \tag{8.9}$$

By adding a suitable linear combination of $\xi_1, \ldots, \xi_p$ to each ξ_j, $p < j \leqslant s$, we may take $\xi_{j\restriction L} = 0$ for $p < j \leqslant s$. After a linear change of coordinates, we may suppose that $\xi_j(z) = z_j$, $L = \mathbb{C}^p \times \{0\}$ and $A = \Omega \cap (\mathbb{C}^p \times \{0\})$. Let $\xi = (\xi_1, \ldots, \xi_s) : \mathbb{C}^n \longrightarrow \mathbb{C}^s$ be the projection onto the first s variables. As Z is closed in $\Omega \smallsetminus A$, $Z \cup A$ is closed in Ω. Moreover, our construction gives $(Z \cup A) \cap \xi^{-1}(0) = \big(Z \cap \xi^{-1}(0)\big) \cup \{0\}$ and the case $j = s$ of (8.9) shows that $Z \cap \xi^{-1}(0)$ is a locally finite sequence in $\Omega \cap (\{0\} \times \mathbb{C}^{n-s}) \smallsetminus \{0\}$. Therefore, we can find a small ball $\overline{B}''$ of center 0 in $\mathbb{C}^{n-s}$ such that $Z \cap (\{0\} \times \partial B'') = \emptyset$. As $\{0\} \times \partial B''$ is compact and disjoint from the closed set $Z \cup A$, there is a small ball B' of center 0 in $\mathbb{C}^s$ such that $(Z \cup A) \cap (\overline{B}' \times \partial B'') = \emptyset$. This implies that the projection $\xi : (Z \cup A) \cap (B' \times B'') \longrightarrow B'$ is proper. Set $A' = B' \cap (\mathbb{C}^p \times \{0\})$.

Then the restriction

$$\pi = \xi : Z \cap (B' \times B'') \smallsetminus (A' \times B'') \longrightarrow B' \smallsetminus A'$$

is proper, and $Z \cap (B' \times B'')$ is analytic in $(B' \times B'') \smallsetminus A$, so π has finite fibers by Th. 5.9. By definition of the rank we have $\overline{\rho}(\pi) = s$. Let $S_1 = Z_{\mathrm{sing}} \cap \pi^{-1}(B' \smallsetminus A')$ and $S_1' = \pi(S_1)$; further, let S_2 be the set of points $x \in Z \cap \pi^{-1}\big(B' \smallsetminus (A' \cup S_1')\big) \subset Z_{\mathrm{reg}}$ such that $d\pi_x$ has rank $< s$ and $S_2' = \pi(S_2)$. We have $\dim S_j \leqslant s-1 \leqslant m-1$. Hypothesis $(8.8)_{m-1}$ implies that S_1' is analytic in $B' \smallsetminus A'$ and that S_2' is analytic in $B' \smallsetminus (A' \cup S_1')$. By Remark 4.2, $B' \smallsetminus (A' \cup S_1' \cup S_2')$ is connected and every bounded holomorphic function on this set extends to B'. As π is a (non ramified) covering over $B' \smallsetminus (A' \cup S_1' \cup S_2')$, the sheet number is a constant q.

Let $\lambda(z) = \sum_{j>s} \lambda_j z_j$ be a linear form on $\mathbb{C}^n$ in the coordinates of index $j > s$. For $z' \in B' \smallsetminus (A' \cup S_1' \cup S_2')$, we let $\sigma_j(z')$ be the elementary symmetric functions in the q complex numbers $\lambda(z)$ corresponding to $z \in \pi^{-1}(z')$. Then these functions can be extended as bounded holomorphic functions on B' and we get a polynomial $P_\lambda(z'\,;T)$ such that $P_\lambda\big(z'\,;\lambda(z'')\big)$ vanishes identically on $Z \smallsetminus \pi^{-1}(A' \cup S_1' \cup S_2')$. Since π is finite, $Z \cap \pi^{-1}(A' \cup S_1' \cup S_2')$ is a union of three (non necessarily closed) analytic subsets of dimension $\leqslant s-1$, thus has empty interior in Z. It follows that the closure $\overline{Z} \cap (B' \times B'')$ is contained in the analytic set $W \subset B' \times B''$ equal to the common zero set of all functions $P_\lambda\big(z'\,;\lambda(z'')\big)$. Moreover, by construction,

$$Z \smallsetminus \pi^{-1}(A' \cup S_1' \cup S_2') = W \smallsetminus \pi^{-1}(A' \cup S_1' \cup S_2').$$

As in the proof of Cor. 5.4, we easily conclude that $\overline{Z} \cap (B' \times B'')$ is equal to the union of all irreducible components of W that are not contained in $\pi^{-1}(A' \cup S_1' \cup S_2')$. Hence $\overline{Z}$ is analytic. □

Finally, we give two interesting applications of the Remmert-Stein theorem. We assume here that the reader knows what is the complex projective space $\mathbb{P}^n$. For more details, see Sect. **??**

(8.10) Chow's theorem ([Chow 1949]). *Let A be an analytic subset of the complex projective space $\mathbb{P}^n$. Then A is algebraic, i.e. A is the common zero set of finitely many homogeneous polynomials $P_j(z_0, \ldots, z_n)$, $1 \leqslant j \leqslant N$.*

Proof. Let $\pi : \mathbb{C}^{n+1} \smallsetminus \{0\} \longrightarrow \mathbb{P}^n$ be the natural projection and $Z = \pi^{-1}(A)$. Then Z is an analytic subset of $\mathbb{C}^{n+1} \smallsetminus \{0\}$ which is invariant by homotheties and $\dim Z = \dim A + 1 \geqslant 1$. The Remmert-Stein theorem implies that $\overline{Z} = Z \cup \{0\}$ is an analytic subset of $\mathbb{C}^{n+1}$. Let $f_1, \ldots, f_N$ be holomorphic functions on a small polydisk $\Delta \subset \mathbb{C}^{n+1}$ of center 0 such that $\overline{Z} \cap \Delta = \bigcap f_j^{-1}(0)$. The Taylor series at 0 gives an expansion $f_j = \sum_{k=0}^{+\infty} P_{j,k}$ where $P_{j,k}$ is a homogeneous polynomial of degree k. We claim that $\overline{Z}$ coincides with the common zero W set of the polynomials $P_{j,k}$. In fact, we clearly have $W \cap \Delta \subset \bigcap f_j^{-1}(0) = \overline{Z} \cap \Delta$. Conversely, for $z \in \overline{Z} \cap \Delta$, the invariance of Z by homotheties shows that $f_j(tz) = \sum P_{j,k}(z) t^k$ vanishes for every complex number t of modulus < 1, so all coefficients $P_{j,k}(z)$ vanish and $z \in W \cap \Delta$. By homogeneity $\overline{Z} = W$; since $\mathbb{C}[z_0, \ldots, z_n]$ is Noetherian, W can be defined by finitely many polynomial equations. □

(8.11) E.E. Levi's continuation theorem. *Let X be a normal complex space and A an analytic subset such that $\dim(A,x) \leqslant \dim(X,x) - 2$ for all $x \in A$. Then every meromorphic function on $X \smallsetminus A$ has a meromorphic extension to X.*

Proof. We may suppose X irreducible, $\dim X = n$. Let f be a meromorphic function on $X \smallsetminus A$. By Th. 6.13, the pole set P_f has pure dimension $(n-1)$, so the Remmert-Stein theorem implies that $\overline{P}_f$ is analytic in X. Fix a point $x \in A$. There is a connected neighborhood V of x and a non zero holomorphic function $h \in \mathscr{O}_X(V)$ such that $\overline{P}_f \cap V$ has finitely many irreducible components $\overline{P}_{f,j}$ and $\overline{P}_f \cap V \subset h^{-1}(0)$. Select a point x_j in $\overline{P}_{f,j} \smallsetminus (X_{\text{sing}} \cup (\overline{P}_f)_{\text{sing}} \cup A)$. As x_j is a regular point on X and on $\overline{P}_f$, there is a local coordinate $z_{1,j}$ at x_j defining an equation of $\overline{P}_{f,j}$, such that $z_{1,j}^{m_j} f \in \mathscr{O}_{X,x_j}$ for some integer m_j. Since h vanishes along P_f, we have $h^{m_j} f \in \mathscr{O}_{X,x}$. Thus, for $m = \max\{m_j\}$, the pole set P_g of $g = h^m f$ in $V \smallsetminus A$ does not contain x_j. As P_g is $(n-1)$-dimensional and contained in $P_f \cap V$, it is a union of irreducible components $\overline{P}_{f,j} \smallsetminus A$. Hence P_g must be empty and g is holomorphic on $V \smallsetminus A$. By Cor. 7.8, g has an extension to a holomorphic function $\tilde{g}$ on V. Then $\tilde{g}/h^m$ is the required meromorphic extension of f on V. □

§ 9. Complex Analytic Schemes

Our goal is to introduce a generalization of the notion of complex space given in Def. 5.2. A complex space is a space locally isomorphic to an analytic set A in an open subset $\Omega \subset \mathbb{C}^n$, together with the sheaf of rings $\mathscr{O}_A = (\mathscr{O}_\Omega/\mathscr{I}_A)_{\restriction A}$. Our desire is to enrich the structure sheaf $\mathscr{O}_A$ by replacing $\mathscr{I}_A$ with a possibly smaller ideal $\mathscr{J}$ defining the same zero variety $V(\mathscr{J}) = A$. In this way holomorphic functions are described not merely by their values on A, but also possibly by some "transversal derivatives" along A.

§ 9.A. Ringed Spaces

We start by an abstract notion of ringed space on an arbitrary topological space.

(9.1) Definition. *A ringed space is a pair $(X, \mathscr{R}_X)$ consisting of a topological space X and of a sheaf of rings $\mathscr{R}_X$ on X, called the structure sheaf. A morphism*

$$F : (X, \mathscr{R}_X) \longrightarrow (Y, \mathscr{R}_Y)$$

of ringed spaces is a pair $(f, F^\star)$ where $f : X \longrightarrow Y$ is a continuous map and

$$F^\star \ : \ f^{-1}\mathscr{R}_Y \longrightarrow \mathscr{R}_X, \qquad F^\star_x \ : \ (\mathscr{R}_Y)_{f(x)} \longrightarrow (\mathscr{R}_X)_x$$

a homomorphism of sheaves of rings on X, called the comorphism of F.

If $F : (X, \mathscr{R}_X) \longrightarrow (Y, \mathscr{R}_Y)$ and $G : (Y, \mathscr{R}_Y) \longrightarrow (Z, \mathscr{R}_Z)$ are morphisms of ringed spaces, the composite $G \circ F$ is the pair consisting of the map $g \circ f : X \longrightarrow Z$ and of the comorphism $(G \circ F)^\star = F^\star \circ f^{-1}G^\star$:

$$\begin{aligned} F^\star \circ f^{-1}G^\star \ &: \ f^{-1}g^{-1}\mathscr{R}_Z \xrightarrow{f^{-1}G^\star} f^{-1}\mathscr{R}_Y \xrightarrow{F^\star} \mathscr{R}_X \\ F^\star_x \circ G^\star_{f(x)} \ &: \ (\mathscr{R}_Z)_{g\circ f(x)} \longrightarrow (\mathscr{R}_Y)_{f(x)} \longrightarrow (\mathscr{R}_X)_x \end{aligned} \tag{9.2}$$

§ 9.B. Definition of Complex Analytic Schemes

We begin by a description of what will be the local model of an analytic scheme. Let $\Omega \subset \mathbb{C}^n$ be an open subset, $\mathcal{J} \subset \mathcal{O}_\Omega$ a coherent sheaf of ideals and $A = V(\mathcal{J})$ the analytic set in Ω defined locally as the zero set of a system of generators of $\mathcal{J}$. By Hilbert's Nullstellensatz 4.22 we have $\mathcal{I}_A = \sqrt{\mathcal{J}}$, but $\mathcal{I}_A$ differs in general from $\mathcal{J}$. The sheaf of rings $\mathcal{O}_\Omega/\mathcal{J}$ is supported on A, i.e. $(\mathcal{O}_\Omega/\mathcal{J})_x = 0$ if $x \notin A$. Ringed spaces of the type $(A, \mathcal{O}_\Omega/\mathcal{J})$ will be used as the local models of analytic schemes.

(9.3) Definition. *A morphism*

$$F = (f, F^\star) : (A, \mathcal{O}_\Omega/\mathcal{J}_{\restriction A}) \longrightarrow (A', \mathcal{O}_{\Omega'}/\mathcal{J}'_{\restriction A'})$$

is said to be analytic if for every point $x \in A$ there exists a neighborhood W_x of x in Ω and a holomorphic function $\Phi : W_x \longrightarrow \Omega'$ such that $f_{\restriction A \cap W_x} = \Phi_{\restriction A \cap W_x}$ and such that the comorphism

$$F_x^\star : (\mathcal{O}_{\Omega'}/\mathcal{J}')_{f(x)} \longrightarrow (\mathcal{O}_\Omega/\mathcal{J})_x$$

is induced by $\Phi^\star : \mathcal{O}_{\Omega',f(x)} \ni u \longmapsto u \circ \Phi \in \mathcal{O}_{\Omega,x}$ with $\Phi^\star \mathcal{J}' \subset \mathcal{J}$.

(9.4) Example. Take $\Omega = \mathbb{C}^n$ and $\mathcal{J} = (z_n^2)$. Then A is the hyperplane $\mathbb{C}^{n-1} \times \{0\}$, and the sheaf $\mathcal{O}_{\mathbb{C}^n}/\mathcal{J}$ can be identified with the sheaf of rings of functions $u + z_n u'$, $u, u' \in \mathcal{O}_{\mathbb{C}^{n-1}}$, with the relation $z_n^2 = 0$. In particular, z_n is a nilpotent element of $\mathcal{O}_{\mathbb{C}^n}/\mathcal{J}$. A morphism F of $(A, \mathcal{O}_{\mathbb{C}^n}/\mathcal{J})$ into itself is induced (at least locally) by a holomorphic map $\Phi = (\widetilde{\Phi}, \Phi_n)$ defined on a neighborhood of A in $\mathbb{C}^n$ with values in $\mathbb{C}^n$, such that $\Phi(A) \subset A$, i.e. $\Phi_{n\restriction A} = 0$. We see that F is completely determined by the data

$$\begin{aligned} f(z_1, \ldots, z_{n-1}) &= \widetilde{\Phi}(z_1, \ldots, z_{n-1}, 0), \quad f &&: \mathbb{C}^{n-1} \longrightarrow \mathbb{C}^{n-1}, \\ f'(z_1, \ldots, z_{n-1}) &= \frac{\partial \Phi}{\partial z_n}(z_1, \ldots, z_{n-1}, 0), \quad f' &&: \mathbb{C}^{n-1} \longrightarrow \mathbb{C}^{n}, \end{aligned}$$

which can be chosen arbitrarily.

(9.5) Definition. *A complex analytic scheme is a ringed space $(X, \mathcal{O}_X)$ over a separable Hausdorff topological space X, satisfying the following property: there exist an open covering (U_λ) of X and isomorphisms of ringed spaces*

$$G_\lambda : (U_\lambda, \mathcal{O}_{X\restriction U_\lambda}) \longrightarrow (A_\lambda, \mathcal{O}_{\Omega_\lambda}/\mathcal{J}_{\lambda\restriction A_\lambda})$$

where A_λ is the zero set of a coherent sheaf of ideals $\mathcal{J}_\lambda$ on an open subset $\Omega_\lambda \subset \mathbb{C}^{N_\lambda}$, such that every transition morphism $G_\lambda \circ G_\mu^{-1}$ is a holomorphic isomorphism from $g_\mu(U_\lambda \cap U_\mu) \subset A_\mu$ onto $g_\lambda(U_\lambda \cap U_\mu) \subset A_\lambda$, equipped with the respective structure sheaves $\mathcal{O}_{\Omega_\mu}/\mathcal{J}_{\mu\restriction A_\mu}$, $\mathcal{O}_{\Omega_\lambda}/\mathcal{J}_{\lambda\restriction A_\lambda}$.

We shall often consider the maps G_λ as identifications and write simply $U_\lambda = A_\lambda$. A morphism $F : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$ of analytic schemes obtained by gluing patches $(A_\lambda, \mathcal{O}_{\Omega_\lambda}/\mathcal{J}_{\lambda\restriction A_\lambda})$ and $(A'_\mu, \mathcal{O}_{\Omega'_\mu}/\mathcal{J}'_{\mu\restriction A'_\mu})$, respectively, is a morphism F of ringed spaces such that for each pair (λ, μ), the restriction of F from $A_\lambda \cap f^{-1}(A'_\mu) \subset X$ to $A'_\mu \subset Y$ is holomorphic in the sense of Def. 9.3.

§ 9.C. Nilpotent Elements and Reduced Schemes

Let $(X, \mathscr{O}_X)$ be an analytic scheme. The set of *nilpotent elements* is the sheaf of ideals of $\mathscr{O}_X$ defined by

$$\mathscr{N}_X = \{u \in \mathscr{O}_X \,;\, u^k = 0 \ \text{ for some } \ k \in \mathbb{N}\}. \tag{9.6}$$

Locally, we have $\mathscr{O}_{X\restriction A_\lambda} = (\mathscr{O}_{\Omega_\lambda}/\mathscr{J}_\lambda)_{\restriction A_\lambda}$, thus

$$\mathscr{N}_{X\restriction A_\lambda} = (\sqrt{\mathscr{J}_\lambda}/\mathscr{J}_\lambda)_{\restriction A_\lambda}, \tag{9.7}$$

$$(\mathscr{O}_X/\mathscr{N}_X)_{\restriction A_\lambda} \simeq (\mathscr{O}_{\Omega_\lambda}/\sqrt{\mathscr{J}_\lambda})_{\restriction A_\lambda} = (\mathscr{O}_{\Omega_\lambda}/\mathscr{I}_{A_\lambda})_{\restriction A_\lambda} = \mathscr{O}_{A_\lambda}. \tag{9.8}$$

The scheme $(X, \mathscr{O}_X)$ is said to be *reduced* if $\mathscr{N}_X = 0$. The associated ringed space $(X, \mathscr{O}_X/\mathscr{N}_X)$ is reduced by construction; it is called the *reduced scheme* of $(X, \mathscr{O}_X)$. We shall often denote the original scheme by the letter X merely, the associated reduced scheme by X_{red}, and let $\mathscr{O}_{X,\text{red}} = \mathscr{O}_X/\mathscr{N}_X$. There is a canonical morphism $X_{\text{red}} \to X$ whose comorphism is the reduction morphism

$$\mathscr{O}_X(U) \longrightarrow \mathscr{O}_{X,\text{red}}(U) = (\mathscr{O}_X/\mathscr{N}_X)(U), \qquad \forall U \text{ open set in } X. \tag{9.9}$$

By (9.8), the notion of reduced scheme is equivalent to the notion of complex space introduced in Def. 5.2. It is easy to see that a morphism F of reduced schemes X, Y is completely determined by the set-theoretic map $f : X \longrightarrow Y$.

§ 9.D. Coherent Sheaves on Analytic Schemes

If $(X, \mathscr{O}_X)$ is an analytic scheme, a sheaf $\mathscr{S}$ of $\mathscr{O}_X$-modules is said to be *coherent* if it satisfies the same properties as those already stated when X is a manifold:

(9.10) $\mathscr{S}$ is locally finitely generated over $\mathscr{O}_X$;

(9.10′) for any open set $U \subset X$ and any sections $G_1, \dots, G_q \in \mathscr{S}(U)$, the relation sheaf $\mathscr{R}(G_1, \dots, G_q) \subset \mathscr{O}_{X\restriction U}^{\oplus q}$ is locally finitely generated.

Locally, we have $\mathscr{O}_{X\restriction A_\lambda} = \mathscr{O}_{\Omega_\lambda}/\mathscr{J}_\lambda$, so if $i_\lambda : A_\lambda \to \Omega_\lambda$ is the injection, the direct image $\mathscr{S}_\lambda = (i_\lambda)_\star(\mathscr{S}_{\restriction A_\lambda})$ is a module over $\mathscr{O}_{\Omega_\lambda}$ such that $\mathscr{J}_\lambda.\mathscr{S}_\lambda = 0$. It is clear that $\mathscr{S}_{\restriction \Omega_\lambda}$ is coherent if and only if $\mathscr{S}_\lambda$ is coherent as a module over $\mathscr{O}_{\Omega_\lambda}$. It follows immediately that the Oka theorem and its consequences 3.19–22 are still valid over analytic schemes.

§ 9.E. Subschemes

Let X be an analytic scheme and $\mathscr{G}$ a coherent sheaf of ideals in $\mathscr{O}_X$. The image of $\mathscr{G}$ in $\mathscr{O}_{X,\text{red}}$ is a coherent sheaf of ideals, and its zero set Y is clearly an analytic subset of X_{red}. We can make Y into a scheme by introducing the structure sheaf

$$\mathscr{O}_Y = (\mathscr{O}_X/\mathscr{G})_{\restriction Y}, \tag{9.11}$$

and we have a scheme morphism $F : (Y, \mathscr{O}_Y) \longrightarrow (X, \mathscr{O}_X)$ such that f is the inclusion and $F^\star : f^{-1}\mathscr{O}_X \longrightarrow \mathscr{O}_Y$ the obvious map of $\mathscr{O}_{X\restriction Y}$ onto its quotient $\mathscr{O}_Y$. The scheme $(Y, \mathscr{O}_Y)$ will be denoted $V(\mathscr{G})$. When the analytic set Y is given, the structure sheaf of $V(\mathscr{G})$ depends of course on the choice of the equations of Y in the ideal $\mathscr{G}$; in general $\mathscr{O}_Y$ has nilpotent elements.

§ 9.F. Inverse Images of Coherent Sheaves

Let $F : (X, \mathscr{O}_X) \longrightarrow (Y, \mathscr{O}_Y)$ be a morphism of analytic schemes and $\mathscr{S}$ a coherent sheaf over Y. The sheaf theoretic inverse image $f^{-1}\mathscr{S}$, whose stalks are $(f^{-1}\mathscr{S})_x = \mathscr{S}_{f(x)}$, is a sheaf of modules over $f^{-1}\mathscr{O}_Y$. We define the *analytic inverse image* $F^\star\mathscr{S}$ by

$$F^\star\mathscr{S} = \mathscr{O}_X \otimes_{f^{-1}\mathscr{O}_Y} f^{-1}\mathscr{S}, \qquad (F^\star\mathscr{S})_x = \mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,f(x)}} \mathscr{S}_{f(x)}. \tag{9.12}$$

Here the tensor product is taken with respect to the comorphism $F^\star : f^{-1}\mathscr{O}_Y \to \mathscr{O}_X$, which yields a ring morphism $\mathscr{O}_{Y,f(x)} \to \mathscr{O}_{X,x}$. If $\mathscr{S}$ is given over $U \subset Y$ by a local presentation

$$\mathscr{O}_{Y\restriction U}^{\oplus p} \xrightarrow{A} \mathscr{O}_{Y\restriction U}^{\oplus q} \longrightarrow \mathscr{S}_{\restriction U} \longrightarrow 0$$

where A is a $(q \times p)$-matrix with coefficients in $\mathscr{O}_Y(U)$, our definition shows that $F^\star\mathscr{S}$ is a coherent sheaf over $\mathscr{O}_X$, given over $f^{-1}(U)$ by the local presentation

$$\mathscr{O}_{X\restriction f^{-1}(U)}^{\oplus p} \xrightarrow{F^\star A} \mathscr{O}_{X\restriction f^{-1}(U)}^{\oplus q} \longrightarrow F^\star\mathscr{S}_{\restriction f^{-1}(U)} \longrightarrow 0. \tag{9.13}$$

§ 9.G. Products of Analytic Schemes

Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be analytic schemes, and let $(A_\lambda, \mathscr{O}_{\Omega_\lambda}/\mathscr{J}_\lambda)$, $(B_\mu, \mathscr{O}_{\Omega'_\mu}/\mathscr{J}'_\mu)$ be local models of X, Y, respectively. The *product scheme* $(X \times Y, \mathscr{O}_{X\times Y})$ is obtained by gluing the open patches

$$\Big(A_\lambda \times B_\mu \ , \ \mathscr{O}_{\Omega_\lambda\times\Omega'_\mu}\big/\big(\mathrm{pr}_1^{-1}\mathscr{J}_\lambda + \mathrm{pr}_2^{-1}\mathscr{J}'_\mu\big)\mathscr{O}_{\Omega_\lambda\times\Omega'_\mu}\Big). \tag{9.14}$$

In other words, if A_λ, B_μ are the subschemes of Ω_λ, Ω'_μ defined by the equations $g_{\lambda,j}(x) = 0$, $g'_{\mu,k}(y) = 0$, where $(g_{\lambda,j})$ and $(g'_{\mu,k})$ are generators of $\mathscr{J}_\lambda$ and $\mathscr{J}'_\mu$ respectively, then $A_\lambda \times B_\mu$ is equipped with the structure sheaf $\mathscr{O}_{\Omega_\lambda\times\Omega'_\mu}\big/\big(g_{\lambda,j}(x), g'_{\mu,k}(y)\big)$.

Now, let $\mathscr{S}$ be a coherent sheaf over $\mathscr{O}_X$ and let $\mathscr{S}'$ be a coherent sheaf over $\mathscr{O}_Y$. The (analytic) *external tensor product* $\mathscr{S} \boxtimes \mathscr{S}'$ is defined to be

$$\mathscr{S} \boxtimes \mathscr{S}' = \mathrm{pr}_1^\star\mathscr{S} \otimes_{\mathscr{O}_{X\times Y}} \mathrm{pr}_2^\star\mathscr{S}'. \tag{9.15}$$

If we go back to the definition of the inverse image, we see that the stalks of $\mathscr{S} \boxtimes \mathscr{S}'$ are given by

$$(\mathscr{S} \boxtimes \mathscr{S}')_{(x,y)} = \mathscr{O}_{X\times Y,(x,y)} \otimes_{\mathscr{O}_{X,x}\otimes\mathscr{O}_{Y,y}} \big(\mathscr{S}_x \otimes_{\mathbb{C}} \mathscr{S}'_y\big) \ , \tag{9.15$'$}$$

in particular $(\mathscr{S} \boxtimes \mathscr{S}')_{(x,y)}$ *does not coincide* with the sheaf theoretic tensor product $\mathscr{S}_x \otimes \mathscr{S}'_y$ which is merely a module over $\mathscr{O}_{X,x} \otimes \mathscr{O}_{Y,y}$. If $\mathscr{S}$ and $\mathscr{S}'$ are given by local presentations

$$\mathscr{O}_{X\restriction U}^{\oplus p} \xrightarrow{A} \mathscr{O}_{X\restriction U}^{\oplus q} \longrightarrow \mathscr{S}_{\restriction U} \longrightarrow 0, \qquad \mathscr{O}_{Y\restriction U'}^{p'} \xrightarrow{B} \mathscr{O}_{Y\restriction U'}^{q'} \longrightarrow \mathscr{S}'_{\restriction U'} \longrightarrow 0,$$

then $\mathscr{S} \boxtimes \mathscr{S}'$ is the coherent sheaf given by

$$\mathscr{O}_{X\times Y\restriction U\times U'}^{pq'\oplus qp'} \xrightarrow{(A(x)\otimes\mathrm{Id},\mathrm{Id}\otimes B(y))} \mathscr{O}_{X\times Y\restriction U\times U'}^{qq'} \longrightarrow (\mathscr{S} \boxtimes \mathscr{S}')_{\restriction U\times U'} \longrightarrow 0.$$

§ 9.H. Zariski Embedding Dimension

If x is a point of an analytic scheme $(X, \mathscr{O}_X)$, the *Zariski embedding dimension* of the germ (X, x) is the smallest integer N such that (X, x) can be embedded in $\mathbb{C}^N$, i.e. such that there exists a patch of X near x isomorphic to $(A, \mathscr{O}_\Omega/\mathscr{J})$ where Ω is an open subset of $\mathbb{C}^N$. This dimension is denoted

$$\text{embdim}(X, x) = \text{ smallest such } N. \tag{9.16}$$

Consider the maximal ideal $\mathfrak{m}_{X,x} \subset \mathscr{O}_{X,x}$ of functions which vanish at point x. If (X, x) is embedded in $(\Omega, x) = (\mathbb{C}^N, 0)$, then $\mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2$ is generated by $z_1, \ldots, z_N$, so $d = \dim \mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2 \leqslant N$. Let $s_1, \ldots, s_d$ be germs in $\mathfrak{m}_{\Omega,x}$ which yield a basis of $\mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2 \simeq \mathfrak{m}_{\Omega,x}/(\mathfrak{m}_{\Omega,x}^2 + \mathscr{J}_x)$. We can write

$$z_j = \sum_{1 \leqslant k \leqslant d} c_{jk} s_k + u_j + f_j, \quad c_{jk} \in \mathbb{C}, \quad u_j \in \mathfrak{m}_{\Omega,x}^2, \quad f_j \in \mathscr{J}_x, \quad 1 \leqslant j \leqslant n.$$

Then we find $dz_j = \sum c_{jk}\, ds_k(x) + df_j(x)$, so that the rank of the system of differentials $\big(df_j(x)\big)$ is at least $N - d$. Assume for example that $df_1(x), \ldots, df_{N-d}(x)$ are linearly independant. By the implicit function theorem, the equations $f_1 = \ldots = f_{N-d} = 0$ define a germ of smooth subvariety $(Z, x) \subset (\Omega, x)$ of dimension d which contains (X, x). We have $\mathscr{O}_Z = \mathscr{O}_\Omega/(f_1, \ldots, f_{N-d})$ in a neighborhood of x, thus

$$\mathscr{O}_X = \mathscr{O}_\Omega/\mathscr{J} \simeq \mathscr{O}_Z/\mathscr{J}' \quad \text{ where } \mathscr{J}' = \mathscr{J}/(f_1, \ldots, f_{N-d}).$$

This shows that (X, x) can be imbedded in $\mathbb{C}^d$, and we get

$$\text{embdim}(X, x) = \dim \mathfrak{m}_{X,x}/\mathfrak{m}_{X,x}^2. \tag{9.17}$$

(9.18) Remark. For a given dimension $n = \dim(X, x)$, the embedding dimension d can be arbitrarily large. Consider for example the curve $\Gamma \subset \mathbb{C}^N$ parametrized by $\mathbb{C} \ni t \longmapsto (t^N, t^{N+1}, \ldots, t^{2N-1})$. Then $\mathscr{O}_{\Gamma,0}$ is the ring of convergent series in $\mathbb{C}\{t\}$ which have no terms $t, t^2, \ldots, t^{N-1}$, and $\mathfrak{m}_{\Gamma,0}/\mathfrak{m}_{\Gamma,0}^2$ admits precisely $z_1 = t^N, \ldots, z_N = t^{2N-1}$ as a basis. Therefore $n = 1$ but $d = N$ can be as large as we want.

§ 10. Bimeromorphic maps, Modifications and Blow-ups

It is a very frequent situation in analytic or algebraic geometry that two complex spaces have isomorphic dense open subsets but are nevertheless different along some analytic subset. These ideas are made precise by the notions of modification and bimeromorphic map. This will also lead us to generalize the notion of meromorphic function to maps between analytic schemes. If $(X, \mathscr{O}_X)$ is an analytic scheme, $\mathscr{M}_X$ denotes the sheaf of meromorphic functions on X, defined at the beginning of § 6.B.

(10.1) Definition. *Let $(X, \mathscr{O}_X)$, $(Y, \mathscr{O}_Y)$ be analytic schemes. An analytic morphism $F : X \to Y$ is said to be a modification if F is proper and if there exists a nowhere dense closed analytic subset $B \subset Y$ such that the restriction $F : X \smallsetminus F^{-1}(B) \to Y \smallsetminus B$ is an isomorphism.*

(10.2) Proposition. *If $F : X \to Y$ is a modification, then the comorphism $F^\star : f^\star \mathscr{O}_Y \to \mathscr{O}_X$ induces an isomorphism $F^\star : f^\star \mathscr{M}_Y \to \mathscr{M}_X$ for the sheaves of meromorphic functions on X and Y.*

Proof. Let $v = g/h$ be a section of $\mathscr{M}_Y$ on a small open set Ω where u is actually given as a quotient of functions $g, h \in \mathscr{O}_Y(\Omega)$. Then $F^\star u = (g \circ F)/(h \circ F)$ is a section of $\mathscr{M}_X$ on $F^{-1}(\Omega)$, for $h \circ F$ cannot vanish identically on any open subset W of $F^{-1}(\Omega)$ (otherwise h would vanish on the open subset $F(W \smallsetminus F^{-1}(B))$ of $\Omega \smallsetminus B$). Thus the extension of the comorphism to sheaves of meromorphic functions is well defined. Our claim is that this is an isomorphism. The injectivity of $F^\star$ is clear: $F^\star u = 0$ implies $g \circ F = 0$, which implies $g = 0$ on $\Omega \smallsetminus B$ and thus $g = 0$ on Ω because B is nowhere dense. In order to prove surjectivity, we need only show that every section $u \in \mathscr{O}_X(F^{-1}(\Omega))$ is in the image of $\mathscr{M}_Y(\Omega)$ by $F^\star$. For this, we may shrink Ω into a relatively compact subset $\Omega' \subset\subset \Omega$ and thus assume that u is bounded (here we use the properness of F through the fact that $F^{-1}(\Omega')$ is relatively compact in $F^{-1}(\Omega)$). Then $v = u \circ F^{-1}$ defines a bounded holomorphic function on $\Omega \smallsetminus B$. By Cor. I.5.25 and Def. 7.1, it follows that v is weakly holomorphic for the reduced structure of Y. Our claim now follows from the following Lemma. □

(10.3) Lemma. *If $(Y, \mathscr{O}_Y)$ is an analytic scheme, then every holomorphic function v in the complement of a nowhere dense analytic subset $B \subset Y$ which is weakly holomorphic on Y_{red} is meromorphic on Y.*

Proof. It is enough to argue with the germ of v at any point $x \in Y$, and thus we may suppose that $(Y, \mathscr{O}_Y) = (A, \mathscr{O}_\Omega/\mathscr{I})$ is embedded in $\mathbb{C}^N$. Because v is weakly holomorphic, we can write $v = g/h$ in Y_{red}, for some germs of holomorphic functions g, h. Let $\widetilde{g}$ and $\widetilde{h}$ be extensions of g, h to $\mathscr{O}_{\Omega,x}$. Then there is a neighborhood U of x such that $\widetilde{g} - v\widetilde{h}$ is a nilpotent section of $\mathscr{O}_\Omega(U \smallsetminus B)$ which is in $\mathscr{I}$ on

(10.4) Definition. *A meromorphic map $F : X \dashrightarrow Y$ is a scheme morphism $F : X \smallsetminus A \to Y$ defined in the complement of a nowhere dense analytic subset $A \subset X$, such that the closure of the graph of F in $X \times Y$ is an analytic subset (for the reduced complex space structure of $X \times Y$).*

§ 11. Exercises

§ 11.1 Let $\mathscr{A}$ be a sheaf on a topological space X. If the sheaf space $\widetilde{\mathscr{A}}$ is Hausdorff, show that $\mathscr{A}$ satisfies the following *unique continuation principle*: any two sections $s, s' \in \mathscr{A}(U)$ on a connected open set U which coincide on some non empty open subset $V \subset U$ must coincide identically on U. Show that the converse holds if X is Hausdorff and locally connected.

§ 11.2 Let $\mathscr{A}$ be a sheaf of abelian groups on X and let $s \in \mathscr{A}(X)$. The support of s, denoted Supp s, is defined to be $\{x \in X\,;\, s(x) \neq 0\}$. Show that Supp s is a closed subset of X. The support of $\mathscr{A}$ is defined to be Supp $\mathscr{A} = \{x \in X\,;\, \mathscr{A}_x \neq 0\}$. Show that Supp $\mathscr{A}$ is not necessarily closed: if Ω is an open set in X, consider the sheaf $\mathscr{A}$ such that $\mathscr{A}(U)$ is the set of continuous functions $f \in \mathscr{C}(U)$ which vanish on a neighborhood of $U \cap (X \smallsetminus \Omega)$.

§ 11.3 Let $\mathscr{A}$ be a sheaf of rings on a topological space X and let $\mathscr{F}$, $\mathscr{G}$ be sheaves of $\mathscr{A}$-modules. We define a presheaf $\mathscr{H} = \mathscr{H}om_{\mathscr{A}}(\mathscr{F}, \mathscr{G})$ such that $\mathscr{H}(U)$ is the module of all sheaf-homomorphisms $\mathscr{F}_{\restriction U} \to \mathscr{G}_{\restriction U}$ which are $\mathscr{A}$-linear.

a) Show that $\mathcal{H}om_{\mathcal{A}}(\mathcal{F},\mathcal{G})$ is a sheaf and that there exists a canonical homomorphism $\varphi_x : \mathcal{H}om_{\mathcal{A}}(\mathcal{F},\mathcal{G})_x \longrightarrow \hom_{\mathcal{A}_x}(\mathcal{F}_x,\mathcal{G}_x)$ for every $x \in X$.

b) If $\mathcal{F}$ is locally finitely generated, then φ_x is injective, and if $\mathcal{F}$ has local finite presentations as in (3.12), then φ_x is bijective.

c) Suppose that $\mathcal{A}$ is a coherent sheaf of rings and that $\mathcal{F}$, $\mathcal{G}$ are coherent modules over $\mathcal{A}$. Then $\mathcal{H}om_{\mathcal{A}}(\mathcal{F},\mathcal{G})$ is a coherent $\mathcal{A}$-module.

Hint: observe that the result is true if $\mathcal{F} = \mathcal{A}^{\oplus p}$ and use a local presentation of $\mathcal{F}$ to get the conclusion.

§ 11.4 Let $f : X \to Y$ be a continuous map of topological spaces. Given sheaves of abelian groups $\mathcal{A}$ on X and $\mathcal{B}$ on Y, show that there is a natural isomorphism

$$\hom_X(f^{-1}\mathcal{B},\mathcal{A}) = \hom_Y(\mathcal{B}, f_\star\mathcal{A}).$$

Hint: use the natural morphisms (1.17).

§ 11.5 Show that the sheaf of polynomials over $\mathbb{C}^n$ is a coherent sheaf of rings (with either the ordinary topology or the Zariski topology on $\mathbb{C}^n$). Extend this result to the case of regular algebraic functions on an algebraic variety.
Hint: check that the proof of the Oka theorem still applies.

§ 11.6 Let P be a non zero polynomial on $\mathbb{C}^n$. If P is irreducible in $\mathbb{C}[z_1,\dots,z_n]$, show that the hypersurface $H = P^{-1}(0)$ is globally irreducible as an analytic set. In general, show that the irreducible components of H are in a one-to-one correspondence with the irreducible factors of P.
Hint: for the first part, take coordinates such that $P(0,\dots,0,z_n)$ has degree equal to P; if H splits in two components H_1, H_2, then P can be written as a product P_1P_2 where the roots of $P_j(z',z_n)$ correspond to points in H_j.

§ 11.7 Prove the following facts:

a) For every algebraic variety A of pure dimension p in $\mathbb{C}^n$, there are coordinates $z' = (z_1,\dots,z_p)$, $z'' = (z_{p+1},\dots,z_n)$ such that $\pi : A \to \mathbb{C}^p$, $z \mapsto z''$ is proper with finite fibers, and such that A is entirely contained in a cone

$$|z''| \leqslant C(|z'|+1).$$

Hint: imitate the proof of Cor. 4.11.

b) Conversely if an analytic set A of pure dimension p in $\mathbb{C}^n$ is contained in a cone $|z''| \leqslant C(|z'|+1)$, then A is algebraic.
Hint: first apply (5.9) to conclude that the projection $\pi : A \to \mathbb{C}^p$ is finite. Then repeat the arguments used in the final part of the proof of Th. 4.19.

c) Deduce from a), b) that an algebraic set in $\mathbb{C}^n$ is irreducible if and only if it is irreducible as an analytic set.

§ 11.8 Let $\Gamma : f(x,y) = 0$ be a germ of analytic curve in $\mathbb{C}^2$ through $(0,0)$ and let $(\Gamma_j,0)$ be the irreducible components of $(\Gamma,0)$. Suppose that $f(0,y) \not\equiv 0$. Show that the roots y of $f(x,y) = 0$ corresponding to points of Γ near 0 are given by *Puiseux expansions* of the form $y = g_j(x^{1/q_j})$, where $g_j \in \mathcal{O}_{\mathbb{C},0}$ and where q_j is the sheet number of the projection $\Gamma_j \to \mathbb{C}$, $(x,y) \mapsto x$.
Hint: special case of the parametrization obtained in (4.27).

§ 11.9 The goal of this exercise is to prove the existence and the analyticity of the *tangent cone* to an arbitrary analytic germ $(A,0)$ in $\mathbb{C}^n$. Suppose that A is defined by holomorphic equations $f_1 = \dots = f_N = 0$ in a ball $\Omega = B(0,r)$. Then the (set theoretic) tangent cone to A at 0 is the set $C(A,0)$ of all limits of sequences $t_\nu^{-1}z_\nu$ with $z_\nu \in A$ and $\mathbb{C}^\star \ni t_\nu$ converging to 0.

a) Let E be the set of points $(z,t) \in \Omega \times \mathbb{C}^\star$ such that $z \in t^{-1}A$. Show that the closure $\overline{E}$ in $\Omega \times \mathbb{C}$ is analytic.
Hint: observe that $E = A \smallsetminus (\Omega \times \{0\})$ where $A = \{f_j(tz) = 0\}$ and apply Cor. 5.4.

b) Show that $C(A,0)$ is a conic set and that $\overline{E} \cap (\Omega \times \{0\}) = C(A,0) \times \{0\}$ and conclude. Infer from this that $C(A,0)$ is an algebraic subset of $\mathbb{C}^n$.

§ 11.10 Give a new proof of Theorem 5.5 based on the coherence of ideal sheaves and on the strong noetherian property.

§ 11.11 Let X be an analytic space and let A, B be analytic subsets of pure dimensions. Show that $\operatorname{codim}_X(A \cap B) \leqslant \operatorname{codim}_X A + \operatorname{codim}_X B$ if A or B is a local complete intersection, but that the equality does not necessarily hold in general.
Hint: see Remark 6.5.

§ 11.12 Let Γ be the curve in $\mathbb{C}^3$ parametrized by $\mathbb{C} \ni t \longmapsto (x, y, z) = (t^3, t^4, t^5)$. Show that the ideal sheaf $\mathscr{I}_\Gamma$ is generated by the polynomials $xz - y^2$, $x^3 - yz$ and $x^2y - z^2$, and that the germ $(\Gamma, 0)$ is not a (sheaf theoretic) complete intersection.
Hint: Γ is smooth except at the origin. Let $f(x, y, z) = \sum a_{\alpha\beta\gamma} x^\alpha y^\beta z^\gamma$ be a convergent power series near 0. Show that $f \in \mathscr{I}_{\Gamma,0}$ if and only if all weighted homogeneous components

$$f_k = \sum_{3\alpha+4\beta+5\gamma=k} a_{\alpha\beta\gamma} x^\alpha y^\beta z^\gamma$$

are in $\mathscr{I}_{\Gamma,0}$. By means of suitable substitutions, reduce the proof to the case when $f = f_k$ is homogeneous with all non zero monomials satisfying $\alpha \leqslant 2$, $\beta \leqslant 1$, $\gamma \leqslant 1$; then check that there is at most one such monomial in each weighted degree $\leqslant 15$ the product of a power of x by a homogeneous polynomial of weighted degree $\leqslant 8$.

Chapter III

Positive Currents and Lelong Numbers

Chapter IV

Sheaf Cohomology and Spectral Sequences

Chapter V

Vector Bundles

Chapter VI

Hodge Theory

The goal of this chapter is to prove a number of basic facts in the Hodge theory of real or complex manifolds. The theory rests essentially on the fact that the De Rham (or Dolbeault) cohomology groups of a compact manifold can be represented by means of spaces of harmonic forms, once a Riemannian metric has been chosen. At this point, some knowledge of basic results about elliptic differential operators is required. The special properties of compact Kähler manifolds are then investigated in detail: Hodge decomposition theorem, hard Lefschetz theorem, Jacobian and Albanese variety, ...; the example of curves is treated in detail. Finally, the Hodge-Frölicher spectral sequence is applied to get some results on general compact complex manifolds, and it is shown that Hodge decomposition still holds for manifolds in the Fujiki class ($\mathscr{C}$).

§ 1. Differential Operators on Vector Bundles

We first describe some basic concepts concerning differential operators (symbol, composition, adjunction, ellipticity), in the general setting of vector bundles. Let M be a $\mathscr{C}^\infty$ differentiable manifold, $\dim_{\mathbb{R}} M = m$, and let E, F be $\mathbb{K}$-vector bundles over M, with $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$, $\operatorname{rank} E = r$, $\operatorname{rank} F = r'$.

(1.1) Definition. *A (linear) differential operator of degree δ from E to F is a $\mathbb{K}$-linear operator $P : \mathscr{C}^\infty(M, E) \to \mathscr{C}^\infty(M, F)$, $u \mapsto Pu$ of the form*

$$Pu(x) = \sum_{|\alpha| \leqslant \delta} a_\alpha(x) D^\alpha u(x),$$

where $E_{\restriction\Omega} \simeq \Omega \times \mathbb{K}^r$, $F_{\restriction\Omega} \simeq \Omega \times \mathbb{K}^{r'}$ are trivialized locally on some open chart $\Omega \subset M$ equipped with local coordinates $(x_1, \ldots, x_m)$, and where $a_\alpha(x) = \big(a_{\alpha\lambda\mu}(x)\big)_{1 \leqslant \lambda \leqslant r', 1 \leqslant \mu \leqslant r}$ are $r' \times r$-matrices with $\mathscr{C}^\infty$ coefficients on Ω. Here $D^\alpha = (\partial/\partial x_1)^{\alpha_1} \ldots (\partial/\partial x_m)^{\alpha_m}$ as usual, and $u = (u_\mu)_{1 \leqslant \mu \leqslant r}$, $D^\alpha u = (D^\alpha u_\mu)_{1 \leqslant \mu \leqslant r}$ are viewed as column matrices.

If $t \in \mathbb{K}$ is a parameter and $f \in C^\infty(M, \mathbb{K})$, $u \in C^\infty(M, E)$, a simple calculation shows that $e^{-tf(x)} P(e^{tf(x)} u(x))$ is a polynomial of degree δ in t, of the form

$$e^{-tf(x)} P(e^{tf(x)}) = t^\delta \sigma_P(x, df(x)) \cdot u(x) + \text{lower order terms } c_j(x) t^j,\ j < \delta,$$

where σ_P is the polynomial map from $T^\star_M \to \operatorname{Hom}(E, F)$ defined by

$$T^\star_{M,x} \ni \xi \mapsto \sigma_P(x, \xi) \in \operatorname{Hom}(E_x, F_x), \qquad \sigma_P(x, \xi) = \sum_{|\alpha| = \delta} a_\alpha(x) \xi^\alpha. \tag{1.2}$$

The formula involving $e^{-tf}P(e^{tf}u)$ shows that $\sigma_P(x,\xi)$ actually does not depend on the choice of coordinates nor on the trivializations used for E, F. It is clear that $\sigma_P(x,\xi)$ is smooth on $T^\star_M$ as a function of (x,ξ), and is a homogeneous polynomial of degree δ in ξ. We say that σ_P is *the principal symbol* of P. Now, if E, F, G are vector bundles and

$$P : \mathscr{C}^\infty(M,E) \to \mathscr{C}^\infty(M,F), \qquad Q : \mathscr{C}^\infty(M,F) \to \mathscr{C}^\infty(M,G)$$

are differential operators of respective degrees δ_P, δ_Q, it is easy to check that $Q \circ P : \mathscr{C}^\infty(M,E) \to \mathscr{C}^\infty(M,G)$ is a differential operator of degree $\delta_P + \delta_Q$ and that

$$\sigma_{Q\circ P}(x,\xi) = \sigma_Q(x,\xi)\sigma_P(x,\xi). \tag{1.3}$$

Here the product of symbols is computed as a product of matrices.

Now, assume that M is oriented and is equipped with a smooth volume form $dV(x) = \gamma(x)dx_1 \wedge \ldots dx_m$, where $\gamma(x) > 0$ is a smooth density. If E is a euclidean or hermitian vector bundle, we have a Hilbert space $L^2(M,E)$ of global sections u of E with measurable coefficients, satisfying the L^2 estimate

$$\|u\|^2 = \int_M |u(x)|^2\, dV(x) < +\infty. \tag{1.4}$$

We denote by

$$\langle\!\langle u,v\rangle\!\rangle = \int_M \langle u(x), v(x)\rangle\, dV(x), \qquad u,v \in L^2(M,E) \tag{1.4$'$}$$

the corresponding L^2 inner product.

(1.5) Definition. *If $P : \mathscr{C}^\infty(M,E) \to \mathscr{C}^\infty(M,F)$ is a differential operator and both E, F are euclidean or hermitian, there exists a unique differential operator*

$$P^\star : \mathscr{C}^\infty(M,F) \to \mathscr{C}^\infty(M,E),$$

called the formal adjoint of P, such that for all sections $u \in \mathscr{C}^\infty(M,E)$ and $v \in \mathscr{C}^\infty(M,F)$ there is an identity

$$\langle\!\langle Pu,v\rangle\!\rangle = \langle\!\langle u,P^\star v\rangle\!\rangle, \qquad \text{whenever } \operatorname{Supp} u \cap \operatorname{Supp} v \subset\subset M.$$

Proof. The uniqueness is easy, using the density of the set of elements $u \in \mathscr{C}^\infty(M,E)$ with compact support in $L^2(M,E)$. Since uniqueness is clear, it is enough, by a partition of unity argument, to show the existence of $P^\star$ locally. Now, let $Pu(x) = \sum_{|\alpha|\leqslant\delta} a_\alpha(x)D^\alpha u(x)$ be the expansion of P with respect to trivializations of E, F given by orthonormal frames over some coordinate open set $\Omega \subset M$. Assuming $\operatorname{Supp} u \cap \operatorname{Supp} v \subset\subset \Omega$, an integration by parts yields

$$\begin{aligned}
\langle\!\langle Pu,v\rangle\!\rangle &= \int_\Omega \sum_{|\alpha|\leqslant\delta,\lambda,\mu} a_{\alpha\lambda\mu} D^\alpha u_\mu(x)\overline{v}_\lambda(x)\,\gamma(x)\,dx_1,\ldots,dx_m \\
&= \int_\Omega \sum_{|\alpha|\leqslant\delta,\lambda,\mu} (-1)^{|\alpha|} u_\mu(x)\overline{D^\alpha(\gamma(x)\,\overline{a}_{\alpha\lambda\mu}v_\lambda(x)}\,dx_1,\ldots,dx_m \\
&= \int_\Omega \langle u, \sum_{|\alpha|\leqslant\delta} (-1)^{|\alpha|}\gamma(x)^{-1}D^\alpha\big(\gamma(x)\,{}^t\overline{a}_\alpha v(x)\big)\rangle\, dV(x).
\end{aligned}$$

Hence we see that $P^\star$ exists and is uniquely defined by

$$P^\star v(x) = \sum_{|\alpha|\leqslant\delta} (-1)^{|\alpha|}\gamma(x)^{-1} D^\alpha\big(\gamma(x)\,{}^t\overline{a}_\alpha v(x)\big). \tag{1.6}$$ □

It follows immediately from (1.6) that the principal symbol of $P^\star$ is

$$\sigma_{P^\star}(x,\xi) = (-1)^\delta \sum_{|\alpha|=\delta} {}^t\overline{a}_\alpha \xi^\alpha = (-1)^\delta \sigma_P(x,\xi)^\star. \tag{1.7}$$

(1.8) Definition. *A differential operator P is said to be elliptic if*

$$\sigma_P(x,\xi) \in \mathrm{Hom}(E_x, F_x)$$

is injective for every $x \in M$ and $\xi \in T^\star_{M,x} \smallsetminus \{0\}$.

§ 2. Formalism of PseudoDifferential Operators

We assume throughout this section that (M,g) is a compact Riemannian manifold. For any positive integer k and any hermitian bundle $F \to M$, we denote by $W^k(M,F)$ the Sobolev space of sections $s : M \to F$ whose derivatives up to order k are in L^2. Let $\|\ \|_k$ be the norm of the Hilbert space $W^k(M,F)$. Let P be an elliptic differential operator of order d acting on $\mathscr{C}^\infty(M,F)$. We need the following basic facts of elliptic *PDE* theory, see e.g. [Hörmander 1963] or [Demailly 2019].

(2.1) Sobolev lemma. *For $k > l + \frac{m}{2}$, $W^k(M,F) \subset C^l(M,F)$.*

(2.2) Rellich lemma. *For every integer k, the inclusion*

$$W^{k+1}(M,F) \hookrightarrow W^k(M,F)$$

is a compact linear operator.

(2.3) Gårding's inequality. *Let $\widetilde{P}$ be the extension of P to sections with distribution coefficients. For any $u \in W^0(M,F)$ such that $\widetilde{P}u \in W^k(M,F)$, then $u \in W^{k+d}(M,F)$ and*

$$\|u\|_{k+d} \leqslant C_k(\|\widetilde{P}u\|_k + \|u\|_0),$$

where C_k is a positive constant depending only on k.

(2.4) Corollary. *The operator $P : \mathscr{C}^\infty(M,F) \to \mathscr{C}^\infty(M,F)$ has the following properties:*

i) $\ker P$ *is finite dimensional.*

ii) $P\big(\mathscr{C}^\infty(M,F)\big)$ *is closed and of finite codimension; furthermore, if $P^\star$ is the formal adjoint of P, there is a decomposition*

$$\mathscr{C}^\infty(M,F) = P\big(\mathscr{C}^\infty(M,F)\big) \oplus \ker P^\star$$

as an orthogonal direct sum in $W^0(M,F) = L^2(M,F)$.

Proof. (i) Gårding's inequality shows that $\|u\|_{k+d} \leqslant C_k \|u\|_0$ for any u in $\ker P$. Thanks to the Sobolev lemma, this implies that $\ker P$ is closed in $W^0(M,F)$. Moreover, the unit closed $\| \ \|_0$-ball of $\ker P$ is contained in the $\| \ \|_d$-ball of radius C_0, thus compact by the Rellich lemma. Riesz' theorem implies that $\dim \ker P < +\infty$.

(ii) We first show that the extension

$$\widetilde{P} : W^{k+d}(M,F) \to W^k(M,F)$$

has a closed range for any k. For every $\varepsilon > 0$, there exists a finite number of elements $v_1, \ldots, v_N \in W^{k+d}(M,F)$, $N = N(\varepsilon)$, such that

$$\|u\|_0 \leqslant \varepsilon \|u\|_{k+d} + \sum_{j=1}^{N} |\langle\!\langle u, v_j \rangle\!\rangle_0| \ ; \tag{2.5}$$

indeed the set

$$K_{(v_j)} = \Big\{ u \in W^{k+d}(M,F) \ ; \ \varepsilon \|u\|_{k+d} + \sum_{j=1}^{N} |\langle\!\langle u, v_j \rangle\!\rangle_0| \leqslant 1 \Big\}$$

is relatively compact in $W^0(M,F)$ and $\bigcap_{(v_j)} \overline{K}_{(v_j)} = \{0\}$. It follows that there exist elements (v_j) such that $\overline{K}_{(v_j)}$ is contained in the unit ball of $W^0(M,F)$, *QED*. Substitute $||u||_0$ by the upper bound (2.5) in Gårding's inequality; we get

$$(1 - C_k \varepsilon) \|u\|_{k+d} \leqslant C_k \Big(\|\widetilde{P} u\|_k + \sum_{j=1}^{N} |\langle\!\langle u, v_j \rangle\!\rangle_0| \Big).$$

Define $G = \big\{ u \in W^{k+d}(M,F) \ ; \ u \perp v_j, \ 1 \leqslant j \leqslant n \big\}$ and choose $\varepsilon = 1/2C_k$. We obtain

$$\|u\|_{k+d} \leqslant 2C_k \|\widetilde{P} u\|_k, \quad \forall u \in G.$$

This implies that $\widetilde{P}(G)$ is closed. Therefore

$$\widetilde{P}\big(W^{k+d}(M,F)\big) = \widetilde{P}(G) + \mathrm{Vect}\big(\widetilde{P}(v_1), \ldots, \widetilde{P}(v_N)\big)$$

is closed in $W^k(M,F)$. Take in particular $k = 0$. Since $\mathscr{C}^\infty(M,F)$ is dense in $W^d(M,F)$, we see that in $W^0(M,F)$

$$\Big(\widetilde{P}\big(W^d(M,F)\big)\Big)^\perp = \Big(P\big(\mathscr{C}^\infty(M,F)\big)\Big)^\perp = \ker \widetilde{P^\star}.$$

We have proved that

$$W^0(M,F) = \widetilde{P}\big(W^d(M,F)\big) \oplus \ker \widetilde{P^\star}. \tag{2.6}$$

Since $P^\star$ is also elliptic, it follows that $\ker \widetilde{P^\star}$ is finite dimensional and that $\ker \widetilde{P^\star} = \ker P^\star$ is contained in $\mathscr{C}^\infty(M,F)$. Thanks to Gårding's inequality, the decomposition formula (2.6) yields

$$W^k(M,F) = \widetilde{P}\big(W^{k+d}(M,F)\big) \oplus \ker P^\star, \tag{2.7}$$

$$\mathscr{C}^\infty(M,F) = P\big(\mathscr{C}^\infty(M,F)\big) \oplus \ker P^\star. \tag{2.8}$$

Chapter VII

Positive Vector Bundles and Vanishing Theorems

Chapter VIII

L^2 Estimates on Pseudoconvex Manifolds

Chapter IX

Finiteness Theorems for q-Convex Spaces and Stein Spaces

References

Abdelkader, O. [**1980**] — *Annulation de la cohomologie d'une variété kählérienne faiblement* 1*-complète à valeurs dans un fibré vectoriel holomorphe semi-positif*, C. R. Acad. Sci. Paris **290** (1980) 75–78 ; Thèse de 3e cycle Université Paris VI (1980).

Akizuki, Y., Nakano, S. [**1954**] — *Note on Kodaira-Spencer's proof of Lefschetz theorems*, Proc. Jap. Acad. **30** (1954) 266–272.

Andreotti, A., Grauert, H. [**1962**] — *Théorèmes de finitude pour la cohomologie des espaces complexes*, Bull. Soc. Math. France **90** (1962) 193–259.

Andreotti, A., Vesentini, E. [**1965**] — *Carleman estimates for the Laplace-Beltrami equation in complex manifolds*, Publ. Math. I.H.E.S. **25** (1965) 81–130.

Aronszajn, N. [**1957**] — *A unique continuation theorem for solutions of elliptic partial differential equations or inequalities of second order*, J. Math. Pures Appl. **36** (1957) 235–249.

Atiyah, M.F., Bott, R., Patodi, V.K. [**1973**] — *On the heat equation and the index theorem*, Invent. Math. **19** (1973) 279–330.

Atiyah, M.F., Singer, I.M. [**1968**] — *The index of elliptic operators III, IV*, Ann. of Math. **87** (1968) 546–604 ; **93** (1971) 119–138.

Avron, J., Herbst, I., Simon, B. [**1978**] — *Schrödinger operators with magnetic fields I*, Duke Math. J. **45** (1978) 847–883.

Azhari, A. [**1990**] — *Sur la conjecture de Chudnovsky-Demailly et les singularités des hypersurfaces algébriques*, Ann. Inst. Fourier (Grenoble) **40** (1990) 106–117.

Bedford, E., Taylor, B.A. [**1976**] — *The Dirichlet problem for a complex Monge-Ampère equation*, Invent. Math. **37** (1976) 1–44.

Bedford, E., Taylor, B.A. [**1982**] — *A new capacity for plurisubharmonic functions*, Acta Math. **149** (1982) 1–41.

Bismut, J.M. [**1987**] — *Demailly's asymptotic inequalities: a heat equation proof*, J. Funct. Anal. **72** (1987) 263–278.

Bochner, S. [**1948**] — *Curvature and Betti numbers* (I) *and* (II), Ann. of Math. **49** (1948) 379–390 ; **50** (1949) 77–93.

Bombieri, E. [**1970**] — *Algebraic values of meromorphic maps*, Invent. Math. **10** (1970) 267–287 ; Addendum, Invent. Math. **11** (1970) 163–166.

Borel, A., Serre, J.P. [**1958**] — *Le théorème de Riemann-Roch*, Bull. Soc. Math. France **86** (1958) 97–136.

Borel, A., Weil, A. [**1954**] — *Représentations linéaires et espaces homogènes kählériens des groupes de Lie compacts*, Séminaire Bourbaki exposé no **100** par J.-P. Serre, (mai 1954), 8p.

Bott, R. [**1957**] — *Homogeneous vector bundles*, Ann. of Math. **66** (1957) 203–248.

Bott, R., Chern S.S. [1965] — *Hermitian vector bundles and the equidistribution of the zeroes of their holomorphic sections*, Acta Math. **114** (1965) 71–112.

Bremermann, H. [1954] — *Über die Äquivalenz der pseudokonvexen Gebiete und der Holomorphiegebiete im Raum von n komplexen Veränderlichen*, Math. Ann. **128** (1954) 63–91.

Briançon, J., Skoda, H. [1974] — *Sur la clôture intégrale d'un idéal de germes de fonctions holomorphes en un point de* $\mathbb{C}^n$, C. R. Acad. Sci. série A **278** (1974) 949–951.

Cartan, H. [1954] — *Séminaire de l'École Normale Supérieure 1950/54*, W.A. Benjamin, New York (1967).

Cartan, H., Serre, J.P. [1953] — *Un théorème de finitude concernant les variétés analytiques compactes*, C. R. Acad. Sci. Paris **237** (1953) 128–130.

Chern, S.S. [1946] — *Characteristic classes of hermitian manifolds*, Ann. of Math. **71** (1946) 85–121.

Chern, S.S., Levine, H.I., Nirenberg, L. [1969] — *Intrinsic norms on a complex manifold*, Global Analysis (papers in honor of K. Kodaira), Univ. of Tokyo Press, Tokyo (1969) 119–139.

Chow, W.L. [1949] — *On compact complex analytic varieties*, Amer. J. Math. **71** (1949) 893–914.

Chudnovsky, G.V. [1979] — *Singular points on complex hypersurfaces and multidimensional Schwarz lemma*, Sém. Delange-Pisot-Poitou, année 1979/80, Progress in Math. no **12**, Birkhäuser (1981) 29–69.

Deligne, P., Griffiths, P.A., Morgan, J., Sullivan, D. [1975] — *Real homotopy theory of Kähler manifolds*, Invent. Math. **29** (1975) 245–274.

Demailly, J.P. [1982a] — *Formules de Jensen en plusieurs variables et applications arithmétiques*, Bull. Soc. Math. France **110** (1982) 75–102.

Demailly, J.P. [1982b] — *Sur les nombres de Lelong associés à l'image directe d'un courant positif fermé*, Ann. Inst. Fourier (Grenoble) **32** (1982) 37–66.

Demailly, J.P. [1982c] — *Estimations* L^2 *pour l'opérateur* $\overline{\partial}$ *d'un fibré vectoriel holomorphe semi-positif au dessus d'une variété kählérienne complète*, Ann. Sci. École Norm. Sup. **15** (1982) 457–511.

Demailly, J.P. [1984] — *Sur l'identité de Bochner-Kodaira-Nakano en géométrie hermitienne*, Séminaire P. Lelong - P. Dolbeault - H. Skoda (Analyse) 1983/84, Lecture Notes in Math. no **1198**, Springer-Verlag (1985) 88–97.

Demailly, J.P. [1985a] — *Mesures de Monge-Ampère et caractérisation géométrique des variétés algébriques affines*, Mém. Soc. Math. France (N.S.) **19** (1985) 1–124.

Demailly, J.P. [1985b] — *Sur les théorèmes d'annulation et de finitude de T. Ohsawa et O. Abdelkader*, Séminaire P. Lelong - P. Dolbeault - H. Skoda (Analyse) 1985/86, Lecture Notes in Math. no **1295**, Springer-Verlag (1987) 48–58.

Demailly, J.P. [1985c] — *Une preuve simple de la conjecture de Grauert-Riemenschneider*, Séminaire P. Lelong - P. Dolbeault - H. Skoda (Analyse) 1985/86, Lecture Notes in Math. no **1295**, Springer-Verlag (1987) 24–47.

Demailly, J.P. [1985d] — *Champs magnétiques et inégalités de Morse pour la* d''*-cohomologie*, Ann. Inst. Fourier (Grenoble) **35** (1985) 189–229.

Demailly, J.P. [**1987a**] — *Nombres de Lelong généralisés*, théorèmes d'intégralité et d'analyticité, Acta Math. **159** (1987) 153–169.

Demailly, J.P. [**1987b**] — *Mesures de Monge-Ampère et mesures pluriharmoniques*, Math. Zeit. **194** (1987) 519–564.

Demailly, J.P. [**1988a**] — *Vanishing theorems for tensor powers of a positive vector bundle*, Proceedings of the Conference "Geometry and Analysis on Manifolds" held in Katata-Kyoto (august 1987), edited by T. Sunada, Lecture Notes in Math. no **1339**, Springer-Verlag (1988).

Demailly, J.P. [**1988b**] — *Vanishing theorems for tensor powers of an ample vector bundle*, Invent. Math. **91** (1988) 203–220.

Demailly, J.P. [**2019**] — *Differential and pseudodifferential operators on manifolds*, Lecture Note, available at https://www-fourier.univ-grenoble-alpes.fr/~demailly/analytic_geometry_2019/pseudodiff.pdf.

Demailly, J.P., Skoda, H. [**1979**] — *Relations entre les notions de positivité de P.A. Griffiths et de S. Nakano*, Séminaire P. Lelong-H. Skoda (Analyse), année 1978/79, Lecture notes in Math. no **822**, Springer-Verlag, Berlin (1980) 304–309.

Demazure, M. [**1976**] — *A very simple proof of Bott's theorem*, Invent. Math. **33** (1976) 271–272.

De Rham, G. [**1931**] — *Sur l'Analysis situs des variétés à n dimensions* (*Thèse*), J. Math. Pures Appl. **10** (1931) 115–200.

De Rham, G. [**1952**] — *Sur la réductibilité d'un espace de Riemann*, Comment. Math. Helv. **26** (1952) 328–344.

De Rham, G. [**1955**] — *Variétés différentiables*, Hermann, Paris (1955).

Diederich, K., Fornaess, J.E. [**1977**] — *Pseudoconvex domains: bounded strictly plurisubharmonic functions*, Invent. Math. **39** (1977) 129–141.

Diederich, K., Pflug, P. [**1981**] — *Über Gebiete mit vollständiger Kählermetrik*, Math. Ann. **257** (1981) 191–198.

Dolbeault, P. [**1953**] — *Sur la cohomologie des variétés analytiques complexes*, C. R. Acad. Sc. Paris **236** (1953) 175–177.

Draper, R.N. [**1969**] — *Intersection theory in analytic geometry*, Math. Ann. **180** (1969) 175–204.

EGA, Grothendieck, G., Dieudonné, J.A. [**1960-67**] — *Elements de géométrie algébrique I*, II, III, IV, Publ. Math. I.H.E.S. no **4, 8, 11, 17, 20, 24, 28, 32** (1960–1967).

El Mir, H. [**1984**] — *Sur le prolongement des courants positifs fermés*, Acta Math. **153** (1984) 1–45.

Esnault, H., Viehweg, E. [**1983**] — *Sur une minoration du degré d'hypersurfaces s'annulant en certains points*, Math. Ann. **263** (1983) 75–86.

Farkas, H.M., Kra, I. [**1980**] — *Riemann surfaces*, Graduate Texts in Math. **71**, Springer-Verlag, Berlin (1980).

Federer, H. [**1969**] — *Geometric measure theory*, Springer Verlag, Grundlehren der math. Wissenschaften, Band **153**, Berlin (1969).

Fornaess, J.E. [**1976**] — *An increasing sequence of Stein manifolds whose limit is not Stein*, Math. Ann. **223** (1976), no. 3, 275–277.

Frölicher, A. [**1955**] — *Relations between the cohomology groups of Dolbeault and topological invariants*, Proc. Nat. Acad. Sci. U.S.A. **41** (1955) 641–644.

Fujiki, A. [**1978**] — *Closedness of the Douady spaces of compact Kähler spaces*, Publ. R.I.M.S., Kyoto Univ. **14** (1978) 1–52.

Girbau, J. [**1976**] — *Sur le théorème de Le Potier d'annulation de la cohomologie*, C. R. Acad. Sci. Paris, série A **283** (1976) 355–358.

Godement, R. [**1958**] — *Théorie des faisceaux*, Hermann, Paris (1958).

Grauert, H. [**1958**] — *On Levi's problem and the embedding of real-analytic manifolds*, Ann. Math. **68** (1958) 460–472.

Grauert, H. [**1960**] — *Ein Theorem der analytischen Garbentheorie und die Modulräume komplexer Strukturen*, Publ. Math. I.H.E.S. **5** (1960) 233–292.

Grauert, H. [**1962**] — *Über Modifikationen und exzeptionelle analytische Mengen*, Math. Ann. **146** (1962) 331–368.

Grauert, H., Remmert, R. [**1984**] — *Coherent analytic sheaves*, Grundlehren der math. Wissenschaften **265**, Springer-Verlag, Berlin (1984).

Grauert, H., Riemenschneider, O. [**1970**] — *Kählersche Mannigfaltigkeiten mit hyper-q-konvexem Rand*, Problems in Analysis: a Symposium in honor of Salomon Bochner, Princeton, N.J., Princeton University Press 1970.

Grauert, H., Riemenschneider, O. [**1971**] — *Verschwindungssätze für analytische Kohomologiegruppen auf komplexen Räumen*, Invent. Math. **11** (1970) 263–292.

Griffiths, P.A. [**1966**] — *The extension problem in complex analysis II; embeddings with positive normal bundle*, Amer. J. Math. **88** (1966) 366–446.

Griffiths, P.A. [**1969**] — *Hermitian differential geometry*, Chern classes and positive vector bundles, Global Analysis, papers in honor of K. Kodaira, Princeton Univ. Press, Princeton (1969) 181–251.

Griffiths, P.A., Harris, J. [**1978**] — *Principles of algebraic geometry*, Wiley, New York (1978).

Grothendieck, A. [**1957**] — *Sur quelques points d'algèbre homologique*, Tôhoku Math. J. **9** (1957) 119–221.

Grothendieck, A. [**1958**] — *La théorie des classes de Chern*, Bull. Soc. Math. France **86** (1958) 137–154.

Grothendieck, A. [**1966**] — *On the De Rham cohomology of algebraic varieties*, Publ. Math. I.H.E.S. **29** (1966) 351–359.

Grothendieck, A. [**1968**] — *Standard conjectures on algebraic cycles*, Bombay Colloquium on Algebraic Geometry, 1968, Oxford Univ. Press (1969).

Greene, R.E., Wu, H.H. [**1975**] — *Embedding of open riemannian manifolds by harmonic functions*, Ann. Inst. Fourier (Grenoble) **25** (1975) 215–235.

Gunning, R.C., Rossi, H. [**1965**] — *Analytic functions of several complex variables*, Prentice-Hall, Englewood Cliffs, N.J. (1965).

Hartshorne, R. [**1966**] — *Ample vector bundles*, Publ. Math. I.H.E.S. **29** (1966) 319–350.

Hartshorne, R. [**1970**] — *Ample subvarieties of algebraic varieties*, Lecture Notes in Math. no **156**, Springer-Verlag, Berlin (1970).

Hartshorne, R. [**1977**] — *Algebraic geometry*, Springer-Verlag, Berlin (1977).

Harvey, R. [**1975**] — *Holomorphic chains and their boundaries*, Proc. of Symposia in Pure Math., Several Complex Variables, held in Williamstown in 1975, vol. XXX, part I, ed. by R.O. Wells (1977) 309–382.

Hironaka, H. [**1964**] — *Resolution of singularities of an algebraic variety over a field of characteristic zero*, Ann. of Math. **79** (1964) 109–326.

Hirzebruch, F. [**1966**] — *Topological methods in algebraic geometry*, Springer-Verlag, Berlin (1966).

Hodge, W.V.D. [**1941**] — *The theory and applications of harmonic integrals*, Cambridge University Press (1941).

Hörmander, L. [**1963**] — *Linear partial differential operators Grundlehren der math. Wissenschaften*, Band **116**, Springer-Verlag, Berlin (1963).

Hörmander, L. [**1965**] — *L^2 estimates and existence theorems for the $\overline{\partial}$ operator*, Acta Math. **113** (1965) 89–152.

Hörmander, L. [**1966**] — *An introduction to Complex Analysis in several variables*, 1st edition, Elsevier Science Pub., New York, 1966, 3rd revised edition, North-Holland Math. library, Vol 7, Amsterdam (1990).

Hopf, H., Rinow, W. [**1931**] — *Über den Begriff der vollständigen differentialgeometrischen Fläche*, Comment. Math. Helv. **3** (1931) 209–255.

Jennane, B. [**1976**] — *Extension d'une fonction holomorphe définie sur une sous-variété avec contrôle de la croissance*, Séminaire P. Lelong - H. Skoda (Analyse) 1976/77, Lecture Notes in Math. no **694**, Springer-Verlag, Berlin (1978) 126–133.

Kawamata, Y. [**1981**] — *Characterization of abelian varieties*, Compositio Math. **43** (1981) 253–276.

Kawamata, Y. [**1982**] — *A generalization of Kodaira-Ramanujam's vanishing theorem*, Math. Ann. **261** (1982) 43–46.

Kiehl, R., Verdier, J.L. [**1971**] — *Ein einfacher Beweis des Kohärenzsatzes von Grauert*, Math. Ann. **195** (1971) 24–50.

King, J.R. [**1970**] — *A residue formula for complex subvarieties*, Proc. Carolina Conf. on Holomorphic mappings and minimal surfaces, University of North Carolina, Chapel Hill (1970) 43–56.

Kiselman, C.O. [**1978**] — *The partial Legendre transformation for plurisubharmonic functions*, Invent. Math. **49** (1978) 137–148.

Kiselman, C.O. [**1979**] — *Densité des fonctions plurisousharmoniques*, Bull. Soc. Math. France **107** (1979) 295–304.

Kiselman, C.O. [**1983**] — *Sur la définition de l'opérateur de Monge-Ampère complexe*, Analyse Complexe, Proceedings of Journées Fermat (SMF), Toulouse, May 1983, Lecture Notes in Math., vol. 1094, Springer-Verlag, Berlin (1984) 139—150.

Kiselman, C.O. [**1986**] — *Un nombre de Lelong raffiné*, Preprint Uppsala 1986, Sém. d'Analyse Complexe et Géométrie 1985–87, Faculté des Sciences de Monastir, Tunisie (1987).

Kleiman, S. [1966] — *Toward a numerical theory of ampleness*, Ann. of Math. **84** (1966) 293–344.

Kobayashi, S., Nomizu, K. [1963] — *Foundations of differential geometry*, Vol. I, II, Interscience Publishers, John Wiley & Sons, New York (1963).

Kodaira, K. [1953a] — *On cohomology groups of compact analytic varieties with coefficients in some analytic faisceaux*, Proc. Nat. Acad. Sci. U.S.A. **39** (1953) 868–872.

Kodaira, K. [1953b] — *On a differential geometric method in the theory of analytic stacks*, Proc. Nat. Acad. Sci. U.S.A. **39** (1953) 1268–1273.

Kodaira, K. [1954] — *On Kähler varieties of restricted type*, Ann. of Math. **60** (1954) 28–48.

Kodaira, K. [1975] — *Collected works*, Vol. I, II, III, Iwanami Shoten Publishers and Princeton University Press (1975).

Kohn, J.J. [1963] — *Harmonic integrals on strongly pseudo-convex manifolds I*, Ann. Math. **78** (1963) 206–213.

Kohn, J.J. [1964] — *Harmonic integrals on strongly pseudo-convex manifolds II*, Ann. Math. **79** (1964) 450–472.

Lang, S. [1965] — *Algebra*, Addison-Wesley (1965).

Lelong, P. [1942] — *Définition des fonctions plurisousharmoniques*, C. R. Acad. Sci. Paris **215** (1942) p. 398 and p. 454.

Lelong, P. [1957] — *Intégration sur un ensemble analytique complexe*, Bull. Soc. Math. France **85** (1957) 239–262.

Lelong P. [1967] — *Fonctionnelles analytiques et fonctions entières* (n *variables*), Sém. de Math. Supérieures, 6e session, été 1967, Presses Univ. Montréal (1968).

Lelong, P. [1968] — *Fonctions plurisousharmoniques et formes différentielles positives*, Dunod, Paris, Gordon & Breach, New York (1968).

Le Potier, J. [1975] — *Annulation de la cohomologie à valeurs dans un fibré vectoriel holomorphe de rang quelconque*, Math. Ann. **218** (1975) 35–53.

Le Potier, J. [1977] — *Cohomologie de la grassmannienne à valeurs dans les puissances extérieures et symétriques du fibré universel*, Math. Ann. **226** (1977) 257–270.

Leray, J. [1950] — *L'anneau spectral et l'anneau filtré d'homologie d'un espace localement compact et d'une application continue*, J. Math. Pures Appl. **29** (1950) 1–139.

Levi, E.E. [1910] — *Studii sui punti singolari essenziali delli funzioni analitiche di due o più variabili complesse*, Ann. Mat. Pura Appl. (3), **17** (1910) 61–87.

Malgrange, B. [1955] — *Existence et approximation des solutions des équations aux dérivées partielles et des équations de convolution*, Ann. Inst. Fourier (Grenoble) **6** (1955–1956) 271–355.

Moishezon, B.G. [1967] — *On n-dimensional compact varieties with n algebraically independent meromorphic functions*, Amer. Math. Soc. Translations **63** (1967) 51–177.

Nakai, Y. [1963] — *A criterion of an ample sheaf on a projective scheme*, Amer. J. Math. **85** (1963) 14–26.

Nakano, S. [1955] — *On complex analytic vector bundles*, J. Math. Soc. Japan **7** (1955) 1–12.

Nakano, S. [**1973**] — *Vanishing theorems for weakly* 1*-complete manifolds*, Number Theory, Algebraic Geometry and Commutative Algebra, in honor of Y. Akizuki, Kinokuniya, Tokyo (1973) 169–179.

Nakano, S. [**1974**] — *Vanishing theorems for weakly* 1*-complete manifolds II*, Publ. R.I.M.S., Kyoto Univ. **10** (1974) 101–110.

Narasimhan, R. [**1966**] — *Introduction to the theory of analytic spaces*, Lecture Notes in Math. no **25**, Springer-Verlag (1966).

Narasimhan, R. [**1971**] — *Analysis on real and complex manifolds*, Masson, New York (1971).

Newlander, A., Nirenberg, N. [**1957**] — *Complex analytic coordinates in almost complex manifolds*, Ann. Math. (2), **65** (1957) 391–404.

Norguet, F. [**1954**] — *Sur les domaines d'holomorphie des fonctions uniformes de plusieurs variables complexes*, Bull. Soc. Math. France **82** (1954) 137–159.

Ohsawa, T. [**1979**] — *Finiteness theorems on weakly* 1*-complete manifolds*, Publ. R.I.M.S., Kyoto Univ. **15** (1979) 853–870.

Ohsawa, T. [**1981**] — *On* $H^{p,q}(X,B)$ *of weakly* 1*-complete manifolds*, Publ. R.I.M.S., Kyoto Univ. **17** (1981) 113–126.

Ohsawa, T. [**1982**] — *A reduction theorem for cohomology groups of very strongly* q*-convex Kähler manifolds*, Invent. Math. **63** (1981) 335–354 ; **66** (1982) 391–393.

Ohsawa, T. [**1984**] — *Completeness of noncompact analytic spaces*, Publ. R.I.M.S., Kyoto Univ. **20** (1984) 683–692.

Oka, K. [**1942**] — *Sur les fonctions de plusieurs variables VI*, Domaines pseudoconvexes, Tôhoku Math. J. **49** (1942) 15–52.

Oka, K. [**1950**] — *Sur les fonctions de plusieurs variables VII*, Sur quelques notions arithmétiques, Bull. Soc. Math. France **78** (1950) 1–27.

Oka, K. [**1953**] — *Sur les fonctions de plusieurs variables IX*, Domaines finis sans point critique intérieur, Jap. J. Math. **23** (1953) 97–155.

Peternell, Th., Le Potier, J., Schneider, M. [**1987a**] — *Vanishing theorems*, linear and quadratic normality, Invent. Math. **87** (1987) 573–586.

Peternell, Th., Le Potier, J., Schneider, M. [**1987b**] — *Direct images of sheaves of differentials and the Atiyah class*, Math. Zeit. **196** (1987) 75–85.

Ramanujam, C.P. [**1974**] — *Remarks on the Kodaira vanishing theorem*, J. Indian Math. Soc. **36** (1972) 41–50 ; **38** (1974) 121–124.

Remmert, R. [**1956**] — *Projectionen analytischer Mengen*, Math. Ann. **130** (1956) 410–441.

Remmert, R. [**1957**] — *Holomorphe und meromorphe Abbildungen komplexer Räume*, Math. Ann. **133** (1957) 328–370.

Remmert, R., Stein, K. [**1953**] — *Über die wesentlichen Singularitäten analytischer Mengen*, Math. Ann. **126** (1953) 263–306.

Richberg, R. [**1968**] — *Stetige streng pseudokonvexe Funktionen*, Math. Ann. **175** (1968) 257–286.

Rothstein, W. [**1955**] — *Zur Theorie der Analytischen Mannigfaltigkeiten im Raume von n komplexen Veränderlichen*, Math. Ann. **129** (1955) 96–138.

Rudin, W. [**1966**] — *Real and complex analysis*, McGraw-Hill, New York (1966).

Schneider, M. [**1974**] — *Ein einfacher Beweis des Verschwindungssatzes für positive holomorphe Vektorraumbündel*, Manuscripta Math. **11** (1974) 95–101.

Schwartz, L. [**1953**] — *Homomorphismes et applications complètement continues*, C. R. Acad. Sci **236** (1953) 2472–2473.

Serre, J.P. [**1954**] — *Fonctions automorphes: quelques majorations dans le cas où X/G est compact*, Sém. Cartan (1953–54) 2-1 à 2-9.

Serre, J.P. [**1955a**] — *Un théorème de dualité*, Comment. Math. Helv. **29** (1955) 9–26.

Serre, J.P. [**1955b**] — *Faisceaux algébriques cohérents*, Ann. of Math. **61** (1955) 197–278.

Serre, J.P. [**1956**] — *Géométrie algébrique et géométrie analytique*, Ann. Inst. Fourier (Grenoble) **6** (1956) 1–42.

Shiffman, B., Sommese, A.J. [**1985**] — *Vanishing theorems on complex manifolds*, Progress in Math. no **56**, Birkhäuser (1985).

Sibony, N. [**1985**] — *Quelques problèmes de prolongement de courants en analyse complexe*, Duke Math. J. **52** (1985) 157–197.

Siegel, C.L. [**1955**] — *Meromorphic Funktionen auf kompakten Mannigfaltigkeiten*, Nachrichten der Akademie der Wissenschaften in Göttingen, Math.-Phys. Klasse **4** (1955) 71–77.

Siu, Y.T. [**1969**] — *Analytic sheaf cohomology groups of dimension n of n-dimensional complex spaces*, Trans. Amer. Math. Soc. **143** (1969) 77–94.

Siu, Y.T. [**1974**] — *Analyticity of sets associated to Lelong numbers and the extension of closed positive currents*, Invent. Math. **27** (1974) 53–156.

Siu, Y.T. [**1976**] — *Every Stein subvariety has a Stein neighborhood*, Invent. Math. **38** (1976) 89–100.

Siu, Y.T. [**1984**] — *A vanishing theorem for semi-positive line bundles over non-Kähler manifolds*, J. Differential Geom. **19** (1984) 431–452.

Siu, Y.T. [**1985**] — *Some recent results in complex manifold theory related to vanishing theorems for the semi-positive case*, Proceedings of the Bonn Arbeitstagung 1984, Lecture Notes in Math. no **1111**, Springer (1985) 169–192.

Siu, Y.T. [**1986**] — *Asymptotic Morse inequalities for analytic sheaf cohomology*, Sém. Bourbaki, exposé no **666**, juin 1986, 16 p.

Siu, Y.T. [**1990**] — *Calculus inequalities derived from holomorphic Morse inequalities*, Math. Ann. **286** (1990) 549–558.

Skoda, H. [**1972a**] — *Sous-ensembles analytiques d'ordre fini ou infini dans $\mathbb{C}^n$*, Bull. Soc. Math. France **100** (1972) 353–408.

Skoda, H. [**1972b**] — *Application des techniques L^2 à l'étude des idéaux d'une algèbre de fonctions holomorphes avec poids*, Ann. Sci. École Norm. Sup. **5** (1972) 545–579.

Skoda, H. [**1976**] — *Estimations L^2 pour l'opérateur $\overline{\partial}$ et applications arithmétiques*, Séminaire P. Lelong (Analyse), année 1975/76, Lecture Notes in Math. no **538**, Springer-Verlag, Berlin (1977) 314–323.

Skoda, H. [**1978**] — *Morphismes surjectifs de fibrés vectoriels semi-positifs*, Ann. Sci. École Norm. Sup. **11** (1978) 577–611.

Skoda, H. [**1982**] — *Prolongement des courants positifs fermés de masse finie*, Invent. Math. **66** (1982) 361–376.

Sommese, A.J. [**1978**] — *Submanifolds of abelian varieties*, Math. Ann. **233** (1978) 229–256.

Spanier, E.H. [**1966**] — *Algebraic topology*, McGraw-Hill, New York (1966).

Stein, K. [**1951**] — *Analytische Funktionen mehrerer komplexer Veränderlichen und das zweite Cousin'sche Problem*, Math. Ann. **123** (1951) 201–222.

Stoll, W. [**1966**] — *The multiplicity of a holomorphic map*, Invent. Math. **2** (1966) 15–58.

Thie, P. [**1967**] — *The Lelong number of a point of a complex analytic set*, Math. Ann. **172** (1967) 269–312.

Umemura, H. [**1973**] — *Some results in the theory of vector bundles*, Nagoya Math. J. **52** (1973) 97–128.

Viehweg, E. [**1982**] — *Vanishing theorems*, J. Reine Angew. Math. **335** (1982) 1–8.

Von Neumann, J. [**1929**] — *Zur Algebra der Funktionaloperationen und Theorie der normalen Operatoren*, Math. Ann. **102** (1929) 370-427.

Waldschmidt, M. [**1976**] — *Propriétés arithmétiques des fonctions de plusieurs variables II*, Séminaire P. Lelong (Analyse), année 1975/76, Lecture Notes in Math. no **538**, Springer-Verlag (1977) 108–135.

Waldschmidt, M. [**1978**] — *Nombres transcendants et groupes algébriques*, Astérisque no **69-70** (1979).

Warner, F. [**1971**] — *Foundations of differentiable manifolds and Lie groups*, Academic Press, New York (1971).

Weil, A. [**1952**] — *Sur les théorèmes de De Rham*, Comment. Math. Helv. **26** (1952) 119–145.

Weil, A. [**1957**] — *Variétés kählériennes*, Hermann, Paris (1957).

Wells, R.O. [**1980**] — *Differential analysis on complex manifolds*, Graduate Texts in Math. **65**, 2nd edition, Springer-Verlag, Berlin (1980).

Weyl, H. [**1913**] — *Die Idee der Riemannschen Fläche*, Teubner, Leipzig, (1913), 3. Auflage, 1955.

Witten, E. [**1982**] — *Supersymmetry and Morse theory*, J. Differential Geom. **17** (1982) 661–692.

Witten, E. [**1984**] — *Holomorphic Morse inequalities*, Teubner-Texte zur Math. **70**, Algebraic and Differential Topology, Ed. G.M. Rassias (1984) 318–333.